



Real Estate Investment Trusts (REITs): The Definitive Guide

This Real Estate Modeling course covers both **individual properties** and **real estate investment trusts (REITs)**. These two topics are related because you must know about individual properties if you want to understand REITs.

However, they are also different in terms of the analysis and financial modeling, and so we have separated them into two guides. This one covers **real estate investment trusts (REITs)**.

There are dozens of hours of video tutorials in the course, so it's impossible to cover everything in a single PDF, but this guide summarizes the key points:

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Equity REITs: The Crash-Course Overview

A **real estate investment trust (REIT)** is a company that buys, sells, develops, and operates **properties** or **other real estate assets**.

It's similar to a private equity firm, but for *real estate* rather than *normal companies*.



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Just as private equity firms raise capital from outside investors (Limited Partners) and then deploy it by acquiring, expanding, and improving companies, and eventually selling them, REITs also raise capital from outside investors and do the same thing with individual properties.

However, this description could match many types of companies – for example, real estate operating companies and real estate private equity firms operate similarly.

What sets REITs apart are the **special requirements** they must comply with and the **unique attributes** they gain as a result, which make their accounting, financial statements, and valuation different.

If a REIT complies with these requirements, **it will incur no corporate income taxes** – or at least, very low corporate income taxes.

The exact requirements differ by country, but here are the high-level rules that apply to REITs anywhere in the world:

Requirement #1:	Requirement #2:	Requirement #3:
REITs must distribute a high percentage of their Taxable Income as Dividends.	A high percentage of Revenue must be from real estate-related sources.	A high percentage of Assets must be real estate-related.
<i>Examples:</i>	<i>Examples:</i>	<i>Examples:</i>
The required percentage is 90% in the U.S., Hong Kong, and Germany, 95% in Brazil and Mexico, 75% in South Africa, and 100% in various other countries (as of the time this guide was written). This requirement ensures that REITs act as “pass-through entities” that distribute most of their earnings directly to the shareholders; it prevents normal companies from calling themselves “REITs” just to avoid taxes.	The required percentage is 75% in Canada, Germany, and South Africa; it’s 80% of earnings excluding capital gains in Spain, and it’s 75% of gross income in the U.S. and 75% of pre-tax profits in the U.K. This requirement also ensures that REITs are not standard companies evading taxes. If it’s an Equity REIT that acquires and develops properties, its Revenue will come from Rental Income; if it’s a Mortgage REIT, its Revenue will come from Net Interest Income.	In the U.S., U.K., and Hong Kong, 75% of the REIT’s Total Assets must be related to real estate; 70% must be RE-related in Japan, and 80% must be RE-related in India. This rule means that the company must be a REIT <i>on a long-term basis</i> rather than just an “overnight conversion.” It might take years or decades for a company to change its Assets to meet this requirement, which eliminates companies that are trying to game the system.



There may also be requirements related to leverage, total capital, foreign investments, legal structure, number of shareholders, the concentration of ownership (e.g., no fewer than 5 shareholders can own more than 50% of a REIT's stock in the U.S.), and so on.

These requirements are not important for financial modeling in most cases, so we do not go into them in detail in this guide. But if you want the exact numbers and rules in every region, you can find them with a quick Google search.

The U.S. government passed the legislation that created the first REITs in 1960 to allow “normal people” to invest in properties without having to pay for entire properties.

Before REITs existed, if you wanted to invest in apartments, you had to spend millions or tens of millions of your own cash to acquire multifamily properties.

But after REITs came into existence, you could pay \$100 to buy a multifamily REIT's shares and gain exposure to its property portfolio.

Since REITs must constantly pay out a high percentage of their Taxable Income, they cannot accumulate much Cash for acquisitions, renovations, and developments.

As a result, they need to raise Debt and Equity **constantly**, which means their financial statements look quite different from those of a normal company.

Also, since REITs frequently buy and sell properties (and under IFRS, revalue their existing properties), their Net Income figures fluctuate significantly because of all the Gains and Losses.

EBITDA is still meaningful because it excludes Gains and Losses, but Net Income and the accompanying P / E multiples don't produce consistent results due to these fluctuations.

As a result, REITs use alternative financial metrics and (some) alternative valuation methodologies.

Categories of REITs

You can classify REITs based on 3 main criteria:

1. **Structure:** Traditional vs. UPREIT vs. DownREIT.
2. **Strategy:** Equity vs. Mortgage vs. Hybrid.



3. **Sector:** Office vs. Industrial vs. Retail vs. Single-Family vs. Multifamily vs. Hotels vs. Storage vs. Data Center vs. Specialized.

Regarding **Structure**, “Traditional” REITs own their Assets directly. By contrast, UPREITs and DownREITs form “Operating Partnerships” that own and manage the properties.

With an UPREIT (“Umbrella Partnership REIT”), a property investor transfers ownership of his/her property to the Operating Partnership in exchange for a percentage ownership in that partnership.

The investor receives “Operating Partnership Units” or “OP Units” that represent this ownership, and these units have the same value as the REIT’s common shares – but they cannot be sold or otherwise cashed out.

With a DownREIT, the property investor forms a joint venture (JV) with the REIT and receives a type of “put option” that can be converted into REIT shares.

We are not going into these definitions in-depth because **they do not matter much for modeling purposes**. You just need to be aware that OP and DownREIT “Units” exist and factor them into the Equity Value and Enterprise Value calculations accordingly.

The second criterion – **Strategy** – is exactly what it sounds like. The majority of REITs are Equity REITs, meaning they acquire, develop, operate, and sell individual properties.

Mortgage REITs invest in real estate loans (and variants such as mortgage-backed securities) and are much less common than Equity REITs.

They are more similar to commercial banks than Equity REITs because they build and manage loan portfolios, earn revenue primarily from Net Interest Income, and fund their loan portfolios with Debt; metrics such as Book Value, P / BV, and P / E are also important for them.

Finally, Hybrid REITs invest in both real estate loans and properties and are even less common than Mortgage REITs.

The third criterion – **Sector** – matters because lease terms differ tremendously among different property types.

For example, the risk profile of an office REIT with primarily Fortune 500 tenants on 10-year leases is far different from that of a multifamily REIT with individual tenants on 1-year leases. And that risk profile, in turn, is far different from that of a hotel REIT with no long-term leases at all.



As a result, you project revenue and expenses differently and also use different Discount Rates and Cap Rates to value different types of REITs.

You'll see *similar* line items on the financial statements of all REITs, but some line items will differ because of the underlying property type.

For example, there is often an adjustment for the "Straight-Lining of Rent" on the statements of office, industrial, and retail REITs because accounting rules require REITs to recognize rental income based on the average annual rate over the lease term.

So, if the rent increases from \$10 to \$12 to \$14 to \$16 to \$18 to \$20 over 6 years, the REIT will record \$15 for rental income each year on its Income Statement.

In reality, however, the REIT collects only \$10 in Cash Rent in Year 1, so it needs to adjust for this difference on the Cash Flow Statement.

By contrast, you won't see this line item for multifamily REITs because the leases almost always last for 1 year or less, so this concept doesn't apply.

In this course, we cover Equity REITs in the multifamily, hotel, office, retail, industrial, and data center sectors, and the main case study is based on a multifamily REIT (AvalonBay).

If you want to learn more about Mortgage REITs, take a look at our Bank & Financial Institution Modeling course. Commercial banks are a close proxy, and the accounting and valuation are similar.

REIT Financial Statements

Each major line item on an Equity REIT's financial statements corresponds to the **major actions** it takes each year:

- Maintain or Raise Rents.
- Acquire Properties.
- Develop New Properties.
- Renovate or Redevelop Properties.
- Sell Properties.
- Raise Debt and Equity.
- Issue Dividends.

To illustrate, here is a quick overview of AvalonBay's financial statements, with the most important items highlighted and noted:

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ASSETS

Real estate:

Land and improvements	\$ 4,237,318
Buildings and improvements	15,708,666
Furniture, fixtures and equipment	615,288
	<u>20,561,272</u>
Less accumulated depreciation	<u>(4,218,379)</u>
Net operating real estate	16,342,893
Construction in progress, including land	1,306,300
Land held for development	68,364
Real estate assets held for sale, net	—
Total real estate, net	17,717,557

The most significant Asset for any REIT; U.S.-based ones record Depreciation, while IFRS-based REITs mark this item to Fair Market Value periodically and do not depreciate it at all.

Cash tends to be very low for REITs because of the Dividend requirements.

Cash and cash equivalents	67,088
Cash in escrow	134,818
Resident security deposits	32,686
Investments in unconsolidated real estate entities	163,475
Deferred development costs	45,819
Prepaid expenses and other assets	253,378
Total assets	\$ 18,414,821

Equity Investments or Associate Companies are very common for REITs because they often set up Joint Ventures with other parties to fund developments and acquisitions.

Debt is a major funding source for REITs, so it's always the biggest Liability.

LIABILITIES AND EQUITY

Unsecured notes, net	\$ 5,852,764
Variable rate unsecured credit facility	—
Mortgage notes payable, net	1,476,706
Dividends payable	196,094
Payables for construction	85,377
Accrued expenses and other liabilities	308,189
Accrued interest payable	43,116
Resident security deposits	58,473
Liabilities related to real estate assets held for sale	—
Total liabilities	8,020,719

Redeemable NCI relate to the "OP Units" the REIT has and the UPREIT or DownREIT structure.

Commitments and contingencies

Redeemable noncontrolling interests	6,056
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REITs raise significant Equity consistently, so this item is also very large; however, their Net Incomes are often relatively low or even negative, so APIC is often the largest component within Equity.

Equity:

Preferred stock, \$0.01 par value; \$25 liquidation shares issued and outstanding at December 31, 2019	—
Common stock, \$0.01 par value; 280,000,000 shares issued and outstanding at December 31, 2019	1,381
Additional paid-in capital	10,235,475
Accumulated earnings less dividends	188,609
Accumulated other comprehensive loss	(37,419)
Total equity	10,388,046

Total liabilities and equity	\$ 18,414,821
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Revenue:

Rental and other income	\$	2,154,481
Management, development and other fees		4,147
Total revenue		2,158,628

Most of this is from maintaining and raising Rents on existing properties.

Expenses:

Operating expenses, excluding property taxes	503,946
Property taxes	221,375
Interest expense, net	199,661
Loss (gain) on extinguishment of debt, net	25,472
Depreciation expense	584,150
General and administrative expense	50,673
Expensed acquisition, development and other pursuit costs, net of recoveries	2,736
Casualty and impairment loss (gain), net	6,250
Total expenses	1,594,263

These major expense line items also come primarily from existing properties.

Interest Expense is massive for most REITs because of their high Debt balances.

U.S.-based REITs record a high Depreciation expense; IFRS-based REITs record a Fair Value Gain or Loss and mark their properties to market value periodically.

Income before equity in income of unconsolidated real estate entities, gain on sale of other real estate transactions, and income taxes

564,365

Most Equity REITs engage in significant JV activity, so there's usually something for Equity Investment Income.

Equity in income of unconsolidated real estate entities	70,744
Gain on sale of communities	252,599
(Loss) gain on other real estate transactions	(10,907)

Gains and Losses are always significant for REITs; these are *Realized* Gains and Losses from selling properties.

Income before income taxes	876,801
Income tax expense	141

Technically, the Income Tax Expense should be 0, but it might be some very small number due to state/local taxes.

Net income	876,660
Net loss attributable to noncontrolling interests	261

Most REITs have Noncontrolling Interests due to majority ownership in JVs and the UPREIT or DownREIT structure.

Net income attributable to common stockholders	\$	876,921
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Cash flows from operating activities:

Net income	\$	876,660
Adjustments to reconcile net income to cash provided by operating activities:		
Depreciation expense		584,150
Amortization of deferred financing costs		7,657
Amortization of debt premium		(5,915)
Loss (gain) on extinguishment of debt, net		25,472
Amortization of stock-based compensation		17,920
Equity in (income) loss of, and return on, unconsolidated real estate entities and noncontrolling interests, net of eliminations		(19,798)
Casualty and impairment loss (gain), net		8,568
Abandonment of development pursuits		388
Cash flow hedge losses reclassified to earnings		7,070
Gain on sale of real estate assets		(281,745)
Decrease (increase) in resident security deposits, prepaid expenses and other assets		3,076
Increase in accrued expenses, other liabilities and accrued interest payable		32,754
Net cash provided by operating activities		1,256,257

Depreciation is always a huge add-back on the CFS for U.S.-based REITs.

And they reverse Gains and Losses that appeared on the Income Statement as well.

The major CFI line items are Developments / Redevelopments, Acquisitions, Proceeds from Sales, and CapEx for existing properties or other RE Asset ("Maintenance CapEx").

Cash flows from investing activities:

Development/redevelopment of real estate assets including land acquisitions and deferred development costs	(979,947)
Acquisition of real estate assets, including partnership interest	(462,317)
Capital expenditures - existing real estate assets	(65,181)
Capital expenditures - non-real estate assets	(8,809)
Proceeds from sale of real estate, net of selling costs	503,039
Insurance proceeds for property damage claims	16,233
Mortgage note receivable lending	(17,590)
(Decrease) increase in payables for construction	(15,621)
Distributions from unconsolidated real estate entities	89,305
Investments in unconsolidated real estate entities	(24,493)
Net cash used in investing activities	(965,381)

REITs are constantly issuing Equity and Debt, and sometimes repaying the Debt upon maturity (repurchasing Stock is extremely rare).

Cash flows from financing activities:

Issuance of common stock, net	111,093
Dividends paid	(772,657)
Issuance of mortgage notes payable	206,800
Repayments of mortgage notes payable, including prepayment penalties	(1,313,025)
Issuance of unsecured notes	1,696,826
Repayment of unsecured notes, including prepayment penalties	(300,000)
Payment of deferred financing costs	(17,552)
Redemption of noncontrolling interest and units for cash by minority partners	—
Payment of capital lease obligation	(18,951)
Receipts (payments) for termination of forward interest rate swaps	391
Payments related to tax withholding for share-based compensation	(10,450)
Distributions to DownREIT partnership unitholders	(42)
Contributions from joint venture and profit-sharing partners	1,038
Distributions to joint venture and profit-sharing partners	(418)
Preferred interest obligation redemption and dividends	(2,000)
Net cash (used in) provided by financing activities	(418,947)
Net decrease in cash and cash equivalents	(128,071)

And they're also paying out significant Dividends each year to maintain their status as REITs and avoid corporate income taxes.

Cash and cash equivalents and restricted cash, beginning of year

329,977

Cash and cash equivalents and restricted cash, end of year

\$ 201,906



Many of the other, smaller items on the financial statements above relate to “corporate overhead” and the expenses that the REIT needs to run its business, as well as fees on activities such as acquisitions, developments, and capital raises.

You tend to project these line items with simple methods, such as using percentage growth rates in the low single digits or holding them constant across all the years in the projection period.

REIT Key Metrics

When you analyze REITs, traditional metrics such as EBITDA and Revenue Growth still apply.

But since REITs have high Depreciation and Gains/Losses, they also use a few new metrics, such as Funds from Operations (FFO):

Funds from Operations (FFO) = Net Income + Real Estate-Related Depreciation & Amortization + Losses / (Gains) + Impairments.

FFO is intended to be an “improved version” of Net Income that removes large but non-recurring items, such as Gains and Losses, and large but non-cash items, such as Depreciation. It is *not* a replacement for Free Cash Flow or any variation of FCF.

You can use the same calculation for both U.S. GAAP-based REITs and IFRS-based REITs, but the numbers differ because IFRS-based REITs record Realized Gains and Losses and Unrealized Gains and Losses (due to the annual revaluation of properties) on their Income Statements – but they do not depreciate their investment properties at all.

So, IFRS-based REITs have almost nothing for D&A, but a significant adjustment for Gains and Losses, while U.S. GAAP-based REITs have a large D&A adjustment but a smaller one for Gains and Losses:

AvalonBay (U.S. GAAP-based REIT) – FFO Calculation:

Net income attributable to common stockholders	\$	1,034,002
Depreciation—real estate assets, including discontinued operations and joint venture adjustments		538,606
Distributions to noncontrolling interests, including discontinued operations		41
Gain on sale of unconsolidated entities holding previously depreciated real estate assets		(58,069)
Gain on sale of previously depreciated real estate assets (1)		(374,623)
Gain on acquisition of unconsolidated real estate entity		—
Casualty and impairment (recovery) loss, net on real estate (2) (6)		(4,195)
FFO attributable to common stockholders	\$	1,135,762

Westfield (Australian REIT) – FFO Calculation:

Profit after tax	1,366.1
Adjusted for:	
– Amortisation of leasing incentives and related leasing costs	54.9
– Mark to market of derivatives and preference shares	30.2
– Property revaluations	(1,005.0)
– (Gain)/loss in respect of capital transactions	(1.7)
– Deferred tax expense	255.9
FFO ^(a)	700.4



You often use the P / FFO multiple (i.e., Equity Value / Funds from Operations, or Share Price / FFO per Share) to value REITs via comparable public companies and precedent transactions.

Another metric is Adjusted Funds from Operations, or AFFO:

Adjusted Funds from Operations (AFFO) = FFO – Recurring Maintenance CapEx +/- Amortization of Lease Intangibles and Straight-Lining of Rent +/- Other Adjustments.

“Recurring Maintenance CapEx” should also include Leasing Commissions and Tenant Improvements if they are listed separately on the Cash Flow Statement (for REITs that have these items).

In theory, AFFO should better reflect the ongoing cash costs of operating a REIT’s property portfolio.

In practice, AFFO often proves confusing and arbitrary because different REITs define it differently and include different items in the calculation.

Financial metrics are best when they are **simple and easily defined** – if you need to add or subtract 27 different items to calculate a certain metric, the metric is not that useful.

Some REITs do not even list AFFO in their filings because of these problems – they know that their AFFO figures won’t be comparable to the AFFO figures of other REITs, so there’s not much point in disclosing them.

Many of the key metrics for individual properties, such as Net Operating Income (NOI), Cap Rates, and Occupancy Rates, also apply to REITs.

NOI should be fairly close to the REIT’s EBITDA since both metrics represent the same idea – cash flow from core-business operations on a capital structure-neutral basis – but NOI will be a bit higher because it does not include corporate overhead expenses.

You can calculate a REIT’s portfolio-wide Cap Rate by dividing its forward NOI by its Gross Real Estate Asset Value, or you can calculate the Cap Rate *implied* by the REIT’s share price.

To do that, you divide the REIT’s forward NOI by its (Equity Value – Fair Market Value of All Assets Except Gross Real Estate Assets + Fair Market Value of All Liabilities).

If the Implied Cap Rate is higher than the REIT’s portfolio-wide Cap Rate, the REIT may be undervalued, and vice versa if it’s lower.

Dividends, Dividend Yields (Dividends per Share / Share Price), and Payout Ratios (Dividends / Net Income) are also important for evaluating REITs, and you'll see variations such as "Distributable Income" used in regions such as Singapore.

If you're analyzing an office, retail, or industrial REIT, you can look at the individual leases and tenants to determine the lease-expiration risk as well.

You can use the disclosures in the REIT's filings to calculate the **Weighted Average Lease Expiration**, or WALE, which equals % Rental Income or Rentable Area * Remaining Term in Lease, Across All Leases:

Lease Expirations (1)(2)

Year of Lease Expiration	Rentable Square Feet Subject to Expiring Leases	Current Annualized Contractual Rent Under Expiring Leases Without Future Step-Ups (3)	Current Annualized Contractual Rent Under Expiring Leases Without Future Step-Ups p.s.f. (3)	Current Annualized Contractual Rent Under Expiring Leases With Future Step-Ups (4)	Current Annualized Contractual Rent Under Expiring Leases With Future Step-Ups p.s.f. (4)	Percentage of Total Square Feet
2016 (5)	115,331	\$5,755,938	\$49.91	\$5,755,938	\$49.91	0.3%
2017	2,328,197	151,100,135	64.90	152,513,741	65.51	5.7%
2018	1,541,680	103,841,839	67.36	105,931,206	68.71	3.8%
2019	3,524,261	186,301,876	52.86	190,824,771	54.15	8.7%
2020	4,454,917	284,094,049	63.77	293,865,720	65.96	11.0%
2021	3,820,575	213,053,299	55.76	228,417,659	59.79	9.4%
2022	4,244,368	246,333,201	58.04	272,361,921	64.17	10.5%
2023	1,643,788	95,716,163	58.23	109,314,767	66.50	4.1%
2024	2,766,152	165,609,504	59.87	183,522,732	66.35	6.8%
2025	2,608,773	150,380,359	57.64	172,691,441	66.20	6.4%
Thereafter	11,267,301	797,434,531	70.77	1,029,945,825	91.41	27.7%

Lower WALE values mean there is more lease-expiration risk, but also more potential upside if market rents rise rapidly since the REIT can take advantage of this trend to increase its rates.

Credit stats and ratios such as Debt / EBITDA, EBITDA / Interest, and Debt / Total Assets are also important because most REITs are highly leveraged and issue new Debt and Equity frequently.

To make financing decisions, REITs forecast these metrics in different market and operational scenarios and determine their likelihood of compliance with the **covenants** on the Debt (requirements such as always staying above 3x EBITDA / Interest).

These credit stats and ratios are also important because they may tie into REITs' requirements in some countries.



For example, in Singapore, the maximum “Gearing Ratio” (Debt / Total Assets, with Gross Real Estate Assets marked to market value) is 45%, and many REITs maintain ratios in the 30-35% range.

So, even if you’re not considering a Debt vs. Equity financing decision, specifically, it may still be important to evaluate these metrics.

REIT Valuation

You still use Public Comps, Precedent Transactions, and the DCF to value Equity REITs, but the setup, assumptions, and key metrics and multiples are a bit different.

You can also use a Dividend Discount Model (DDM), as well as a new methodology specific to REITs called the **Net Asset Value (NAV) Model** (which applies best to U.S.-based REITs).

With Public Comps and Precedent Transactions, you still screen based on geography, industry, and financial size, with “time” as an additional criterion for Precedent Transactions. It’s just that the screening metrics are a bit different – here are a few examples:

- **Public Comps:** U.S.-Based Multifamily REITs with Gross Real Estate Assets Over \$1 Billion
- **Public Comps:** European Office REITs with Fair Value of Investment Properties Over €5 Billion
- **Precedent Transactions:** Canadian Industrial REITs Acquired for Equity Purchase Prices Over CAD\$ 500 Million in the Past Three Years
- **Precedent Transactions:** Japanese Retail REITs Acquired for Equity Purchase Prices Over JPY ¥1 Billion in the Past Five Years

Regardless of the accounting system used by the REITs in your set, you can use metrics such as Equity Value, Enterprise Value, NOI, Cap Rates, Real Estate Assets, Implied Cap Rates, and EV / EBITDA.

For REITs following U.S. GAAP, you will use FFO and AFFO as key metrics and the corresponding P / FFO and P / AFFO as the key multiples.

For REITs following IFRS, you will use Book Value (BV) and P / BV more often since properties are marked to market value.

You’ll sometimes see FFO and AFFO even for IFRS-based REITs, but as you saw above, the calculations are a bit different, and the reversal of the Changes in Fair Value is a major component.



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There are other variations as well, such as “EPRA Earnings,” (EPRA = “European Public Real Estate Association”) which is Net Income adjusted for Changes in Fair Value, Realized Gains and Losses, Deferred Taxes, and other, smaller items. It’s similar to FFO but more specific to IFRS-based REITs.

Here are two examples of Public Comps for REITs, starting with Park Hotels in the U.S.:

Comparable Companies - Independent, U.S.-Based Hotel REITs with Gross Real Estate Assets Above \$3 Billion

(\$ USD in Millions Except Per Share and Per Unit Amounts in USD as Stated)

Operating Statistics: Company Name	Capitalization		Gross Real Estate Assets	Portfolio Cap Rate	Leverage (Debt to Total Assets)	EBITDA			Funds from Operations (FFO)			Projected EBITDA Growth	Projected FFO Growth
	Equity Value	Enterprise Value				FY16	FY17	FY18	FY16	FY17	FY18		
Host Hotels & Resorts, Inc.	\$ 13,743	\$ 17,224	\$ 17,993	8.8%	32.0%	\$ 1,424	\$ 1,470	\$ 1,455	\$ 1,257	\$ 1,279	\$ 1,251	(1.0%)	(2.2%)
Apple Hospitality REIT, Inc.	4,561	5,895	5,381	9.2%	26.9%	375	436	443	298	366	367	1.5%	0.3%
Ashford Hospitality Trust, Inc.	795	4,303	5,055	9.6%	76.1%	410	414	420	130	137	143	1.5%	4.6%
LaSalle Hotel Properties	3,548	4,582	5,049	7.5%	29.6%	389	346	328	281	260	249	(5.3%)	(4.0%)
RLJ Lodging Trust	2,994	4,138	4,324	10.4%	39.6%	383	419	575	318	352	489	37.2%	38.8%
Sunstone Hotel Investors, Inc.	3,339	4,155	4,257	8.6%	25.3%	336	334	342	259	248	257	2.3%	3.7%
DiamondRock Hospitality Company	2,366	3,044	3,382	8.1%	30.0%	253	246	253	203	195	203	2.8%	4.1%
Ryman Hospitality Properties, Inc.	3,191	4,634	3,319	11.5%	62.5%	319	356	378	268	301	321	6.2%	6.8%
Xenia Hotels & Resorts, Inc.	2,019	2,897	3,064	9.7%	37.7%	276	263	285	218	195	214	8.4%	9.9%
Pebblebrook Hotel Trust	2,261	3,227	3,031	8.5%	35.5%	252	231	236	191	173	178	2.3%	3.0%
Maximum	\$ 13,743	\$ 17,224	\$ 17,993	11.5%	76.1%	\$ 1,424	\$ 1,470	\$ 1,455	\$ 1,257	\$ 1,279	\$ 1,251	37.2%	38.8%
75th Percentile	3,495	4,621	5,053	9.6%	39.1%	387	418	437	294	339	356	5.3%	6.3%
Median	\$ 3,092	\$ 4,229	\$ 4,291	9.0%	33.7%	\$ 355	\$ 351	\$ 360	\$ 263	\$ 254	\$ 253	2.3%	3.9%
25th Percentile	2,287	3,455	3,335	8.5%	29.7%	287	281	296	207	195	205	1.5%	1.0%
Minimum	795	2,897	3,031	7.5%	25.3%	252	231	236	130	137	143	(5.3%)	(4.0%)
Park Hotels & Resorts Inc.	\$ 5,487	\$ 8,367	\$ 10,413	8.0%	32.8%	\$ 733	\$ 750	\$ 763	\$ 482	\$ 659	\$ 675	1.8%	2.3%

Valuation Statistics: Company Name	Capitalization		Gross Real Estate Assets	Portfolio Cap Rate	Leverage (Debt to Total Assets)	Enterprise Value / EBITDA			Equity Value / FFO		
	Equity Value	Enterprise Value				FY16	FY17	FY18	FY16	FY17	FY18
Host Hotels & Resorts, Inc.	\$ 13,743	\$ 17,224	\$ 17,993	8.8%	32.0%	12.1 x	11.7 x	11.8 x	10.9 x	10.7 x	11.0 x
Apple Hospitality REIT, Inc.	4,561	5,895	5,381	9.2%	26.9%	15.7 x	13.5 x	13.3 x	15.3 x	12.4 x	12.4 x
Ashford Hospitality Trust, Inc.	795	4,303	5,055	9.6%	76.1%	10.5 x	10.4 x	10.2 x	6.1 x	5.8 x	5.6 x
LaSalle Hotel Properties	3,548	4,582	5,049	7.5%	29.6%	11.8 x	13.2 x	14.0 x	12.6 x	13.7 x	14.2 x
RLJ Lodging Trust	2,994	4,138	4,324	10.4%	39.6%	10.8 x	9.9 x	7.2 x	9.4 x	8.5 x	6.1 x
Sunstone Hotel Investors, Inc.	3,339	4,155	4,257	8.6%	25.3%	12.4 x	12.4 x	12.1 x	12.9 x	13.5 x	13.0 x
DiamondRock Hospitality Company	2,366	3,044	3,382	8.1%	30.0%	12.1 x	12.4 x	12.0 x	11.7 x	12.2 x	11.7 x
Ryman Hospitality Properties, Inc.	3,191	4,634	3,319	11.5%	62.5%	14.5 x	13.0 x	12.3 x	11.9 x	10.6 x	9.9 x
Xenia Hotels & Resorts, Inc.	2,019	2,897	3,064	9.7%	37.7%	10.5 x	11.0 x	10.2 x	9.3 x	10.4 x	9.4 x
Pebblebrook Hotel Trust	2,261	3,227	3,031	8.5%	35.5%	12.8 x	14.0 x	13.7 x	11.9 x	13.1 x	12.7 x
Maximum	\$ 13,743	\$ 17,224	\$ 17,993	11.5%	76.1%	15.7 x	14.0 x	14.0 x	15.3 x	13.7 x	14.2 x
75th Percentile	3,495	4,621	5,053	9.6%	39.1%	12.7 x	13.2 x	13.1 x	12.4 x	12.9 x	12.6 x
Median	\$ 3,092	\$ 4,229	\$ 4,291	9.0%	33.7%	12.1 x	12.4 x	12.1 x	11.8 x	11.4 x	11.3 x
25th Percentile	2,287	3,455	3,335	8.5%	29.7%	11.1 x	11.2 x	10.6 x	9.8 x	10.4 x	9.6 x
Minimum	795	2,897	3,031	7.5%	25.3%	10.5 x	9.9 x	7.2 x	6.1 x	5.8 x	5.6 x
Park Hotels & Resorts Inc.	\$ 5,487	\$ 8,367	\$ 10,413	8.0%	32.8%	11.4 x	11.2 x	11.0 x	11.4 x	8.3 x	8.1 x

Based on these numbers, Park Hotels seems appropriately valued. Its EV / EBITDA and P / FFO multiples are below the set medians, but so are its Projected EBITDA and FFO Growth.

Here's the European example for Inmobiliaria Colonial SOCIMI, S.A. ("Colonial"), which operates in Spain and France:

Comparable Companies - European Office, Industrial, and Retail REITs with Fair Value of Investment Properties Between €3 and €15 Billion
(€ EUR in Millions Except Per Share and Per Unit Amounts in EUR as Stated)

Operating Statistics:		Capitalisation		Total Assets	Fair Value of Investment Properties	Leverage (Debt to Total Assets)	EBITDA			Book Value FY16	Projected EBITDA Growth
Company Name		Equity Value	Enterprise Value				FY16	FY17	FY18		
intu properties plc	€	4,286	€ 9,963	€ 12,128	€ 10,800	48.6%	€ 434	€ 459	€ 460	€ 5,823	0.3%
British Land Company Plc		7,065	11,555	15,751	10,722	27.5%	582	572	564	10,785	(1.4%)
Derwent London Plc		3,226	4,543	5,912	5,666	21.3%	141	158	159	4,599	0.3%
Société Foncière Lyonnaise		2,236	4,596	5,764	5,632	34.9%	171	177	180	3,123	1.8%
Hammerson plc		5,088	9,229	11,452	5,622	36.1%	392	378	380	6,755	0.6%
SEGRO Plc		4,512	6,384	7,145	5,514	26.9%	199	271	286	4,892	5.7%
Altarea Cogedim		2,554	6,358	7,081	4,256	44.6%	214	321	375	1,621	16.8%
Beni Stabili SpA SIIQ		1,202	3,467	4,240	3,995	54.1%	138	141	140	1,863	(1.2%)
Wereldhave N.V.		1,604	3,312	3,948	3,804	39.7%	179	182	185	1,979	1.6%
Shaftesbury PLC		2,860	3,856	3,983	3,764	25.5%	82	89	103	2,888	16.2%
Eurocommercial Properties N.V.		1,658	3,219	3,619	3,461	43.9%	150	155	157	1,832	1.1%
alstria office REIT-AG		1,748	2,987	3,383	3,006	43.9%	155	152	157	1,728	3.4%
Maximum	€	7,065	€ 11,555	€ 15,751	€ 10,800	54.1%	€ 582	€ 572	€ 564	€ 10,785	16.8%
75th Percentile		4,343	7,096	8,222	5,640	44.1%	259	335	376	5,124	4.0%
Median	€	2,707	€ 4,569	€ 5,838	€ 4,885	37.9%	€ 175	€ 179	€ 182	€ 3,005	1.3%
25th Percentile		1,726	3,428	3,974	3,794	27.3%	148	154	157	1,855	0.3%
Minimum		1,202	2,987	3,383	3,006	21.3%	82	89	103	1,621	(1.4%)
Inmobiliaria Colonial SOCIMI, S.A.	€	2,256	€ 7,553	€ 8,228	€ 7,770	44.9%	€ 216	€ 232	€ 242	€ 2,302	4.4%

Valuation Statistics:		Capitalisation		Total Assets	Fair Value of Investment Properties	Leverage (Debt to Total Assets)	Enterprise Value / EBITDA			P / BV FY16
Company Name		Equity Value	Enterprise Value				FY16	FY17	FY18	
intu properties plc	€	4,286	€ 9,963	€ 12,128	€ 10,800	48.6%	23.0 x	21.7 x	21.7 x	0.7 x
British Land Company Plc		7,065	11,555	15,751	10,722	27.5%	19.9 x	20.2 x	20.5 x	0.7 x
Derwent London Plc		3,226	4,543	5,912	5,666	21.3%	32.1 x	28.7 x	28.6 x	0.7 x
Société Foncière Lyonnaise		2,236	4,596	5,764	5,632	34.9%	27.0 x	26.0 x	25.6 x	0.7 x
Hammerson plc		5,088	9,229	11,452	5,622	36.1%	23.5 x	24.4 x	24.3 x	0.8 x
SEGRO Plc		4,512	6,384	7,145	5,514	26.9%	32.1 x	23.6 x	22.3 x	0.9 x
Altarea Cogedim		2,554	6,358	7,081	4,256	44.6%	29.7 x	19.8 x	17.0 x	1.6 x
Beni Stabili SpA SIIQ		1,202	3,467	4,240	3,995	54.1%	25.1 x	24.6 x	24.8 x	0.6 x
Wereldhave N.V.		1,604	3,312	3,948	3,804	39.7%	18.5 x	18.2 x	17.9 x	0.8 x
Shaftesbury PLC		2,860	3,856	3,983	3,764	25.5%	46.8 x	43.5 x	37.4 x	1.0 x
Eurocommercial Properties N.V.		1,658	3,219	3,619	3,461	43.9%	21.4 x	20.8 x	20.5 x	0.9 x
alstria office REIT-AG		1,748	2,987	3,383	3,006	43.9%	19.3 x	19.7 x	19.0 x	1.0 x
Maximum	€	7,065	€ 11,555	€ 15,751	€ 10,800	54.1%	46.8 x	43.5 x	37.4 x	1.6 x
75th Percentile		4,343	7,096	8,222	5,640	44.1%	30.3 x	24.9 x	25.0 x	0.9 x
Median	€	2,707	€ 4,569	€ 5,838	€ 4,885	37.9%	24.3 x	22.6 x	22.0 x	0.8 x
25th Percentile		1,726	3,428	3,974	3,794	27.3%	21.0 x	20.1 x	20.1 x	0.7 x
Minimum		1,202	2,987	3,383	3,006	21.3%	18.5 x	18.2 x	17.0 x	0.6 x
Inmobiliaria Colonial SOCIMI, S.A.	€	2,256	€ 7,553	€ 8,228	€ 7,770	44.9%	35.0 x	32.6 x	31.2 x	1.0 x

In this case, Colonial may be slightly overvalued because its EBITDA growth is near the 75th percentile of the set, but it trades *above* the 75th percentile multiples.

In addition to the Public Comps and Precedent Transactions, you can also use a DCF to value REITs. The setup is similar to a standard Unlevered DCF, but there are a few differences:

- 1) Ongoing CapEx and Dispositions** – Unlike in a standard DCF, where you might assume modest CapEx that stays slightly ahead of Depreciation going into the future, in a REIT DCF you assume that the company continues to acquire, develop, redevelop, and dispose of properties.
- 2) Continual Debt and Equity Issuances** – To support these ongoing capital costs, the REIT must also continue to issue Debt and Equity. These issuances are significant because they will change the REIT's share count and, therefore, its Implied Share Price in the DCF. And these issuances continue into the Terminal Period, further complicating the share-count calculation.

Here's an example of a DCF for Park Hotels, the U.S.-based REIT described above, starting with the Unlevered FCF projections:

Free Cash Flow Projections:		Units:	Projected:									
			FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26
Revenue:	\$ M	\$	2,819	2,903	3,045	3,194	3,340	3,510	3,686	3,866	4,052	4,247
Revenue Growth Rate:	%		3.4%	3.0%	4.9%	4.9%	4.6%	5.1%	5.0%	4.9%	4.8%	4.8%
(-) Normalized Operating Expenses:	\$ M		(2,385)	(2,471)	(2,604)	(2,748)	(2,890)	(3,019)	(3,133)	(3,248)	(3,363)	(3,525)
% Revenue:	%		(84.6%)	(85.1%)	(85.5%)	(86.0%)	(86.5%)	(86.0%)	(85.0%)	(84.0%)	(83.0%)	(83.0%)
Operating Income (EBIT):	\$ M		434	432	440	446	450	491	553	619	689	722
Operating Margin:	%		15.4%	14.9%	14.5%	14.0%	13.5%	14.0%	15.0%	16.0%	17.0%	17.0%
Growth Rate:	%		0.2%	(0.5%)	1.9%	1.4%	0.8%	9.2%	12.5%	11.9%	11.4%	4.8%
Adjustments for Non-Cash Charges:												
(+) Depreciation & Amortization:	\$ M		316	331	353	377	401	425	450	476	502	531
% Revenue:	%		11.2%	11.4%	11.6%	11.8%	12.0%	12.1%	12.2%	12.3%	12.4%	12.5%
(+/-) Change in Working Capital:	\$ M		(6)	(8)	(24)	(31)	(48)	(55)	(52)	(49)	(46)	(49)
% Change in Revenue:	%		(6.6%)	(9.8%)	(17.0%)	(20.4%)	(33.1%)	(32.0%)	(29.5%)	(27.0%)	(25.0%)	(25.0%)
% Revenue:	%		(0.2%)	(0.3%)	(0.8%)	(1.0%)	(1.4%)	(1.6%)	(1.4%)	(1.3%)	(1.1%)	(1.1%)
(-) Recurring Maintenance CapEx:	\$ M		(237)	(247)	(262)	(278)	(294)	(305)	(317)	(329)	(344)	(361)
% Revenue:	%		(8.4%)	(8.5%)	(8.6%)	(8.7%)	(8.8%)	(8.7%)	(8.6%)	(8.5%)	(8.5%)	(8.5%)
(-) Acquisitions, Net of Cash Acquired:	\$ M		-	(250)	(275)	(300)	(300)	(263)	(203)	(193)	(203)	(212)
% Revenue:	%		0.0%	(8.6%)	(9.0%)	(9.4%)	(9.0%)	(7.5%)	(5.5%)	(5.0%)	(5.0%)	(5.0%)
(-) Developments & Redevelopments:	\$ M		(50)	(60)	(70)	(80)	(90)	(91)	(88)	(85)	(81)	(85)
% Revenue:	%		(1.8%)	(2.1%)	(2.3%)	(2.5%)	(2.7%)	(2.6%)	(2.4%)	(2.2%)	(2.0%)	(2.0%)
(-) Proceeds from Asset Dispositions:	\$ M		150	150	150	150	150	147	144	139	134	127
% Revenue:	%		5.3%	5.2%	4.9%	4.7%	4.5%	4.2%	3.9%	3.6%	3.3%	3.0%
Unlevered Free Cash Flow:	\$ M	\$	607	348	312	285	269	349	486	578	651	673
Growth Rate:	%		29.4%	(42.7%)	(10.2%)	(8.8%)	(5.7%)	29.9%	39.3%	18.8%	12.6%	3.5%

Continual Acquisitions, Development and Redevelopment, and Dispositions Assumed!

Free Cash Flow Projections:		Units:	Projected:									
			FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26
Stock Issuances:	\$ M		-	18	135	166	552	179	158	154	158	165
% Total Capital Costs:	%		0.0%	4.5%	29.5%	32.6%	103.4%	35.0%	34.0%	33.0%	32.0%	31.0%
Growth Rate:	%		N/A	N/A	638.2%	22.7%	233.4%	(67.5%)	(12.0%)	(2.2%)	2.5%	4.0%
Cost of Equity:	%		9.46%	9.46%	9.46%	9.46%	9.46%	9.46%	9.46%	9.46%	9.46%	9.46%
Cost of Equity - Cumulative Discount Factor:	#		0.914	0.835	0.762	0.696	0.636	0.581	0.531	0.485	0.443	0.405
PV of Future Stock Issuances:	\$ M		-	15	103	115	351	=+O77*O82	84	75	70	67

The company keeps spending on acquisitions, developments, and redevelopments, so we must also assume continued Debt and Equity funding.

The Equity funding will create additional shares. We can estimate the exact number by taking the Present Value of these Future Stock Issuances, based on the Cost of Equity and the year number we're in.

The Terminal Value calculation and the bridge from Implied Enterprise Value to Implied Equity Value are standard, but these Future Stock Issuances create additional shares that we must account for when calculating the Implied Share Price:

DCF Assumptions & Output:		Units:
Risk-Free Rate:	%	2.49%
Equity Risk Premium:	%	5.91%
Median Levered Beta from Comps:	#	1.18
Cost of Equity:	%	9.46%
Planned Debt-to-Total-Capital Ratio:	%	50.0%
Cost of Debt:	%	3.50%
Discount Rate (WACC):	%	6.48%
PV of Future Stock Issuances:	\$ M	\$ 984
PV of Terminal Value of Stock Issuances:	\$ M	852
Estimated # of Future Shares to Be Issued:	# Millions	=SUM(H17:H18)/Share_Price
Total Shares Outstanding:	# Millions	263.717
Current Equity Value:	\$	5,487
(-) Cash & Cash-Equivalents:		(293)
(+) Total Debt:		3,222
(+) Preferred Stock:		-
(+) Noncontrolling Interests:		(49)
Current Enterprise Value:		8,367

Calculated using a formula similar to the one in the standard Terminal Value formula: (Final Year Stock Issuances * (1 + Terminal Growth Rate) / (Cost of Equity - Terminal Growth Rate)) * Cumulative Discount Factor in Final Year.

Today's share count is ~66 million higher as a result of these future issuances.

Once we have an accurate share count, we can calculate the Implied Share Price and create sensitivities to analyze the output in different cases:



Real Estate & REIT Modeling

Quick Reference Guide for REITs

Terminal Value - Multiples Method:

Median EV / EBITDA of Comps:	12.1 x
Baseline Terminal EBITDA Multiple:	11.0 x
Baseline Terminal Value:	\$ 13,780
Implied Terminal FCF Growth Rate:	1.5%
(+) PV of Terminal Value:	\$ 7,354
(+) Sum of PV of Free Cash Flows:	3,185
Implied Enterprise Value:	10,538
<i>% of Implied EV from Terminal Value:</i>	<i>69.8%</i>
(+) Cash & Cash-Equivalents:	293
(-) Total Debt:	(3,222)
(-) Preferred Stock:	-
(-) Noncontrolling Interests:	49
Implied Equity Value:	7,658
Implied Share Price from DCF:	\$ 29.04
Premium / (Discount) to Current:	4.6%

Terminal Value - Perpetuity Growth Method:

Expected Long-Term GDP Growth:	2.0%
Baseline Terminal FCF Growth Rate:	1.5%
Baseline Terminal Value:	\$ 13,780
Implied Terminal EBITDA Multiple:	11.0 x
(+) PV of Terminal Value:	\$ 7,354
(+) Sum of PV of Free Cash Flows:	3,185
Implied Enterprise Value:	10,538
<i>% of Implied EV from Terminal Value:</i>	<i>69.8%</i>
(+) Cash & Cash-Equivalents:	\$ 293
(-) Total Debt:	(3,222)
(-) Preferred Stock:	-
(-) Noncontrolling Interests:	49
Implied Equity Value:	7,658
Implied Share Price from DCF:	\$ 29.04
Premium / (Discount) to Current:	4.6%

Sensitivity Analyses:

		Weighted Average Cost of Capital (WACC):											
		5.25%	5.50%	5.75%	6.00%	6.25%	6.50%	6.75%	7.00%	7.25%	7.50%	7.75%	
Terminal EV	12.00 x	\$ 36.12	\$ 35.15	\$ 34.21	\$ 33.29	\$ 32.39	\$ 31.51	\$ 30.66	\$ 29.82	\$ 29.01	\$ 28.21	\$ 27.44	
/ EBITDA	11.75 x	35.41	34.46	33.53	32.62	31.74	30.88	30.04	29.22	28.42	27.64	26.88	
Multiple	11.50 x	34.70	33.76	32.85	31.96	31.09	30.25	29.42	28.61	27.83	27.06	26.31	
(Terminal	11.25 x	33.98	33.07	32.17	31.30	30.44	29.61	28.80	28.01	27.24	26.49	25.75	
Value	11.00 x	33.27	32.37	31.49	30.63	29.80	28.98	28.18	27.41	26.65	25.91	25.19	
Calculated	10.75 x	32.56	31.67	30.81	29.97	29.15	28.35	27.57	26.80	26.06	25.33	24.62	
Using the	10.50 x	31.85	30.98	30.13	29.31	28.50	27.72	26.95	26.20	25.47	24.76	24.06	
Multiples	10.25 x	31.14	30.28	29.45	28.64	27.85	27.08	26.33	25.60	24.88	24.18	23.50	
Method):	10.00 x	30.42	29.59	28.78	27.98	27.21	26.45	25.71	24.99	24.29	23.60	22.94	

		Weighted Average Cost of Capital (WACC):											
		5.25%	5.50%	5.75%	6.00%	6.25%	6.50%	6.75%	7.00%	7.25%	7.50%	7.75%	
Terminal FCF	1.70%	\$ 45.66	\$ 41.66	\$ 38.16	\$ 35.07	\$ 32.33	\$ 29.88	\$ 27.67	\$ 25.67	\$ 23.86	\$ 22.21	\$ 20.70	
Growth Rate	1.65%	45.07	41.16	37.73	34.70	32.00	29.59	27.42	25.45	23.67	22.04	20.54	
(Terminal	1.60%	44.49	40.67	37.31	34.33	31.68	29.31	27.17	25.23	23.47	21.86	20.39	
Value	1.55%	43.93	40.19	36.89	33.97	31.37	29.03	26.93	25.02	23.28	21.69	20.23	
Calculated	1.50%	43.39	39.72	36.49	33.62	31.06	28.76	26.69	24.80	23.09	21.52	20.08	
Using the	1.45%	42.86	39.26	36.09	33.28	30.76	28.50	26.45	24.60	22.90	21.35	19.93	
Gordon	1.40%	42.34	38.82	35.71	32.94	30.47	28.24	26.22	24.39	22.72	21.19	19.78	
Growth	1.35%	41.83	38.38	35.33	32.61	30.18	27.98	26.00	24.19	22.54	21.03	19.63	
Method):	1.30%	41.34	37.96	34.96	32.29	29.90	27.73	25.78	23.99	22.36	20.87	19.49	

Park Hotels' share price at the time of this valuation was \$27.77, so based on these results, the company was appropriately valued or *slightly* undervalued by about 5%. However, 5% is such a small difference that it's more reasonable to say, "appropriately valued."

You can also use a Dividend Discount Model (DDM) to value REITs; the setup is nearly identical to the setup for a DCF, so we are not going to repeat all the steps above.

The DDM takes more work to set up because you can't stop at EBIT and then move to Unlevered FCF – instead, you must calculate FFO (or equivalent metric) and make Dividends a percentage of FFO. And that means you must also project the Debt and Interest Expense.

As in the DCF shown above, you also assume continued Debt and Equity issuances to support the REIT's activities, which means additional shares and potentially a different Debt / Equity ratio over time.

We recommend **against** the DDM in most cases; it takes more time and effort to set up, it requires more assumptions, and it doesn't offer substantial advantages over the DCF.

For your reference, we will show below an example DDM for Colonial, the European REIT:

Dividend Projections:	Units:	Projected:									
		FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26
Fair Value of Investment Properties:	€ M	8,148	8,601	9,510	10,093	10,379	10,832	11,307	11,802	12,320	12,861
Total Debt:	€ M	3,755	3,812	4,070	4,236	4,254	4,368	4,486	4,611	4,741	4,876
Revenue:	€ M	293	305	315	341	359	377	395	413	432	451
Revenue Growth Rate:	%	6.9%	4.1%	3.3%	8.3%	5.1%	5.0%	4.8%	4.6%	4.5%	4.5%
(-) Operating Expenses:	€ M	(64)	(66)	(67)	(71)	(75)	(78)	(81)	(85)	(88)	(92)
% Revenue:	%	(21.9%)	(21.6%)	(21.2%)	(20.9%)	(20.8%)	(20.7%)	(20.6%)	(20.5%)	(20.5%)	(20.5%)
Operating Income (EBIT):	€ M	229	239	248	270	284	299	313	328	343	359
Operating Margin:	%	78.1%	78.4%	78.8%	79.1%	79.2%	79.3%	79.4%	79.5%	79.5%	79.5%
Growth Rate:	%	7.4%	4.4%	3.8%	8.8%	5.3%	5.1%	4.9%	4.7%	4.5%	4.5%
(+/-) Change in Fair Value of Properties:	€ M	199	339	530	319	204	208	217	226	236	246
(-) Net Interest Expense:	€ M	(70)	(71)	(71)	(75)	(78)	(79)	(81)	(83)	(85)	(88)
(-) Impairments & Other Items:	€ M	-	-	-	-	-	-	-	-	-	-
Profit Before Tax:	€ M	358	507	707	514	410	428	449	471	494	517
(-) Income Tax Expense:	€ M	(36)	(51)	(71)	(51)	(41)	(43)	(45)	(47)	(49)	(52)
Consolidated Net Profit:	€ M	322	456	636	463	369	385	404	424	444	466
(-) Net Profit Attributable to NCI:	€ M	(148)	(210)	(293)	(213)	(170)	(177)	(186)	(195)	(204)	(214)
Net Income Attributable to Parent:	€ M	174	246	344	250	199	208	218	229	240	251
(+/-) Change in Fair Value of Properties:	€ M	(199)	(339)	(530)	(319)	(204)	(208)	(217)	(226)	(236)	(246)
(+/-) Other Adjustments:	€ M	36	36	36	36	36	36	36	36	36	36
(+/-) Deferred Income Taxes:	€ M	20	34	53	32	20	21	22	23	24	25
(+) Noncontrolling Interest Portion:	€ M	116	164	229	166	133	138	145	152	160	167
EPRA Earnings:	€ M	146	141	132	165	184	195	205	214	223	233
Change in Fair Value of Properties % Previous Value:	%	2.6%	4.2%	6.2%	3.4%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%
Weighted Average Cost of Debt:	%	1.90%	1.90%	1.85%	1.85%	1.85%	1.85%	1.85%	1.85%	1.85%	1.85%
Income Tax Rate:	%	10.0%	10.0%	10.0%	10.0%	10.0%	10.0%	10.0%	10.0%	10.0%	10.0%
Net Profit to NCI % Consolidated Net Profit:	%	46.0%	46.0%	46.0%	46.0%	46.0%	46.0%	46.0%	46.0%	46.0%	46.0%
Deferred Taxes % Income Tax Expense:	%	55.7%	66.9%	74.9%	62.1%	49.8%	48.5%	48.2%	48.0%	47.8%	47.6%
NCI Portion % / Net Profit Attrib. to NCI:	%	78.2%	78.2%	78.2%	78.2%	78.2%	78.2%	78.2%	78.2%	78.2%	78.2%
Dividends % (EPRA Earnings + NCI Interests):	%	35.0%	37.5%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%	40.0%
Dividends:	€ M	92	114	144	132	127	147	153	160	160	160
Growth Rate:	%	(5.4%)	24.7%	26.1%	(8.2%)	(4.3%)	5.3%	4.9%	4.7%	4.5%	4.5%

We must track Total Debt to calculate the Interest Expense; we track the FV of Properties since this REIT follows IFRS.

Income Taxes are low, but *not 0* due to the tax regime in Spain and France.

Very similar to FFO, but the main adjustment is the Change in Fair Value of Properties rather than Depreciation.

Just like Dividends are normally a % of FFO for U.S.-based REITs, they are a % of EPRA Earnings here.

The Terminal Value and Implied Share Price calculations are simpler in a DDM because there is no “bridge” – you calculate Implied Equity Value directly by summing up the PV of Dividends and the PV of the Terminal Value:

Terminal Value - Multiples Method:

Median P / EPRA Earnings of Comps:	20.0 x
Baseline Terminal P / EPRA E Multiple:	15.0 x
Baseline Terminal Value:	€ 3,495
Implied Terminal EPRA Earnings Growth:	0.4%
(+) PV of Terminal Value:	€ 1,770
(+) Sum of PV of Dividends:	919
Implied Equity Value:	2,689

% of Implied Equity Value from TV: 65.8%

Implied Share Price from DDM	€ 5.92
Premium / (Discount) to Current:	(11.8%)

Terminal Value - Perpetuity Growth Method:

Expected Long-Term GDP Growth:	1.0%
Baseline Terminal EPRA Earnings Growth:	0.4%
Baseline Terminal Value:	€ 3,495
Implied Terminal P / EPRA E Multiple:	15.0 x
(+) PV of Terminal Value:	€ 1,770
(+) Sum of PV of Dividends:	919
Implied Equity Value:	2,689

% of Implied Equity Value from TV: 65.8%

Implied Share Price from DDM:	€ 5.92
Premium / (Discount) to Current:	(11.8%)

The sensitivity tables here indicate that Colonial is likely overvalued:

Colonial's Current Share Price - it's greater than the majority of these tables, indicating that the REIT's Implied Value is below its Current Value.

Sensitivity Analyses:

		Cost of Equity:											
		5.75%	6.00%	6.25%	6.50%	6.75%	7.00%	7.25%	7.50%	7.75%	8.00%	8.25%	
Terminal P / EPRA Earnings Multiple (Terminal Value with the Multiples Method):	17.0 x	€ 6.71	€ 6.67	€ 6.62	€ 6.57	€ 6.51	€ 6.45	€ 6.39	€ 6.32	€ 6.25	€ 6.18	€ 6.11	
	16.5 x	6.57	6.53	6.49	6.43	6.38	6.32	6.26	6.19	6.13	6.06	5.99	
	16.0 x	6.43	6.39	6.35	6.30	6.25	6.19	6.13	6.07	6.00	5.94	5.87	
	15.5 x	6.30	6.26	6.22	6.17	6.12	6.06	6.00	5.94	5.88	5.81	5.75	
	15.0 x	6.16	6.12	6.08	6.03	5.98	5.93	5.87	5.81	5.75	5.69	5.63	
	14.5 x	6.02	5.99	5.95	5.90	5.85	5.80	5.75	5.69	5.63	5.57	5.50	
	14.0 x	5.88	5.85	5.81	5.77	5.72	5.67	5.62	5.56	5.50	5.44	5.38	
	13.5 x	5.74	5.71	5.68	5.63	5.59	5.54	5.49	5.43	5.38	5.32	5.26	
	13.0 x	5.61	5.58	5.54	5.50	5.46	5.41	5.36	5.31	5.25	5.20	5.14	

		Cost of Equity:											
		5.75%	6.00%	6.25%	6.50%	6.75%	7.00%	7.25%	7.50%	7.75%	8.00%	8.25%	
Terminal EPRA Earnings Growth Rate (Terminal Value with the Gordon Growth Method):	0.60%	€ 7.34	€ 7.05	€ 6.78	€ 6.53	€ 6.30	€ 6.08	€ 5.87	€ 5.68	€ 5.50	€ 5.32	€ 5.16	
	0.55%	7.30	7.01	6.75	6.50	6.27	6.06	5.85	5.66	5.47	5.30	5.14	
	0.50%	7.26	6.98	6.72	6.47	6.25	6.03	5.83	5.64	5.45	5.28	5.12	
	0.45%	7.22	6.94	6.69	6.44	6.22	6.00	5.80	5.61	5.43	5.26	5.10	
	0.40%	7.18	6.91	6.65	6.42	6.19	5.98	5.78	5.59	5.41	5.25	5.09	
	0.35%	7.15	6.88	6.62	6.39	6.16	5.95	5.76	5.57	5.39	5.23	5.07	
	0.30%	7.11	6.84	6.59	6.36	6.14	5.93	5.73	5.55	5.37	5.21	5.05	
	0.25%	7.07	6.81	6.56	6.33	6.11	5.91	5.71	5.53	5.36	5.19	5.04	
	0.20%	7.04	6.78	6.53	6.30	6.09	5.88	5.69	5.51	5.34	5.17	5.02	



NAV Model Valuation

The final REIT valuation methodology is arguably the most important one: **The Net Asset Value (NAV) Model.**

The basic idea is that you calculate the Fair Market Value of all the REIT's Assets and then subtract the Fair Market Value of all its Liabilities. Then, you divide by its share count to calculate its Implied Share Price, which you compare to the REIT's Current Share Price.

This methodology is **not** that useful for IFRS-based REITs because the major Asset on their Balance Sheets – Investment Properties – should already be marked to market value.

You could argue that it's marked inaccurately, or that there are other required adjustments, but this methodology rarely produces dramatically different results for IFRS-based REITs.

It's more interesting and useful for U.S.-based REITs because you must revalue their Gross Real Estate Assets by projecting their Net Operating Income and dividing it by an appropriate Cap Rate.

We defined **Cap Rates** in the written guide to individual properties, but to review, they are **the reciprocals** of valuation multiples.

If a company has an EV / EBITDA of 20x, that corresponds to a Cap Rate of $1 / 20 = 5\%$.

Investors use Cap Rates to value properties because they view properties as an asset class similar to fixed income (e.g., government or corporate bonds).

You value a property by projecting its forward, stabilized "Net Operating Income" (NOI), which is similar to EBITDA but which may partially reflect capital costs, and then dividing this NOI figure by current Cap Rates for similar properties in the area.

So, if you're analyzing a REIT and have forecast its forward NOI as \$200 million, and it operates in areas where the Cap Rates for similar properties are between 6% and 7%, you might estimate this REIT's portfolio value as \$2.9 billion to \$3.3 billion ($\$200 \text{ million} / 7\%$ to $\$200 \text{ million} / 6\%$).

The NAV Model assumes that there are pricing discrepancies between local real estate markets and the stock market and that you can quantify these discrepancies by dividing the REIT's NOI by local Cap Rates to revalue its Real Estate Assets.

For example, Cap Rates in a REIT's local markets might have fallen to 6% recently, but the Cap Rate *implied by the company's stock price* might be closer to 7%.



<http://breakingintowallstreet.com>

We then found prevailing market Cap Rates for similar-quality hotels in the geographies that Park Hotels operates in (primarily coastal areas of the U.S.). The range seemed to be 7-8%:

Park Hotels & Resorts Inc. - Net Asset Value (NAV) Model

(\$ USD in Millions Except Per Share and Per Unit Amounts in USD as Stated)

Balance Sheet:	Units:	Historical:		
		FY14	FY15	FY16
ASSETS:				
Forward Property Net Operating Income (NOI):	\$ M			\$ 838
(÷) Assumed Cap Rate:	%			7.5%
Market Value of Gross Real Estate Operating Assets:	\$ M			\$ 11,168

We ignored Accumulated Depreciation here, and as a result of that and the revaluation at this 7.5% Cap Rate, the Market Value of Gross Real Estate Operating Assets increases to \$11.2 billion vs. a book value of \$10.4 billion.

Step 3: Project Non-Rental Income and Divide by a (Higher) Cap Rate to Value It

Many REITs also earn income from management fees, development work, and other sources.

These income sources are cancellable on short notice, so you tend to value them at much higher Cap Rates, such as 10-15%, to reflect the additional risk.

Park Hotels is a pure-play hotel REIT, so it has no income in these categories.

You will see an example of a REIT that does earn income from these sources in [the more detailed NAV Model in the Valuation section](#).

Step 4: Make Minor Adjustments to All Other Assets and Add Them to the Step 2 and 3 Results

You usually count Assets such as Cash, Accounts Receivable, and Other Assets in full and assume that their book values are proxies for their fair market values.

And you always eliminate Goodwill & Other Intangibles since their fair market values are zero – which can be a significant adjustment for highly acquisitive REITs.

If the REIT has Joint Ventures or Equity Investments, you'll have to re-value them at the appropriate Cap Rates and then multiply by the REIT's ownership percentage to estimate their fair market values.

Almost all REITs have Construction-in-Progress as well, which you often adjust upward based on its expected fair market value upon completion vs. its historical cost:

Balance Sheet:	Units:	Historical:		
		FY14	FY15	FY16
ASSETS:				
Forward Property Net Operating Income (NOI):	\$ M			\$ 838
(÷) Assumed Cap Rate:	%			7.5%
Market Value of Gross Real Estate Operating Assets:	\$ M			\$ 11,168
Construction-in-Progress:	\$ M			79
(x) Market Value Adjustment:	%			120.0%
Market Value of Construction-in-Progress:	\$ M			95
Cash & Cash-Equivalents:	\$ M			293
Accounts Receivable:	\$ M			130
Goodwill & Other Intangibles:	\$ M			648
(x) Market Value Adjustment:	%			-
Market Value of Goodwill & Other Intangibles:	\$ M			-
Other Assets:	\$ M			222
Total Market Value of Assets:	\$ M			\$ 11,908

Step 5: Mark Liabilities to Fair Market Value and Subtract Them from the Market Value of All Assets

The fair market values of many Liabilities tend to be close to their book values, but there are still differences.

For example, if prevailing market interest rates for similar Debt issuances have risen from 4% to 5%, and your REIT's Debt has a fixed coupon rate of 4%, it's worth less than book value now because investors must receive a discount on it to earn a Yield to Maturity (YTM) of 5% – equivalent to the yield they could earn on new issuances.

You might also assume some benefit from tax-exempt bonds, and you'll have to revalue Debt and other Liabilities held by JVs and Equity Investments and then multiply by your REIT's ownership percentage to factor those in.

Here are the adjustments for Park Hotels:

	A	B	C	D	E	F	G	H	I
24									
25			Total Market Value of Assets:	\$ M				\$ 11,908	
26									
27			LIABILITIES & EQUITY:						
28			Debt & Other Borrowings:	\$ M				(3,222)	
29			(x) Market Value Adjustment:	%				100.5%	
30			Market Value of Debt & Other Borrowings:	\$ M				(3,238)	
31									
32			Accounts Payable:	\$ M				(167)	
33			Other Liabilities:	\$ M				(2,622)	
34			Noncontrolling Interests (NCI):	\$ M				49	
35									
36			Net Asset Value (NAV):	\$ M				=+H25+H30+SUM(H32:H34)	

Interest rates have *fallen* since issuance, so the market value of this Debt exceeds its book value.

Step 6: Divide Net Asset Value by the REIT's Share Count to Determine NAV per Share

LIABILITIES & EQUITY:			
Debt & Other Borrowings:	\$ M		(3,222)
(x) Market Value Adjustment:	%		100.5%
Market Value of Debt & Other Borrowings:	\$ M		(3,238)
Accounts Payable:	\$ M		(167)
Other Liabilities:	\$ M		(2,622)
Noncontrolling Interests (NCI):	\$ M		49
Net Asset Value (NAV):	\$ M		\$ 5,930
NAV per Share:	\$ as Stated		\$ 30.01
Current Share Price:	\$ as Stated		\$ 27.77
NAV per Share Premium / (Discount) to Current:	%		8.1%
Cap Rate Implied by Current Share Price:	%		7.8%

The REIT seems slightly undervalued based on these results.

The "Cap Rate Implied by Current Share Price" is based on Forward NOI / (Current Equity Value – Market Value of All Assets Except Gross Real Estate Operating Assets + Market Value of All Liabilities).

The fact that it's 7.8% vs. our Assumed Cap Rate of 7.5% confirms that Park Hotels may be *slightly undervalued* at the moment (since a higher Cap Rate represents a *lower* valuation).

The sensitivity table confirms that finding as well:



Real Estate & REIT Modeling

Quick Reference Guide for REITs

Sensitivity Analysis - NAV per Share:

		Assumed Cap Rate:											
		7.0%	7.1%	7.2%	7.3%	7.4%	7.5%	7.6%	7.7%	7.8%	7.9%	8.0%	
Forward Property Net Operating Income (NOI):	\$ 850	\$ 34.94	\$ 34.08	\$ 33.23	\$ 32.42	\$ 31.62	\$ 30.84	\$ 30.09	\$ 29.35	\$ 28.64	\$ 27.94	\$ 27.26	
	845	34.58	33.72	32.88	32.07	31.28	30.51	29.76	29.03	28.31	27.62	26.94	
	840	34.22	33.36	32.53	31.72	30.94	30.17	29.42	28.70	27.99	27.30	26.63	
	838	34.04	33.19	32.36	31.56	30.77	30.01	29.26	28.54	27.83	27.15	26.48	
	835	33.86	33.01	32.18	31.38	30.59	29.83	29.09	28.37	27.67	26.98	26.31	
	830	33.50	32.65	31.83	31.03	30.25	29.49	28.76	28.04	27.34	26.66	25.99	
	825	33.13	32.29	31.48	30.68	29.91	29.16	28.43	27.71	27.02	26.34	25.68	
	820	32.77	31.94	31.13	30.34	29.57	28.82	28.09	27.38	26.69	26.02	25.36	
	815	32.41	31.58	30.77	29.99	29.23	28.48	27.76	27.05	26.37	25.70	25.05	
	810	32.05	31.22	30.42	29.64	28.88	28.15	27.43	26.73	26.04	25.38	24.73	

This region is the only part of the table, roughly, that's below Park Hotels' share price at the time of this valuation, so we can say that the REIT seems slightly undervalued.

You can also calculate a P / NAV multiple (Equity Value / Net Asset Value, or Share Price / NAV per Share) for this REIT, calculate the same multiple for its peer REITs, and use that to value this REIT, under the logic that Equity Value / NAV should be similar for similar REITs.

We don't do that in this course because it takes a lot of time and effort to calculate NAV for each REIT, and it wouldn't necessarily tell us anything new.

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Industry-Specific Differences and U.S. GAAP vs. IFRS

In the previous section, you saw example financial statements for an apartment or "multifamily" REIT (AvalonBay).

These statements include many of the expected items: Real Estate Assets, Debt, and Equity on the Balance Sheet; Rent, Operating Expenses, Property Taxes, Interest, Depreciation, and Gains/Losses on the Income Statement; and Net Income, non-cash adjustments, Developments, Acquisitions, Maintenance CapEx, Asset Sales, Dividends, and Equity/Debt Issuances on the Cash Flow Statement.

If you look at an **office, retail, industrial, or data center REIT**, such as Boston Properties (a major office REIT), you'll find the following differences:

- 1) Income Statement:** There will be more categories of revenue, such as "Recoveries" or "Expense Reimbursements" and Parking Income, due to the greater variety of lease types. Data center REITs may earn revenue based on power usage, and healthcare REITs may earn revenue based on "Service Fees" charged to patients.

- 2) Cash Flow Statement:** There will be line items for Tenant Improvements, Leasing Commissions, and the Straight-Lining of Rent.

These types of properties tend to have fewer tenants with more variable lease terms, and tenants can request “customizations” that the owner covers (Tenant Improvements). Also, real estate brokers are usually required to find tenants (Leasing Commissions).

We discussed [the rationale for the Straight-Lining of Rent adjustment in the Crash Course section above](#). In short, it reflects the difference between Book Rent and Cash Rent Received, which may be significant for multi-year leases.

- 3) Balance Sheet:** There are no huge industry-specific differences, but you’ll often see items like “Tenant Receivables” instead of “Resident Deposits” because the tenants are companies, not individuals.

Many REITs also have Balance Sheet line items such as “Acquired Lease Intangibles,” which are created when they acquire properties or entire REITs with leases that are above or below market rents. Like all Definite-Lived Intangibles, this one amortizes over time.

This line item is *not* specific to certain REITs; all REITs that have made acquisitions, except for hotel REITs, could have it (though it would be much smaller for multifamily REITs due to the 1-year leases).

Here are a few examples of these differences on the financial statements, based on extracts from Boston Properties’ filings:

ASSETS	
Real estate, at cost (amounts related to variable interest entities (“VIEs”) of \$6,760,078 at December 31, 2016)	\$ 20,147,263
Less: accumulated depreciation (amounts related to VIEs of (\$758,640) at December 31, 2016)	(4,222,235)
Total real estate	15,925,028
Cash and cash equivalents (amounts related to VIEs of \$253,999 at December 31, 2016)	356,914
Cash held in escrows (amounts related to VIEs of \$4,955 at December 31, 2016)	63,174
Investments in securities	23,814
Tenant and other receivables (amounts related to VIEs of \$23,525 at December 31, 2016)	92,548
Accrued rental income (amounts related to VIEs of \$224,185 at December 31, 2016)	799,138
Deferred charges, net (amounts related to VIEs of \$290,436 at December 31, 2016)	686,163
Prepaid expenses and other assets (amounts related to VIEs of \$42,718 at December 31, 2016)	129,666
Investments in unconsolidated joint ventures	775,198
Total assets	\$ 18,851,643

In place of "Resident Security Deposits" for multifamily REITs.

Revenue	
Rental	
Base rent	\$ 2,017,767
Recoveries from tenants	358,975
Parking and other	100,910
Total rental revenue	2,477,652
Hotel revenue	44,884
Development and management services	28,284
Total revenue	2,550,820
Expenses	
Operating	
Rental	889,768
Hotel	31,466
General and administrative	105,229
Transaction costs	2,387
Impairment loss	1,783
Depreciation and amortization	694,403
Total expenses	1,725,036
Operating income	825,784

More revenue categories since office/retail/industrial tenants are responsible for significant property expenses and taxes; parking revenue can also be significant at some of these properties.

Cash flows from operating activities:	
Net income	\$ 569,977
Adjustments to reconcile net income to net cash provided by operating activities:	
Depreciation and amortization	694,403
Impairment loss	1,783
Non-cash compensation expense	32,911
Income from unconsolidated joint ventures	(8,074)
Gain on sale of investment in unconsolidated joint venture	(59,370)
Distributions of net cash flow from operations of unconsolidated joint ventures	24,955
Losses (gains) from investments in securities	(2,273)
Non-cash portion of interest expense	(35,052)
Settlement of accreted debt discount on repurchases/repayments of unsecured senior notes and unsecured exchangeable senior notes	—
Losses from early extinguishments of debt	371
Gains on sales of real estate	(80,606)
Change in assets and liabilities:	
Cash held in escrows	2,277
Tenant and other receivables, net	3,688
Accrued rental income, net	(28,127)
Prepaid expenses and other assets	52,923
Accounts payable and accrued expenses	15,666
Accrued interest payable	53,547
Other liabilities	(106,022)
Tenant leasing costs	(96,103)
Total adjustments	466,897
Net cash provided by operating activities	1,036,874

These items would not exist in the same way for multifamily REITs.

Cash flows from investing activities:

Acquisitions of real estate	(78,000)
Construction in progress	(500,350)
Building and other capital improvements	(150,640)
Tenant improvements	(230,298)
Proceeds from sales of real estate	122,750
Proceeds from sales of real estate and sales of interests in property partnerships placed in escrow	(122,647)
Proceeds from sales of real estate and sales of interests in property partnerships released from escrow	122,647
Cash placed in escrow for land sale contracts	—
Cash released from escrow for land sale contracts	1,596
Cash released from escrow for investing activities	6,694
Capital contributions to unconsolidated joint ventures	(575,795)
Capital distributions from unconsolidated joint ventures	20,440
Proceeds from sale of investment in unconsolidated joint venture	55,707
Investments in marketable securities	—
Investments in securities, net	(1,161)
Net cash used in investing activities	(1,329,057)

This item only exists for office / industrial / retail (and similar) REITs because tenants in that sector have significantly more power to request customizations and other incentives.

The following table reconciles GAAP net income attributable to Boston Properties, Inc. to taxable

	2016	Fo
Net income attributable to Boston Properties, Inc.	\$ 512,785	
Straight-line rent and net "above-" and "below-market" rent adjustments	(65,861)	
Book/Tax differences from depreciation and amortization	235,819	
Book/Tax differences from interest expense	(36,223)	
Book/Tax differences on gains/(losses) from capital transactions	(70,880)	
Book/Tax differences from stock-based compensation	33,463	
Tangible Property Regulations (2)	(104,783)	
Other book/tax differences, net	(6,121)	
Taxable income	\$ 498,199	

The Straight-Lining adjustment here is significant because of the multi-year leases used by office / industrial / retail properties; it wouldn't even exist for multifamily or hotel REITs.

With **hotel and lodging REITs**, you will see the following differences on the financial statements:

- 1) Income Statement:** Revenue and expenses are split into categories such as Rooms, Food & Beverage, and Other; other expense categories could be "Variable," "Fixed" (i.e., no variance with revenue or occupancy rates), or "Corporate-Level." Management Fees are also common.
- 2) Cash Flow Statement:** Maintenance CapEx is often labeled the "FF&E Reserve" (Furniture, Fixtures & Equipment) or "Replacement Fund."
- 3) Balance Sheet:** The FF&E Replacement Fund may be listed separately from the Real Estate Assets, and there may be Assets and Liabilities related to the hotel managers.

Here's a quick screenshot summary:

ASSETS		
Property and equipment, net	\$	10,145
Assets held for sale		150
Due from managers		55
Advances to and investments in affiliates		286
Furniture, fixtures and equipment replacement fund		173
Other		225
Restricted cash		2
Cash and cash equivalents		372
Total assets	\$	11,408
REVENUES		
Rooms	\$	3,492
Food and beverage		1,599
Other		339
Total revenues		5,430
EXPENSES		
Rooms		893
Food and beverage		1,114
Other departmental and support expenses		1,306
Management fees		236
Other property-level expenses		382
Depreciation and amortization		724
Corporate and other expenses		106
Gain on insurance and business interruption settlements		(15)
Total operating costs and expenses		4,746
OPERATING PROFIT		684
INVESTING ACTIVITIES		
Proceeds from sales of assets, net		467
Return of investments in affiliates		23
Advances to and investments in affiliates		(5)
Acquisitions		(63)
Capital expenditures:		
Renewals and replacements		(293)
Redevelopment and acquisition-related investments		(226)
New development		—
Change in furniture, fixtures and equipment ("FF&E") replacement fund		(31)
Change in restricted cash for investing activities		13
Property insurance proceeds		—
Cash used in investing activities		(115)

Due to the agreements with individual management teams in charge of hotels.

Funds set aside for the equivalent of Maintenance CapEx across the portfolio of hotels.

Different business model with primarily short-term stays, high daily rates, and high ancillary charges.

Due to the agreements with individual management teams and incentive fees paid to them; branding fees owed to major hotel chains could also show up here.

CapEx is split up a bit differently here because of the FF&E replacements / reserves and the need to maintain the quality of rooms and public areas of hotels.

You should already know the main **U.S. GAAP vs. IFRS differences** if you've made it this far in the guide, but to summarize:

- Difference #1:** IFRS-based REITs record Fair Value Gains & Losses on their financial statements because they mark their Properties to Market Value each year – and there's no Depreciation (or minimal Depreciation, and only for non-RE Assets) on their statements.
- Difference #2:** NAV is much easier to calculate because you can base it on the company's most recent Balance Sheet and adjust for a few non-marked items.



- 3. Difference #3:** Different metrics may be important depending on the region (e.g., EPRA Earnings in Europe or Distributions to Unitholders in Singapore). EPRA Earnings is very similar to FFO for U.S.-based REITs, with a few minor differences, such as an add-back for Deferred Taxes.
- 4. Difference #4:** The Cash Flow Statement often uses the Direct Method, which shows Cash Received and Cash Spent at the top, rather than the Indirect Method, which starts with Net Income and then adjusts for non-cash items.

The first difference makes an impact, but it is not *as* significant as you might think because it just means you'll have to project Cap Rates or Yields and record changes in property values.

But you'll also have to project Cap Rates for U.S.-based REITs if you create 3-statement models for them because you will have to forecast Acquisitions, Developments, and Dispositions and the NOI that REITs will receive or lose on all those.

So, this difference does not create much extra work in practice.

Also, Depreciation vs. Fair Value Gains & Losses does not make a difference for REITs' Dividends or Unlevered FCF because these are non-cash items that get added back, and there is no real tax impact because REITs have very low or no corporate taxes.

The second difference – NAV being much easier to calculate – means that you can use REITs' Book Values and P / BV multiples as-is in the Public Comps. Effectively, Book Value is very close to NAV if a REIT's major Asset is marked to market value already.

However, if market conditions have suddenly changed, or you believe that the REIT did not use appropriate Cap Rates to mark its properties, then a NAV Model might still be worthwhile.

The third difference about alternative metrics being important means that you have to pay attention to the figures a REIT uses in its filings and follow its practices.

For example, European REITs often use EPRA Earnings instead of FFO and EPRA NAV (a variation of Book Value with minor adjustments) instead of standard NAV.

Therefore, if you're valuing a European REIT, you should use these metrics and follow the calculations in the company's filings.

The last difference means that if the REIT's Cash Flow Statement uses the Direct Method, you should *ideally* convert it to the Indirect Method to make financial modeling easier.

The problem with the Direct Method is that “Cash Received” and “Cash Spent” do not match up to specific Balance Sheet line items, so it’s nearly impossible to link the statements.

If the company presents its statements like this, you will have to find a reconciliation in its filings that shows the individual components of “Cash Received” and “Cash Spent.”

If you can’t find it, then you might simplify the setup and make “Cash Received” and “Cash Spent” percentages of Net Income or FFO and not even attempt to link them to the Balance Sheet.

To illustrate the differences above, we’ll briefly go through the financial statements of Klépierre, a French REIT:

EPRA Earnings

In millions of euros (group share)	12/31/2016	12/31/2015
Earnings per IFRS income statement	1,191.2	274.6
<i>Adjustments to calculate EPRA Earnings, exclude:</i>		
(i) Changes in value of investment properties, development properties held for investment and other interests	828.8	883.3
(ii) Profit or losses on disposal of investment properties, development properties held for investment and other interests	23.5	-1.1
(iii) Profit or losses on sales of trading properties including impairment charges in respect of trading properties	-	-
(iv) Tax on profits or losses on disposals	-	-
(v) Negative goodwill/goodwill impairment	-	-922.6
(vi) Changes in fair value of financial instruments and associated close-out costs	-28.3	-33.8
(vii) Acquisition costs on share deals and non-controlling joint venture interests	-	-24.4
(viii) Deferred tax in respect of EPRA adjustments	-199.0	-171.7
(ix) Adjustments (i) to (viii) above in respect of joint ventures (unless already included under proportional consolidation)	28.5	35.0
(x) Non-controlling interests in respect of the above	-156.8	-135.1
EPRA EARNINGS	694.4	645.2

Similar to FFO, EPRA Earnings is a “Cash Net Income” metric; besides the big adjustment for the Change in Fair Value of properties, there are some others here related to disposals, financial instruments, deferred taxes, and noncontrolling interests, all of which are non-cash in some way.

2.7.2.7 EPRA NAV and NNNAV calculation

EPRA NAV & NNNAV

In millions of euros	12/31/2016	06/30/2016	12/31/2015
Consolidated shareholders’ equity (group share)	10,107	9,534	9,526
Unrealized capital gains on other assets	300	293	293
Goodwill restatement	-647	-625	-608
Fair value of financial instruments	48	46	38
Deferred taxes on asset values on the balance sheet	1,270	1,223	1,159
Optimization of duties and fees on the sale of assets	368	388	384
EPRA NAV	11,446	10,859	10,792
Optimized deferred taxes on unrealized capital gains	-245	-227	-257
Fair value of financial instruments	-48	-46	-38
Fair value of fixed-rate debt	-185	-321	-146
EPRA NNNAV	10,967	10,265	10,351
Number of shares, end of period ⁽¹⁾	311,827,611	311,773,309	311,457,530
Per share (in euros)			
EPRA NAV PER SHARE	36.7	34.8	34.7
EPRA NNNAV PER SHARE	35.2	32.9	33.2

The adjustments for EPRA NAV and EPRA NNNAV are quite minor compared with the ones you’d see in a NAV Model for a U.S.-based REIT.

Deferred Taxes are added back under the logic that they’re unlikely to be paid out in the future.

EPRA NNNAV makes further adjustments for more deferred taxes and the fair value of debt and financial instruments, which a standard NAV Model includes from the start.

3.1.1 Consolidated statements of comprehensive income

In millions of euros	Notes	12/31/2016 Fair value
Gross rental income	6.1	1,214.0
Land expenses (real estate)	6.2	-14.9
Non-recovered rental expenses	6.3	-73.9
Building expenses (owner)	6.4	-41.9
Net rental income		1,083.4
Management, administrative and related income		86.5
Other operating revenue	6.5	18.4
Survey and research costs		-2.8
Payroll expenses	10.1	-131.4
Other general expenses		-63.4
Depreciation and impairment allowance on intangible assets and properties, plant and equipment	6.6	-14.8
Provisions		-5.2
Change in value of investment properties	6.7	828.8
Proceeds from disposal of investment properties and equity investments	6.8	416.1
Net book value of investment properties and equity investments sold	6.8	-392.5
Income from the disposal of investment properties and equity investments		23.5
Goodwill impairment		0.0
Operating income		1,822.8
Net dividends and provisions on non-consolidated investments		0.1
Financial income		109.0
Financial expenses		-306.7
Net cost of debt	6.9	-197.7
Change in the fair value of financial instruments		-12.1
Share in earnings of equity method investments	5.5	89.5
Profit before tax		1,702.5
Corporate income tax	7	-225.6
Net income of consolidated entity		1,476.9

Depreciation is tiny here and is only recorded on *non-real-estate Assets*; the Change in Value of Investment Properties is huge, though.

And in France, REITs still pay some small amount of corporate income tax.

3.1.2 Consolidated statements of financial position

In millions of euros	Notes	12/31/2016 Fair value	12/31/2015 Fair value	01/01/2015 Fair value
Goodwill	5.1	648.4	612.5	125.9
Intangible assets	5.2	45.2	45.7	46.5
Property, plant and equipment	5.3	16.0	20.6	13.0
Investment properties at fair value	5.4	20,390.2	18,750.5	12,362.2
Investment properties at cost	5.4	282.6	1,076.1	304.1
Equity method investments	5.5	1,067.5	1,455.9	663.8
Other non-current assets	5.6	350.8	371.8	173.0
Non-current derivatives	5.12	74.0	96.5	118.1
Deferred tax assets	7	40.7	67.6	54.1
Non-current assets		22,915.4	22,497.1	13,860.7

Vast majority of properties are marked to market value.



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3.1.3 Consolidated cash-flow statements

In millions of euros	12/31/2016 Fair value	12/31/2015 Fair value
CASH-FLOWS FROM OPERATING ACTIVITIES		
Net income from consolidated companies	1,476.9	528.3
Elimination of expenditure and income with no cash effect or not related to operating activities		
— Depreciation, amortization and provisions	20.1	14.2
— Change in value of investment properties	-828.8	-883.3
— Goodwill impairment		922.6
— Capital gains and losses on asset disposals	-23.6	1.4
— Income taxes	225.6	204.6
— Share in earnings of equity method investees	-89.5	-97.6
— Reclassification of financial interests and other items	253.4	280.2
Gross cash-flow from consolidated companies	1,034.1	970.4
Paid taxes	-61.8	-25.5
Change in operating working capital	-16.1	-19.0
Cash-flows from operating activities	956.2	926.0

The company is not using the Direct Method, but its CFS is still slightly different - it adds back Book Income Taxes and then records a negative for Cash Taxes Paid, for example.

The Change in Value of Investment Properties is also huge.

Let's now take a look at extracts from the financial statements of RioCan, a Canadian REIT, and see how they differ from the statements of a U.S.-based REIT:

Funds from Operations (FFO) and Operating Funds from Operations (OFFO)

The following table presents a reconciliation of IFRS net income attributable to unitholders to FFO and OFFO on both a continuing and discontinued operations basis:

(thousands of dollars, except per unit amounts)	Three months ended December 31,		Year ended December 31,	
	2016	2015	2016	2015
Net income from continuing operations attributable to unitholders	\$ 178,472	\$ 199,796	\$ 683,060	\$ 416,892
Add back/(Deduct):				
Fair value (gains) losses, net	(44,371)	(1,183)	(182,888)	91,548
Non-controlling interest relating to fair value losses	—	43	91	674
Fair value (gains) losses included in equity accounted investments	(1,476)	468	846	676
Deferred income tax expense (recovery)	(3,000)	1,350	(3,850)	1,290
Internal leasing costs	2,663	2,340	10,931	9,750
Transaction (gains) losses on investment properties, net (i)	—	(4,608)	(6,075)	2,632
Transaction costs (i)	1,978	4,574	8,165	8,459
Preferred trust unit redemption (ii)	—	—	(4,304)	—
Preferred unit distributions	(1,757)	(3,397)	(8,667)	(13,590)
Foreign exchange loss	—	—	—	131
FFO from continuing operations	\$ 132,509	\$ 199,383	\$ 497,309	\$ 518,462

This Canadian REIT still calculates FFO, but the individual components of it differ significantly, with the Fair Value Gain or Loss as the largest adjustment now.

Adjusted Funds from Operations (AFFO)

The following table is a reconciliation of FFO to AFFO prepared based on the results of our continuing and discontinued operations for all periods shown:

(thousands of dollars, except per unit amounts)	Three months ended December 31,		Year ended December 31,	
	2016	2015	2016	2015
FFO (i)	\$ 132,174	\$ 220,978	\$ 547,879	\$ 622,364
Add back/(Deduct):				
Costs not capitalized on non-active developments (ii):				
Property operating costs	479	354	1,603	1,175
Interest costs	1,459	1,833	5,800	6,811
Demolition costs (iii)	—	487	891	2,164
Deduction of straight-line rents (iv)	(4,187)	(1,285)	(9,995)	(9,328)
Non-cash unit based compensation expense	706	1,765	1,640	5,135
Normalized capital expenditures:				
Leasing commissions and tenant improvements (v)	(6,250)	(6,250)	(25,000)	(25,000)
Capital expenditures recoverable from tenants	(3,750)	(3,750)	(15,000)	(15,000)
Capital expenditures not recoverable from tenants	(2,500)	(2,500)	(10,000)	(10,000)
(Gain) loss on sale of residential inventory	1,197	(1,285)	(74)	(2,594)
Target settlement proceeds, net	—	(88,267)	—	(88,267)
Preferred trust unit redemption (vi)	—	—	4,304	—
Long-term debt redemption costs	—	—	—	9,929
Transaction (gains) losses, net (vii)	—	8,478	(2,309)	8,478
Other transaction (gains) losses, net (viii)	99	(1,932)	1,444	(4,891)
AFFO	\$ 119,427	\$ 128,626	\$ 501,183	\$ 500,976

This REIT also calculates AFFO; the straight-lining of rent adjustment and the deduction for normalized CapEx are both standard.

As with all AFFO calculations, the rest ranges from "questionable" to "non-standard and highly variable among different REITs," which is why we tend not to use AFFO much.

RIOCAN REAL ESTATE INVESTMENT TRUST CONSOLIDATED STATEMENTS OF INCOME

(In thousands of Canadian dollars, except per unit amounts)

Year ended December 31,	Note	2016	2015
Revenue			
Rental revenue	16	\$ 1,103,884	\$ 1,039,068
Residential inventory sales		16,262	31,937
Property and asset management fees		13,186	16,731
		1,133,332	1,087,736
Operating costs			
Rental operating costs			
Recoverable under tenant leases		397,776	373,698
Non-recoverable costs		19,684	20,465
Residential inventory cost of sales		16,188	29,343
		433,648	423,506
Operating income		699,684	664,230
Other income			
Interest income		5,744	5,370
Income from equity accounted investments		9,972	10,378
Fair value gain (loss) on investment properties, net		182,888	(91,548)
Investment and other income	17	33,268	98,426
		231,872	22,626

This REIT is showing the Fair Value Gain / Loss in "Other Income" instead; this position for Net Income from Equity Investments is also a bit non-standard, but it still makes the same impact on Net Income all the way at the bottom.



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Year ended December 31,	Note	2016	2015
Operating activities			
Net income (loss) from:			
Continuing operations		\$ 683,151	\$ 417,566
Discontinued operations	4	147,687	(275,129)
Net income		830,838	142,437
Items not affecting cash:			
Depreciation and amortization		4,398	4,655
Amortization of straight-line rent		(8,006)	(9,328)
Unit-based compensation expense	13	1,640	5,135
Income from equity accounted investments	6	(9,972)	(6,233)
Fair value (gains) losses on investment properties, net	5	(199,787)	238,608
Deferred income taxes (recovery)		(234,525)	231,764
Transaction (gain) loss, net on disposition of:			
Available-for-sale securities		(14,040)	—
Canadian investment properties		(6,075)	2,631
U.S. investment properties		(65,116)	(7,529)
Adjustments for other changes in working capital items	28	156,069	12,676
Net operating cash flow activities		455,424	614,816

The D&A add-back is only for this REIT's *non-real-estate* Assets; the Fair Value adjustment is much bigger.

Despite these accounting differences, the *projection methods* for REITs remain fairly similar – in other words, you do not project the 3 statements much differently for a European vs. U.S. REIT.

As you'll see in the next section, you tend to split a REIT up by geographic or business segment and assume Acquisition, Development, and Disposition activity, Yields or Cap Rates on that activity, and modest changes to the REIT's same-store properties.

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REIT Financial Statement Projections (3-Statement Models)

We build many examples of REIT 3-statement models in this course, but the projection methodology is always quite similar: split up the REIT by geography or business segment, assume growth in Same-Store Rents, and then assume Acquisition, Development, and Redevelopment spending (and Disposition activity), and Yields and Margins on that spending to project the resulting NOI and Revenue.

The ideal setup depends on your available time, the purpose of the model, and the REIT's operations.

For example, if it's a retail REIT that owns only 2 properties, you would project revenue and expenses for those 2 properties, and you *might* forecast some Acquisition or Development activity if the company has indicated plans to pursue it.

On the other hand, if it's an industrial REIT that operates 50 properties in 10 cities, it would not make sense to split it up by geography or individual property because there would be too many numbers, and you wouldn't be able to find good data.

Instead, you would probably split this REIT into Same-Store, Acquisition, Development, Redevelopment, and Disposition segments and forecast the overall activity in each one.

If you are building a quick model that your MD wants to see in a few hours, you might do something very simple and project NOI and revenue for each segment in a few rows of Excel.

But if you're building a model for a client near the end of a deal, or you're valuing the company in a Fairness Opinion, then you might use hundreds of rows to forecast the segments, build in scenarios, and add other bells and whistles.

We'll walk through a few examples from the course below, including segment-level models, city or property-level models, and then examples of quarterly and half-year models:

Models by Business Segment

In the simplified case study for Park Hotels, we start by forecasting revenue and expenses for its Existing Hotels:

Existing Hotels - Revenue and Expenses:	Units:	Historical:			Projected:				
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21
Average Room Count:	# Rooms	30,668	30,652	30,485	30,485	30,485	30,485	30,485	30,485
Occupancy Rate:	%	80.9%	81.9%	81.0%	81.2%	81.4%	81.6%	81.8%	82.0%
Days in Period:	# Days	365	365	366	365	365	365	366	365
Average Daily Rate (ADR):	\$ / Room Night	\$ 185.41	\$ 194.59	\$ 198.62	\$ 204.58	\$ 209.69	\$ 214.93	\$ 219.23	\$ 223.62
ADR Growth %:	%	4.2%	5.0%	2.1%	3.0%	2.5%	2.5%	2.0%	2.0%
Revenue per Available Room (RevPAR):	\$ / Room Night	\$ 149.99	\$ 159.37	\$ 160.88	\$ 166.12	\$ 170.69	\$ 175.39	\$ 179.33	\$ 183.37
Available Room Nights:	# Room Nights	11,193,820	11,187,798	11,157,327	11,126,843	11,126,843	11,126,843	11,157,327	11,126,843
Occupied Room Nights:	# Room Nights	9,055,800	9,162,806	9,037,435	9,034,996	9,057,250	9,079,503	9,126,693	9,124,011
F&B Revenue per Occupied Room Night:	\$ / Room Night	\$ 71.11	\$ 75.41	\$ 79.56	\$ 83.14	\$ 86.46	\$ 89.49	\$ 92.17	\$ 94.94
F&B Revenue Growth %:	%	3.5%	6.0%	5.5%	4.5%	4.0%	3.5%	3.0%	3.0%
Other Revenue per Occupied Room Night:	\$ / Room Night	\$ 20.98	\$ 23.36	\$ 23.57	\$ 24.28	\$ 24.88	\$ 25.50	\$ 26.01	\$ 26.54
Other Revenue Growth %:	%	5.6%	11.3%	0.9%	3.0%	2.5%	2.5%	2.0%	2.0%
Room Margin:	%	72.8%	74.4%	74.0%	74.1%	74.2%	74.3%	74.4%	74.5%
Food & Beverage Margin:	%	29.5%	29.5%	30.0%	30.0%	30.0%	30.0%	30.0%	30.0%
Other Property-Level Expenses per Available Room:	\$ / Room	\$ 25,108	\$ 27,079	\$ 27,850	\$ 28,964	\$ 30,123	\$ 31,177	\$ 32,268	\$ 33,236
Property-Level Expenses Growth %:	%	4.9%	7.9%	2.8%	4.0%	4.0%	3.5%	3.5%	3.0%
Management Fees % Revenue:	%	3.1%	3.3%	3.3%	3.3%	3.3%	3.3%	3.3%	3.3%

Obviously, the rooms stay the same...

Very slight improvement, in-line with trends.

Growth rate in-line with inflation.

Slightly above inflation due to higher price elasticity.

In-line with inflation.

Very slight improvements, or nothing at all.

Above inflation to be more conservative.

Always the same %.



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This hotel REIT's revenue and expenses for Existing Hotels **barely change** over the 5-year period, which is the expected outcome – it would be odd if existing, stabilized properties were experiencing major changes. Next, we project the other segments:

Acquisitions:	Units:	Historical:			Projected:				
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21
Acquisitions, Net of Cash Acquired:	\$ M	\$ -	\$ 1,410	\$ -	\$ -	\$ 250	\$ 275	\$ 300	\$ 300
Cap Rate:	%				6.1%	6.1%	6.2%	6.3%	6.3%
Incremental Annual NOI:	\$ M				-	-	16	17	19
NOI Margin:	%				30.0%	30.0%	30.0%	30.0%	30.0%
Incremental Annual Revenue:	\$ M				-	-	52	58	63

Development & Redevelopment:	Units:	Historical:			Projected:				
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21
Renovation Spending:	\$ M	\$ -	\$ -	\$ -	\$ 50	\$ 60	\$ 70	\$ 80	\$ 90
Yield on Renovation Spending:	%				17.5%	17.5%	17.5%	17.5%	17.5%
Incremental Annual NOI:	\$ M				-	9	11	12	14
NOI Margin:	%				30.0%	30.0%	30.0%	30.0%	30.0%
Incremental Annual Revenue:	\$ M				-	29	35	41	47

Dispositions:	Units:	Historical:			Projected:				
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21
Proceeds from Asset Dispositions:	\$ M	\$ 15	\$ 1,866	\$ -	\$ 150	\$ 150	\$ 150	\$ 150	\$ 150
Gain / (Loss) on Sale of Real Estate Assets:	\$ M	-	143	-	(30)	(29)	(27)	(26)	(24)
Gain / (Loss) % Asset Proceeds:	%	0.0%	7.7%	0.0%	(20.0%)	(19.0%)	(18.0%)	(17.0%)	(16.0%)
Book Value of Dispositions:	\$ M	15	1,723	-	180	179	177	176	174
Cap Rate:	%				5.6%	5.6%	5.6%	5.6%	5.6%
Incremental Annual NOI Reduction:	\$ M				-	10	10	10	10
NOI Margin:	%				30.0%	30.0%	30.0%	30.0%	30.0%
Incremental Annual Revenue Reduction:	\$ M				-	34	33	33	33

These types of numbers might be based on spending in past years, comments in the company's investor presentation or earnings call transcripts, or the spending and Yields or Cap Rates of similar companies in this market.

The general idea is that the company should dispose of lower-yielding assets and develop, redevelop, and acquire higher-yielding ones, though that may or may not happen depending on the market environment.

"Incremental" means "in this year only," so we'll need to take the cumulative sum of these items later.

The significance of the Acquisition, Development, Redevelopment, and Disposition activity depends heavily on the specific REIT and the market cycle.

Some REITs focus on Developments, while others focus on Acquisitions; others might do little of either because they own only a few properties or because they don't feel market conditions are ideal.

The only "rule" is that your projections should be consistent with the company's plans, its historical performance, your industry research, and the metrics of peer companies.

For example, if we looked at Park Hotels' acquisitions going back ~10 years and recent hotel acquisitions by other companies, and the average Cap Rate was always between 7.0% and 9.0%, then the 6.1% to 6.4% range above would be difficult to justify.

We aggregate each segment into the total revenue and property-level expenses for this REIT:



Real Estate & REIT Modeling

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Consolidated Hotels:	Units:	Historical:			Projected:					
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21	
Revenue:										
(+) Rooms:	\$ M	\$ 1,679	\$ 1,783	\$ 1,795	\$ 1,848	\$ 1,899	\$ 1,951	\$ 2,001	\$ 2,040	
(+) Food & Beverage:	\$ M	644	691	719	751	783	813	841	866	
(+) Other:	\$ M	190	214	213	219	225	232	237	242	
(+) Revenue from Acquired Hotels:	\$ M	-	-	-	-	-	52	109	172	
(+) Revenue from Developed/Redeveloped Hotels:	\$ M	-	-	-	-	29	64	105	152	
(-) Lost Revenue from Disposed Hotels:	\$ M	-	-	-	-	(34)	(67)	(100)	(133)	
Total Revenue:	\$ M	2,513	2,688	2,727	2,819	2,903	3,045	3,194	3,340	
Property-Level Expenses:										
(-) Rooms:	\$ M	(457)	(456)	(466)	(479)	(490)	(502)	(512)	(520)	
(-) Food & Beverage:	\$ M	(454)	(487)	(503)	(525)	(548)	(568)	(589)	(606)	
(-) Other Property-Level:	\$ M	(770)	(830)	(849)	(883)	(918)	(950)	(984)	(1,013)	
(-) Management Fees:	\$ M	(77)	(89)	(91)	(94)	(97)	(102)	(107)	(111)	
(-) Expenses from Acquired Hotels:	\$ M	-	-	-	-	-	(36)	(77)	(121)	
(-) Expenses from Developed/Redeveloped Hotels:	\$ M	-	-	-	-	(20)	(45)	(74)	(106)	
(+) Expenses from Disposed Hotels:	\$ M	-	-	-	-	24	47	70	93	
Total Property-Level Expenses:	\$ M	(1,758)	(1,862)	(1,909)	(1,981)	(2,050)	(2,156)	(2,271)	(2,385)	
Net Operating Income (NOI):	\$ M	755	826	818	838	853	888	923	955	
NOI Margin:	%	30.0%	30.7%	30.0%	29.7%	29.4%	29.2%	28.9%	28.6%	
Gross Real Estate Assets:	\$ M	8,318	10,353	10,413	10,520	10,898	11,328	11,810	12,320	
Cap Rate:	%	9.9%	7.9%	8.0%	8.1%	8.2%	8.1%	8.1%		

We then use these projections to create a basic Income Statement for the REIT, and we project the missing items using simple percentages or straight-line assumptions:

Income Statement:	Units:	Historical:			Projected:				
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21
Total Revenue:	\$ M	\$ 2,513	\$ 2,688	\$ 2,727	\$ 2,819	\$ 2,903	\$ 3,045	\$ 3,194	\$ 3,340
Operating Expenses:									
(-) Property-Level Expenses:	\$ M	(1,758)	(1,862)	(1,909)	(1,981)	(2,050)	(2,156)	(2,271)	(2,385)
(-) Impairment Loss:	\$ M	-	-	(15)	-	-	-	-	-
(-) Depreciation & Amortization:	\$ M	(248)	(287)	(300)	(316)	(331)	(353)	(377)	(401)
(-) Corporate & Other:	\$ M	(67)	(96)	(85)	(88)	(91)	(95)	(100)	(104)
Total Operating Expenses:	\$ M	(2,073)	(2,245)	(2,309)	(2,385)	(2,471)	(2,604)	(2,748)	(2,890)
(+) Gain / (-) Loss on Sale of Assets, Net:	\$ M	-	143	-	(30)	(29)	(27)	(26)	(24)
Operating Income:	\$ M	440	586	418	404	403	413	421	426
(-) Net Interest Expense:	\$ M	(185)	(185)	(179)	(119)	(117)	(115)	(120)	(122)
(+/-) Other Items:	\$ M	43	16	(19)	-	-	-	-	-
Pre-Tax Income:	\$ M	298	417	220	285	286	299	301	305
(-) Income Taxes:	\$ M	(117)	(118)	(82)	-	-	-	-	-
Net Income:	\$ M	181	299	138	285	286	299	301	305
(-) Net Income Attributable to NCI:	\$ M	(5)	(7)	(6)	(6)	(6)	(6)	(6)	(6)
Net Income Attributable to Parent:	\$ M	\$ 176	\$ 292	\$ 132	\$ 279	\$ 280	\$ 293	\$ 295	\$ 299

Direct links from the segment-by-segment schedule for revenue and property-level expenses.

Projected using simple percentages of revenue; it's rarely worth the effort to create a detailed schedule for Depreciation unless the REIT owns very few properties.

Income Statement Drivers:	Units:	Historical:			Projected:				
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21
Depreciation & Amortization % Revenue:	%	9.9%	10.7%	11.0%	11.2%	11.4%	11.6%	11.8%	12.0%
Corporate & Other % Revenue:	%	2.7%	3.6%	3.1%	3.1%	3.1%	3.1%	3.1%	3.1%
Weighted Average Interest Rate on Debt:	%	4.4%	4.3%	4.4%	3.7%	3.7%	3.6%	3.6%	3.5%
Income Tax Rate:	%	39.3%	28.3%	37.3%	-	-	-	-	-



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Next, we project the Working Capital line items on the Balance Sheet, such as Accounts Receivable and Accounts Payable, typically using percentage-of-revenue or percentage-of-expense assumptions:

Balance Sheet Drivers:	Units:	Historical:			Projected:				
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21
Accounts Receivable % Revenue:	%	3.9%	4.5%	4.8%	5.0%	5.1%	5.2%	5.3%	5.4%
Other Assets % Revenue:	%	71.8%	9.2%	8.1%	8.0%	7.9%	7.8%	7.7%	7.6%
Accounts Payable % Operating Expenses:	%	7.0%	7.6%	7.2%	7.3%	7.3%	7.3%	7.3%	7.3%
Other Liabilities % Operating Expenses:	%	131.3%	120.7%	113.6%	110.0%	106.0%	100.0%	94.0%	88.0%

We simply follow historical trends here and extend them forward, or if no trend exists, we use an average, as with Accounts Payable.

We then fill in these items on the Balance Sheet, but we leave much of it blank because items such as Cash, Gross Real Estate Assets, Debt, and Shareholders' Equity depend on the Cash Flow Statement.

As usual, the Cash Flow Statement starts with Net Income, adjusts for non-cash charges and the Change in Working Capital, and then reflects investing and financing activities.

Most items are direct links to the Income Statement or the segment-by-segment schedule, but the Debt and Equity Issuances and Dividends take more time and effort:



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Cash Flow Statement:	Units:	Historical:			Projected:												
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21								
Cash Flow from Operations:																	
(+) Net Income Attributable to Parent:	\$ M	\$	176	\$	292	\$	132	\$	279	\$	280	\$	293	\$	295	\$	299
(+) Net Income Attributable to NCI:	\$ M		5		7		6		6		6		6		6		6
(+) Depreciation & Amortization:	\$ M		248		287		300		316		331		353		377		401
(+) Impairment Loss:	\$ M		-		-		15		-		-		-		-		-
(-) Gain / (+) Loss on Sale of Assets, Net:	\$ M		-		(143)		-		30		29		27		26		24
(+/-) Changes in Goodwill & Intangibles:	\$ M		-		-		-		-		-		-		-		-
(+/-) Other Items, Net:	\$ M		75		15		(17)		-		-		-		-		-
(+/-) Change in Working Capital:	\$ M		12		61		(37)		(6)		(8)		(24)		(31)		(48)
Cash Flow from Operations:	\$ M	\$	516	\$	519	\$	399	\$	624	\$	637	\$	655	\$	673	\$	681
Cash Flow from Investing:																	
(-) Recurring Maintenance CapEx:	\$ M	\$	(171)	\$	(226)	\$	(227)		(237)		(247)		(262)		(278)		(294)
(-) Acquisitions, Net of Cash Acquired:	\$ M		-		(1,410)		-		-		(250)		(275)		(300)		(300)
(-) Developments & Redevelopments:	\$ M		-		-		-		(50)		(60)		(70)		(80)		(90)
(-) Proceeds from Asset Dispositions:	\$ M		15		1,866		-		150		150		150		150		150
(+/-) Other Items, Net:	\$ M		36		-		17		-		-		-		-		-
Cash Flow from Investing:	\$ M	\$	(120)	\$	230	\$	(210)		(137)		(407)		(457)		(508)		(534)
Cash Flow from Financing:																	
(+/-) Change in Debt:	\$ M	\$	(14)	\$	(612)	\$	(765)		(55)		18		135		154		(198)
(+) Stock Issuances / (-) Repurchases:	\$ M		-		-		-		-		18		135		166		552
(+) Cash Contribution from Parent:	\$ M		-		-		987		-		-		-		-		-
(-) Cash Dividends Paid to Parent:	\$ M		(351)		(81)		(180)		-		-		-		-		-
(-) Cash Dividends to Shareholders:	\$ M		-		-		-		(429)		(439)		(460)		(476)		(493)
(+/-) Other Items, Net:	\$ M		(36)		(22)		(12)		-		-		-		-		-
Cash Flow from Financing:	\$ M	\$	(401)	\$	(715)	\$	30	\$	(484)	\$	(402)	\$	(190)	\$	(157)	\$	(139)
Effect of FX Rate Effects:	\$ M		(1)		(4)		2		(1)		(1)		(1)		(1)		(1)
(+) Net Change in Cash & Cash-Equivalents:	\$ M		(6)		30		221		3		(172)		7		7		7
(+) Beginning Cash:	\$ M		48		42		72		293		296		124		130		137
Ending Cash:	\$ M	\$	42	\$	72	\$	293	\$	296	\$	124	\$	130	\$	137	\$	144

Direct links to the Income Statement; in some cases, reversals of line items there.

Simple percentage of revenue assumption.

Direct links from the segment-by-segment schedule.

And these flow in from the CFS Drivers or Debt / Equity / Dividend schedule (see the next screenshot).

With the Debt, Equity, and Dividends, the idea is simple: this REIT has a minimum Cash balance equal to some percentage of its revenue or expenses (5% of operating expenses here). It also plans to issue Dividends as a percentage of its FFO (65% here).

We calculate the Cash balance *before* Debt and Equity financing based on the company's Beginning Cash balance + its Cash Flow Statement, excluding Debt/Equity issuances, but including Dividends.

If this Preliminary Ending Cash balance is below the minimum Cash balance, we assume the REIT issues Debt and Equity to boost its Cash up to the minimum level:

Cash Flow Statement Drivers:	Units:	Historical:			Projected:				
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21
Recurring Maintenance CapEx % Revenue:	%	6.8%	8.4%	8.3%	8.4%	8.5%	8.6%	8.7%	8.8%
(+) Net Income Attributable to Parent:	\$ M	\$ 176	\$ 292	\$ 132	\$ 279	\$ 280	\$ 293	\$ 295	\$ 299
(+) Depreciation & Amortization:	\$ M	\$ 248	\$ 287	\$ 300	\$ 316	\$ 331	\$ 353	\$ 377	\$ 401
(+) Impairment Loss:	\$ M	\$ -	\$ -	\$ 15	\$ -	\$ -	\$ -	\$ -	\$ -
(-) Gain / (+) Loss on Sale of Assets, Net:	\$ M	\$ -	\$ (143)	\$ -	\$ 30	\$ 29	\$ 27	\$ 26	\$ 24
(+/-) Other Adjustments:	\$ M	\$ -	\$ 18	\$ 35	\$ 35	\$ 35	\$ 35	\$ 35	\$ 35
Funds from Operations (FFO):	\$ M	\$ 424	\$ 454	\$ 482	\$ 659	\$ 675	\$ 708	\$ 733	\$ 758
Dividends % FFO:	%				65.0%	65.0%	65.0%	65.0%	65.0%
Planned Dividends:	\$ M				429	439	460	476	493
Scheduled Debt Maturities:	\$ M				55	-	-	12	750
Minimum Cash Balance:	\$ M				119	124	130	137	144
Cash Balance Before Debt/Equity Financing:	\$ M				296	87	(140)	(194)	(959)
Debt/Equity Funding Required:	\$ M				-	37	270	331	1,104
Debt Issued:	\$ M				-	18	135	166	552
Equity Issued:	\$ M				-	18	135	166	552
Share Count:	# Millions				197.605	198.263	203.122	209.083	228.959

Based on company disclosures, historical trends, etc.; Dividends are mandatory for a REIT so it maintains its status and avoids paying corporate taxes! Both these items reduce the preliminary Cash balance here, increasing the required Debt and Equity funding.

MAX(Min Cash - Cash Balance Before Debt/Equity, 0)

Simple 50/50 Debt/Equity split... and increase the share count for the Equity.



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With these items projected, we can return to the Cash Flow Statement and link them all in.

REITs tend to have low Cash balances because of the Dividend requirements, so it is not unusual to see a declining Cash balance or one that matches the minimum balance throughout the entire period.

With the Cash Flow Statement done, we can return to the Balance Sheet and finish the model:

Balance Sheet:	Units:	Historical:			Projected:				
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21
ASSETS:									
Gross Real Estate Assets:	\$ M	\$ 8,318	\$ 10,353	\$ 10,413	\$ 10,520	\$ 10,898	\$ 11,328	\$ 11,810	\$ 12,320
(-) Accumulated Depreciation:	\$ M	(1,407)	(1,677)	(1,872)	(2,188)	(2,519)	(2,872)	(3,249)	(3,650)
Net Real Estate Assets:	\$ M	6,911	8,676	8,541	8,332	8,379	8,456	8,562	8,671
Cash & Cash-Equivalents:	\$ M	42	72	293	296	124	130	137	144
Accounts Receivable:	\$ M	99	122	130	141	148	158	169	180
Goodwill & Other Intangibles:	\$ M	857	669	648	648	648	648	648	648
Other Assets:	\$ M	1,805	248	222	226	229	237	246	254
TOTAL ASSETS:	\$ M	\$ 9,714	\$ 9,787	\$ 9,834	\$ 9,642	\$ 9,528	\$ 9,630	\$ 9,762	\$ 9,897
LIABILITIES & EQUITY:									
Debt & Other Borrowings:	\$ M	4,253	4,109	3,222	3,167	3,185	3,320	3,474	3,276
Accounts Payable:	\$ M	146	171	167	174	180	190	201	211
Other Liabilities:	\$ M	2,722	2,710	2,622	2,623	2,620	2,604	2,583	2,543
Total Liabilities:	\$ M	7,121	6,990	6,011	5,964	5,985	6,115	6,257	6,030
Common Shareholders' Equity:	\$ M	2,617	2,821	3,872	3,721	3,580	3,546	3,530	3,887
(+) Noncontrolling Interests (NCI):	\$ M	(24)	(24)	(49)	(43)	(37)	(31)	(25)	(19)
Total Equity:	\$ M	2,593	2,797	3,823	3,678	3,543	3,515	3,505	3,868
TOTAL LIABILITIES & EQUITY:	\$ M	\$ 9,714	\$ 9,787	\$ 9,834	\$ 9,642	\$ 9,528	\$ 9,630	\$ 9,762	\$ 9,897
Balance Check:		OK!	OK!	OK!	OK!	OK!	OK!	OK!	OK!

Old balances and then all relevant CFS line items - Depreciation, Impairments, Gains / Losses, and All CapEx.

Ending line on CFS.

Old balance and Change in Debt from CFS.

Old balance and Net Income, "Other Items," Stock Issuances, Dividends, and FX Rate Effects.

Old balance and Net Income to NCI.

Old balances and then all relevant CFS line items - Depreciation, Impairments, Gains / Losses, and All CapEx.

Ending line on CFS.

Old balance and Change in Debt from CFS.

Old balance and Net Income, "Other Items," Stock Issuances, Dividends, and FX Rate Effects.

Old balance and Net Income to NCI.

This step takes us to the end of the 3-statement model. After this, we value the company with a DCF, NAV, and Public Comps, but you've seen all of that in the [Crash Course section above](#).

This same methodology applies to any REIT, so we're not going to walk through all the other models in the course step-by-step; instead, we'll point out the key differences for each one.

Added Complexities: Scenarios

When you build scenarios into a REIT projection model, you must understand that **everything is interrelated and based on overall market conditions**.

For example, if the property market is **strong**:

1. Rents will rise more quickly.
2. Margins will be higher.

3. The company will spend more on its main strategy (Acquisitions or Developments).
4. The company will dispose of more properties since it can get higher prices for them.
5. And Cap Rates will also be lower.

By contrast, in a **weaker** property market:

1. Rents will stagnate or even fall.
2. Margins will be lower.
3. The company will spend less on its core strategy.
4. The company will dispose of fewer properties since prices will be lower.
5. And Cap Rates will be higher.

Here's an example for Pure Industrial Real Estate Trust (PIRET), an industrial REIT based in Canada that grows by acquiring warehouse facilities:

Income Statement Drivers:	Units:	Historical:			Projected:				
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22
Same-Store Rent and Recovery Growth:	%	(0.7%)	2.6%	2.8%	2.5%	2.5%	2.0%	2.0%	2.0%
Upside	%				3.5%	4.0%	3.5%	3.0%	2.5%
Base	%				2.5%	2.5%	2.0%	2.0%	2.0%
Downside	%				(2.0%)	(2.0%)	4.0%	3.0%	2.0%
All Segments - Expenses % Rent and Recoveries:	%	30.0%	28.9%	27.3%	27.0%	26.5%	26.5%	26.0%	26.0%
Upside	%				26.0%	25.5%	25.0%	24.5%	24.0%
Base	%				27.0%	26.5%	26.5%	26.0%	26.0%
Downside	%				29.0%	31.0%	29.0%	28.0%	27.0%
Acquisition Spending:	\$ M	88.4	289.7	665.6	340.0	350.0	360.0	370.0	380.0
Upside	\$ M				500.0	475.0	450.0	425.0	400.0
Base	\$ M				340.0	350.0	360.0	370.0	380.0
Downside	\$ M				150.0	200.0	250.0	300.0	350.0
Weighted Average Going-In Cap Rate:	%	7.4%	6.5%	5.4%	5.7%	6.0%	6.3%	6.5%	6.8%
Upside	%				5.2%	5.5%	5.8%	6.2%	6.5%
Base	%				5.7%	6.0%	6.3%	6.5%	6.8%
Downside	%				8.0%	7.7%	7.5%	7.3%	7.0%
Incremental Annual NOI from Acquisitions:	\$ M				35.9	19.4	21.0	22.7	24.1
Incremental Annual Revenue from Acquisitions:	\$ M				49.2	26.4	28.6	30.6	32.5

Strongest rental growth in the Upside Case, and weakest growth in the Downside Case.

Slightly higher margins in the Upside Case, but not massive differences between cases.

Highest Acquisition spending in the Upside Case since that's the company's "core activity" - but the prices are also highest in the Upside Case because the overall market is strong.

Continuing to the Development Spending and Disposition Activity:

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Income Statement Drivers:	Units:	Historical:			Projected:				
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22
Development Spending:	\$ M	110.3	42.6	25.2	65.0	60.0	55.0	50.0	50.0
Upside	\$ M				100.0	90.0	80.0	70.0	60.0
Base	\$ M				65.0	60.0	55.0	50.0	50.0
Downside	\$ M				20.0	25.0	30.0	35.0	40.0
Weighted Average Stabilized Yield on Cost:	%	8.5%	7.2%	6.5%	6.5%	6.6%	6.7%	6.8%	6.9%
Upside	%				8.0%	7.8%	7.6%	7.4%	7.0%
Base	%				6.5%	6.6%	6.7%	6.8%	6.9%
Downside	%				5.5%	6.0%	6.3%	6.5%	6.7%
Incremental Annual NOI from Developments:	\$ M				1.6	4.2	4.0	3.7	3.4
Incremental Annual Revenue from Developments:	\$ M				2.2	5.7	5.4	5.0	4.6
Net Proceeds from Asset Dispositions:	\$ M	33.9	36.1	177.5	50.0	55.0	60.0	60.0	60.0
Upside	\$ M				100.0	90.0	80.0	70.0	60.0
Base	\$ M				50.0	55.0	60.0	60.0	60.0
Downside	\$ M				30.0	35.0	40.0	45.0	50.0
(+) Gain / (-) Loss on Sale of Assets:	\$ M	(2.5)	(0.9)	(4.6)	(1.5)	(1.7)	(1.5)	(1.5)	(1.5)
% Net Proceeds:	%	(7.4%)	(2.4%)	(2.6%)	(3.0%)	(3.0%)	(2.5%)	(2.5%)	(2.5%)
Weighted Average Exit Cap Rate:	%	5.6%	4.6%	5.7%	5.7%	5.6%	5.4%	5.3%	5.2%
Upside	%				4.5%	4.7%	4.9%	5.1%	5.2%
Base	%				5.7%	5.6%	5.4%	5.3%	5.2%
Downside	%				7.5%	6.7%	6.0%	5.5%	5.2%
Incremental Annual NOI Lost from Dispositions:	\$ M				(10.4)	(2.9)	(3.2)	(3.3)	(3.3)
Incremental Annual Revenue Lost from Dispositions:	\$ M				(14.3)	(4.0)	(4.3)	(4.5)	(4.4)

Developments are more of an afterthought for PIRET, but they still spend the most in the Upside Case and realize the highest yields in that case as well - stronger market with higher rents.

The company will be far more motivated to sell properties when the market is strong and prices are high (meaning low Cap Rates), and far less motivated to sell when the market is weaker and prices are lower.

Finally, we select overall Portfolio Cap Rates to mark this REIT's Investment Properties to fair market value in each year:

Income Statement Drivers:	Units:	Historical:			Projected:				
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22
General & Administrative % Revenue:	%	3.8%	4.3%	4.6%	4.8%	5.0%	5.2%	5.4%	5.5%
Implied Cap Rate on Overall Portfolio:	%	6.4%	6.7%	6.3%	6.2%	6.1%	6.0%	6.0%	6.0%
Upside	%				5.9%	5.5%	5.7%	5.8%	5.9%
Base	%				6.2%	6.1%	6.0%	6.0%	6.0%
Downside	%				7.5%	7.0%	6.5%	6.3%	6.1%
Fair Value of Investment Properties:	\$ M				3,579.6	4,052.6	4,583.9	5,047.9	
Old Fair Value +/- Changes in Investment Properties:	\$ M				3,508.5	3,961.9	4,438.4	4,978.5	
Fair Value Adjustments to Investment Properties:	\$ M				71.1	90.7	145.6	69.4	
Income Tax Rate:	%	8.9%	2.8%	(0.2%)	3.0%	3.0%	3.0%	3.0%	3.0%
NCI Net Earnings % Total Net Earnings:	%	1.1%	3.6%	1.5%	2.1%	2.1%	2.1%	2.1%	2.1%

This REIT follows IFRS, so we must mark its Investment Properties to fair market value each year; the value will rise modestly in the Base Case due to rising NOI and falling Cap Rates, it will increase more quickly in the Upside Case, and it will decline in the Downside Case.

As a result of these assumptions, this company looks different across the scenarios, but not *that much* different:



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Income Statement:	Units:	Historical:			Projected:				
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22
Revenue:									
(+) Same-Store Rent and Recoveries:	\$ M	\$ 171.4	\$ 186.9	\$ 225.5	\$ 233.4	\$ 242.7	\$ 251.2	\$ 258.7	\$ 265.2
(+) Rent and Recoveries from Acquisitions:	\$ M				48.6	83.5	118.3	152.9	187.5
(+) Rent and Recoveries from Developments:	\$ M				2.2	12.9	22.3	30.4	37.2
(-) Rent and Recoveries from Dispositions:	\$ M				(14.1)	(20.3)	(26.1)	(31.5)	(36.3)
Total Revenue:	\$ M	171.4	186.9	225.5	270.1	318.8	365.7	410.5	453.7
Property-Level Operating Expenses:									
(-) Same-Store Expenses:	\$ M	(51.3)	(54.1)	(61.5)	(60.7)	(61.9)	(62.8)	(63.4)	(63.7)
(-) Expenses from Acquisitions:	\$ M				(12.6)	(21.5)	(30.2)	(38.7)	(47.0)
(-) Expenses from Developments:	\$ M				(0.6)	(3.3)	(5.7)	(7.6)	(9.3)
(+) Expenses from Dispositions:	\$ M				3.7	5.3	6.7	8.0	9.2
Total Property-Level Operating Expenses:	\$ M	(51.3)	(54.1)	(61.5)	(70.2)	(81.5)	(92.0)	(101.7)	(110.8)
Net Operating Income (NOI):	\$ M	120.0	132.8	164.0	199.8	237.3	273.7	308.8	342.9
Revenue:									
(+) Same-Store Rent and Recoveries:	\$ M	\$ 171.4	\$ 186.9	\$ 225.5	\$ 231.1	\$ 236.9	\$ 241.6	\$ 246.5	\$ 251.4
(+) Rent and Recoveries from Acquisitions:	\$ M				49.2	75.6	104.2	134.8	167.3
(+) Rent and Recoveries from Developments:	\$ M				2.2	8.0	13.4	18.4	23.0
(-) Rent and Recoveries from Dispositions:	\$ M				(14.3)	(18.3)	(22.6)	(27.1)	(31.5)
Total Revenue:	\$ M	171.4	186.9	225.5	268.3	302.2	336.6	372.6	410.2
Property-Level Operating Expenses:									
(-) Same-Store Expenses:	\$ M	(51.3)	(54.1)	(61.5)	(62.4)	(62.8)	(64.0)	(64.1)	(65.4)
(-) Expenses from Acquisitions:	\$ M				(13.3)	(20.3)	(27.9)	(35.8)	(44.3)
(-) Expenses from Developments:	\$ M				(0.6)	(2.1)	(3.6)	(4.9)	(6.0)
(+) Expenses from Dispositions:	\$ M				3.9	4.9	6.1	7.2	8.4
Total Property-Level Operating Expenses:	\$ M	(51.3)	(54.1)	(61.5)	(72.4)	(80.3)	(89.4)	(97.5)	(107.3)
Net Operating Income (NOI):	\$ M	120.0	132.8	164.0	195.9	221.9	247.2	275.0	302.9
Revenue:									
(+) Same-Store Rent and Recoveries:	\$ M	\$ 171.4	\$ 186.9	\$ 225.5	\$ 221.0	\$ 216.6	\$ 225.2	\$ 232.0	\$ 236.6
(+) Rent and Recoveries from Acquisitions:	\$ M				50.6	68.0	89.7	115.7	145.7
(+) Rent and Recoveries from Developments:	\$ M				2.3	3.9	6.0	8.6	11.8
(-) Rent and Recoveries from Dispositions:	\$ M				(14.7)	(18.1)	(21.5)	(24.9)	(28.3)
Total Revenue:	\$ M	171.4	186.9	225.5	259.2	270.4	299.5	331.5	365.8
Property-Level Operating Expenses:									
(-) Same-Store Expenses:	\$ M	(51.3)	(54.1)	(61.5)	(64.1)	(67.1)	(65.3)	(65.0)	(63.9)
(-) Expenses from Acquisitions:	\$ M				(14.7)	(20.1)	(26.4)	(33.7)	(41.8)
(-) Expenses from Developments:	\$ M				(0.7)	(1.2)	(1.8)	(2.5)	(3.4)
(+) Expenses from Dispositions:	\$ M				4.3	5.3	6.3	7.2	8.2
Total Property-Level Operating Expenses:	\$ M	(51.3)	(54.1)	(61.5)	(75.2)	(83.1)	(87.2)	(93.9)	(100.8)
Net Operating Income (NOI):	\$ M	120.0	132.8	164.0	184.0	187.4	212.3	237.6	265.0

Upside Case:

Year 5 Revenue of \$454 million and NOI of \$343 million

Base Case:

Year 5 Revenue of \$410 million and NOI of \$303 million

Downside Case:

Year 5 Revenue of \$366 million and NOI of \$265 million

This is the basic idea behind scenarios in 3-statement models for REITs, but you can always make them more complex.

For example, you might split up the REIT's properties by *both* region *and* business segment, as we do in the 3-statement modeling and valuation case study for AvalonBay:



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Established (Same-Store) Communities:	Units:	Historical:			Projected:					CAGR:
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22	
Revenue:										
New England:	\$ M	\$ 170.3	\$ 226.7	\$ 233.1	\$ 238.9	\$ 244.7	\$ 250.3	\$ 255.8	\$ 261.2	2.3%
Metro NY/NJ:	\$ M	342.8	359.4	362.3	371.0	379.5	387.8	396.0	403.9	2.2%
Mid-Atlantic:	\$ M	209.0	233.7	221.1	225.5	229.8	233.9	237.9	241.7	1.8%
Pacific Northwest:	\$ M	67.9	72.5	84.2	87.6	90.9	94.2	97.5	100.8	3.7%
Northern California:	\$ M	273.4	319.1	336.8	345.5	354.2	362.7	371.0	379.2	2.4%
Southern California:	\$ M	252.5	291.6	337.9	353.1	368.3	383.4	398.3	413.1	4.1%
Total Revenue:	\$ M	1,315.9	1,503.0	1,575.3	1,621.5	1,667.2	1,712.2	1,756.4	1,799.7	2.7%
Revenue Growth Rates:										
New England:	%	4.5%	3.4%	2.4%	2.5%	2.4%	2.3%	2.2%	2.1%	Each market differs so the rental rates differ the strength of each
Upside	%				3.5%	3.4%	3.3%	3.2%	3.1%	
Base	%				2.5%	2.4%	2.3%	2.2%	2.1%	
Downside	%				(1.0%)	(0.5%)	2.5%	2.0%	1.5%	
Metro NY/NJ:	%	3.4%	2.9%	2.1%	2.4%	2.3%	2.2%	2.1%	2.0%	
Upside	%				3.0%	2.9%	2.8%	2.7%	2.6%	
Base	%				2.4%	2.3%	2.2%	2.1%	2.0%	
Downside	%				(1.5%)	(1.0%)	2.5%	2.0%	1.5%	

Each region differs historically, so the future rental growth rates should also differ based on the strength and competitiveness of each market.

We then assume a different NOI margin for each region and tweak those margins based on the selected scenario:

Established (Same-Store) Communities:	Units:	Historical:			Projected:					CAGR:
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22	
Net Operating Income (NOI):										
New England:	\$ M	107.2	145.7	150.3	154.8	159.3	163.4	167.5	171.1	2.6%
Metro NY/NJ:	\$ M	244.3	245.7	247.7	253.7	259.6	265.7	271.3	276.7	2.2%
Mid-Atlantic:	\$ M	145.5	162.2	153.8	156.7	159.7	162.8	165.6	168.5	1.8%
Pacific Northwest:	\$ M	48.8	52.4	61.5	64.2	66.8	69.4	72.0	74.5	3.9%
Northern California:	\$ M	210.2	244.5	257.7	264.0	270.2	276.3	282.3	288.5	2.3%
Southern California:	\$ M	173.9	207.5	241.5	253.2	264.8	276.4	287.6	298.6	4.3%
Total Net Operating Income (NOI):	\$ M	929.9	1,058.0	1,112.5	1,146.6	1,180.3	1,214.1	1,246.3	1,277.8	2.8%
Total NOI Margin:	%	70.7%	70.4%	70.6%	70.7%	70.8%	70.9%	71.0%	71.0%	
Baseline NOI Margins by Region:										
New England:	%	62.9%	64.2%	64.5%	64.8%	65.1%	65.3%	65.5%	65.5%	All bas histo most about but s up on
Metro NY/NJ:	%	71.3%	68.4%	68.4%	68.4%	68.4%	68.5%	68.5%	68.5%	
Mid-Atlantic:	%	69.6%	69.4%	69.5%	69.5%	69.5%	69.6%	69.7%	69.7%	
Pacific Northwest:	%	71.9%	72.3%	73.1%	73.3%	73.5%	73.7%	73.9%	73.9%	
Northern California:	%	76.9%	76.6%	76.5%	76.4%	76.3%	76.2%	76.1%	76.1%	
Southern California:	%	68.9%	71.2%	71.5%	71.7%	71.9%	72.1%	72.2%	72.3%	
Actual NOI Margins in Selected Scenario:										
New England:	%				64.8%	65.1%	65.3%	65.5%	65.5%	Slight the Up the sa Base slight Down
Metro NY/NJ:	%				68.4%	68.4%	68.5%	68.5%	68.5%	
Mid-Atlantic:	%				69.5%	69.5%	69.6%	69.6%	69.7%	
Pacific Northwest:	%				73.3%	73.5%	73.7%	73.9%	73.9%	
Northern California:	%				76.4%	76.3%	76.2%	76.1%	76.1%	
Southern California:	%				71.7%	71.9%	72.1%	72.2%	72.3%	

All based on historical trends; most regions stay about the same, but some move up or down.

Slightly higher in the Upside Case, the same in the Base Case, and slightly lower in the Downside Case.

The other parts of this model are fairly similar to the ones above for PIRET. For example, Disposition Activity is highest in the Upside Case, and Cap Rates are the lowest, as the company is incentivized to sell in strong markets.

One **difference** in this model is that the Development and Redevelopment projections are more complex since we assume that each new Development will:

1. Take several years to finish;



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2. Generate a certain amount of NOI and Revenue upon completion; and
3. Be reclassified into the "Other Stabilized" segment after stabilization, where its Yields and NOI Margins will rise.

Here's an example:

Development Properties:	Units:	Historical:			Projected:					
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22	
Average # of Development Years:	# Years	3.0								
Development Starts (Capitalized Spending):	\$ M	\$ 1,200.0	\$ 1,588.0	\$ 808.0	\$ 950.0	\$ 940.0	\$ 930.0	\$ 920.0	\$ 910.0	
Upside	\$ M				1,300.0	1,250.0	1,200.0	1,150.0	1,100.0	
Base	\$ M				950.0	940.0	930.0	920.0	910.0	
Downside	\$ M				500.0	650.0	750.0	800.0	850.0	
Annual Development Spending (Amortization of Starts):										
Year 1:	\$ M	400.0	400.0	400.0	-	-	-	-	-	
Year 2:	\$ M		529.3	529.3	529.3	-	-	-	-	
Year 3:	\$ M			269.3	269.3	269.3	-	-	-	
Year 4:	\$ M				316.7	316.7	316.7	-	-	
Year 5:	\$ M					313.3	313.3	313.3	-	
Year 6:	\$ M						310.0	310.0	310.0	
Year 7:	\$ M							306.7	306.7	
Year 8:	\$ M								303.3	
Total Annual Development Spending:	\$ M	400.0	929.3	1,198.7	1,115.3	899.3	940.0	930.0	920.0	
Cost of Completed Deliveries:	\$ M	1,313.0	510.0	1,900.0	700.0	808.0	950.0	940.0	930.0	
NOI from Completed Deliveries Before Stabilization:	\$ M	68.1	56.9	62.8	52.0	40.4	48.5	48.9	49.3	
Pre-Stabilized Yield on Cost of Completed Deliveries:	%	5.2%	11.2%	3.3%	7.4%	5.0%	5.1%	5.2%	5.3%	
Upside	%					6.5%	6.1%	5.7%	5.5%	
Base	%					5.0%	5.1%	5.2%	5.3%	
Downside	%					4.0%	4.3%	4.7%	5.0%	
Baseline NOI Margin:	%	63.9%	63.3%	65.8%	64.3%	64.3%	64.3%	64.3%	64.3%	
Actual NOI Margin in Selected Scenario:	%				64.3%	64.3%	64.3%	64.3%	64.3%	
Revenue from Completed Deliveries Before Stabilization:	\$ M	\$ 106.7	\$ 89.9	\$ 95.4	\$ 80.8	\$ 62.8	\$ 75.3	\$ 76.0	\$ 76.6	

AVB is primarily a Developer, so it focuses more on Development when the market is strong.

But each new Development takes 3 years to complete, so we create a "waterfall schedule" to show the total annual spending based on the current year + prior years' activities.

Upon completion, each Development will generate a certain amount of NOI based on market conditions and rents at that time.

It takes time for any new Development to stabilize; we assume 1 year between completion and stabilization in this model. At that point, we reclassify it into the "Other Stabilized Communities" segment:

Other Stabilized Communities:	Units:	Historical:			Projected:					CAGR:
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22	
Revenue:	\$ M	211.6	233.0	278.9	288.6	297.3	306.2	313.9	321.7	2.9%
Net Operating Income (NOI):	\$ M	140.2	165.1	196.7	199.8	205.8	212.0	217.3	222.7	2.5%
Revenue Growth Rate:	%	N/A	10.1%	19.7%	3.5%	3.0%	3.0%	2.5%	2.5%	
NOI Margin:	%	66.2%	70.9%	70.5%	69.2%	69.2%	69.2%	69.2%	69.2%	
Beginning Stabilized Development Assets:	\$ M				-	1,900.0	2,600.0	3,408.0	4,358.0	
(+) Additions from Development Properties:	\$ M				1,900.0	700.0	808.0	950.0	940.0	
Ending Stabilized Development Assets:	\$ M				1,900.0	2,600.0	3,408.0	4,358.0	5,298.0	29.2%
Revenue:	\$ M				170.2	236.6	315.1	409.2	497.5	30.8%
Net Operating Income (NOI):	\$ M				117.8	163.8	218.1	283.3	344.4	30.8%
Yield on Stabilized Development Assets:	%	6.7%	6.7%	6.1%	6.2%	6.3%	6.4%	6.5%	6.5%	
Upside	%				7.5%	7.2%	6.9%	6.8%	6.7%	
Base	%				6.2%	6.3%	6.4%	6.5%	6.5%	
Downside	%				5.0%	5.3%	5.7%	6.0%	6.3%	
Baseline NOI Margin:	%				69.2%	69.2%	69.2%	69.2%	69.2%	
Actual NOI Margin in Selected Scenario:	%				69.2%	69.2%	69.2%	69.2%	69.2%	

The new Developments go here as they are delivered and completed, boosting the total Assets in this segment.

And their Yields and Margins also increase as they stabilize.



As a result, the NOI and Revenue in the “Other Stabilized” segment grow significantly over this period, and the Other Stabilized Revenue grows from 13% of Total Revenue to 29% by Year 5 in the Base Case.

Besides these added complexities, the rest of the model is very similar to the example above for Park Hotels: link to the segment-level projections for rental income, property-level expenses, and CapEx, make Dividends a percentage of FFO, and assume a mix of Debt and Equity issuances to meet the company’s funding requirements and maintain its minimum Cash balance.

Models by Individual City or Property

Another option is to base the 3-statement model on the individual cities in which the REIT operates, or the individual properties it owns.

This option tends to work best if the REIT discloses the data this way and the REIT’s business is **not** expected to change significantly over the next few years.

For example, maybe the REIT owns 25 properties in 2 cities and it is planning to acquire 2-3 new properties over the next few years, but it’s not expecting any *massive* changes, such as an additional 50-100% of gross leasable area.

This REIT might also plan to focus on just one type of activity, such as Acquisitions or Developments, and might not plan to do much in the other business segments.

If the REIT is planning to engage fully in all possible activities (Same-Store, Acquisitions, Developments, Redevelopments, and Dispositions), then it will be cumbersome to create city-by-city projections because you’ll have to forecast *each activity for each city*.

If that’s the case, then you’re better off [following the first section here](#) and splitting up the REIT by business segment rather than geography.

If the REIT operates only a handful of properties (e.g., 2-3 rather than 10-15+), then it’s viable to forecast each property separately and aggregate the results.

Even if the REIT wants to acquire or develop another property, it’s still viable to project each property separately as long as it’s just *one* other property... or maybe 1-2.

Here are a few examples of each method, starting with city-by-city projections for Colonial (the Spanish and French REIT):

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Barcelona:	Units:	Historical:			Projected:					
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22
Previous Year - Annual Gross Rental Income:	€ M		28	27	30	31	36	37	43	44
(+) Like-for-Like:	€ M		1	3	1	1	1	1	1	1
(+) Acquisitions, Developments, and Divestments:	€ M		1	-	-	4	-	5	-	-
(+/-) Others and Compensation:	€ M		(3)	-	-	-	-	-	-	-
Current Year - Annual Gross Rental Income:	€ M	28	27	30	31	36	37	43	44	45
Net Rental Income:	€ M	23	23	28	29	34	35	40	41	42
NRI % Gross Rental Income:	%	82.1%	85.2%	93.3%	93.5%	93.7%	93.9%	94.0%	94.0%	94.0%
Like-for-Like Increase in Rental Income:	%	(4.6%)	1.9%	10.0%	2.7%	2.7%	2.7%	2.4%	2.4%	2.2%
Gross Leasable Area (GLA) - Stabilised Properties:	sq. m.	184,342	186,780	188,766	188,766	188,766	188,766	188,766	188,766	188,766
Occupancy Rate:	%	77.0%	89.0%	97.0%	97.2%	97.4%	97.6%	97.8%	98.0%	98.0%
Occupied Square Metres:	sq. m.	141,943	166,234	183,103	183,481	183,858	184,236	184,613	184,991	184,991
Average Gross Rent per Square Metre per Month:	€ / sq. m. / Mo.	16.44	13.54	13.65	13.99	14.34	14.70	15.03	15.37	15.72
Change in Average Gross Rent per Square Metre per Month:	%		(17.66%)	0.87%	2.50%	2.50%	2.50%	2.25%	2.25%	2.25%
Acquired/Developed Gross Square Metres Put Into Operation:	sq. m.				0	24,551	0	28,737	0	0
Acquisition/Development Costs per Square Metre:	€ / sq. m.				3,136	-	2,257	-	-	-
Total Acquisition/Development Costs (Growth CapEx):	€ M				77	-	65	-	-	-
Yield on Cost:	%				N/A	5.5%	N/A	8.0%	N/A	N/A
Asset Valuation:	€ M	582	676	761	897	933	1,101	1,127	1,152	
Gross Yield:	%	4.64%	4.44%	4.05%	4.00%	3.95%	3.90%	3.90%	3.90%	
Change in Fair Value of Investment Property:	€ M				59	36	103	27	25	

The setup here is a bit different, and we plot the change in Gross Rental Income each year based on the Same-Store (or "Like-for-Like") increase and the change from Acquisitions, Developments, and Divestments.

And there's only a modest amount of Acquisition / Development activity, so the Rental Income doesn't change *that much* as a result of this.

And since it's an IFRS-based REIT, we must mark the Assets in each segment to fair market value each year.

The trickiest part of this one is the "Like-for-Like" row at the top, which we calculate with:

Like-for-Like Rental Income = (Occupied Square Meters This Year in Same-Store Properties + Cumulative Acquired and Developed Square Meters Put into Operation So Far) * Average Gross Rent Per Square Meter per Year / 1,000,000 – Previous Year's Gross Rental Income

This formula is the only way to capture *all* the items that affect Gross Rental Income in the current year: The Occupancy Rate change, additional space from Acquisitions and Developments, and the change in the Average Gross Rent.

We do the same thing for the company's other segments – "Madrid and the Rest of Spain" and Paris – and then sum up the results. These other segments use the same setup, but different numbers.

One small difference is that in France, properties are valued based on *Net Yields*, i.e., (Rental Income minus Expenses) / Property Value, while in Spain, they are valued based on *Gross Yields*, or Rental Income / Property Value.

That doesn't change much, but you have to be careful to tweak the formulas for the Asset Valuation at the bottom instead of just copying and pasting them all the way down.

Here are the consolidated results for the REIT:



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Consolidated Property Results:	Units:	Historical:			Projected:						
		FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY21	
Revenue:											
(+) Barcelona:	€ M	€ 28	€ 27	€ 30	€ 31	€ 36	€ 37	€ 43	€ 44	€ 45	
(+) Madrid and Rest of Spain:	€ M	32	35	43	48	52	56	58	60	61	
(+) Paris:	€ M	152	169	198	211	214	219	237	252	257	
Total Revenue:	€ M	212	231	271	290	302	312	338	356	363	
Property-Level Expenses:											
(-) Barcelona:	€ M	(5)	(4)	(2)	(2)	(2)	(2)	(3)	(3)	(3)	
(-) Madrid and Rest of Spain:	€ M	(4)	(4)	(5)	(5)	(5)	(6)	(6)	(5)	(6)	
(-) Paris:	€ M	(13)	(14)	(10)	(11)	(11)	(11)	(12)	(13)	(13)	
Total Property-Level Expenses:	€ M	(22)	(22)	(17)	(18)	(18)	(19)	(20)	(21)	(21)	
Net Rental Income (NRI):	€ M	190	209	254	272	283	293	318	335	342	
NRI Margin:	%	89.6%	90.5%	93.7%	93.8%	93.9%	94.0%	94.1%	94.2%	94.2%	
Fair Value of Investment Property:	€ M	5,294	6,824	7,770	8,148	8,601	9,510	10,093	10,379		
Gross Yield:	%	4.4%	4.0%	3.7%	3.7%	3.6%	3.6%	3.5%	3.5%		
Net Yield:	%	3.9%	3.7%	3.5%	3.5%	3.4%	3.3%	3.3%	3.3%		
Change in Fair Value of Investment Property:	€ M				199	339	530	319	204		
Total Acquisition/Development Costs (Growth CapEx):	€ M				122	51	311	189	-		

We need to link in the Fair Value of Total Investment Property from the Balance Sheet because of Recurring Maintenance CapEx and a few other smaller items that do not appear in these segment-by-segment projections.

And then we use these results to create a 3-statement model similar to all the others, with a few minor differences because of metrics like EPRA Earnings and EPRA NAV rather than FFO and NAV.

For an example of property-by-property projections for a REIT, we'll walk through the Debt vs. Equity case study on SPH REIT in Singapore, which owned only 2 properties at the time of the case study.

Here are the projections for the company's first property, Paragon:

A B		C	D	K	L	M	N	O	P	Q	R	S
Paragon:		Units:	Projected:									
			2H18	1H19	2H19	1H20	2H20	1H21	2H21	1H22	2H22	
Gross Revenue:	\$ M	\$ 86.7	\$ 87.0	=+K32*(1-M40)+M40*(1+M42)	\$ 87.9	\$ 88.0	\$ 88.0	\$ 88.0	\$ 88.0	\$ 88.1		
Net Property Income:	\$ M	69.2	69.5	69.5	70.0	70.1	70.3	70.4	70.4	70.5		
NPI Margin:	%	79.8%	79.9%	79.9%	80.0%	80.0%	80.0%	80.0%	80.0%	80.0%		
Base	%	79.8%	79.9%	79.9%	80.0%	80.0%	80.0%	80.0%	80.0%	80.0%		
Downside	%	78.5%	78.0%	78.0%	78.5%	78.5%	79.0%	79.0%	79.5%	79.5%		
Extreme Downside	%	77.5%	77.5%	78.0%	78.0%	78.5%	78.5%	79.0%	79.0%	79.0%		
% New/Renewal Leases:	%	8.8%	15.2%	7.4%	16.1%	17.5%	12.0%	17.0%	4.0%	2.0%		
Rental Reversion for New/Renewal Leases:	%	5.0%	4.5%	4.5%	4.0%	4.0%	3.0%	3.0%	2.5%	2.5%		
Base	%	5.0%	4.5%	4.5%	4.0%	4.0%	3.0%	3.0%	2.5%	2.5%		
Downside	%	(5.0%)	(5.0%)	(5.0%)	(3.0%)	(2.0%)	4.0%	3.5%	3.0%	2.5%		
Extreme Downside	%	(7.5%)	(7.5%)	(5.0%)	(5.0%)	(3.0%)	5.0%	4.0%	3.0%	2.5%		
Asset Valuation:	\$ M	2,605.9	2,541.1	2,497.8	2,447.6							
Applicable Forward Net Operating Income (NOI):	\$ M	116.0	116.9	117.4	117.5							
Capitalisation Rate:	%	4.45%	4.60%	4.70%	4.80%							
Base	%	4.45%	4.60%	4.70%	4.80%							
Downside	%	5.25%	5.10%	4.95%	4.80%							
Extreme Downside	%	5.75%	5.40%	5.15%	4.80%							
Change in Fair Value of Investment Property:	\$ M	(89.1)	(64.9)	(43.2)	(50.2)							

Most of the Rental Income is "locked in" because of long-term contracts, so only a small percentage comes up for renewal each year and might potentially change. Therefore, this property's rental income stays remarkably stable even in a recession.

Margins are also extremely stable because this is a huge shopping mall in a high-demand area.

So, changing Cap Rates based on markets conditions actually make the biggest impact here.

We use the same approach, but with different numbers, for the other segments, including a new property that SPH REIT plans to acquire (the Seletar Mall), and then we sum up everything at the top:



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SPH REIT - Revenue and Expense Projections

(\$ SGD in Millions Except Per Share and Per Unit Amounts in SGD as Stated)

Consolidated Property Results:	Units:	Projected:									
		2H18	1H19	2H19	1H20	2H20	2H21	2H21	2H21	2H21	2H21
Gross Revenue:											
(+) Paragon:	\$S M	\$S 86.7	\$S 87.0	\$S 87.0	\$S 87.5	\$S 87.6	\$S 87.9	\$S 88.0	\$S 88.0	\$S 88.1	\$S 88.1
(+) Clementi:	\$S M	19.9	20.7	19.9	21.0	20.2	21.0	20.3	21.0	20.3	20.3
(+) Seletar Mall:	\$S M	-	18.0	19.0	18.2	19.2	18.3	19.3	18.5	19.5	19.5
Total Revenue:	\$S M	106.6	125.7	125.9	126.7	127.0	127.2	127.7	127.4	127.9	127.9
Property-Level Expenses:											
(-) Paragon:	\$S M	(17.5)	(17.5)	(17.5)	(17.5)	(17.5)	(17.6)	(17.6)	(17.6)	(17.6)	(17.6)
(-) Clementi:	\$S M	(5.0)	(5.2)	(5.0)	(5.2)	(5.1)	(5.3)	(5.1)	(5.3)	(5.1)	(5.1)
(-) Seletar Mall:	\$S M	-	(5.3)	(5.7)	(5.3)	(5.6)	(5.1)	(5.4)	(5.0)	(5.3)	(5.3)
Total Property-Level Expenses:	\$S M	(22.5)	(28.0)	(28.2)	(28.0)	(28.1)	(28.0)	(28.1)	(27.8)	(28.0)	(28.0)
Net Property Income (NPI):	\$S M	84.1	97.7	97.7	98.7	98.9	99.3	99.6	99.6	99.9	99.9
NPI Margin:	%	78.9%	77.8%	77.6%	77.9%	77.8%	78.0%	78.0%	78.2%	78.1%	78.1%
Fair Value of Investment Property:	\$S M	3,701.9	3,653.4	3,621.5	3,571.4	3,571.4	3,571.4	3,571.4	3,571.4	3,571.4	3,571.4
Gross Yield:	%	6.8%	6.9%	7.0%	7.1%	7.1%	7.1%	7.1%	7.1%	7.1%	7.1%
Net Yield:	%	5.3%	5.4%	5.5%	5.6%	5.6%	5.6%	5.6%	5.6%	5.6%	5.6%
Change in Fair Value of Investment Property:	\$S M	(76.1)	(48.6)	(31.8)	(50.2)	(50.2)	(50.2)	(50.2)	(50.2)	(50.2)	(50.2)

Revenue and NPI jump up immediately following the acquisition in 2H 18, but stay remarkably stable after that because of this property's long-term leases.

After we have these results, we then build a 3-statement model for SPH REIT that's similar to all the others, with a few minor differences in its FFO-equivalent metric, "Income Available for Distribution."

Quarterly and Half-Year Models

Finally, you can also build 3-statement models for REITs based on non-annual periods, such as quarters, half-years, or months.

These models are most useful when the REIT has high concentrations of lease expirations in certain periods, and you need to analyze the REIT's compliance with credit stats and ratios in each period.

For example, if a REIT wants to issue Debt, but the lenders are concerned that it might fall below 3x EBITDA / Interest in certain periods due to lease-expiration risk, a quarterly model could help assess the real risk.

You would create Base, Downside, and Extreme Downside Cases to show the possible outcomes of these lease expirations (e.g., renewal vs. non-renewal and a short amount of time required to find a new tenant vs. non-renewal and significant time required to find a new tenant), and you would evaluate the EBITDA / Interest in each case.

An annual model would not be granular enough to capture what happens in specific quarters; the company's EBITDA / Interest might look fine at the end of the year, but maybe not in Q2.

In the SPH REIT model above, the half-year periods matter because the percentage of expiring leases differs in each period for each property. As a result, some periods are riskier than others.



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However, since the margins stay very consistent and the properties are marked to fair market value only once per year, the half-year nature doesn't make *that much* of a difference.

With some items on the full financial statements, the percentages vary from one period to the next:

Income Statement Drivers:	Units:	Historical:					Projected:						
		1H16	2H16	1H17	2H17	1H18	2H18	1H19	2H19	1H20	2H20	1H21	2H21
Management Fees % NPI:	%	10.1%	10.2%	9.9%	10.0%	9.9%	10.1%	9.9%	10.1%	9.9%	10.1%	9.9%	10.1%
Trust Expenses % Gross Revenue:	%	0.9%	0.6%	0.8%	0.6%	0.8%	0.6%	0.8%	0.6%	0.8%	0.6%	0.8%	0.6%
Finance Income % Prior Cash Balance:	%			1.05%	1.10%	1.17%	1.17%	1.17%	1.17%	1.17%	1.17%	1.17%	1.17%
Finance Costs % Prior Debt Balance:	%			2.81%	2.83%	2.85%	2.85%	3.00%	3.00%	3.00%	3.00%	3.00%	3.00%
Income Tax Rate:	%	-	-	-	-	-	-	-	-	-	-	-	-

We reflect this variation between half-year periods (1H of each year is consistently higher for the Management Fees and consistently lower for Trust Expenses) in the projections. But many other items on the statements barely change or do not follow a "seasonal" trend in the same way.

In the Digital Realty / DuPont Fabros merger model, we build quarterly models for the Buyer and Seller, and the merger model itself is also quarterly, with a transaction close date in Q3.

We didn't have to do it this way, especially since the quarterly view of these companies does not differ that much from the annual one. However, there are a few differences worth noting:

First, the Same-Store Rental Growth, Property Expenses % Rental Income, and Tenant Reimbursements % Property Expenses differ slightly from quarter to quarter, primarily because most of Digital Realty's lease expirations and new tenant move-ins happen in Q1 of each year.

The company's Development Spending and Yields, by contrast, increase modestly over time but do not follow a trend linked to specific quarters:

Income Statement Drivers:	Units:	Historical:					Projected:											
		1Q 16	2Q 16	3Q 16	4Q 16	1Q 17	2Q 17	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19	2Q 19	3Q 19	4Q 19	
Same-Store, Same-Capital Rent - YoY Growth:	%	1.8%	(0.3%)	(1.0%)	(1.3%)	(0.1%)	(0.2%)	(0.5%)	(1.0%)	1.5%	(0.1%)	(0.3%)	(0.8%)	1.3%	(0.1%)	(0.3%)	(0.6%)	
Property Expenses % Rental Income:	%	50.1%	51.1%	52.2%	52.0%	50.0%	51.0%	52.0%	52.0%	50.0%	51.0%	52.0%	52.0%	50.0%	51.0%	52.0%	52.0%	
Tenant Reimbursements % Property Expenses:	%	45.3%	45.7%	46.4%	42.3%	43.2%	45.0%	46.0%	42.0%	43.0%	45.0%	46.0%	42.0%	43.0%	45.0%	46.0%	42.0%	
NOI Margin on Existing Properties:	%	59.2%	58.6%	58.0%	57.3%	58.9%	58.5%	58.0%	57.3%	58.8%	58.5%	58.0%	57.3%	58.8%	58.5%	58.0%	57.3%	
Development Spending:	\$ M	These growth rates and expense margins all differ because of differing lease expirations in each quarter...					180.0	180.0	180.0	190.0	190.0	200.0	200.0	215.0	215.0	215.0	215.0	
Completed & Stabilized Developments:	\$ M						120.0	130.0	140.0	135.0	180.0	180.0	180.0	190.0	190.0	200.0	200.0	
Average Stabilized Yield:	%						5.0%	5.0%	5.0%	5.5%	5.5%	5.5%	5.5%	6.0%	6.0%	6.0%	6.0%	
NOI from Stabilized Development:	\$ M						6.0	6.5	7.0	7.4	9.9	9.9	9.9	11.4	11.4	12.0	12.0	
Rent + Tenant Reimbursements from Developments:	\$ M						10.3	11.2	12.2	12.6	16.9	17.1	17.3	19.4	19.5	20.7	20.9	
Property Expenses from Developments:	\$ M						4.3	4.7	5.2	5.2	7.0	7.2	7.4	8.0	8.1	8.7	8.9	
Tenant Reimbursements from Developments:	\$ M						1.9	2.2	2.2	2.2	3.2	3.3	3.1	3.4	3.6	4.0	3.8	
Rental Income from Developments:	\$ M						8.3	9.0	10.0	10.4	13.8	13.8	14.2	15.9	15.8	16.7	17.2	

These growth rates and expense margins all differ because of differing lease expirations in each quarter...

But the Development Spending, Completed & Stabilized Developments, and Stabilized Yields just tick up modestly over time as the company expands.



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We do **not** have to multiply or divide these numbers by 4 because the Development Deliveries are *only for the current quarter*. So, these assumptions are already in “quarterly form.”

Digital Realty at this time was primarily a developer, so we focused on Same-Store and Development activities and ignored Acquisitions and Dispositions.

Income Statement Drivers:	Units:	Historical:					Projected:										
		1Q 16	2Q 16	3Q 16	4Q 16	1Q 17	2Q 17	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19	2Q 19	3Q 19	4Q 19
Property Management Fees % Rental Income:	%	0.5%	0.3%	0.4%	0.4%	0.5%	0.3%	0.4%	0.4%	0.5%	0.3%	0.4%	0.4%	0.5%	0.3%	0.4%	0.4%
Interconnection Fees - YoY Growth Rate:	%	3348.1%	3205.7%	3164.5%	32.0%	21.9%	15.0%	15.0%	12.5%	12.5%	10.0%	10.0%	7.5%	7.5%	5.0%	5.0%	5.0%
General & Administrative % Revenue:	%	6.2%	6.6%	8.4%	7.1%	6.3%	6.5%	8.0%	7.0%	6.0%	6.5%	8.0%	7.0%	6.0%	6.5%	8.0%	7.0%
Depreciation & Amortization % Revenue:	%	33.5%	34.1%	32.6%	30.6%	32.1%	34.0%	33.0%	31.0%	32.5%	34.0%	33.0%	31.0%	32.5%	34.0%	33.0%	31.0%
Annualized Interest Rate on Debt:	%		4.1%	4.1%	3.8%	3.8%	3.8%	3.9%	4.0%	4.0%	4.1%	4.1%	4.2%	4.3%	4.3%	4.4%	4.4%
Annualized Preferred Stock Coupon Rate:	%		7.0%	6.7%	6.9%	6.9%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%
Income Tax Rate:	%	3.3%	4.2%	1.6%	2.3%	2.6%	4.2%	1.6%	2.3%	2.6%	4.2%	1.6%	2.3%	2.6%	4.2%	1.6%	2.3%
Equity Investment Income - YoY Growth Rate:	%	(11.7%)	22.1%	(0.4%)	42.8%	30.6%	20.0%	15.0%	10.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%
NCI Net Income - YoY Growth Rate:	%	(62.7%)	(77.2%)	260.8%	(277.5%)	30.7%	20.0%	15.0%	10.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%

Some of these percentages clearly vary by quarter due to the timing of leases and certain business activities, but others do not. For example, we assume simple declining growth rates for Interconnection Fees (specific to data center REITs) and Equity Investment and NCI Net Income, and rising interest rates on the Debt and a constant Preferred Stock Coupon Rate.

There is no universal rule for which revenue and expenses vary from quarter to quarter, so you should review the company’s historical performance to find the trends.

Another difference is that when we project Balance Sheet items, we use **annualized** Income Statement numbers in the assumptions – either use the LTM figures or multiply the current quarter’s results by 4.

We multiply the current quarter’s results by 4 here to avoid gathering even more data:

Balance Sheet Drivers:	Units:	Historical:					Projected:										
		1Q 16	2Q 16	3Q 16	4Q 16	1Q 17	2Q 17	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19	2Q 19	3Q 19	4Q 19
Accounts Receivable % Annualized Revenue:	%	8.9%	8.1%	9.5%	8.8%	8.9%	8.1%	9.5%	8.8%	8.9%	8.1%	9.5%	8.8%	8.9%	8.1%	9.5%	8.8%
Other Assets % Annualized Revenue:	%	10.9%	16.2%	11.3%	11.3%	10.0%	11.5%	11.3%	11.3%	11.0%	11.5%	11.3%	11.3%	11.0%	11.5%	11.3%	11.3%
AP & Accruals % Annualized Cash Expenses:	%	65.7%	65.9%	81.6%	97.4%	84.9%	85.0%	83.0%	90.0%	87.5%	85.0%	83.0%	90.0%	87.5%	85.0%	83.0%	90.0%
Other Liabilities % Annualized Cash Expenses:	%	17.6%	=G130/(SUM(G542:G544)*Num_Periods)				15.6%	16.5%	17.2%	18.4%	15.6%	16.5%	17.2%	18.4%	15.6%	16.5%	17.2%

We annualize by multiplying the current quarter's numbers by 4 ("Num_Periods")...

And then in future periods, we reflect the quarterly variations in the percentages. To project these items on the Balance Sheet, we multiply the percentage by the corresponding Income Statement line item for the quarter by 4.

This setup is **not** unique to REITs – it’s standard practice in any quarterly model in any industry.

One final difference is that REITs do not necessarily pay fixed Dividends each quarter. Some might pay Dividends only once or twice per year, or they might opt for higher Dividends in some quarters and lower Dividends in others.

As a result, the Dividend payout patterns differ, and the REIT may need to raise a different amount of Debt and Equity depending on the quarter:

Cash Flow Statement Drivers:	Units:	Historical:					Projected:										
		1Q 16	2Q 16	3Q 16	4Q 16	1Q 17	2Q 17	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19	2Q 19	3Q 19	4Q 19
(+) Net Income to Common:	\$ M	\$ 39.1	\$ 28.0	\$ 187.3	\$ 77.7	\$ 66.1	\$ 43.6	\$ 47.0	\$ 56.2	\$ 70.9	\$ 46.8	\$ 53.0	\$ 66.7	\$ 77.3	\$ 53.6	\$ 58.1	\$ 75.8
(+) Noncontrolling Interests in Op. Partnership:	\$ M	0.7	0.5	3.0	1.2	0.9	0.9	0.7	3.6	1.1	1.1	1.0	0.7	3.8	1.2	1.1	1.0
(+) Real Estate-Related Depreciation & Amortization:	\$ M	166.9	167.0	175.3	173.5	173.4	176.7	185.9	176.3	195.3	196.1	206.6	196.0	221.5	221.4	232.3	220.8
(+/-) Other Adjustments:	\$ M	-	6.1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
(+) Unconsolidated JV Depreciation:	\$ M	2.8	2.8	2.8	2.8	2.8	2.6	2.4	2.6	2.8	2.7	2.6	2.7	2.9	2.8	2.7	2.9
(-) Gain / (+) Loss on Real Estate Transactions:	\$ M	(1.1)	-	(169.0)	0.2	0.5	-	-	-	-	-	-	-	-	-	-	-
Funds from Operations (FFO):	\$ M	208.4	204.4	199.5	255.4	243.8	223.8	236.0	238.7	270.0	246.7	263.2	266.1	305.5	279.0	294.2	300.5
Funds from Operations (FFO):	\$ M	208.4	204.4	199.5	255.4	243.8	223.8	236.0	238.7	270.0	246.7	263.2	266.1	305.5	279.0	294.2	300.5
(+) Non-RE Depreciation:	\$ M	2.1	2.4	2.8	3.1	3.0	3.4	3.0	3.1	3.4	3.8	3.3	3.5	3.9	4.3	3.7	3.9
(+/-) Straight-Lining of Rent & Expenses, Net:	\$ M	(1.8)	0.4	0.4	(0.1)	0.1	(1.9)	0.4	0.4	(0.1)	0.1	(2.3)	0.5	0.5	(0.1)	0.2	(2.6)
(+/-) Above/Below-Market Rent Amortization:	\$ M	(2.3)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)	(2.0)
(-) Recurring Maintenance CapEx:	\$ M	(21.1)	(17.9)	(15.3)	(21.2)	(29.6)	(18.4)	(16.0)	(21.3)	(32.9)	(20.5)	(17.8)	(23.7)	(37.3)	(23.1)	(20.0)	(26.7)
(-) Capitalized Leasing Commissions:	\$ M	(4.7)	(4.1)	(4.6)	(5.5)	(4.1)	(4.3)	(4.8)	(5.5)	(4.6)	(4.7)	(5.3)	(6.1)	(5.2)	(5.3)	(6.0)	(6.9)
Adjusted Funds from Operations (AFFO):	\$ M	\$ 180.7	\$ 183.1	\$ 180.8	\$ 229.5	\$ 211.2	\$ 200.7	\$ 216.6	\$ 213.3	\$ 233.9	\$ 223.5	\$ 239.1	\$ 238.2	\$ 265.4	\$ 252.8	\$ 270.1	\$ 266.2
Dividends % AFFO:	%	143.1%	71.8%	72.8%	-	139.7%	71.8%	72.8%	-	139.7%	71.8%	72.8%	-	139.7%	71.8%	72.8%	-
Planned Dividends:	\$ M						144.1	157.6	-	326.8	160.5	174.0	-	370.8	181.5	196.6	-
Minimum Cash Balance:	\$ M						92.3	103.7	104.0	104.5	102.5	115.3	115.9	119.3	116.7	130.4	131.1
Cash Balance Before Debt/Equity Financing:	\$ M						(148.1)	11.8	229.8	(82.8)	(72.6)	12.6	258.3	(74.8)	(83.9)	22.0	299.5
Debt/Equity Funding Required:	\$ M						240.4	91.9	-	187.2	175.2	102.7	-	194.1	200.5	108.5	-
Debt Issued:	\$ M						216.3	82.7	-	168.5	157.7	92.4	-	174.7	180.5	97.6	-
Equity Issued:	\$ M						24.0	9.2	-	18.7	17.5	10.3	-	19.4	20.1	10.8	-
Shares Issued:	# Millions						0.204	0.078	0.000	0.159	0.148	0.087	0.000	0.164	0.170	0.092	0.000

For whatever reason, the company skips Dividends in Q4 each year, pays the normal amounts in Q2 and Q3, and pays twice the normal amount in Q1.

And as a direct result, the company's funding requirements vary significantly from quarter to quarter, with no changes in Q4 and large issuances in the other quarters.

Finally, we can review the results and see why a quarterly model might be useful in some cases:

Income Statement:	Units:	Historical:					Projected:										
		1Q 16	2Q 16	3Q 16	4Q 16	1Q 17	2Q 17	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19	2Q 19	3Q 19	4Q 19
Revenue:																	
(+) Rental Income - Existing Properties:	\$ M	\$ 371.1	\$ 377.1	\$ 395.2	\$ 399.1	\$ 404.1	\$ 376.4	\$ 393.2	\$ 395.1	\$ 410.2	\$ 376.0	\$ 392.1	\$ 391.9	\$ 415.5	\$ 375.6	\$ 390.9	\$ 389.6
(+) Rental Income - Developments:	\$ M	-	-	-	-	-	8.3	17.4	27.4	37.8	51.5	65.3	79.5	95.4	111.3	128.0	145.1
(+) Tenant Reimbursements - Existing Properties:	\$ M	84.2	88.2	95.7	87.8	87.3	86.4	94.1	86.3	88.2	86.3	93.8	85.6	89.3	86.2	93.5	85.1
(+) Tenant Reimbursements - Developments:	\$ M	-	-	-	-	-	1.9	4.1	6.3	8.5	11.7	14.9	18.0	21.5	25.1	29.1	32.9
(+) Property Management Fees:	\$ M	1.8	1.3	1.5	1.7	1.9	1.2	1.6	1.7	2.2	1.3	1.8	1.9	2.6	1.5	2.1	2.1
(+) Interconnection Fees:	\$ M	47.0	48.4	53.9	55.1	57.2	55.6	62.0	62.0	64.4	61.2	68.2	66.6	69.2	64.2	71.6	70.0
(+) Other Income:	\$ M	0.1	-	0.0	33.1	0.0	-	-	-	-	-	-	-	-	-	-	-
Total Revenue:	\$ M	504.2	514.9	546.3	576.8	550.6	529.8	572.4	578.7	611.3	587.9	636.1	643.5	693.5	663.9	715.1	724.7
Revenue generally ticks up, but there's a drop in Q2 of each year followed by a recovery after that.																	
Operating Expenses:																	
(-) Property Expenses - Existing Properties:	\$ M	(185.8)	(192.9)	(206.3)	(207.7)	(202.2)	(191.9)	(204.5)	(205.4)	(205.1)	(191.7)	(203.9)	(203.8)	(207.8)	(191.6)	(203.3)	(202.6)
(-) Property Expenses - Developments:	\$ M	-	-	-	-	-	(4.3)	(9.0)	(14.2)	(19.4)	(26.4)	(33.5)	(40.9)	(48.9)	(57.0)	(65.6)	(74.6)
(-) General & Administrative:	\$ M	(31.3)	(34.2)	(46.1)	(41.2)	(34.6)	(34.4)	(45.8)	(40.5)	(36.7)	(38.2)	(50.9)	(45.0)	(41.6)	(43.2)	(57.2)	(50.7)
(-) Depreciation & Amortization:	\$ M	(169.0)	(175.6)	(178.1)	(176.6)	(176.5)	(180.1)	(188.9)	(179.4)	(198.7)	(199.9)	(209.9)	(199.5)	(225.4)	(225.7)	(236.0)	(224.7)
(-) Other Items:	\$ M	-	-	0.0	(0.2)	-	-	-	-	-	-	-	-	-	-	-	-
Total Operating Expenses:	\$ M	(386.1)	(402.6)	(430.5)	(425.7)	(413.3)	(410.7)	(448.1)	(439.5)	(459.8)	(456.2)	(498.2)	(489.2)	(523.6)	(517.4)	(562.1)	(552.5)
Operating Income:	\$ M	118.1	112.3	115.8	151.1	137.3	119.0	124.3	139.2	151.5	131.7	137.9	154.3	169.9	146.5	153.0	172.2

Revenue generally ticks up, but there's a drop in Q2 of each year followed by a recovery after that.

And the same trend applies to Operating Income.



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Key Metrics and Ratios:	Units:	Historical:					Projected:										
		1Q 16	2Q 16	3Q 16	4Q 16	1Q 17	2Q 17	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19	2Q 19	3Q 19	4Q 19
EBITDA / Net Interest:	x	4.96 x	4.55 x	4.68 x	5.73 x	5.67 x	5.08 x	5.00 x	4.90 x	5.39 x	4.85 x	4.97 x	4.87 x	5.31 x	4.88 x	4.86 x	4.89 x
Minimum Coverage Required:	x																
EBITDA / Fixed Charges:	x	3.58 x	3.36 x	3.49 x	4.40 x	4.32 x	3.90 x	3.90 x	3.85 x	4.23 x	3.85 x	3.97 x	3.91 x	4.29 x	3.96 x	3.98 x	4.02 x
Minimum Coverage Required:	x																
Not necessarily a "big deal," but flipping from < 5x to > 5x or < 4x to > 4x could make a difference.																	
Debt / EBITDA:	x	5.36 x	5.33 x	5.13 x	4.45 x	4.94 x	5.36 x	5.19 x	5.10 x	4.76 x	5.15 x	4.97 x	4.89 x	4.49 x	4.89 x	4.74 x	4.64 x
Net Debt / EBITDA:	x	5.32 x	5.28 x	5.08 x	4.43 x	4.92 x	5.29 x	5.11 x	4.92 x	4.69 x	5.07 x	4.89 x	4.71 x	4.41 x	4.81 x	4.65 x	4.46 x
(Net Debt + Preferred) / EBITDA:	x	6.44 x	6.40 x	5.95 x	5.21 x	5.73 x	6.13 x	5.92 x	5.72 x	5.41 x	5.83 x	5.62 x	5.42 x	5.05 x	5.49 x	5.30 x	5.09 x
Cap Rate Applied to Annualized EBITDA:	%																
Total Asset Value:	\$ M	13,921.6	13,958.4	14,247.8	15,900.1	15,212.1	14,503.0	15,183.0	15,446.3	16,977.0	16,077.0	16,863.7	17,154.0	19,164.7	18,046.9	18,860.7	19,241.8
Total Debt / Total Assets:	%	44.2%	44.0%	42.3%	36.7%	40.8%	44.3%	42.8%	42.1%	39.3%	42.5%	41.0%	40.3%	37.0%	40.3%	39.1%	38.3%
Maximum Level Allowed:	%																

Not necessarily a "big deal," but flipping from < 5x to > 5x or < 4x to > 4x could make a difference.

An annual model might not show these types of fluctuations clearly since the Debt / Total Assets ratio would appear to be over 40% by year end.

And, as always, we create annual summary columns because it's much harder to see the big picture in a quarterly model.

Here's the annual Income Statement summary:

Income Statement:	Units:	Historical:	Projected:		
		FY16	FY17	FY18	FY19
Revenue:					
(+) Rental Income - Existing Properties:	\$ M	\$ 1,542.5	\$ 1,568.8	\$ 1,570.1	\$ 1,571.6
(+) Rental Income - Developments:	\$ M	-	53.1	234.1	479.8
(+) Tenant Reimbursements - Existing Properties:	\$ M	355.9	354.0	353.9	354.1
(+) Tenant Reimbursements - Developments:	\$ M	-	12.3	53.1	108.5
(+) Property Management Fees:	\$ M	6.3	6.4	7.2	8.2
(+) Interconnection Fees:	\$ M	204.3	236.8	260.4	275.0
(+) Other Income:	\$ M	33.2	0.0	-	-
Total Revenue:	\$ M	2,142.2	2,231.4	2,478.9	2,797.2
Operating Expenses:					
(-) Property Expenses - Existing Properties:	\$ M	(792.7)	(804.0)	(804.5)	(805.1)
(-) Property Expenses - Developments:	\$ M	-	(27.4)	(120.2)	(246.0)
(-) General & Administrative:	\$ M	(152.7)	(155.4)	(170.8)	(192.7)
(-) Depreciation & Amortization:	\$ M	(699.3)	(724.9)	(808.0)	(911.8)
(-) Other Items:	\$ M	(0.2)	-	-	-
Total Operating Expenses:	\$ M	(1,644.9)	(1,711.6)	(1,903.5)	(2,155.6)
Operating Income:	\$ M	497.3	519.7	575.4	641.6
(+) Equity Investment Income:	\$ M	17.1	20.3	21.3	22.4
(+) Gain / (-) Loss on Real Estate Transactions:	\$ M	169.9	(0.5)	-	-
(-) Net Interest Expense:	\$ M	(241.0)	(241.8)	(276.0)	(311.7)
(-) Loss from Debt Extinguishment:	\$ M	(1.0)	-	-	-
Pre-Tax Income:	\$ M	442.2	297.7	320.7	352.2
(-) Income Taxes:	\$ M	(10.4)	(7.9)	(8.5)	(9.4)
Net Income:	\$ M	431.9	289.8	312.2	342.8
(-) Net Income Attributable to NCI:	\$ M	(5.7)	(6.2)	(3.9)	(7.1)
Net Income Attributable to Parent:	\$ M	426.2	283.6	308.3	335.8
(-) Preferred Stock Dividends:	\$ M	(83.8)	(70.6)	(70.9)	(70.9)
(-) Preferred Stock Issuance Costs:	\$ M	(10.3)	-	-	-
Net Income to Common:	\$ M	\$ 332.1	\$ 213.0	\$ 237.4	\$ 264.8

This annual summary makes it much more obvious that the REIT is growing significantly over several years, despite quarterly fluctuations.

It's also easier to see that the vast majority of its growth comes from Development activities.

Finally, the REIT's Net Income to Common increases by less than expected because of rising expenses, a rising Debt balance, and rising interest rates on the Debt.



Quarterly and half-year models have their uses, but they also require more time and effort to set up, check, and summarize – as well as more data gathering in the beginning.

Since a REIT's value depends on its current Balance Sheet (with adjustments) as well as its trading multiples and long-term DCF profile, these more granular models are not necessarily that helpful.

So, we recommend using them only when your boss or the case study instructions tell you to do so, or when they are useful because of a specific issue, such as a REIT with seasonal financial performance or high percentages of lease expirations in certain quarters.

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REIT Valuation

We covered [the basics of REIT Valuation in the summary section above](#), so we will focus on the more advanced nuances here, where applicable.

NAV Model

The NAV Model is arguably the most important valuation methodology for U.S.-based REITs, and you follow the same basic set of steps regardless of the model's complexity:

- **Step 1:** Project 12-Month Forward NOI for the REIT (or Project NOI by Segment)
- **Step 2:** Divide the Forward NOI by Appropriate Cap Rate(s) to Calculate the Market Value of Gross Real Estate Assets
- **Step 3:** Project Non-Rental Income and Divide by a (Higher) Cap Rate to Value It
- **Step 4:** Make Minor Adjustments to All Other Assets and Add Them to the Step 2 and 3 Results
- **Step 5:** Mark Liabilities to Fair Market Value and Subtract Them from the Market Value of All Assets
- **Step 6:** Divide Net Asset Value by the REIT's Share Count to Determine NAV per Share

We'll look at a few added complexities here based on NAV Models for AvalonBay and Digital Realty.

For AvalonBay, we split up the Same-Store NOI by geography and apply a different Cap Rate to each region; properties in suburban areas of Pennsylvania and Maryland should be worth less than ones in downtown New York City or Los Angeles.

We also deduct a small percentage of each segment's NOI, under the argument that many REITs do not account for the required **Replacement Reserves** (i.e., the amounts allocated each year to fund future CapEx) directly in their financial statements.

That may or may not be true, depending on the REIT, but it seems justified in AVB's case since there are no references to the Replacement Reserves in its filings.

We do the same thing for the company's other business segments, such as Developments, Redevelopments, Acquisitions, and Dispositions, and then we divide the Economic NOI in each of those segments by the appropriate Cap Rate. Here's the first part of this schedule:

ASSETS:							
Operating Real Estate & Other Income:							
Selected Scenario:	Name	Base	<--- Cut and paste scenario drop-down box here if required for sensitivity table.				
Cap Rate Adjustment for Sensitivities:	%	-					
Replacement Reserves % NOI:	%	5.0%					
Established Properties by Region:		Nominal Forward NOI:	Repl. Reserve Deduction:	Economic Forward NOI:	Economic Cap Rate:	Sensitized Cap Rate:	Implied Value:
New England:	\$ M	154.8	(7.7)	147.1	5.0%	5.0%	2,941.6
Metro NY/NJ:	\$ M	253.7	(12.7)	241.1	4.0%	4.0%	6,026.4
Mid-Atlantic:	\$ M	156.7	(7.8)	148.9	5.3%	5.3%	2,809.0
Pacific Northwest:	\$ M	64.2	(3.2)	61.0	4.3%	4.3%	1,417.9
Northern California:	\$ M	264.0	(13.2)	250.8	4.5%	4.5%	5,572.9
Southern California:	\$ M	253.2	(12.7)	240.5	4.2%	4.2%	5,726.2
Other Business Segments:							
Other Stabilized Properties:	\$ M	317.6	(15.9)	301.7	6.1%	6.1%	4,946.0
Development Properties:	\$ M	52.0	(2.6)	49.4	7.0%	7.0%	705.7
Redevelopment Properties:	\$ M	88.8	(4.4)	84.3	5.0%	5.0%	1,686.6
Acquired Properties:	\$ M	23.1	(1.2)	22.0	5.0%	5.0%	439.2
Disposed Properties:	\$ M	(25.2)	1.3	(23.9)	5.3%	5.3%	(451.7)

These Cap Rates are based on local market data from JLL at the time of this valuation. Put simply, some markets are more expensive than others.

The "Sensitized Cap Rate" column here lets us tweak all these rates up and down for use in the sensitivity tables at the end. That way, we can see what happens if the entire market shifts in one direction or the other.

You might deduct Replacement Reserves if it seems like the REIT has not already done so in its NOI calculations.

We use a higher Cap Rate for the Development segment to reflect greater risk, and lower Cap Rates for the others to reflect more stability and less risk.

Next, we need to value AvalonBay's **Equity Investments** – partially-owned subsidiaries and joint ventures where it owns less than a 50% stake.

Equity Investment accounting is a mess, so we need to adjust and add/remove various items.

On AvalonBay's Balance Sheet, it lists the Equity Investments at **historical cost**: the original invested amount + Net Income from Equity Investments – Dividends from Equity Investments, cumulatively over time.

But the company does not show the underlying Assets and Liabilities of the Equity Investments on its Balance Sheet – instead, we need to hunt through its filings to find the numbers, re-value them, and then multiply them by AvalonBay's approximate ownership percentage.

In this case, we have enough information to create a forecast for the Equity Investments:

Equity Investments (Unconsolidated Real Estate):	Units:	Historical:			Projected:				
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22
Assets:									
Real Estate Assets, Net:	\$ M	\$ 1,392.8	\$ 954.5	\$ 695.1	\$ 699.1	\$ 695.3	\$ 683.6	\$ 664.0	\$ 636.5
Other Assets:	\$ M	57.0	49.5	40.0	41.9	43.1	44.4	44.5	44.5
Total Assets:	\$ M	1,449.9	1,004.0	735.1	741.1	738.4	728.0	708.5	681.1
Liabilities & Equity:									
Total Debt:	\$ M	947.2	689.6	523.8	531.3	535.4	533.2	523.0	509.2
Other Liabilities:	\$ M	20.5	16.5	10.5	11.0	10.9	10.8	10.4	10.0
Partners' Capital:	\$ M	482.2	297.9	200.7	198.7	192.1	184.1	173.5	161.9
Total Liabilities & Equity:	\$ M	1,449.9	1,004.0	735.1	741.1	738.4	728.0	708.5	681.1
Balance Check:		OK!	OK!	OK!	OK!	OK!	OK!	OK!	OK!
Weighted Average Percentage Ownership:	%			25.1%	25.1%	25.1%	25.1%	25.1%	25.1%
(+) Rental & Other Income:	\$ M	173.6	131.9	101.6	104.4	105.7	105.8	105.4	101.0
(-) Property-Level Expenses:	\$ M	(68.0)	(50.9)	(38.6)	(39.3)	(39.6)	(39.4)	(39.7)	(37.4)
(+) Gain / (-) Loss on Sale of Communities:	\$ M	98.9	196.7	136.3	34.0	36.2	38.3	40.4	42.5
(-) Net Interest Expense:	\$ M	(45.5)	(45.9)	(27.1)	(21.0)	(21.3)	(21.4)	(21.3)	(21.0)
(-) Depreciation:	\$ M	(45.3)	(34.5)	(25.9)	(26.8)	(27.4)	(27.6)	(27.7)	(26.8)
Net Income:	\$ M	113.7	197.3	146.4	51.3	53.6	55.6	57.8	58.4
Equity Investment Income on Income Statement:	\$ M	70.0	65.0	70.7	12.9	13.5	14.0	14.5	14.7
Net Operating Income (NOI):	\$ M	105.6	81.0	63.0	65.0	66.1	66.3	66.4	63.7

We'll need to adjust the values of these Assets and Liabilities, multiply by AVB's 25.1% ownership percentage, and include them in the NAV Model.

And we'll use the Forward NOI of the Equity Investments here, divide it by an appropriate Cap Rate, and use that to represent the fair market value of the Real Estate Assets.

We then factor in the Equity Investments' Assets, as well as the remaining items on the Assets side of AVB's Balance Sheet:

Established Properties by Region:	Nominal Forward NOI:	Repl. Reserve Deduction:	Economic Forward NOI:	Economic Cap Rate:	Sensitized Cap Rate:	Implied Value:
Equity Investments (Unconsolidated Real Estate):	\$ M	65.0	(3.3)	61.8	5.0%	1,235.3
(x) Pro-Rata Allocation Percentage:	%					25.1%
Market Value of Pro-Rata Portion of Unconsol. RE:	\$ M					310.4
Other JV Assets:	\$ M					40.0
(x) Pro-Rata Allocation Percentage:	%					25.1%
Market Value of Pro-Rata Portion of Unconsol. RE:	\$ M					10.0
Management, Development & Other Fees:	\$ M	4.6	-	4.6	12.0%	38.0
Other Assets:						
Construction-in-Progress:	\$ M					525.6
(x) Market Value Adjustment:	%					110.0%
Market Value of Construction-in-Progress:	\$ M					578.2
Land Held for Development:	\$ M					68.4
(x) Market Value Adjustment:	%					105.0%
Market Value of Land Held for Development:	\$ M					71.8
Cash & Cash-Equivalents:	\$ M					201.9
Resident Security Deposits:	\$ M					32.7
Other Assets:	\$ M					299.2
Total Market Value of Assets:	\$ M					\$ 33,361.9

In-line with the Cap Rates in other segments, and we multiply by the 25.1% ownership.

No adjustments required here; smaller item, so we just multiply by the ownership percentage.

We almost always apply a much higher Cap Rate to Management Fee Income because it is cancellable on short notice and does not represent actual leases.

Usually a very modest markup to Land Held for Development.

We rarely adjust the company's Cash and Other Assets because book value is a reasonable proxy for fair market value. If the company had any Goodwill & Other Intangibles, we would exclude them or list them at 0.

Next, we make a set of adjustments to the Liabilities & Equity side of the Balance Sheet:

LIABILITIES & EQUITY:				
Debt & Other Borrowings:	\$ M			(7,329.5)
(x) Market Value Adjustment:	%			99.6%
(+) Benefit of Tax-Exempt Bonds:	\$ M	0.7	8.0%	9.4
Market Value of Debt & Other Borrowings:	\$ M			(7,290.1)
Pro-Rata Portion of JV Debt:	\$ M			(523.8)
(x) Market Value Adjustment:	%			104.8%
(x) Pro-Rata Allocation Percentage:	%			25.1%
Market Value of Pro-Rata Portion of JV Debt:	\$ M			(138.0)
Pro-Rata Portion of Other JV Liabilities:	\$ M			(10.5)
(x) Pro-Rata Allocation Percentage:	%			25.1%
Pro-Rata Portion of Other JV Liabilities:	\$ M			(2.6)
Resident Security Deposits:	\$ M			(58.5)
Other Liabilities:	\$ M			(632.8)
Net Asset Value (NAV):	\$ M			\$ 25,240.0

If prevailing interest rates have changed since the company's Debt was issued, we'll adjust the value up or down here. In this case, we found a fair value estimate for the Debt in AVB's filings.

This is a VERY small item, but some REITs have "Tax-Exempt Bonds," which reduce the Income Tax Burden. Of course, most REITs have a tax rate close to 0%, so these barely matter. We apply a higher Cap Rate as industry convention.

Interest rates have fallen since this Debt was issued, so we adjust its value UP based on the estimates in the filings and the PRICE function in Excel. And then we multiply by AVB's ownership % in the Equity Investments.

No real adjustments to these other items except for the Ownership % multiplication for the Other JV Liabilities. Make sure you exclude Redeemable NCI if you're counting the REIT's DownREIT or OP Units in the share count!

The **Net Asset Value (NAV)** at the bottom equals the Market Value of All Assets minus the Market Value of All Liabilities.

Next, we need to calculate the NAV per Share and compare it to the REIT's current share price.

This calculation will be **circular** if the REIT has options or warrants because the dilution from those options or warrants depends on the NAV per Share, but the NAV per Share depends on the diluted share count.

You could skip this circular reference and use a constant share count, but it's slightly more accurate to make it circular.

Also, you must be careful with DownREIT and OP Units – these correspond to *Redeemable Noncontrolling Interests*, so you cannot count both these "Units" and the Redeemable NCI.

In other words, if you do **not** include Redeemable NCI in Total Liabilities, you **should** include DownREIT and OP Units in the share count; but if you **do** include Redeemable NCI in Total Liabilities, you **should not** include DownREIT and OP Units in the share count.

Here's our calculation for AVB:



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	A	B	C	D	E	F	G	H	I	J	K
71											
72			Net Asset Value (NAV):	\$ M						\$ 25,240.0	Treasury Stock Method based on NAV per Share and the Exercise Price.
73											
74			Share Count Calculation:								
75											
76			Basic Shares Outstanding:	Millions				138.096		138.096	
77			Options A:	Millions				0.150	\$ 126.77	=IF((I77<\$J\$84,H77-((H77*177)/\$J\$84),0)	
78			Options B:	Millions				0.008	48.60	0.006	
79			Performance Awards:	Millions				0.252		0.252	
80			DownREIT Units:	Millions				0.008		0.008	We're excluding Redeemable NCI from Total Liabilities, so we must count the DownREIT Units here.
81			RSUs:	Millions				0.000		0.000	
82			Diluted Shares Outstanding @ NAV per Share:	Millions						138.406	
83											
84			NAV per Share:	\$ as Stated						\$ 182.36	
85			Current Share Price:	\$ as Stated						165.09	
86											
87			NAV per Share Premium / (Discount) to Current Price:	%						10.5%	The REIT seems at least modestly undervalued based on these results.
88											
89			12-Month Forward NOI (Nominal):	\$ M						\$ 1,602.9	
90			12-Month Forward NOI (Economic):	\$ M						1,522.7	
91											
92			Nominal Cap Rate Implied by Current Share Price:	%						5.4%	
93			Economic Cap Rate Implied by Current Share Price:	%						5.2%	

As always, valuation is about the **range of possible implied values**, so we also create a sensitivity table based on the Cap Rate differentials and operational scenarios:

Sensitivity Analysis - NAV per Share:													
		Cap Rate Differentials to Baseline Estimates:											
		(0.5%)	(0.4%)	(0.3%)	(0.2%)	(0.1%)	-	0.1%	0.2%	0.3%	0.4%	0.5%	
Operating Scenario:	Upside	\$ 219.03	\$ 212.74	\$ 206.74	\$ 201.02	\$ 195.55	\$ 190.32	\$ 185.31	\$ 180.51	\$ 175.91	\$ 171.49	\$ 167.24	
	Base	210.22	204.12	198.30	192.74	187.44	182.36	177.50	172.85	168.39	164.10	159.98	
	Downside	189.65	184.02	178.66	173.54	168.65	163.97	159.49	155.21	151.09	147.14	143.35	

This is the only region of the table where the NAV per Share figures are **below** the company's current share price, so it is very likely undervalued currently.

In this course, we also build NAV Models for Digital Realty and DuPont Fabros, which are slightly different from this example for AvalonBay.

They are data center REITs, so they are more like office, industrial, and retail REITs in terms of non-cash adjustments and the financial statements.

Both companies have nearly identical models, so we will cover the one for Digital Realty here.

Here are the key differences, starting with the Assets side:



Real Estate & REIT Modeling

Quick Reference Guide for REITs

Digital Realty Trust, Inc. - Net Asset Value (NAV) Model

(\$ USD in Millions Except Per Share and Per Unit Amounts in USD as Stated)

ASSETS:					
Operating Real Estate & Other Income:					
Cap Rate Adjustment for Sensitivities:	%	-			
		Forward NOI:	Cap Rate:	Sensitized Cap Rate:	Implied Value:
Annualized GAAP NOI from Most Recent Quarter:	\$ M	\$ 1,386.0			
(+/-) Straight-Lining of Rent & Expenses, Net:	\$ M	0.5			
(+/-) Above/Below-Market Rent Amortization:	\$ M	(7.9)			
(-) Partners' Share of Consolidated JVs:	\$ M	(0.5)			
(-) Annualized NOI from Dispositions & Expirations:	\$ M	(11.1)			
(+) Annualized NOI from Backlog Stabilized in Quarter:	\$ M	51.0			
Total Consolidated Cash NOI:	\$ M	1,418.0			
Expected Growth Rate Over Next 12 Months:	%	4.0%			
Forward Cash NOI:	\$ M	1,474.8	6.6%	6.6%	\$ 22,345.0
Pro-Rata Share of Unconsolidated JV Cash NOI:	\$ M	35.1	7.0%	7.0%	501.4
Management, Development & Other Fees:	\$ M	7.6	12.5%	12.5%	88.8
Other Assets:					
Construction in Progress and Development Land:	\$ M				1,010.4
(-) Investment Associated with Backlog:	\$ M				(248.7)
(x) Market Value Adjustment:	%				120.0%
Market Value of Construction and Development Land:	\$ M				914.0
Pre-Stabilized Inventory at Cost:	\$ M				251.4
Cash & Cash-Equivalents:	\$ M				25.4
Accounts Receivable:	\$ M				195.4
Other Assets:	\$ M				220.8
Partners' Share of Consolidated JV Assets:	\$ M				(0.1)
Total Market Value of Tangible Assets:	\$ M				\$ 24,514.0

We don't have segment-by-segment projections, so the Forward NOI calculation is quite different. We take DLR's NOI from the most recent quarter, multiply by 4, and then adjust for non-cash items and NOI lost and gained from Dispositions and Stabilizations. All these numbers are all disclosed in the company's filings.

Then, we apply a modest growth rate to calculate the Forward NOI over the next 12-month period.

Based on local data center Cap Rates in DLR's main regions of operations.

Based on #s disclosed in the filings and local Cap Rates in the region.

Annualized management fee number and a higher Cap Rate due to the additional risk.

Based on company's disclosures and fairly standard 10-20% adjustment.

Based on the company's disclosures; very few adjustments here.

We **EXCLUDE** Goodwill, Other Intangibles, Deferred Rent, and Above-Market Lease Intangibles because the FMV is 0 for each one.

At this time, Digital Realty provided "NAV Supplements" that listed each component of the company's Net Asset Value.

That saved **a lot** of time in this analysis because we did not have to hunt for information on the JVs (Equity Investments) or many of the other, smaller items here.

The book value of the company's Total Assets at this time was \$12.3 billion, so the Market Value was about twice as high.

The company had \$2.6 billion worth of Goodwill, Other Intangibles, Deferred Rent, and Above-Market Lease Intangibles, all of which we excluded from the NAV Model – but even *with* those exclusions, the Assets' Market Value still greatly exceeded their book value.

The Liabilities side is much simpler and closer to the AvalonBay example:

LIABILITIES & EQUITY:				
Debt & Other Borrowings:	\$ M		\$	(6,201.3)
(x) Market Value Adjustment:	%			104.5%
Market Value of Debt & Other Borrowings:	\$ M			(6,480.4)
Pro-Rata Portion of Unconsolidated JV Debt:	\$ M			(135.8)
Cost to Complete Backlog Assets:	\$ M			(120.4)
Accounts Payable & Other Accruals:	\$ M			(804.4)
Other Liabilities:	\$ M			(174.8)
Preferred Stock:	\$ M			(1,013.0)
Non-OP Noncontrolling Interests:	\$ M			(6.6)
Net Asset Value (NAV):	\$ M		\$	15,778.8
Share Count Calculation:				
		Number:	Exercise Price:	Dilution:
Basic Shares Outstanding:	# Millions	159.775		159.775
Options A:	# Millions	0.006	\$ 41.73	0.003
Incentive Units:	# Millions	1.534		1.534
OP Units:	# Millions	1.012		1.012
RSUs:	# Millions	0.000		0.000
Diluted Shares Outstanding @ NAV per Share:	# Millions			162.325
NAV per Share:	\$ as Stated			\$ 97.21
Current Share Price:	\$ as Stated			118.02
NAV per Share Premium / (Discount) to Current Price:	%			(17.6%)

Modest adjustment due to rising interest rates.

Straight from the company's disclosures; no real insight into these items.

No real adjustments except for the exclusion of the Redeemable Noncontrolling Interests that correspond to the company's OP Units.

A circular calculation once again due to the options; in this case, the company seems modestly overvalued.

When the company discloses this much information, the path of least resistance is to **use it** because the company knows its financials better than you do.

Other than the NOI calculation, this NAV Model is fairly similar to the one for AvalonBay.

The main difference is that we rely on the company's disclosures in its "NAV Supplement" to separate out its JV Assets and Liabilities – whereas we did that manually for AvalonBay.

Comparable Public Companies

You cannot make the Public Comps "more complex" because you always follow the same process when creating them: screen by geography, sub-industry, and asset size, and then display metrics such as the projected FFO and P / FFO multiples for each REIT.

However, you can make *the calculations* more complex by "spreading the comps" – in other words, by reviewing each company's annual and interim reports manually and searching for non-recurring charges and dilutive securities yourself.

This is a time-consuming and painstaking process that rarely adds anything useful, so it is usually not worthwhile.

In practice, you only go into this level of detail in investment banking when you're creating a **Fairness Opinion** for a public client that needs documentation to justify an M&A deal. There, the company is paying your bank a high fee for the valuation, so the effort is justified.

To show you an example of this process, here's an analysis we completed for Equity Residential [EQR], one of AvalonBay's peer companies:

	EQR Equity Residential			
	Calendarization:			
	2017-12-31		2017-12-31	
	Old Partial	New Partial	FY	LTM
Funds from Operations (FFO):			\$ 1,204.9	\$ 1,204.9
Net Operating Income (NOI):			1,729.9	1,729.9
Operating Income (EBIT):			847.5	847.5
(+) Non-Recurring Items in EBIT:			1.7	1.7
(+) Depreciation & Amortization:			747.6	747.6
EBITDA:			\$ 1,596.7	\$ 1,596.7
Effective Tax Rate:				0.1%
Real Estate Data:				
Gross Real Estate Assets:			\$ 26,026.9	
Total Assets:			20,570.6	
Primary Market(s):	Northeast, West Coast, Mid-Atlantic			
# Units Owned:			78,611	
Average Physical Occupancy:			95.0%	
Rental Income (\$ in Millions):			2,470.7	
Average Monthly Rent / Unit:			2,619.1	
Property Quality Grade:			A-	
Development Pipeline (\$ in Millions):			1,451.8	
Dev Pipeline % Gross RE Assets:			5.6%	
Portfolio Yield (Historical Cost):			6.8%	
Leverage (Debt / Total Assets):			44.9%	

As stated in the company's disclosures.

We find these on the company's Income Statement and Cash Flow Statement.

All this information comes from the Balance Sheet, Income Statement, and supplemental real estate data. You can search for the key terms in the 10-K or 10-Q to find most of it.

You can find some of this information in **equity research** as well. For example, Research Analysts often assign a "quality grade" to REITs' property portfolios, which is how we found the "A-" rating for EQR.

Moving on, here's the Balance Sheet data and Diluted Shares:

Balance Sheet Data:			
(-) Cash & Cash-Equivalents:		\$	(100.8)
(-) Equity Investments:			(58.3)
(-) Other Non-Core Assets, Net:			-
(-) Net Operating Losses:			-
(+) Debt & Capital Leases			9,235.1
(+) Preferred Stock:			37.3
(+) Noncontrolling Interests:			4.7
(+) Unfunded Pension Obligations:			-
(+) Restructuring & Other Liabilities:			-

Projected Financials:			
	2018-12-31	2019-12-31	2020-12-31
Funds from Operations (FFO):	\$ 1,238.2	\$ 1,291.8	
Net Operating Income (NOI):	1,770.7	1,830.1	
EBITDA:	1,634.4	1,689.3	

Ideally, we should use the fair market value of all these items, but not all REITs disclose it; it's not a huge deal if you can't find it in the filings, as book value is a reasonable proxy in most cases.

From consensus estimates or individual equity research reports.

[illegible]

Standard Treasury Stock Method for the options.

No convertible bonds, but if there were, we would use the "If Converted" method.

Straight additions; just be careful to exclude the Redeemable NCI in the Enterprise Value calculation if you also count OP and DownREIT Units here!

Can be taken from Bloomberg, Google Finance, etc.

Finally, we calculate the valuation multiples directly on this sheet so that we can link to them on the Public Comps tab and separate the display and the calculations:



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Valuation Multiples:	
	LTM
P / FFO:	19.2 x
EV / NOI:	18.6 x
EV / EBITDA:	20.2 x
Lookup Variables:	
2018-12-31 Funds from Operations (FFO):	\$ 1,238.2
2018-12-31 Net Operating Income (NOI):	1,770.7
2018-12-31 EBITDA:	1,634.4
2019-12-31 Funds from Operations (FFO):	\$ 1,291.8
2019-12-31 Net Operating Income (NOI):	1,830.1
2019-12-31 EBITDA:	1,689.3
LTM P / FFO:	19.2 x
LTM EV / NOI:	18.6 x
LTM EV / EBITDA:	20.2 x
2018-12-31 P / FFO:	18.7 x
2018-12-31 EV / NOI:	18.2 x
2018-12-31 EV / EBITDA:	19.7 x
2019-12-31 P / FFO:	17.9 x
2019-12-31 EV / NOI:	17.6 x
2019-12-31 EV / EBITDA:	19.1 x
Projected EBITDA Growth:	3.4%
Projected FFO Growth:	4.3%
Corporate Website:	http://investors.equityapartments.com/

After all this extra time and effort... not much changes. The conclusions are nearly the same as they would have been with numbers pulled automatically from Capital IQ or FactSet:

Comparable Companies - U.S.-Based Multifamily, Single-Family, and Student Housing REITs with Gross Real Estate Assets Above \$5 Billion

(\$ USD in Millions Except Per Share and Per Unit Amounts in USD as Stated)

Operating Statistics:		Equity Value	Enterprise Value	Gross Real Estate Assets	Portfolio Yield (Hist. Cost)	Leverage (Debt to Total Assets)	LTM	EBITDA FY18	FY19	Funds from Operations (FFO) LTM	FY18	FY19	Projected EBITDA Growth	Projected FFO Growth
Company Name														
Equity Residential		\$ 23,131.7	\$ 32,249.8	\$ 26,026.9	6.8%	44.9%	\$ 1,596.7	\$ 1,634.4	\$ 1,689.3	\$ 1,204.9	\$ 1,238.2	\$ 1,291.8	3.4%	4.3%
Essex Property Trust, Inc.		16,820.6	21,296.5	13,704.6	7.7%	46.1%	915.4	1,018.5	1,062.2	812.1	851.4	890.7	4.3%	4.6%
Mid-America Apartment Communities, Inc.		10,732.6	15,124.3	13,292.0	7.6%	40.0%	849.3	897.6	940.4	699.6	712.3	755.5	4.8%	6.1%
UDR, Inc.		10,591.6	13,506.3	10,177.2	7.2%	48.4%	621.0	658.5	709.4	538.9	574.2	615.6	7.7%	7.2%
American Homes 4 Rent		6,279.5	8,672.3	8,968.9	6.6%	30.0%	491.3	544.3	616.9	260.5	379.7	434.5	13.3%	14.5%
Apartment Investment and Management Company		6,682.5	10,634.0	7,927.8	7.5%	69.6%	596.0	600.2	614.9	384.2	392.8	408.1	2.4%	3.9%
Camden Property Trust		7,991.9	9,946.4	7,667.7	8.0%	36.9%	500.1	539.7	580.6	424.1	460.0	491.5	7.6%	6.8%
American Campus Communities, Inc.		5,131.1	7,741.2	7,159.8	6.4%	44.9%	392.8	445.5	474.0	317.4	331.5	355.7	6.4%	7.3%
Sun Communities, Inc.		7,616.2	10,558.5	6,882.9	9.6%	48.1%	453.1	539.8	574.2	320.1	384.9	425.4	6.4%	10.5%
Maximum		\$ 23,131.7	\$ 32,249.8	\$ 26,026.9	9.6%	69.6%	\$ 1,596.7	\$ 1,634.4	\$ 1,689.3	\$ 1,204.9	\$ 1,238.2	\$ 1,291.8	13.3%	14.5%
75th Percentile		10,732.6	15,124.3	13,292.0	7.7%	48.1%	849.3	897.6	940.4	699.6	712.3	755.5	7.6%	7.3%
Median		\$ 7,991.9	\$ 10,634.0	\$ 8,968.9	7.5%	44.9%	\$ 596.0	\$ 600.2	\$ 616.9	\$ 424.1	\$ 460.0	\$ 491.5	6.4%	6.8%
25th Percentile		6,682.5	9,946.4	7,667.7	6.8%	40.0%	491.3	539.8	580.6	320.1	384.9	425	4.3%	4.6%
Minimum		5,131.1	7,741.2	6,882.9	6.4%	30.0%	392.8	445.5	474.0	260.5	331.5	356	2.4%	3.9%
AvalonBay Communities, Inc.		\$ 22,847.6	\$ 29,772.3	\$ 21,935.9	7.3%	39.6%	\$ 1,348.2	\$ 1,550.4	\$ 1,818.3	\$ 1,167.2	\$ 1,337.4	\$ 1,579.0	17.3%	18.1%

Slightly lower Yield and lower Leverage than the comparables.

...but AVB's growth rates are also much higher, so we'd expect it to trade at premium multiples.



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Valuation Statistics:											
Company Name	Equity Value	Enterprise Value	Gross Real Estate Assets	Portfolio Yield (Hist. Cost)	Leverage (Debt to Total Assets)	Enterprise Value /			Equity Value /		
						LTM	EBITDA FY18	FY19	LTM	FFO FY18	FY19
Equity Residential	\$ 23,131.7	\$ 32,249.8	\$ 26,026.9	6.8%	44.9%	20.2 x	19.7 x	19.1 x	19.2 x	18.7 x	17.9 x
Essex Property Trust, Inc.	16,820.6	21,296.5	13,704.6	7.7%	46.1%	23.3 x	20.9 x	20.0 x	20.7 x	19.8 x	18.9 x
Mid-America Apartment Communities, Inc.	10,732.6	15,124.3	13,292.0	7.6%	40.0%	17.8 x	16.8 x	16.1 x	15.3 x	15.1 x	14.2 x
UDR, Inc.	10,591.6	13,506.3	10,177.2	7.2%	48.4%	21.8 x	20.5 x	19.0 x	19.7 x	18.4 x	17.2 x
American Homes 4 Rent	6,279.5	8,672.3	8,968.9	6.6%	30.0%	17.7 x	15.9 x	14.1 x	24.1 x	16.5 x	14.5 x
Apartment Investment and Management Company	6,682.5	10,634.0	7,927.8	7.5%	69.6%	17.8 x	17.7 x	17.3 x	17.4 x	17.0 x	16.4 x
Camden Property Trust	7,991.9	9,946.4	7,667.7	8.0%	36.9%	19.9 x	18.4 x	17.1 x	18.8 x	17.4 x	16.3 x
American Campus Communities, Inc.	5,131.1	7,741.2	7,159.8	6.4%	44.9%	19.7 x	17.4 x	16.3 x	16.2 x	15.5 x	14.4 x
Sun Communities, Inc.	7,616.2	10,558.5	6,882.9	9.6%	48.1%	23.3 x	19.6 x	18.4 x	23.8 x	19.8 x	17.9 x
Maximum	\$ 23,131.7	\$ 32,249.8	\$ 26,026.9	9.6%	69.6%	23.3 x	20.9 x	20.0 x	24.1 x	19.8 x	18.9 x
75th Percentile	10,732.6	15,124.3	13,292.0	7.7%	48.1%	21.8 x	19.7 x	19.0 x	20.7 x	18.7 x	17.9 x
Median	\$ 7,991.9	\$ 10,634.0	\$ 8,968.9	7.5%	44.9%	19.9 x	18.4 x	17.3 x	19.2 x	17.4 x	16.4 x
25th Percentile	6,682.5	9,946.4	7,667.7	6.8%	40.0%	17.8 x	17.4 x	16.3 x	17.4 x	16.5 x	14.5 x
Minimum	5,131.1	7,741.2	6,882.9	6.4%	30.0%	17.7 x	15.9 x	14.1 x	15.3 x	15.1 x	14.2 x
AvalonBay Communities, Inc.	\$ 22,847.6	\$ 29,772.3	\$ 21,935.9	7.3%	39.6%	22.1 x	19.2 x	16.4 x	19.6 x	17.1 x	14.5 x

But it does **not** trade at higher multiples, for the most part, which is further evidence that it might be slightly undervalued.

There are other examples of Public Comps in this course, but they're very similar to the one above and [the ones for Park Hotels and Colonial in the first section](#).

We recommend that you **avoid over-thinking the concept** because it's pretty simple: if a company is growing more quickly than its peers, its valuation multiple should be higher; if it's growing at a similar rate, its multiples should be in-line; and if it's growing more slowly, its multiple should be lower.

If something different happens, the company may be overvalued or undervalued.

Precedent Transactions

With the Precedent Transactions, we go through the same process of screening by geography, sub-industry, and "size" (purchase price), but we also add **time** to the criteria list.

It doesn't make sense to go back 15 years if market conditions have changed significantly in that time; the data is most meaningful if you include only the past few years.

We do not present a detailed data gathering exercise for Precedent Transactions in this course, but here's the output of our set for AvalonBay:



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Precedent Transactions - U.S.-Based REIT Sellers with Purchase Enterprise Values of At Least \$3 Billion

Announced Between March 19th, 2015 and March 19th, 2018

(\$ USD in Millions Except Per Share and Per Unit Amounts in USD as Stated)

AvalonBay Communities, Inc. - Comparable M&A Transactions

Acquirer Name	Target Name	Date	Purchase Equity Value	Purchase Enterprise Value	Percentage Acquired	Gross Real Estate Assets	LTM FFO	LTM EBITDA	NTM NOI	Portfolio Yield (Hist. Cost)	Valuation Multiples		Premiums Paid		
											P / FFO	EV / EBITDA	1-Day Prior	1-Week Prior	1-Month Prior
Brookfield Property Partners L.P.	GGP Inc.	2017-11-13	\$ 21,046.5	\$ 34,603.6	65.4%	\$ 20,820.4	\$ 1,537.2	\$ 1,781.3	\$1,716.7	8.2%	13.7 x	19.4 x	3.6%	21.0%	6.1%
Greystar Investment Group, LLC	Monogram Residential Trust, Inc.	2017-07-04	2,020.9	3,923.7	100.0%	3,390.1	56.7	138.8	179.3	5.3%	35.6 x	28.3 x	22.4%	22.1%	22.3%
Digital Realty Trust, Inc.	DuPont Fabros Technology, Inc.	2017-06-09	5,831.1	7,395.2	100.0%	3,616.5	245.8	328.9	369.6	10.2%	23.7 x	22.5 x	16.2%	18.4%	24.4%
Sabra Health Care REIT, Inc.	Care Capital Properties, Inc.	2017-05-07	2,626.3	4,076.7	100.0%	3,494.0	238.2	331.5	371.3	10.6%	11.0 x	12.3 x	11.8%	11.5%	10.8%
Regency Centers Corporation	Equity One, Inc.	2016-11-14	4,497.1	5,773.2	100.0%	3,552.3	171.2	234.8	290.4	8.2%	26.3 x	24.6 x	10.8%	10.3%	5.7%
Mid-America Apartment Communities, Inc.	Post Properties, Inc.	2016-08-15	3,973.7	4,913.7	100.0%	3,315.6	169.6	206.9	231.6	7.0%	23.4 x	23.7 x	16.6%	16.5%	16.3%
Cousins Properties Incorporated	Parkway Properties Inc.	2016-04-29	1,976.1	3,763.4	100.0%	3,220.6	156.8	234.7	306.0	9.5%	12.6 x	16.0 x	13.0%	17.3%	17.6%
Weyerhaeuser Company	Plum Creek Timber Co. Inc.	2015-11-08	8,589.8	11,767.3	100.0%	4,633.0	357.0	520.0	537.6	11.6%	24.1 x	22.6 x	20.7%	19.4%	17.7%
Blackstone Real Estate Advisors	Strategic Hotels & Resorts, Inc.	2015-09-08	4,071.2	5,849.2	100.0%	4,078.7	158.5	312.9	439.4	10.8%	25.7 x	18.7 x	4.8%	3.9%	7.8%
Blackstone Real Estate Advisors	BioMed Realty Trust Inc.	2015-08-25	4,988.1	8,002.7	100.0%	6,634.0	360.6	396.2	436.8	6.6%	13.8 x	20.2 x	19.3%	9.2%	12.5%
Global Logistic Properties Limited, Investment Arm	Industrial Income Trust Inc.	2015-07-29	2,210.0	4,268.0	100.0%	3,964.3	125.3	193.7	268.2	6.8%	17.6 x	22.0 x	N/A	N/A	N/A
Lone Star Funds	Home Properties Inc.	2015-06-22	5,132.2	7,883.3	100.0%	5,745.9	300.9	410.5	443.5	7.7%	17.1 x	19.2 x	3.4%	7.1%	0.4%
Maximum			\$ 21,046.5	\$ 34,603.6	100.0%	\$ 20,820.4	\$ 1,537.2	\$ 1,781.3	\$1,716.7	11.6%	35.6 x	28.3 x	22.4%	22.1%	24.4%
75th Percentile			5,306.9	7,913.1	100.0%	4,911.2	314.9	399.8	440.4	10.3%	24.5 x	22.9 x	17.9%	18.9%	17.6%
Median			\$ 4,284.2	\$ 5,811.2	100.0%	\$ 3,790.4	\$ 204.7	\$ 320.9	\$ 370.4	8.2%	20.5 x	21.1 x	13.0%	16.5%	12.5%
25th Percentile			2,522.2	4,220.1	100.0%	3,468.0	158.1	227.8	284.8	7.5%	13.8 x	19.1 x	7.8%	9.8%	6.9%
Minimum			1,976.1	3,763.4	65.4%	3,220.6	56.7	138.8	179.3	5.3%	11.0 x	12.3 x	3.4%	3.9%	0.4%

Mixed results; P/ FFO multiple is higher than AVB's, and the EV / EBITDA multiple is lower.

And premiums indicate that AVB should expect a modest one, at best.

You might sometimes include individual property sales in the Precedent Transactions, but if you do that, you can only look at the NOI and Cap Rate – FFO and EBITDA do not apply to individual properties, and Premiums Paid only apply to public companies.

Ideally, you should **not** mix individual property sales or even property portfolio sales with sales of entire REITs, but if there have been very few transactions in your market, you may have to do this.

Discounted Cash Flow (DCF) Analysis

It is difficult to make a standard Unlevered DCF for a REIT “more complicated,” and the main two differences vs. a DCF for normal companies remain the same:

- 1) Ongoing CapEx and Dispositions** – Unlike in a standard DCF, where you might assume modest CapEx that stays slightly ahead of Depreciation going into the future, in a REIT DCF you assume that the company continues to acquire, develop, redevelop, and dispose of properties.
- 2) Continual Debt and Equity Issuances** – To support these ongoing capital costs, the REIT must also continue to issue Debt and Equity. These issuances are significant because they will change the REIT's share count and, therefore, its Implied Share Price in the DCF. And these issuances continue into the Terminal Period, further complicating the share-count calculation.

One additional difference in this more complex version, shown for AvalonBay below, is that there are **scenarios**, which means that we need different assumptions for the revenue growth rates, margins, and capital costs following the 5 years of projections from the 3-statement model:



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Free Cash Flow Projections:		Units:	Projected:									
			FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27
Revenue:		\$ M	\$ 2,286.9	\$ 2,682.3	\$ 2,536.6	\$ 2,712.2	\$ 2,883.0	\$ 3,041.5	\$ 3,193.6	\$ 3,321.4	\$ 3,437.6	\$ 3,540.7
Revenue Growth Rate:		%	5.9%	17.3%	(5.4%)	6.9%	6.3%	5.5%	5.0%	4.0%	3.5%	3.0%
Upside		%						6.5%	5.5%	4.5%	4.0%	3.5%
Base		%						5.5%	5.0%	4.0%	3.5%	3.0%
Downside		%						5.0%	4.0%	3.5%	3.0%	2.5%
Slightly different outcomes in the last 5 years past the projection period.												
(-) Normalized Operating Expenses:		\$ M	(1,365.4)	(1,615.0)	(1,533.0)	(1,651.2)	(1,768.2)	(1,861.4)	(1,957.7)	(2,042.6)	(2,114.1)	(2,177.6)
% Revenue:		%	59.7%	60.2%	60.4%	60.9%	61.3%	61.2%	61.3%	61.5%	61.5%	61.5%
Upside		%						61.3%	61.5%	61.7%	62.0%	62.0%
Base		%						61.2%	61.3%	61.5%	61.5%	61.5%
Downside		%						61.1%	61.2%	61.3%	61.4%	61.5%
Operating Income (EBIT):		\$ M	921.5	1,067.3	1,003.5	1,061.0	1,114.8	1,180.1	1,235.9	1,278.7	1,323.5	1,363.2
Operating Margin:		%	40.3%	39.8%	39.6%	39.1%	38.7%	38.8%	38.7%	38.5%	38.5%	38.5%
Growth Rate:		%	20.6%	15.8%	(6.0%)	5.7%	5.1%	5.9%	4.7%	3.5%	3.5%	3.0%
Adjustments for Non-Cash Charges:												
(+) Depreciation:		\$ M	628.9	751.0	722.9	786.5	850.5	912.5	958.1	996.4	1,031.3	1,062.2
% Revenue:		%	27.5%	28.0%	28.5%	29.0%	29.5%	30.0%	30.0%	30.0%	30.0%	30.0%
(+/-) Change in Working Capital:		\$ M	(49.6)	44.9	(12.8)	17.8	13.7	11.1	9.1	6.4	5.8	5.2
% Change in Revenue:		%	(38.7%)	11.4%	8.8%	10.2%	8.0%	7.0%	6.0%	5.0%	5.0%	5.0%
% Revenue:		%	(2.2%)	1.7%	(0.5%)	0.7%	0.5%	0.4%	0.3%	0.2%	0.2%	0.1%
(-) Recurring Maintenance CapEx:		\$ M	(77.5)	(83.8)	(89.0)	(94.4)	(99.8)	(103.4)	(108.6)	(116.2)	(120.3)	(123.9)
% Revenue:		%	3.4%	3.1%	3.5%	3.5%	3.5%	3.4%	3.4%	3.5%	3.5%	3.5%
Upside		%						3.2%	3.2%	3.2%	3.2%	3.2%
Base		%						3.4%	3.4%	3.5%	3.5%	3.5%
Downside		%						3.6%	3.6%	3.7%	3.7%	3.7%
CapEx spending also differs, with less spent relative to revenue in the more optimistic cases.												

Similarly, Development, Redevelopment, and Disposition activity differ based on the scenario, with the highest percentages in the Upside Case and the lowest percentages in the Downside Case.

In this model, Acquisition activity follows the opposite trend because we assume that AvalonBay acquires more properties in weaker markets when prices are lower, so the Acquisition *percentage* is highest in the Downside Case.

At the bottom, we calculate Unlevered FCF, discount it to Present Value based on WACC, apply the mid-year discount convention, and assume Stock Issuances to fund a certain percentage of the REIT's Total Capital Costs (around 25% in this case).

These Stock Issuances create additional shares, which we factor in at the top when calculating the REIT's Implied Share Price.

Finally, we also use different assumptions for the Terminal Multiple and Terminal FCF Growth Rate based on the scenario; growth rates and multiples are highest in the Upside Case and lower in the others.

These are linked to the company's FCF growth rate in the last year of the 10-year projections here, as well as the range of numbers from the Public Comps:

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Terminal Value - Multiples Method:

Median Forward EV / EBITDA of Comps: **18.4 x**

Baseline Terminal EBITDA Multiple: **21.7 x**
 Upside **24.4 x**
 Base **21.7 x**
 Downside **17.2 x**

Baseline Terminal Value: \$ 52,640.8
 Implied Terminal FCF Growth Rate: 1.6%

(+) PV of Terminal Value: \$ 34,292.0
 (+) Sum of PV of Free Cash Flows: 8,372.7
Implied Enterprise Value: 42,664.7

% of Implied EV from Terminal Value: 80.4%

(+) Cash & Cash-Equivalents: 201.9
 (+) Equity Investments & Non-Core Assets: 163.5
 (-) Total Debt: (7,290.1)
 (-) Preferred Stock: -
 (-) Noncontrolling Interests: -

Implied Equity Value: 35,740.0

Diluted Shares Outstanding: **176.37**

Implied Share Price from DCF: \$ 202.64
Premium / (Discount) to Current: 22.7%

Dilution:	Number:	Exercise Price:	Dilution:
Options A:	0.150	\$ 126.77	0.056
Options B:	0.008	48.60	0.006
Perf. Awards	0.252		0.252
DownREIT:	0.008		0.008
RSUs:	0.000		0.000
Total:			0.321

Terminal Value - Perpetuity Growth Method:

Expected Long-Term GDP Growth: **2.5%**

Baseline Terminal FCF Growth Rate: **1.6%**
 Upside **2.1%**
 Base **1.6%**
 Downside **0.6%**

Baseline Terminal Value: \$ 52,640.8
 Implied Terminal EBITDA Multiple: 21.7 x

(+) PV of Terminal Value: \$ 34,292.0
 (+) Sum of PV of Free Cash Flows: 8,372.7
Implied Enterprise Value: 42,664.7

% of Implied EV from Terminal Value: 80.4%

(+) Cash & Cash-Equivalents: 201.9
 (+) Equity Investments & Non-Core Assets: 163.5
 (-) Total Debt: (7,290.1)
 (-) Preferred Stock: -
 (-) Noncontrolling Interests: -

Implied Equity Value: 35,740.0

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Options A:	0.150	\$ 126.77	0.056
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Perf. Awards	0.252		0.252
DownREIT:	0.008		0.008
RSUs:	0.000		0.000
Total:			0.321

Different Terminal Multiples and Growth Rates depending on the scenario.

Includes the PV of Future Stock Issuances / Current Share Price.

This entire calculation is also circular since the Option Dilution depends on the Implied Share Price, but the Implied Share Price depends on the Option Dilution.

We create sensitivity tables in the final step, which confirm what we already suspect:

Terminal EV / EBITDA Multiple (Terminal Value Calculated Using the Multiples Method):	Weighted Average Cost of Capital (WACC):										
	4.10%	4.20%	4.30%	4.40%	4.50%	4.60%	4.70%	4.80%	4.90%	5.00%	5.10%
24.40 x	\$ 238.59	\$ 236.25	\$ 233.93	\$ 231.63	\$ 229.36	\$ 227.11	\$ 224.88	\$ 222.67	\$ 220.48	\$ 218.32	\$ 216.18
23.20 x	227.33	225.09	222.87	220.68	218.50	216.35	214.22	212.11	210.02	207.94	205.89
22.00 x	216.07	213.93	211.82	209.72	207.65	205.59	203.56	201.54	199.55	197.57	195.61
20.80 x	204.81	202.77	200.76	198.76	196.79	194.83	192.89	190.98	189.08	187.19	185.33
19.60 x	193.55	191.62	189.70	187.81	185.93	184.07	182.23	180.41	178.61	176.82	175.05
18.40 x	182.29	180.46	178.65	176.85	175.08	173.32	171.57	169.85	168.14	166.44	164.77
17.20 x	171.03	169.30	167.59	165.90	164.22	162.56	160.91	159.28	157.67	156.07	154.48

Only very small portions of the table show Implied Share Prices less than AVB's Current Share Price of \$165.09, indicating that it is likely undervalued.

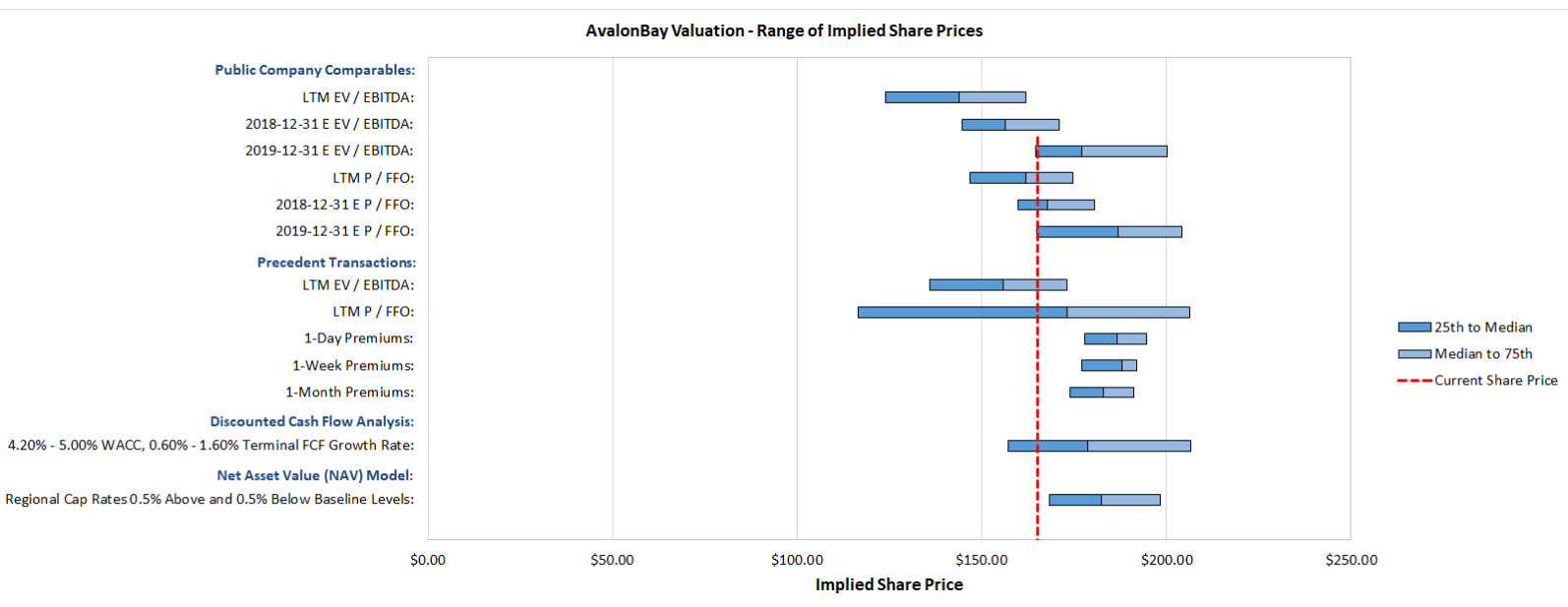
The next tables point to similar conclusions:

		Weighted Average Cost of Capital (WACC):										
		4.10%	4.20%	4.30%	4.40%	4.50%	4.60%	4.70%	4.80%	4.90%	5.00%	5.10%
Terminal FCF Growth Rate (Terminal Value Calculated Using the Gordon Growth Method):	2.10%	\$ 309.70	\$ 292.55	\$ 276.96	\$ 262.73	\$ 249.69	\$ 237.69	\$ 226.62	\$ 216.37	\$ 206.86	\$ 198.01	\$ 189.75
	1.85%	279.68	265.58	252.63	240.70	229.67	219.45	209.95	201.10	192.83	185.08	177.82
	1.60%	255.22	243.36	232.39	222.21	212.73	203.89	195.62	187.87	180.59	173.74	167.29
	1.35%	234.86	224.72	215.28	206.45	198.19	190.44	183.16	176.30	169.83	163.72	157.93
	1.10%	217.63	208.84	200.59	192.85	185.57	178.71	172.23	166.10	160.30	154.79	149.57
	0.85%	202.86	195.13	187.85	180.99	174.50	168.37	162.55	157.03	151.79	146.80	142.05
	0.60%	190.04	183.18	176.69	170.55	164.72	159.19	153.93	148.92	144.15	139.60	135.25

		Weighted Average Cost of Capital (WACC):										
		4.10%	4.20%	4.30%	4.40%	4.50%	4.60%	4.70%	4.80%	4.90%	5.00%	5.10%
Operating Scenario:	Upside	\$ 304.13	\$ 287.31	\$ 272.02	\$ 258.07	\$ 245.28	\$ 233.52	\$ 222.67	\$ 212.62	\$ 203.29	\$ 194.61	\$ 186.52
	Base	255.22	243.36	232.39	222.21	212.73	203.89	195.62	187.87	180.59	173.74	167.29
	Downside	180.38	173.70	167.39	161.41	155.74	150.36	145.24	140.36	135.72	131.29	127.06

It seems like this company could be *overvalued* only if the Downside Case happens and there's a market-wide recession, or if we're quite far off with our basic DCF assumptions; therefore, this company is more likely to be undervalued.

Once we're done with the DCF, the NAV, the Public Comps, and the Precedent Transactions, we create a valuation summary that puts everything side-by-side:



This graph makes it clear that the company is *likely* undervalued at its current share price as of this analysis, especially since the DCF and NAV both produce 25th – 75th percentile ranges of values that are well above AVB's share price.

Dividend Discount Model (DDM)

We [covered the Dividend Discount Model in the valuation section above](#), so we're not going to repeat everything here.



The short version is that the DDM is unlikely to produce more useful valuation results than a standard Unlevered DCF.

It also takes more time and effort to build because you need to forecast the REIT's Debt balance, Net Interest Expense, and other items that affect its Dividends.

You'll see the DDM occasionally, and you can certainly use it in your own analyses, but we don't believe that it adds much value over a traditional Unlevered DCF.

The DDM *may* be useful if, for some reason, the REIT's Dividends fluctuate significantly but its Unlevered FCF does not.

Other Methodologies

You can use other methodologies to value REITs, but many of these other methods are overly simplistic or problematic in other ways.

For example, some people argue that **Dividend Yields** are useful valuation metrics since REITs must issue significant Dividends each year.

That is true, but a single Dividend Yield does not reflect the REIT's growth prospects, leverage, or core strategies in the same way a NAV or DCF does.

For example, a REIT could raise far more Debt than its peers and use high-risk expansion strategies to boost its Dividends, and it might achieve a higher Dividend Yield and become worth more than those peer companies in the process.

However, once the market turns or the REIT can no longer service its Debt properly, its stock price will plummet, and it will have to slash its Dividends.

You could also use a **Levered DCF** (based on Levered Free Cash Flow, AKA Free Cash Flow to Equity) to value a REIT.

Some people argue that it's "better" than an Unlevered DCF because REITs constantly issue Debt and Equity, so it's more realistic to reflect both these activities directly in the DCF.

That argument has some merit, but the Unlevered DCF still tends to produce more consistent results that do not require as many judgment calls or assumptions about the REIT's capital structure.

As a result, we tend to avoid the Levered DCF for the same reasons that we avoid the Dividend Discount Model.

Finally, you can also use the **Replacement Cost** methodology to value entire REITs. With this one, you estimate the cost of reconstructing the REIT's entire property portfolio today, including land and construction costs.

Some REITs even tout this methodology in their earnings reports and investor presentations and claim that they trade, or will trade, at a discount to Replacement Cost:

Assuming a peer EBITDA multiple range⁽¹⁾, Park's assets will likely be valued at a significant discount to replacement cost⁽²⁾



PK EV Range	\$8.0 bn	\$9.0 bn	\$10.0 bn
Portfolio Replacement Cost	\$18.5 bn	\$18.5 bn	\$18.5 bn
Discount to Replacement Cost	57%	51%	46%

	Rooms	Mtg Space (sq. ft.)	Replacement Cost (\$/key)
Top 10	13,949	962k	\$11.0 bn (\$790k/key)
Park Portfolio	35,418	2,300k	\$18.5 bn (\$525k/key)

But as with the Replacement Cost methodology for individual properties, the output varies heavily based on the developers surveyed. A REIT can easily find developers who claim that their property portfolio would be very expensive to reconstruct today.

Also, even if a REIT's Enterprise Value is a discount to its Replacement Cost, that doesn't necessarily mean much because it says more about the overall market (i.e., whether developments or acquisitions are more favorable) than it does about that individual REIT.



It's not as if an investor could use this information to go out and reconstruct all the REIT's buildings instead of buying its shares!

In the example above for Park Hotels, the main conclusion is that construction costs have risen greatly in the company's core areas on the coasts of the U.S.

Our DCF and NAV valuation indicated that the company was *modestly* undervalued by 5-10%, not the 40-60% range in the Replacement Cost output above, so we did not find this argument convincing.

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REIT Debt vs. Equity Decisions

Since Real Estate Investment Trusts are constantly raising capital, financing questions are not about "if" or "when" but rather "Which *type* of capital should we raise this year?"

The trade-offs of Debt and Equity are the same for a REIT as they are for any other company:

- **Debt** tends to be **cheaper** than Equity because lenders target lower returns than common shareholders; also, interest paid on Debt is tax-deductible (although this point matters little for REITs).
- But **Debt** also incurs a **cash cost**, unlike Equity, in the form of interest expense and principal repayments.
- **Debt** also comes with many **restrictions** ("covenants"); lenders won't do a deal if the company's Debt / EBITDA is too high or its EBITDA / Interest is too low, for example.
- So, a company usually prefers to raise the **maximum amount of Debt it can** before issuing Equity. Debt is cheaper, but the company can raise only so much of it.
- If this company **cannot** raise any Debt, or if it can't raise *enough* additional Debt, it will have to issue Equity for some or all of its funding requirements.

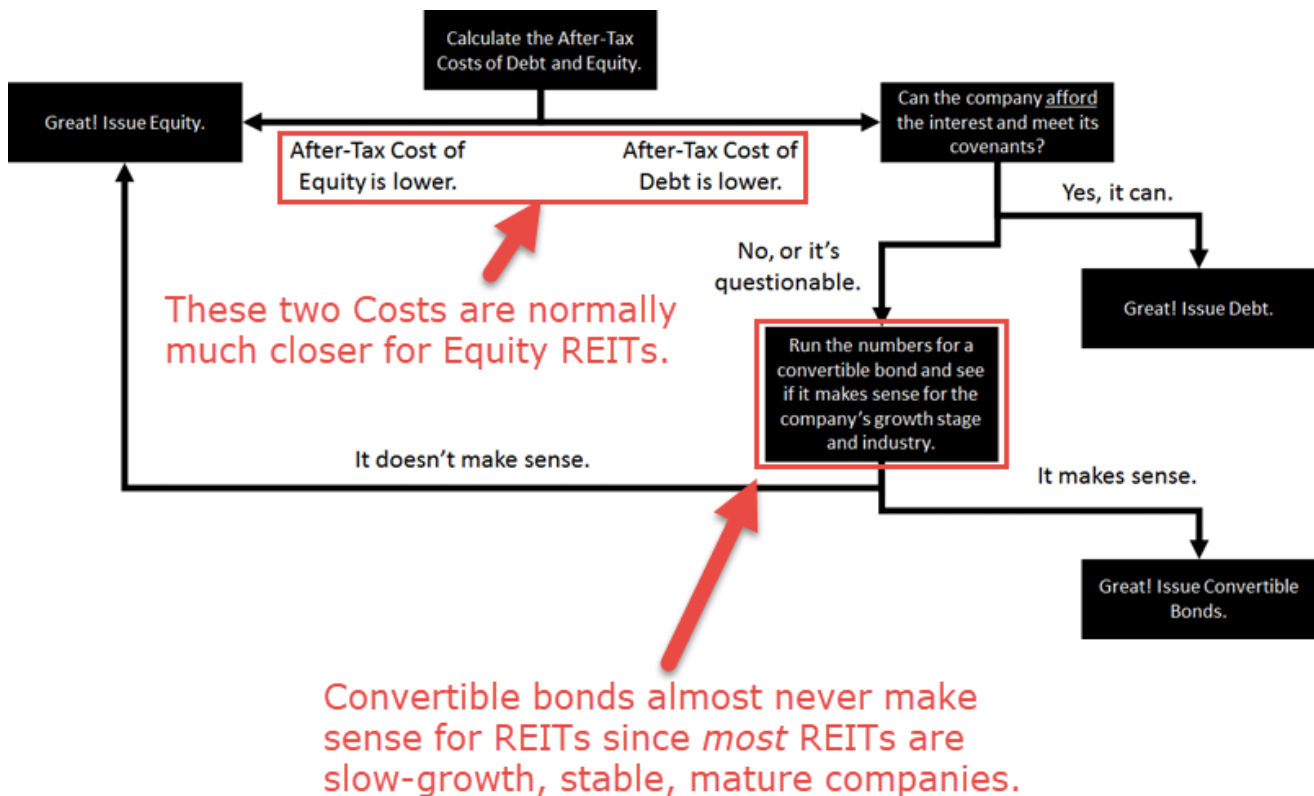
There are some differences because of REITs' business models, tax status, and Dividend requirements:

1. Debt is not necessarily *that much cheaper* than Equity for REITs because they pay no corporate income taxes, or, at least, very low corporate income taxes. So, the Debt vs. Equity decision is sometimes a much closer call.

2. Also, the Cost of Equity tends to be quite **low** for REITs in sectors such as multifamily because these companies have low Betas, meaning low correlation with the overall stock market. This factor also makes the Costs of Debt and Equity closer.
3. The restrictions on Debt might be somewhat different. For example, in addition to the normal constraints on Debt / EBITDA and EBITDA / Interest, REITs might also be concerned about their Dividend Yield, Debt / Total Assets ratio, and EPS or DPU ("Distributions per Unit").

In some cases, REITs might want to avoid outcomes such as EPS dilution or Debt / Total Assets rising above a certain level.

So, the Debt vs. Equity decision-making process is similar, but the numbers and checks differ a bit:



To illustrate this process, we'll walk through the SPH REIT Debt vs. Equity case study in the course, cover the key points, and use the following steps to make a decision:

- 1) Calculate the company's **After-Tax Costs of Equity and Debt** to confirm that Debt is, indeed, cheaper than Equity.



- 2) Create **operational scenarios**, including Base, Downside, and Extreme Downside cases, and reflect them in the cash flow projections or 3-statement model for the company.
- 3) Build a **Debt Schedule** to track the company's Interest Expense, Debt Principal Repayments, and Debt and Cash balances over time (we'll skip this here because it's unnecessary for most REITs – they refinance Debt as it matures).
- 4) Examine the company's **credit stats and ratios** in different scenarios, and pay attention to the more pessimistic ones. Reflect requirements from lenders, such as the maximum leverage and minimum coverage ratios and the maximum Debt / Total Assets ratio.
- 5) If the company **breaches** the requirements from lenders in the more pessimistic cases (e.g., EBITDA / Interest rises to 5.5x, but the maximum is 5.0x), consider alternate structures, such as a higher percentage of Equity.

In this example, SPH REIT currently owns 2 properties in Singapore (the Paragon and Clementi shopping malls) and wants to acquire property #3: the Seletar Mall.

The asking price is S\$ 500 million, and the company wants to use 100% Debt to fund the deal because its Cost of Debt is lower than its Cost of Equity, and using Debt will allow it to maintain its "Distribution Yield" (Dividend Yield) and "Distributions per Unit" (DPU) more easily.

Its specific targets in this deal are as follows:

1. Maintain a Distribution Yield above 5%;
2. Ensure that the deal is neutral or accretive to DPU;
3. Ensure that its Gearing (Debt / Total Assets) stays at or below 35%; and
4. Maintain an Interest Coverage Ratio of at least 5.0x.

We have the company's historical financial statements and data about the performance of their properties in past market cycles.

Additionally, we know that most of the company's tenants are on 3-year leases with fixed rental step-ups each year and that the revenue is recognized on a straight-line basis over the life of the lease.

Since SPH REIT has over 450 tenants, this business model means that the vast majority of its revenue is "locked in" and cannot change in a given period.

Its revenue could change only if there are lease expirations and the new or renewal leases are struck at higher or lower rental rates.

Based on just this information, we should expect relatively small differences between the operational cases, which makes it easier to make a recommendation.

After all, even if rents decline by 10% in the “downturn months” of the Extreme Downside Case, it won’t matter much if only 5% of the company’s leases expire in this downturn period.

The bigger issue is that the value of SPH REIT's Assets will decline substantially in the more pessimistic cases because it's an IFRS-based REIT that marks its properties to market value each year.

Based on this brief description, we already expect that the Gearing Ratio (Debt / Total Assets) may act as the main constraint here.

Step 1: Calculate the After-Tax Costs of Equity and Debt

The case study instructions do not give us the company's Cost of Equity, but we can estimate it by calculating the company's Equity Value, or Market Cap, first:

Operating Assumptions:		Units:	
Company Name:	Name	SPH REIT	Selected Scenario:
Ticker:	Name	SGX:SK6U	Base
Company Type:	Name	Retail REIT	
Last Historical Year:	Date	2017-08-31	
Most Recent Half-Year Period:	Date	2018-02-28	
Valuation Date:	Date	2018-04-19	
Current Share Price:	S\$ as Stated	S\$ 1.00	
Units Outstanding:	Millions	2,567.660	
Months in Year:	#	12	
Conversion Units:	#	100	
Minimum Cash Balance % Annual Property Expenses:	%	90.0%	
Future Funding Requirements - % Debt:	%	100.0%	
Future Funding Requirements - % Equity:	%	-	
Borrowing Costs % Debt Issued:	%	3.0%	

Share price of S\$ 1.00, so Market Cap of approximately S\$ 2.6 billion.

Seletar Mall Acquisition:		%:	S\$:
Forward NOI:			26.0
Capitalisation Rate:	5.20%		
Purchase Price:			500.0
Unit Portion:	-		-
Debt Portion:	100.0%		500.0

Cost of Debt as the company raises different amounts of Debt to fund this S\$ 500 million acquisition.

Range of Debt Raised:		Avg. Interest:
\$ -	S\$ 100.0	2.80%
100.0	200.0	2.90%
200.0	300.0	3.00%
300.0	400.0	3.10%
400.0	500.0	3.20%

Then, we can divide the company's Net Income by its Market Cap to estimate the Cost of Equity in different periods:



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Income Statement:	Units:	Historical:		Projected:			
		FY16	FY17	FY18	FY19	FY20	FY21
(+) Gross Revenue:	\$ M	\$ 209.6	\$ 212.8	\$ 213.6	\$ 251.6	\$ 253.7	\$ 254.9
(-) Property Operating Expenses:	\$ M	(48.7)	(44.7)	(45.1)	(56.1)	(56.2)	(56.1)
Net Property Income (NPI):	\$ M	160.9	168.1	168.5	195.5	197.6	198.8
(-) Income Support:	\$ M	2.4	1.2	-	-	-	-
(-) Amortisation of Intangible Assets:	\$ M	(2.4)	(1.2)	-	-	-	-
(-) Write-Down of Intangible Assets:	\$ M	-	(4.3)	-	-	-	-
(-) Management Fees:	\$ M	(16.3)	(16.7)	(16.8)	(19.6)	(19.8)	(19.9)
(-) Trust Expenses:	\$ M	(1.6)	(1.6)	(1.5)	(1.9)	(1.9)	(1.9)
(+) Finance Income:	\$ M	0.9	0.7	0.7	0.5	0.6	0.5
(-) Finance Costs:	\$ M	(24.0)	(23.9)	(24.1)	(43.2)	(43.3)	(43.4)
Net Income:	\$ M	119.9	122.3	126.9	131.3	133.2	134.2
Cost of Equity - Estimate:	%	4.7%	4.8%	4.9%	5.1%	5.2%	5.2%
(+/-) Change in Fair Value of Investment Properties:	\$ M	7.7	34.9	(76.1)	(48.6)	(31.8)	(50.2)
Total Return Before Taxes & Distribution:	\$ M	127.6	157.2	50.8	82.8	101.4	84.0
(-) Income Tax Expense:	\$ M	-	-	-	-	-	-
Total Return After Taxes & Before Distribution:	\$ M	\$ 127.6	\$ 157.2	\$ 50.8	\$ 82.8	\$ 101.4	\$ 84.0
Cost of Equity - Estimate:	%	5.0%	6.1%	2.0%	3.2%	3.9%	3.3%

We can debate which of these estimates is most accurate, but it is clear that the Cost of Equity is almost certainly higher than the Cost of Debt (2.8% - 3.2% range) for SPH REIT.

Yes, in some cases the Change in Fair Value reduces the Cost of Equity to below the Cost of Debt, but it's better to focus on "normalized" Net Income here.

If we were to calculate the Cost of Equity using the Risk-Free Rate, Beta, and the Equity Risk Premium, we would get similar results, confirming that Equity is indeed more expensive than Debt for SPH REIT.

Step 2: Create Operational Scenarios

Next, we project each property's revenue and expenses based on the percentage of leases expiring in each half-year period, the change in rents, NPI margins ("Net Property Income," the same as Net Operating Income), and the change in prevailing Cap Rates.

We retrieve the lease-expiration data for each period from the company's filings and investor presentations.

Here's a diagram showing how we moved from the historical market data to the projections for Paragon, the first property:

Fig 55: Paragon's rental reversions and occupancy costs

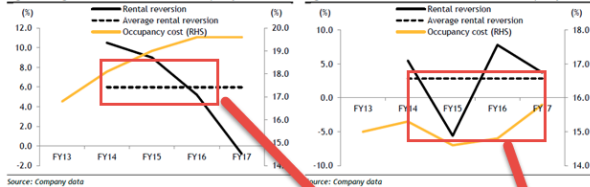
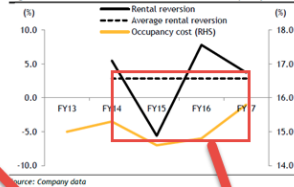


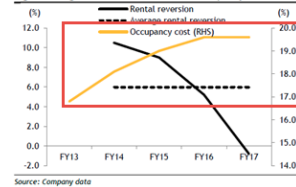
Fig 56: Clementi Mall's rental reversions and occupancy costs



The Base Case Rental Reversions are in-line with the historical averages for both properties; the Downside Case falls to the worst declines for Clementi, and the Extreme Downside Case is slightly below even those.

Occupancy Costs vary by ~3% from peak to trough, so we assume about the same spread between cases here.

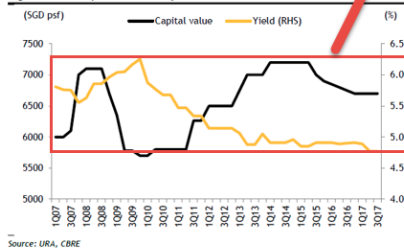
Fig 55: Paragon's rental reversions and occupancy costs



Paragon:		Units:	Project:									
			2H18	1H19	2H19	1H20	2H20	1H21	2H21	1H22	2H22	
Gross Revenue:	\$ M		\$ 86.7	\$ 87.0	\$ 87.0	\$ 87.5	\$ 87.6	\$ 87.9	\$ 88.0	\$ 88.0	\$ 88.1	
Net Property Income:	\$ M		69.2	69.5	69.5	70.0	70.1	70.3	70.4	70.4	70.5	
NPI Margin:	%		79.8%	79.9%	79.9%	80.0%	80.0%	80.0%	80.0%	80.0%	80.0%	
Base	%		79.8%	79.9%	79.9%	80.0%	80.0%	80.0%	80.0%	80.0%	80.0%	
Downside	%		78.5%	78.0%	78.0%	78.5%	78.5%	79.0%	79.0%	79.5%	79.5%	
Extreme Downside	%		77.5%	77.5%	78.0%	78.0%	78.5%	78.5%	79.0%	79.0%	79.0%	
% New/Renewal Leases:	%		8.8%	15.2%	7.4%	16.1%	17.5%	12.0%	17.0%	4.0%	2.0%	
Rental Reversion for New/Renewal Leases:	%		5.0%	4.5%	4.5%	4.0%	4.0%	3.0%	3.0%	2.5%	2.5%	
Base	%		5.0%	4.5%	4.5%	4.0%	4.0%	3.0%	3.0%	2.5%	2.5%	
Downside	%		(5.0%)	(5.0%)	(5.0%)	(3.0%)	(2.0%)	4.0%	3.5%	3.0%	2.5%	
Extreme Downside	%		(7.5%)	(7.5%)	(5.0%)	(5.0%)	(3.0%)	5.0%	4.0%	3.0%	2.5%	
Asset Valuation:	\$ M		2,605.9	2,541.1	2,497.8	2,447.6						
Applicable Forward Net Operating Income (NOI):	\$ M		116.0	116.9	117.4	117.5						
Capitalisation Rate:	%		4.45%	4.60%	4.70%	4.80%						
Base	%		4.45%	4.60%	4.70%	4.80%						
Downside	%		5.25%	5.10%	4.95%	4.80%						
Extreme Downside	%		5.75%	5.40%	5.15%	4.80%						
Change in Fair Value of Investment Property:	\$ M		(89.1)	(64.9)	(43.2)	(50.2)						

From filings and investor presentations.

Fig 60: Retail capital value and yield



Cap Rates spanned the 4.75% - 6.25% range historically, rising to 6.25% in the 2009 downturn. We think they'll move toward the middle of that range, and even if there is a downturn, as in the Extreme Downside Case, it won't be quite as bad as the 2009 one was, resulting in a 5.75% Cap Rate vs. the 6.25% peak back then.

We then repeat this process for Clementi Mall and Seletar Mall, the new property that SPH plans to acquire.

These 3 properties use similar projection methods, but the numbers differ because of different locations and historical trends.

Here are the high-level results in the Base, Downside, and Extreme Downside cases for the property-level revenue and expenses of all 3 properties:



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Consolidated Property Results:		Units:	2H18	1H19	2H19	1H20	Projected:			
							2H20	2H21	2H21	2H21
Net Property Income (NPI):		\$S M	Base Case:							
NPI Margin:		%	84.1	97.7	97.7	98.7	98.9	99.3	99.6	99.9
			78.9%	77.8%	77.6%	77.9%	77.8%	78.0%	78.0%	78.1%
Fair Value of Investment Property:		\$S M	3,701.9	3,653.4	3,621.5	3,571.4				
Gross Yield:		%	6.8%	6.9%	7.0%	7.1%				
Net Yield:		%	5.3%	5.4%	5.5%	5.6%				
Change in Fair Value of Investment Property:		\$S M	(76.1)	(48.6)	(31.8)	(50.2)				
Consolidated Property Results:		Units:	2H18	1H19	2H19	1H20	Projected:			
							2H20	2H21	2H21	2H21
Net Property Income (NPI):		\$S M	Downside Case:							
NPI Margin:		%	82.0	94.8	94.6	94.6	94.7	95.7	96.0	96.6
			77.6%	76.2%	76.1%	76.4%	76.4%	76.9%	77.0%	77.4%
Fair Value of Investment Property:		\$S M	3,162.6	3,219.2	3,362.8	3,493.3				
Gross Yield:		%	7.9%	7.7%	7.4%	7.1%				
Net Yield:		%	6.0%	5.9%	5.7%	5.5%				
Change in Fair Value of Investment Property:		\$S M	(615.4)	56.6	143.6	130.5				
Consolidated Property Results:		Units:	2H18	1H19	2H19	1H20	Projected:			
							2H20	2H21	2H21	2H21
Net Property Income (NPI):		\$S M	Extreme Downside Case:							
NPI Margin:		%	80.7	93.8	94.3	92.7	93.6	94.1	95.0	95.2
			76.5%	75.6%	75.9%	75.5%	76.0%	76.1%	76.6%	76.7%
Fair Value of Investment Property:		\$S M	2,942.7	3,017.1	3,218.2	3,460.8				
Gross Yield:		%	8.4%	8.2%	7.7%	7.2%				
Net Yield:		%	6.4%	6.2%	5.9%	5.5%				
Change in Fair Value of Investment Property:		\$S M	(835.3)	74.4	201.1	242.6				

While the NPI differs in all these cases, there's *more of a difference* in the Fair Value of Investment Property because SPH REIT marks its properties to fair market value.

Next, we complete the rest of the 3-statement model for this company, including an annual summary on the right side.

The setup for this model is virtually identical to every other REIT 3-statement model covered in this course, so we are not going to show the whole thing here.

The only significant difference is that we assume \$S 500 million for the acquisition of Seletar Mall in the first half-year period here, which is funded with some combination of Debt and Equity.

Other than that, it's the same setup as always, with "Distributions" (Dividends) as a percentage of "Income Available for Distribution" (FFO), and the company's capital shortfall funded by 100% Debt (i.e., we assume Debt funding for everything *other than* the Seletar acquisition).

Here's the annual Income Statement in the Base Case with 100% Debt used for the \$S 500 million acquisition in FY 18:



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Income Statement:	Units:	Historical:		Projected:					
		FY16	FY17	FY18	FY19	FY20	FY21		
(+) Gross Revenue:	S\$ M	S\$ 209.6	S\$ 212.8	S\$ 213.6	S\$ 251.6	S\$ 253.7	S\$ 254.9		
(-) Property Operating Expenses:	S\$ M	(48.7)	(44.7)	(45.1)	(56.1)	(56.2)	(56.1)		
Net Property Income (NPI):	S\$ M	160.9	168.1	168.5	195.5	197.6	198.8		
(-) Income Support:	S\$ M	2.4	1.2	-	-	-	-		
(-) Amortisation of Intangible Assets:	S\$ M	(2.4)	(1.2)	-	-	-	-		
(-) Write-Down of Intangible Assets:	S\$ M	-	(4.3)	-	-	-	-		
(-) Management Fees:	S\$ M	(16.3)	(16.7)	(16.8)	(19.6)	(19.8)	(19.9)		
(-) Trust Expenses:	S\$ M	(1.6)	(1.6)	(1.5)	(1.9)	(1.9)	(1.9)		
(+) Finance Income:	S\$ M	0.9	0.7	0.7	0.5	0.6	0.5		
(-) Finance Costs:	S\$ M	(24.0)	(23.9)	(24.1)	(43.2)	(43.3)	(43.4)		
Net Income:	S\$ M	119.9	122.3	126.9	131.3	133.2	134.2		
(+/-) Change in Fair Value of Investment Properties:	S\$ M	7.7	34.9	(76.1)	(48.6)	(31.8)	(50.2)		
Total Return Before Taxes & Distribution:	S\$ M	127.6	157.2	50.8	82.8	101.4	84.0		
(-) Income Tax Expense:	S\$ M	-	-	-	-	-	-		
Total Return After Taxes & Before Distribution:	S\$ M	S\$ 127.6	S\$ 157.2	S\$ 50.8	S\$ 82.8	S\$ 101.4	S\$ 84.0		

The numbers are worse in the Downside and Extreme Downside Cases, but not dramatically so – the main difference is still the Fair Value of Investment Property.

Step 3: Track the Debt, Equity, and Cash Balances

You might create a full Debt Schedule in some analyses, especially if the company has fixed principal repayments and is only raising Debt once in the beginning.

In this case, however, SPH REIT issues Debt every single year, and we assume that whenever there's a maturity, the company refinances and replaces it with new Debt.

Its weighted average interest rate changes based on the amount of Debt used to fund the Seletar acquisition: it increases from 2.80% to 3.20% as the Debt used goes from the S\$0 – \$100 million range to the S\$400 – 500 million range.

These are simplifications, but doing it “the real way” would take far more time and effort and would not yield substantially different results.

The 3-statement model we built in the previous step already tracks the company's Debt, Equity, and Cash balances over time, as well as the Fair Value of Investment Property, so there isn't much to do in this step.

Step 4: Examine the Credit Stats and Ratios in Different Scenarios

Let's start in the Base Case with 100% Debt funding for the acquisition and then tweak the numbers from there:

Key Metrics and Ratios:	Units:	Projected:							
		2H18	1H19	2H19	1H20	2H20	1H21	2H21	
NAV per Unit:	\$S as Stated	\$S	0.91	\$S	0.90	\$S	0.88	\$S	0.88
Earnings per Unit (EPU):	Cents as Stated		(0.50)		2.54		0.66		2.56
Distribution per Unit (DPU):	Cents as Stated		2.85		2.84		2.99		2.86
Distribution Yield:	%		5.7%		5.7%		6.0%		5.7%
(+) NPI from Existing Properties Only:	\$S M		84.1		85.0		84.4		85.8
(-) Income Support:	\$S M		-		-		-		-
(+) Finance Income:	\$S M		0.3		0.2		0.3		0.3
(-) Finance Costs @ Constant Debt and Interest:	\$S M		(12.1)		(12.1)		(12.1)		(12.1)
(-/+) Straight-Line Rental Adjustments:	\$S M		(0.5)		(0.5)		(0.5)		(0.5)
(+) Other Items:	\$S M		(0.5)		0.6		(0.5)		0.6
Income Avail. for Distribution - Pre-Transaction:	\$S M		71.3		73.3		71.5		74.1
Estimated Unit Count - Pre-Transaction:	# Millions:		2,576.147		2,584.633		2,593.120		2,601.606
Estimated DPU - Pre-Transaction:	Cents as Stated		2.85		2.73		2.84		2.74
DPU - Accretion / (Dilution) \$:	Cents as Stated		-		0.12		0.14		0.12
DPU - Accretion / (Dilution) %:	%		-		4.2%		5.1%		4.3%
LTM EBITDA:	\$S M		150.2		161.9		174.0		174.9
LTM EBITDA Growth:	%		0.3%		7.8%		15.9%		8.0%
LTM EBITDA Margin:	%		70.3%		69.7%		69.2%		69.2%
Debt / LTM EBITDA:	x		8.98 x		8.34 x		7.77 x		7.74 x
Net Debt / LTM EBITDA:	x		8.72 x		8.05 x		7.48 x		7.45 x
LTM EBITDA / Interest:	x		6.24 x		4.81 x		4.03 x		4.04 x
LTM EBITDA / Net Interest:	x		6.43 x		4.89 x		4.08 x		4.10 x
Debt / Equity:	%		57.8%		57.8%		59.2%		59.2%
Gearing (Debt / Total Assets):	%		35.9%		35.9%		36.3%		36.3%

Distribution Yield stays above 5.0%, so this is fine.

DPU Accretion stays neutral to positive, so this is fine.

But the Interest Coverage Ratio falls below 5x...

And the Gearing Ratio increases above 35%.

There is no point in looking at the Downside or Extreme Downside Cases in this 100% Debt scenario because if the numbers don't work in the Base Case, they definitely won't work in those cases.

The main problem is the Gearing Ratio, so the only solution is to use **less Debt**.

That's because the Fair Value of Investment Property changes under different market conditions, and this Investment Property is most of the company's Total Assets. Therefore, we can't change the denominator of the Gearing Ratio very much, so we must focus on the numerator instead.

If we try 50% Debt and 50% Equity instead, the numbers work in the Base and Downside Cases (Base Case shown below):

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Key Metrics and Ratios:	Units:	Projected:							
		2H18	1H19	2H19	1H20	2H20	1H21	2H21	
NAV per Unit:	\$S as Stated \$	0.92	\$ 0.91	\$ 0.89	\$ 0.89	\$ 0.88	\$ 0.87	\$ 0.85	
Earnings per Unit (EPU):	Cents as Stated	(0.48)	2.50	0.79	2.51	1.40	2.51	0.77	
Distribution per Unit (DPU):	Cents as Stated	2.72	2.77	2.91	2.78	2.93	2.78	2.93	
Distribution Yield:	%	5.4%	5.5%	5.8%	5.6%	5.9%	5.6%	5.9%	
(+) NPI from Existing Properties Only:	\$S M	84.1	85.0	84.4	85.8	85.3	86.1	85.7	
(-) Income Support:	\$S M	-	-	-	-	-	-	-	
(+) Finance Income:	\$S M	0.3	0.3	0.3	0.3	0.3	0.3	0.3	
(-) Finance Costs @ Constant Debt and Interest:	\$S M	(12.1)	(12.1)	(12.1)	(12.1)	(12.1)	(12.1)	(12.1)	
(-/+) Straight-Line Rental Adjustments:	\$S M	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)	
(+) Other Items:	\$S M	(0.5)	0.6	(0.5)	0.6	(0.5)	0.6	(0.5)	
Income Avail. for Distribution - Pre-Transaction:	\$S M	71.3	73.3	71.6	74.1	72.4	74.4	72.8	
Estimated Unit Count - Pre-Transaction:	# Millions:	2,576.147	2,584.633	2,593.120	2,601.606	2,610.093	2,618.579	2,627.066	
Estimated DPU - Pre-Transaction:	Cents as Stated	2.85	2.73	2.84	2.74	2.86	2.73	2.85	
DPU - Accretion / (Dilution) \$:	Cents as Stated	(0.13)	0.04	0.07	0.04	0.07	0.05	0.08	
DPU - Accretion / (Dilution) %:	%	(4.6%)	1.4%	2.3%	1.4%	2.5%	1.6%	2.8%	
LTM EBITDA:	\$S M	150.2	161.9	174.0	174.9	175.9	176.4	177.0	
LTM EBITDA Growth:	%	0.3%	7.8%	15.9%	8.0%	1.1%	0.9%	0.6%	
LTM EBITDA Margin:	%	70.3%	69.7%	69.2%	69.2%	69.3%	69.4%	69.5%	
Debt / LTM EBITDA:	x	7.32 x	6.80 x	6.33 x	6.30 x	6.27 x	6.26 x	6.25 x	
Net Debt / LTM EBITDA:	x	7.01 x	6.46 x	6.00 x	5.98 x	5.98 x	5.97 x	5.99 x	
LTM EBITDA / Interest:	x	6.24 x	5.66 x	5.28 x	5.30 x	5.32 x	5.33 x	5.35 x	
LTM EBITDA / Net Interest:	x	6.43 x	5.79 x	5.37 x	5.40 x	5.43 x	5.44 x	5.44 x	
Debt / Equity:	%	42.4%	42.4%	43.3%	43.3%	44.0%	44.0%	45.0%	
Gearing (Debt / Total Assets):	%	29.2%	29.1%	29.5%	29.5%	29.8%	29.8%	30.2%	

Yield above 5%.

DPU accretion in most cases; negative in the first half-year period, but on a full-year basis, it's still positive.

Interest Coverage stays above 5x.

And Gearing stays well below 35%.

But the numbers **don't** quite work in the Extreme Downside Case, shown below:

Key Metrics and Ratios:	Units:	Projected:							
		2H18	1H19	2H19	1H20	2H20	1H21	2H21	
NAV per Unit:	\$S as Stated \$	0.65	\$ 0.65	\$ 0.67	\$ 0.67	\$ 0.74	\$ 0.73	\$ 0.81	
Earnings per Unit (EPU):	Cents as Stated	(28.74)	2.37	5.01	2.33	9.38	2.35	10.81	
Distribution per Unit (DPU):	Cents as Stated	2.59	2.63	2.79	2.58	2.74	2.61	2.77	
Distribution Yield:	%	5.2%	5.3%	5.6%	5.2%	5.5%	5.2%	5.5%	
Estimated DPU - Pre-Transaction:	Cents as Stated	2.72	2.58	2.71	2.57	2.69	2.58	2.72	
DPU - Accretion / (Dilution) \$:	Cents as Stated	(0.13)	0.05	0.08	0.02	0.05	0.03	0.05	
DPU - Accretion / (Dilution) %:	%	(4.6%)	2.0%	2.9%	0.7%	1.9%	1.0%	2.0%	
LTM EBITDA:	\$S M	147.2	155.3	167.4	166.4	165.8	167.0	168.3	
LTM EBITDA Growth:	%	(1.7%)	3.4%	13.7%	7.2%	(0.9%)	0.3%	1.5%	
LTM EBITDA Margin:	%	69.2%	67.6%	67.4%	67.4%	67.4%	67.7%	68.0%	
Debt / LTM EBITDA:	x	7.47 x	7.08 x	6.58 x	6.62 x	6.66 x	6.61 x	6.57 x	
Net Debt / LTM EBITDA:	x	7.12 x	6.69 x	6.20 x	6.24 x	6.31 x	6.27 x	6.27 x	
LTM EBITDA / Interest:	x	6.11 x	5.43 x	5.07 x	5.04 x	5.02 x	5.05 x	5.08 x	
LTM EBITDA / Net Interest:	x	6.30 x	5.56 x	5.18 x	5.15 x	5.13 x	5.16 x	5.19 x	
Debt / Equity:	%	60.0%	60.0%	57.8%	57.8%	52.4%	52.4%	47.1%	
Gearing (Debt / Total Assets):	%	36.6%	36.4%	35.5%	35.5%	33.4%	33.4%	31.1%	

The Gearing goes above 35% in some periods; also, while the Interest Coverage technically stays above 5.0x, there's very little "cushion" in case something goes wrong here.



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So, let's tweak this a bit more and try 60% Equity and 40% Debt and review the results in the Extreme Downside Case:

Key Metrics and Ratios:	Units:	Projected:							
		2H18	1H19	2H19	1H20	2H20	1H21	2H21	
NAV per Unit:	<i>S\$ as Stated</i>	S\$ 0.65	S\$ 0.65	S\$ 0.68	S\$ 0.67	S\$ 0.74	S\$ 0.74	S\$ 0.82	
Earnings per Unit (EPU):	<i>Cents as Stated</i>	(28.48)	2.36	4.95	2.31	9.25	2.34	10.65	
Distribution per Unit (DPU):	<i>Cents as Stated</i>	2.57	2.61	2.77	2.56	2.72	2.59	2.75	
Distribution Yield:	%	5.1%	5.2%	5.5%	5.1%	5.4%	5.2%	5.5%	
Estimated DPU - Pre-Transaction:	<i>Cents as Stated</i>	2.72	2.58	2.71	2.57	2.69	2.58	2.72	
DPU - Accretion / (Dilution) \$:	<i>Cents as Stated</i>	(0.15)	0.03	0.06	(0.00)	0.03	0.01	0.03	
DPU - Accretion / (Dilution) %:	%	(5.5%)	1.1%	2.1%	(0.0%)	1.1%	0.2%	1.2%	
LTM EBITDA:	<i>S\$ M</i>	147.2	155.3	167.4	166.4	165.8	167.0	168.3	
LTM EBITDA Growth:	%	(1.7%)	3.4%	13.7%	7.2%	(0.9%)	0.3%	1.5%	
LTM EBITDA Margin:	%	69.2%	67.6%	67.4%	67.4%	67.4%	67.7%	68.0%	
Debt / LTM EBITDA:	x	7.13 x	6.76 x	6.28 x	6.32 x	6.35 x	6.31 x	6.27 x	
Net Debt / LTM EBITDA:	x	6.77 x	6.36 x	5.89 x	5.93 x	6.00 x	5.96 x	5.96 x	
LTM EBITDA / Interest:	x	6.11 x	5.58 x	5.32 x	5.28 x	5.25 x	5.29 x	5.32 x	
LTM EBITDA / Net Interest:	x	6.30 x	5.71 x	5.43 x	5.41 x	5.38 x	5.41 x	5.44 x	
Debt / Equity:	%	55.7%	55.7%	53.7%	53.7%	48.8%	48.8%	44.0%	
Gearing (Debt / Total Assets):	%	34.9%	34.7%	33.9%	33.8%	31.8%	31.8%	29.7%	

There's more of a "cushion" with 60% Equity and 40% Debt, and the Gearing Ratio is fine, never crossing the 35% limit.

If we used more than 60% Equity, the DPU accretion/dilution would move closer to "neutral" rather than "accretive," so we don't think a 70/20 or 80/20 split would work as well.

Therefore, we settle on 60% Equity and 40% Debt as the best financing mix for this S\$ 500 million acquisition.

Step 5: Consider Alternate Structures, Including Higher Percentages of Equity

We already completed this one as part of Step 4 above, and we looked at higher percentages of Equity to find the most appropriate capital structure.

If SPH were a tech or biotech company instead of a REIT, we might consider a convertible bond in this step and examine the credit stats and ratios across the cases there.

However, convertible bonds are rarely viable options for mature, stable REITs, so that's not worth delving into here.



This recommendation is pretty clear because the company's plans to use 100% Debt don't even make sense in the *Base Case*, so clearly, it must use less Debt and more Equity.

It wouldn't necessarily be "wrong" if you recommended 50% Debt or 30% Debt instead of the 40% Debt figure we settled on.

The case study does not give specific numbers for the rental reversions, NPI margins, or Cap Rates, so there is some subjectivity involved.

However, it is safe to say that a 100% Debt or 100% Equity recommendation would be wrong.

Debt vs. Equity Analysis – Other Points

In credit analysis, lenders always focus on the **risk factors** because they have limited upside but the potential to lose everything in pessimistic outcomes.

The biggest risk factor here is that SPH REIT's Gearing Ratio is understated since the company seems to be valuing its properties with below-market Cap Rates.

For reference, see this table that compares SPH REIT to its peer companies at the time of this deal:

	CMT:	FCT:	MCT:	SGREIT:	SPH REIT:
Avg. Interest:	3.2%	2.4%	2.8%	3.1%	2.8%
Gearing:	35%	29%	36%	35%	25%
Int. Coverage:	4.2x – 4.6x	6.6x	4.8x	3.7x – 3.8x	6.2x – 6.4x
Debt % Market Cap:	9%	31%	8%	15%	33%
% Fixed-Rate Borrowings:	100%	55%	74%	99%	86%
WALE (Years):	2.0	1.7	2.0	4.9	2.1
Total NLA (m. sq. ft.)	5.5	1.1	1.1	2.2	0.9
# Malls	16	6	1	11	2
# Tenants	2,900	655	350	316	454
Avg. Shopper Traffic (m)	347	102	56	25	48



CapitaLand Mall Trust (CMT), one of the peer companies, has a Gearing Ratio of 35% and a Debt / Market Cap of 9%. By contrast, SPH has a 25% Gearing Ratio but a Debt / Market Cap of 33%.

This result implies that CMT may be using higher Cap Rates than SPH to value its properties, despite similar locations, leases, and tenant types.

If you look in the company's filings, you'll also see that management used a 4.50% Cap Rate for the retail portion of Paragon, even though the retail average in Singapore at the time was 4.80%, which further supports this argument.

Another risk factor is that SPH is smaller and less diversified than many of its peer REITs – it owns 2 malls vs. 6, 11, or 16 for several of the others, and it has less foot traffic and a smaller tenant base.

Additionally, ~40% of its Gross Rental Income comes from tenants in the luxury and fashion sectors, creating even more risk if those sectors decline.

One final risk factor is that SPH REIT's Weighted Average Lease Expiration (WALE) is 2.1 years; 73% of Clementi Mall's leases expire in 2020, and 34% of Paragon's leases expire in 2020.

A significant portion of the Company's Debt (S\$ 280 million, or 33% of the total) also matures in the same time frame (FY 20), creating significant refinancing risk for the Company in that year. As a result, the Company's interest rates could easily increase in the future, and it might struggle to comply with its targeted Interest Coverage Ratio of 5.0x.

To mitigate these risks, lenders could apply their own Cap Rates to SPH's portfolio and refuse to lend above their internal Gearing Ratio limit (e.g., 35% according to higher Cap Rates), or they could link the interest rates to the occupancy rate, rent levels, or average rental reversions.

It's also worth thinking about the company's other options to complete this deal using Debt and Equity and still comply with its targeted credit stats and ratios.

For example, some REITs may be able to reduce their Dividend Yields and improve some of these figures since they'll retain more Cash, boosting their Total Assets.

Another option might be to change the mix of Debt and Equity for funding ongoing operations and use less than 100% Debt.

In this case, these options are not feasible because SPH REIT also has a minimum Distribution Yield of 5%, so it can't reduce Dividends by much, or it might fall below that.

Also, increasing the percentage of Equity might push the company closer to DPU dilution, which it wants to avoid.

And the main problem, as always, is the 35% Gearing limit. We can cut Dividends or use more Equity, but if the market enters a recession right after the Seletar deal closes, we could easily run into trouble because of that Gearing limit.

So, we don't see many viable alternatives to using less Debt and more Equity for the Seletar acquisition.

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REIT M&A Deals and Merger Models

A REIT might acquire another REIT for the same reason that a normal company might acquire another normal company: it believes it will be better off afterward.

In other words, the Buyer believes that the Purchase Price is less than the Present Value of the Seller's future cash flows.



Equivalently, the Buyer believes that the IRR of the acquisition exceeds its Discount Rate (WACC or the Cost of Equity, depending on how you're measuring the IRR).

REITs' *specific motivations* for mergers and acquisitions often include:

- 1) **Geographic Expansion** – REITs frequently acquire other REITs, and sometimes individual assets, to expand into new regions and diversify their footprint. For example, a REIT that is based in the coastal areas of the U.S. might acquire one with a Sunbelt or Midwest focus to diversify in case there's an economic downturn on the coasts.
- 2) **Enhanced Scale and "Value Creation"** – Often, REITs argue that with greater scale (i.e., more Assets, Revenue, NOI, and FFO) and higher-quality portfolios, they will trade at higher valuation multiples. For example, the Buyer trades at 15x P / FFO and the Seller trades at 13x P / FFO, but after the deal, they will trade at 18x P / FFO, in-line with the multiples of a larger peer company. It's a "1 + 1 = 3" argument, and if you factor in cost synergies, the numbers look even better.
- 3) **Tenant Diversification** – If a REIT earns, say, 80% of its rental income from its top 10 tenants, that's much riskier than earning only 30% of its rental income from those same tenants. Even if

these tenants are “blue-chip companies,” no REIT wants to be overly dependent on a few large customers. Therefore, tenant diversification often motivates REIT M&A.

- 4) **Synergies** – Synergies play a role in almost any REIT M&A deal. Since REITs all have similar business models, it’s feasible to cut costs without impacting the business; a REIT that owns twice as many properties won’t need twice as many employees, especially at the corporate level. Revenue synergies may play a role in higher-growth sectors and ones with income sources beyond rent and tenant reimbursements (e.g., data center and healthcare REITs).

REIT Merger Models: Key Differences



A merger model is still a merger model for REITs: you still combine the financial statements of the Buyer and Seller, adjust for acquisition effects, create Goodwill, and calculate EPS accretion/dilution.

The differences lie in the specific adjustments and the supplemental analyses.

Here’s a summary of the key differences:

Aspect of M&A Deal	How It Works Differently for a REIT
Form of Payment	Most REITs can use only Stock (or mostly Stock) to fund deals; they have low Cash balances due to the high Dividend requirements. Also, REITs already tend to be highly leveraged because of the Debt they raise to fund their everyday operations. They might be able to issue slightly more Debt to fund M&A deals, but 100% Debt deals are highly unlikely. So, REIT M&A deals tend to be 100% Stock or at least > 50% Stock.
Purchase Price Allocation – Elimination of Seller’s Accumulated Depreciation and Deferred Rent	The Buyer must adjust the Seller’s Assets and Liabilities up or down to Fair Market Value in any M&A deal, and for a REIT, this means eliminating the Seller’s Accumulated Depreciation and adjusting its Gross Real Estate Assets up so that Net Real Estate Assets represents Fair Market Value. Items such as Deferred Rent are also written down because all Book vs. Cash differences are reconciled upon transaction close. Under IFRS, these adjustments are much smaller because there is no Accumulated Depreciation. You might just see a slight adjustment to the Fair Value of the Seller’s Investment Properties.

Purchase Price Allocation – New Intangible Assets	Acquirer REITs create new Intangible Assets for the Tenant Relationship Value, Above and Below-Market Leases, and the Acquired In-Place Lease Value; there may also be something for Tenant Origination Costs. These items represent the future benefits (or costs) from properties with leases above or below market rates, and from the fact that existing tenants are more likely to renew their leases, which reduces the REIT's potential future TIs and LCs. Future NOI from these existing tenants is also more likely, which factors into some of these items. Like all Intangibles, these amortize over time on the Income Statement and are reversed on the Cash Flow Statement.
Treatment of Seller's Existing Debt and Preferred Stock	This point can be a deal-breaker because the existing Debt may <i>not</i> be assumable under a Change of Control. If it is not, then the Buyer will have to refinance the existing Debt with a bridge loan, which usually has a higher interest rate, and then eventually replace it with a permanent loan. But a refinancing could also help the combined company if, for example, interest rates have fallen significantly, and the Buyer can replace the bridge loan quickly.
Taxes	There is some complexity around the taxation of OP and DownREIT Units purchased in M&A deals and the Asset Gains that the Seller realizes when its property is sold and revalued. These points don't necessarily play a big role in financial models, but they do affect the Seller's view of the deal.
Cost Savings and Restructuring	Consolidation is predictable in real estate, so cost savings can be quite significant; it's not unusual for a Buyer to cut 50%+ of the Seller's G&A expenses in deals. However, G&A is very small next to Operating Expenses and Property Taxes, and it does take time for the Buyer to realize these cost savings. And if it succeeds in doing that, there are usually restructuring costs associated with the effort.
Accretion/Dilution	You can still calculate EPS accretion/dilution, but since EPS is not that useful a metric for REITs, you tend to focus on FFO PS and AFFO PS accretion/dilution. European REITs might focus on accretion/dilution to EPRA Earnings per Share.

Contribution Analysis	Since most REIT M&A deals use 100% Stock or nearly 100% Stock, the Contribution Analysis is extremely important. For example, it lets you see if the Buyer will own 75% of the combined entity when it contributes 75% of the NAV, EBITDA, and FFO.
Value Creation Analysis	As discussed above, this one is often a key rationale for REIT M&A deals: the combined company should, in theory, trade at higher multiples than the Buyer or Seller alone. At those higher multiples, the Buyer's shareholders should get a boost and end up with shares that are worth more than they were before the deal.
Alternative Deal Assessments	Finally, you can also evaluate a deal by calculating the Internal Rate of Return (IRR) and comparing it to the Buyer's WACC or Cost of Equity . Many REIT executives also think of deals in terms of Replacement Costs and Cap Rates, so you can look at how these metrics change in a deal as well.

The Step-by-Step Process to Build a REIT Merger Model

We'll walk through the Digital Realty / DuPont Fabros model in the course here and cover the main points.

You've already seen many examples of REIT 3-statement models and valuations, so we are **not** going to repeat that material here; we will focus on the *new items* unique to REIT M&A:

Step 1: Build (Simplified) 3-Statement Projection Models and Valuations for the Buyer and Seller

You could follow the standard approach here and project revenue and expenses by geography and business segment, or you could simplify the model by combining some of these items.

In theory, you could skip full Cash Flow Statements for the Buyer and Seller, but in practice, it may be tricky to do that because REITs constantly raise Debt and Equity and issue Dividends.

Here, we create quarterly 3-statement models for the Buyer (Digital Realty or "DLR") and the Seller (DuPont Fabros or "DFT"), but the quarterly setup is not truly necessary.

We also simplify the assumptions and key drivers to make the model easier to manage:



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Income Statement Drivers:	Units:	Projected:										
		2Q 17	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19	2Q 19	3Q 19	4Q 19
Same-Store, Same-Capital Rent - YoY Growth:	%	1.0%	0.5%	2.0%	0.5%	0.8%	0.4%	1.5%	0.4%	0.7%	0.4%	1.2%
Property Expenses % Rental Income:	%	52.0%	50.0%	50.0%	50.0%	52.0%	50.0%	50.0%	50.0%	52.0%	50.0%	50.0%
Tenant Reimbursements % Property Expenses:	%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Other Income - YoY Growth:	%	5.0%	4.5%	4.5%	4.0%	4.0%	3.5%	3.5%	3.0%	3.0%	2.5%	2.5%
Other Expenses % Other Income:	%	82.3%	82.3%	82.3%	82.3%	82.3%	82.3%	82.3%	82.3%	82.3%	82.3%	82.3%
NOI Margin on Existing Properties:	%	65.8%	66.7%	66.7%	66.7%	65.8%	66.7%	66.7%	66.7%	65.8%	66.7%	66.7%
Development Spending:	\$ M	85.0	85.0	90.0	90.0	90.0	90.0	95.0	95.0	100.0	100.0	105.0
Completed & Stabilized Developments:	\$ M	70.0	75.0	80.0	80.0	85.0	85.0	90.0	90.0	90.0	90.0	95.0
Average Stabilized Yield:	%	5.5%	5.5%	5.5%	5.5%	5.5%	5.5%	5.5%	6.0%	6.0%	6.0%	6.0%
NOI from Stabilized Development:	\$ M	3.9	4.1	4.4	4.4	4.7	4.7	5.0	5.4	5.4	5.4	5.7
Rent + Tenant Reimbursements from Developments:	\$ M	5.9	6.2	6.6	6.6	7.1	7.0	7.4	8.1	8.2	8.1	8.6
Property Expenses from Developments:	\$ M	2.0	2.1	2.2	2.2	2.4	2.3	2.5	2.7	2.8	2.7	2.9
Tenant Reimbursements from Developments:	\$ M	2.0	2.1	2.2	2.2	2.4	2.3	2.5	2.7	2.8	2.7	2.9
Rental Income from Developments:	\$ M	3.9	4.1	4.4	4.4	4.7	4.7	5.0	5.4	5.4	5.4	5.7

We're simplifying by assuming all Developments take 1 year to finish and assuming a simple Yield on them. Same-Store Rental Growth, Expenses, and Tenant Reimbursements are all simple percentages as well.

We then create sets of Public Comps and M&A Comps, a DCF, and NAV Models for the Buyer and Seller; [please refer to the Valuation section for more.](#)

Step 2: Make Assumptions for the Purchase Price, Deal Funding, and Fees

Merger Model - Digital Realty Trust, Inc. and DuPont Fabros Technology, Inc.
(\$ USD in Millions Except Per Share and Per Unit Amounts in USD as Stated)

Transaction Assumptions:		Units:
Exchange Ratio:	x	0.545 x
Implied Offer Price for Seller:	\$ as Stated	\$ 64.32
Implied Premium to Seller's Share Price:	%	15.8%
Equity Purchase Price:	\$ M	\$ 5,831.2
Acquisition Close Date:	Date	2017-09-30
% Cash Consideration:	%	-
Total Cash Consideration (via New Debt):	\$ M	-
% Stock Consideration:	%	100.0%
Total Stock Consideration:	\$ M	5,831.2
Synergy Assumptions:		
Annual Cost Synergies:	\$ M	\$ 18.0
% Cost Synergies Realized in Year 1:	%	50.0%
% Cost Synergies Realized in Year 2 and Beyond:	%	100.0%
Merger & Integration Costs % Total Cost Synergies: (Incurred within Year 1 following acquisition close)	%	150.0%

100% Stock deal, so Exchange Ratio drives everything. Offer Price = Exchange Ratio * Buyer Share Price.

Acquisition Funding:		%:	Amount:	Pre-Tax Cost:	After-Tax Cost:
Senior Notes:	-	-	\$ -	3.70%	3.60%
Common Equity:	100.0%	5,831.2			1.24%
New Shares Issued (Millions):		49.408			
Weighted Average Acquisition Cost:					
Target's "Yield":					1.24%
					3.01%
Fees:		%:	Amount:	Treatment:	
Advisory Fee %:	1.75%	\$ 102.0		Expensed.	
Debt Issuance Fee %:	3.00%	45.7		Deducted from BV of Debt.	
Preferred Issuance Fee %:	3.00%	6.0		Deducted from BV of Preferred.	
Legal and Other Fees:		18.0		Expensed.	
Total Fees:			\$ 171.8		

We can guess the deal will be EPS-accretive, but EPS is nearly irrelevant for REITs, so...

Valuation Multiples:	Current Price:	Offer Price:	Public Comps:	Buyer:
2017-12-31 EV / EBITDA:	18.5 x	20.7 x	21.0 x	21.1 x
2017-12-31 P / FFO:	17.5 x	20.3 x	23.7 x	20.3 x
2017-12-31 P / AFFO:	17.8 x	20.6 x	26.2 x	22.8 x
2018-12-31 EV / EBITDA:	15.5 x	17.3 x	18.2 x	19.0 x
2018-12-31 P / FFO:	14.6 x	16.9 x	21.5 x	18.3 x
2018-12-31 P / AFFO:	15.0 x	17.3 x	23.3 x	20.5 x

Seller seems undervalued, even at the Offer Price, and its multiples are still below the Buyer's.

Cost synergies are fairly modest here, but will still take 2 years to realize fully.

We also set up the Sources & Uses schedule, which is moderately important here because the Seller's existing Debt and Preferred Stock **get refinanced** in the deal.

In this case, the refinancing benefits the combined company because the Buyer can get lower coupon rates on these issuances, but it could easily go the other way as well:

Sources & Uses of Funds:

Post-Transaction Minimum Cash Balance:	\$ M	\$ 126.0
Post-Transaction Combined Cash Balance:	\$ M	126.0
Cash Available for Transaction Fees:	\$ M	-
Cash Used for Transaction Fees:	\$ M	-
Debt Issued for Transaction Fees:	\$ M	171.8

If Cash is available to pay for the Transaction Fees, we assume that here; otherwise, we raise more Debt to pay for them.

Sources:	Interest/Coupon		
		Rate:	Differential:
Senior Notes Issued:	\$ 171.8	3.70%	
Refinance Target's Debt:	1,523.5	3.34%	(0.46%)
Refinance Target's Preferred Stock:	201.3	5.25%	(1.37%)
Common Stock Issued:	5,831.2		
Cash for Transaction Fees:	-		
Total Sources:	\$ 7,727.7		

The real difference between "assuming" and "refinancing" Debt is that the interest rates and repayment terms might change if Debt is refinanced. "Refinance" here means "replace."

Uses:	Interest/Coupon	
		Rate:
Equity Purchase Price of Target:	\$ 5,831.2	
Refinance Target's Debt:	1,523.5	3.80%
Refinance Target's Preferred Stock:	201.3	6.62%
Transaction Fees:	120.0	
Financing Fees:	51.7	
Total Uses:	\$ 7,727.7	

CHECK:

OK!

It would be *exceptionally* odd for a REIT Buyer *not* to assume or replace a REIT Seller's Existing Debt. Cash is insufficient for repayment, and issuing Stock to repay Debt makes **no** sense.

Step 3: Write Up Assets and Create the New Intangibles

As discussed at the start of this section, several new Intangible Assets get created in REIT M&A deals: Above- and Below-Market Leases, Acquired In-Place Lease Value, Tenant Relationship Value, and Tenant Origination Costs.

Estimating the values of these items is very difficult without detailed information from the Seller, such as tenant-by-tenant lease schedules and market rents in each region.

We reviewed the corporate filings for this deal, found the real numbers that both sides agreed upon, and then tried to make ours match by using guesstimates.

We choose to combine several of these items into a single "Tenant Relationship Value" item, which represents the future benefit from *not* having to go out and search for new tenants:

Tenant Relationship Value:	
NTM Rental Income - Existing Properties:	358.3
Weighted Average Lease Term (Years):	6.0
Annual Growth Rate:	1.0%
Rental Income At End of Lease Term:	380.4
NOI Margin At End of Lease Term:	65.0%
Net Operating Income:	247.2
Renewal Probability:	66.7%
Renewal NOI:	164.8
Applicable Capitalization Rate:	7.0%
Value of Tenant Relationships:	2,354.7
Discount Rate:	6.8%
PV of Tenant Relationships:	1,589.9
Amortization Period (Years):	14.0

Top part assumes that Existing Rental Income grows by 1% per year over 6 years, and that NOI margin remains at 65% by the end, which gets us the NOI before the "average lease" expires.

Then, we assume there's a 2/3 chance of all the tenants renewing, which we value at a 7.0% Cap Rate, in the middle of the range used in the NAV Model. We then discount that to Present Value based on WACC (from the DCF).

And then we make simple estimates for the percentages of leases at above- and below-market rates, and the differentials to those market rates (by backing into the numbers in the filings):

Above-Market and Below-Market Leases:			
% Rental Income - Above-Market Leases:	30.0%	% Rental Income - Below-Market Leases:	40.0%
NTM Rental Income - Existing Properties:	358.3	NTM Rental Income - Existing Properties:	358.3
Average In-Place Differential to Market Rent:	10.5%	Average In-Place Differential to Market Rent:	(9.0%)
Above-Market Rent Portion:	11.3	Below-Market Rent Portion:	(12.9)
Applicable Capitalization Rate:	7.0%	Applicable Capitalization Rate:	7.0%
Value of Above-Market Leases:	161.2	Value of Below-Market Leases:	(184.3)
Amortization Period (Years):	4.0	Amortization Period (Years):	12.0

Midpoint of the range used in the NAV Model.

	Fair Value	Weighted Average Remaining Intangible Amortization Life (in months)
Land	\$ 312,579	
Buildings and improvements	3,677,497	
Cash and cash equivalents	20,650	
Accounts and other receivables	10,978	
Acquired above-market leases	162,333	47
Goodwill	2,592,181	
Acquired in-place lease value, deferred leasing		
Tenant relationship value	980,267	220
Acquired in-place lease value	557,128	77
Tenant origination costs	44,990	80
Global revolving credit facility, net ⁽¹⁾	(450,697)	
Unsecured term loans ⁽¹⁾	(250,000)	
Unsecured senior notes, net ⁽²⁾	(886,831)	
Mortgage loans ⁽¹⁾	(105,000)	
Acquired below-market leases	(185,543)	137
Accounts payable and other accrued liabilities	(248,259)	
Other working capital, net	(22,640)	
Total equity consideration for DFT merger	\$ 6,209,633	

And our estimates match up fairly well to the ones in the filings from the deal announcement.

These other items add up to around \$1.6 billion total, fairly close to our estimate for everything under "Tenant Relationship Value."

Finally, we also assume write-ups of the Land, Construction in Progress, and Operating Real Estate Assets, based loosely on the company's "Fair Value" estimates in the screenshot above, which indicate about \$4.0 billion in total value post-transaction:

Real Estate Asset Write-Ups:

Estimated Discount to Market Value:	5.0%
Operating Real Estate Asset Market Value:	3,341.5
Write-Up of Operating Real Estate Assets:	167.1
Depreciation Period (Years):	39.0
Market Value Adjustment for Land & CIP:	20.0%
Land & CIP Fair Value:	746.1
Write-Up of Land & CIP:	124.3

We picked these numbers such that the Seller's Net Real Estate increases from \$3.0 billion to \$4.0 billion post-transaction, as in the real deal. Remember that we also write down ~\$750 million of Accumulated Depreciation!

Determining these numbers in real life is usually not the responsibility of the bankers – the company must supply the correct lease and tenant information, and an audit/valuation firm will use that information to estimate these items.

However, you still need to understand the *process* because it affects the rest of the model.

Step 4: Allocate the Purchase Price and Combine the Balance Sheets

The Goodwill calculation is straightforward and, as always, reflects the Equity Purchase Price, the write-off of the Seller's Common Shareholders' Equity, the write-ups, the new Intangibles, and several items that get eliminated:

Purchase Price Allocation:

Goodwill Calculation:

Equity Purchase Price:	\$ 5,831.2
(-) Seller Common Book Value:	(765.6)
(+) Write-Off of Existing Goodwill:	-
Total Allocable Purchase Premium:	5,065.6
(-) Write-Up of Operating Real Estate Assets:	(167.1)
(-) Write-Up of Land & Construction in Progress:	(124.3)
(-) Elimination of Accumulated Depreciation:	(747.3)
(+) Elimination of Deferred Rent:	122.5
(-) Acquired Above-Market Leases:	(161.2)
(+) Acquired Below-Market Leases:	184.3
(-) Acquired In-Place Lease Value:	(1,589.9)
Total Goodwill Created:	\$ 2,582.5

Accumulated Depreciation always gets eliminated because Real Estate Assets must be marked to fair market value upon change of control; Deferred Rent represents timing differences that are reconciled and closed upon transaction close.

There is no new Deferred Tax Liability (DTL) here because REITs have corporate tax rates of 0% or close to 0%. For the same reason, there are also no items related to DTAs.



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Next, we combine the Balance Sheets of the Buyer and Seller:

Balance Sheet:	Units:	Projected:		Transaction Adjustments:			
		2Q 17	3Q 17	Seller	Debit	Credit	3Q 17
ASSETS:							
Real Estate Assets:							
(+) Operating Real Estate Assets:	\$ M	\$ 10,992.9	\$ 11,135.2	\$ 3,174.4	\$ 167.1	\$ -	\$ 14,476.7
(+) Construction in Progress and Development Land:	\$ M	1,070.4	1,120.4	621.7	124.3	-	1,866.5
Gross Real Estate Assets:	\$ M	12,063.3	12,255.6	3,796.2			16,343.2
(-) Accumulated Depreciation:	\$ M	(2,973.0)	(3,161.9)	(747.3)	747.3	-	(3,161.9)
Net Real Estate Assets:	\$ M	9,090.2	9,093.7	3,048.9			13,181.3
Cash & Cash-Equivalents:	\$ M	92.3	103.7	22.3	-	-	116.0
Accounts Receivable:	\$ M	170.6	218.0	14.1	-	-	232.1
Deferred Rent:	\$ M	420.8	420.3	122.5		(122.5)	420.3
Goodwill:	\$ M	757.4	757.4	-	2,582.5		3,340.0
Equity Investments:	\$ M	113.6	113.8	-	-	-	113.8
Other Intangibles & Net Above-Market Leases:	\$ M	1,446.0	1,448.0	5.0	161.2	(184.3)	1,430.0
Acquired In-Place Lease Value:	\$ M				1,589.9	-	1,589.9
Other Assets:	\$ M	243.7	258.7	74.6	-	-	333.3
TOTAL ASSETS:	\$ M	\$ 12,334.6	\$ 12,413.8	\$ 3,287.3			\$ 20,766.7

Write-ups and elimination of Accumulated Depreciation.

Deferred Rent is reconciled upon transaction close.

New Goodwill, as always.

To simplify, we show Above-Market and Below-Market leases in the same line item and report the "Net" amount.

PV of Renewal NOI from existing tenants.

Balance Sheet:	Units:	Projected:		Transaction Adjustments:			
		2Q 17	3Q 17	Seller	Debit	Credit	3Q 17
LIABILITIES & EQUITY:							
Debt & Other Borrowings:	\$ M	\$ 6,417.7	\$ 6,500.4	\$ 1,523.5	\$ (1,569.2)	\$ 1,695.2	\$ 8,149.9
Accounts Payable & Other Accruals:	\$ M	784.1	860.6	41.0	-	-	901.6
Other Liabilities:	\$ M	144.1	171.1	161.6	-	-	332.7
Total Liabilities:	\$ M	\$ 7,345.9	\$ 7,532.1	\$ 1,726.0			\$ 9,384.2
Common Shareholders' Equity:	\$ M	\$ 3,935.8	\$ 3,828.1	\$ 765.6	\$ (885.6)	\$ 5,831.2	\$ 9,539.3
(+) Preferred Stock:	\$ M	1,013.0	1,013.0	201.3	(207.3)	201.3	1,013.0
(+) Noncontrolling Interests in Op. Partnership:	\$ M	32.4	32.4	594.5	-	-	626.9
(+) Noncontrolling Interests in Consolidated JVs:	\$ M	7.5	8.2	-	-	-	8.2
Total Equity:	\$ M	\$ 4,988.7	\$ 4,881.7	\$ 1,561.3			\$ 11,382.5
TOTAL LIABILITIES & EQUITY:	\$ M	\$ 12,334.6	\$ 12,413.8	\$ 3,287.3			\$ 20,766.7
Balance Check:		OK!	OK!	OK!			OK!

Refinanced Debt. New amount is slightly different due to the Senior Notes issued to cover the transaction fees and the financing fees deducted from the book value of the Debt.

Write down Seller's Common SHE, deduct M&A and legal fees, and add new Stock issued to fund the deal.

Refinanced Preferred Stock; financing fees deducted from the book value.

If the Combined Balance Sheet does **not** balance, it means we've forgotten something, double-counted something, or flipped a sign, and we need to go back through the schedules to fix the problem.

Step 5: Project the Combined Balance Sheet

Next, we project the Combined Balance Sheet by:

- 1) Linking to the Buyer + Seller's line items and reflecting the transaction adjustments;



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- 2) Using the brand-new items created in the deal and linking to the Cash Flow Statement; or
- 3) Using a combination of both methods.

Generally, it's best to link to the Buyer + Seller's numbers and reflect the transaction adjustments when the item has not changed in the deal, or when a complex set of changes affect the line item (as with Operating Real Estate Assets, for example).

But it's better to "recalculate" a line item when it is newly created, or it has changed significantly in the deal (as with Debt and Common Shareholders' Equity), or when it changes in a fairly simple way (as with Equity Investments).

Here's the Assets side of the Balance Sheet in the **final version** of our model:

Balance Sheet:	Units:	Projected:		Transaction Adjustments:				Projected:					
		2Q 17	3Q 17	Seller	Debit	Credit	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19
ASSETS:													
Real Estate Assets:													
(+) Operating Real Estate Assets:	\$ M	\$10,992.9	\$11,135.2	\$ 3,174.4	\$ 167.1	\$ -	\$14,476.7	=+'DLR-Op-Model'!M111+'DFT-Op-Model'!M106+SUM(\$N158:\$O158)					
(+) Construction in Progress and Development Land:	\$ M	1,070.4	1,120.4	621.7	124.3	-	1,866.5	1,916.5	1,981.5	1,996.5	2,021.5	2,046.5	2,076.5
Gross Real Estate Assets:	\$ M	12,063.3	12,255.6	3,796.2			16,343.2	16,636.4	16,953.2	17,255.2	17,565.0	17,886.4	18,239.1
(-) Accumulated Depreciation:	\$ M	(2,973.0)	(3,161.9)	(747.3)	747.3	-	(3,161.9)	(3,374.9)	(3,608.1)	(3,842.9)	(4,089.7)	(4,328.8)	(4,594.9)
Net Real Estate Assets:	\$ M	9,090.2	9,093.7	3,048.9			13,181.3	13,261.5	13,345.0	13,412.3	13,475.3	13,557.6	13,644.2
Cash & Cash-Equivalents:	\$ M	92.3	103.7	22.3	-	-	126.0	252.3	121.5	113.5	125.3	276.3	141.9
Accounts Receivable:	\$ M	170.6	218.0	14.1	-	-	232.1	217.9	228.3	201.5	259.0	243.3	259.5
Deferred Rent:	\$ M	420.8	420.3	122.5	-	(122.5)	420.3	419.9	420.1	419.9	422.2	421.7	421.2
Goodwill:	\$ M	757.4	757.4	-	2,582.5	-	3,340.0	3,340.0	3,340.0	3,340.0	3,340.0	3,340.0	3,340.0
Equity Investments:	\$ M	113.6	113.8	-	-	-	113.8	114.4	115.5	116.3	116.5	117.1	118.2
Other Intangibles & Net Above-Market Leases:	\$ M	1,446.0	1,448.0	5.0	161.2	(184.3)	1,430.0	1,425.9	1,421.9	1,417.8	1,413.6	1,409.5	1,405.5
Acquired In-Place Lease Value:	\$ M				1,589.9	-	1,589.9	1,561.5	1,533.1	1,504.7	1,476.3	1,447.9	1,419.5
Other Assets:	\$ M	243.7	258.7	74.6	-	-	333.3	339.0	352.2	349.9	376.3	382.8	403.6
TOTAL ASSETS:	\$ M	\$12,334.6	\$12,413.8	\$ 3,287.3			\$20,766.7	\$20,932.4	\$20,877.5	\$20,875.8	\$21,004.5	\$21,196.3	\$21,153.7

With the items circled in red, we link to the Buyer + Seller's numbers for the period and factor in the Debits and Credits in the Transaction Adjustments area. That's because changes to these items are either complex, or they are not directly shown on the Combined CFS, so we need to link to the separate Balance Sheets instead.

But with the items in green, we link to the figure in the previous quarter and then factor in the line items on the CFS that affect it. So, for example, New Equity Investments = Old Equity Investments + Equity Investment Net Income - Equity Investment Dividends.

And here's the Liabilities & Equity side:



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Balance Sheet:		Projected:		Transaction Adjustments:				Projected:					
	Units:	2Q 17	3Q 17	Seller	Debit	Credit	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19
LIABILITIES & EQUITY:													
Debt & Other Borrowings:	\$ M	\$ 6,417.7	\$ 6,500.4	\$ 1,523.5	\$ (1,569.2)	\$ 1,695.2	\$ 8,149.9	\$ 8,185.6	\$ 8,402.0	\$ 8,622.2	\$ 8,779.7	\$ 8,808.9	\$ 9,030.4
Accounts Payable & Other Accruals:	\$ M	784.1	860.6	41.0	-	-	901.6	987.4	954.9	924.9	1,006.0	1,103.9	1,092.5
Other Liabilities:	\$ M	144.1	171.1	161.6	-	-	332.7	359.0	397.5	348.1	382.9	414.0	462.9
Total Liabilities:	\$ M	\$ 7,345.9	\$ 7,532.1	\$ 1,726.0			\$ 9,384.2	\$ 9,532.0	\$ 9,754.3	\$ 9,895.1	\$ 10,168.6	\$ 10,326.8	\$ 10,585.8
Common Shareholders' Equity:	\$ M	\$ 3,935.8	\$ 3,828.1	\$ 765.6	\$ (885.6)	\$ 5,831.2	\$ 9,539.3	=+P181+Q194+Q200+Q208+Q223+Q224+Q225				\$ 8,984.7	\$ 8,673.5
(+) Preferred Stock:	\$ M	1,013.0	1,013.0	201.3	(207.3)	201.3	1,208.2	1,208.2	1,208.2	1,208.2	1,208.2	1,208.2	1,208.2
(+) Noncontrolling Interests in Op. Partnership:	\$ M	32.4	32.4	594.5	-	-	626.9	632.6	639.0	645.0	651.9	660.9	666.9
(+) Noncontrolling Interests in Consolidated JVs:	\$ M	7.5	8.2	-	-	-	8.2	11.7	12.9	13.9	14.9	15.6	19.4
Total Equity:	\$ M	\$ 4,988.7	\$ 4,881.7	\$ 1,561.3			\$ 11,382.5	\$ 11,400.4	\$ 11,123.2	\$ 10,980.7	\$ 10,835.9	\$ 10,869.4	\$ 10,567.9
TOTAL LIABILITIES & EQUITY:	\$ M	\$ 12,334.6	\$ 12,413.8	\$ 3,287.3			\$ 20,766.7	\$ 20,932.4	\$ 20,877.5	\$ 20,875.8	\$ 21,004.5	\$ 21,196.3	\$ 21,153.7
Balance Check:		OK!	OK!	OK!			OK!	OK!	OK!	OK!	OK!	OK!	OK!

Again, we link to the Buyer + Seller's underlying Balance Sheets plus transaction adjustments for the items in red.

We recalculate the items in green based on the previous balance and all the relevant line items on the CFS. Common Shareholders' Equity is a "catch-all" that includes Net Income, Gains / Losses, "Other Items," Stock Issuances, Common Dividends, and Other Financing Activities.

And the two NCI line items are weird exceptions, where we link to the previous numbers and then the Net Income to NCI on each company's separate Cash Flow Statement. We can't completely combine this line item because we need to keep the NCIs corresponding to OP Units separate for the Enterprise Value calculations.

Step 6: Project the Combined Income Statement and Cash Flow Statement (If Applicable)

In the next step, we combine the Income Statement and Cash Flow Statement for these companies, and the process is nearly the same as the one for normal companies:

- **Add together** items like Rental Income, Tenant Reimbursements, Property Expenses, D&A, G&A, and the Net Interest Expense.
- **Reflect** the acquisition effects, such as the Interest Paid on New Debt, Changes in Interest from Refinancing, Synergies and Restructuring, and new D&A on Above- and Below-Market Leases, Real Estate Asset Write-Ups, and Acquired In-Place Lease Value. Also, add the New Shares Issued to the total share count.
- **Multiply** the Combined Pre-Tax Income by (1 – Buyer's Tax Rate) to get the Combined Net Income; subtract Net Income to Noncontrolling Interests and Preferred Dividends to calculate Net Income to Common.
- **Divide** Net Income to Common by the Post-Deal Share Count to determine the Combined EPS; calculate EPS accretion/dilution based on this number vs. the Buyer's standalone EPS. Do the same for FFO PS and AFFO PS to calculate accretion/dilution for those. If the REIT uses metrics such as EPRA Earnings, you can do the same thing for those on a per-share basis.



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Here's the Combined Income Statement:

Income Statement:	Units:	Projected:		Transaction Adjustments:				Projected:					
		2Q 17	3Q 17	Seller	Debit	Credit	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19
Revenue:													
(+) Rental Income - Existing Properties:	\$ M	\$ 376.4	\$ 393.2	Above-Market Lease Amortization reduces Revenue, while Below-Market Lease Amortization increases it.				\$ 487.4	\$ 501.9	\$ 460.8	\$ 481.5	\$ 485.6	\$ 507.6
(+) Rental Income - Developments:	\$ M	8.3	17.4					39.8	54.6	73.0	91.4	110.6	131.9
(-) Amortization of Acquired Above-Market Leases:	\$ M							(10.1)	(10.1)	(10.1)	(10.1)	(10.1)	(10.1)
(+) Amortization of Acquired Below-Market Leases:	\$ M							3.8	3.8	3.8	3.8	3.8	3.8
(+) Tenant Reimbursements - Existing Properties:	\$ M	86.4	94.1	Cost Synergies are fully phased in in Year 2; restructuring goes away after Year 1.				132.4	134.1	130.4	138.5	132.4	135.4
(+) Tenant Reimbursements - Developments:	\$ M	1.9	4.1					12.5	17.0	22.6	28.2	33.8	39.9
(+) Property Management Fees:	\$ M	1.2	1.6					1.7	2.2	1.3	1.8	1.9	2.6
(+) Interconnection Fees:	\$ M	55.6	62.0					62.0	64.4	61.2	68.2	66.6	69.2
(+) Other Income:	\$ M	-	-					-	6.6	3.0	3.8	1.4	6.8
Total Revenue:	\$ M	529.8	572.4	572.4				736.1	770.9	746.8	804.8	831.4	883.4
Operating Expenses:													
(-) Property Expenses - Existing Properties:	\$ M	(191.9)	(204.5)	In-Place Lease Value Amortization *increases* the expenses, reducing Operating Income.				(251.6)	(251.0)	(235.9)	(248.6)	(250.6)	(253.8)
(-) Property Expenses - Developments:	\$ M	(4.3)	(9.0)					(20.4)	(27.8)	(37.3)	(46.8)	(56.6)	(67.3)
(+) Annual Cost Synergies:	\$ M							2.3	2.3	2.3	2.3	4.5	4.5
(-) Restructuring and Merger Integration Costs:	\$ M							(6.8)	(6.8)	(6.8)	(6.8)	-	-
(-) Amortization of Acquired In-Place Lease Value:	\$ M							(28.4)	(28.4)	(28.4)	(28.4)	(28.4)	(28.4)
(-) Depreciation of Real Estate Asset Write-Ups:	\$ M			Standard straight-line D&A, converted to quarterly figures.				(1.1)	(1.1)	(1.1)	(1.1)	(1.1)	(1.1)
(-) General & Administrative:	\$ M	(34.4)	(45.8)					(48.3)	(44.8)	(45.0)	(58.0)	(54.2)	(51.2)
(-) Depreciation & Amortization:	\$ M	(180.1)	(188.9)					(211.9)	(232.2)	(233.7)	(245.7)	(238.0)	(265.1)
(-) Other Items:	\$ M	-	-					(5.4)	(2.5)	(3.1)	(1.2)	(5.6)	(2.6)
Total Operating Expenses:	\$ M	(410.7)	(448.1)	(448.1)				(571.5)	(592.2)	(588.9)	(634.2)	(630.1)	(664.9)
Operating Income:	\$ M	119.0	124.3	124.3				164.6	178.7	157.9	170.5	201.4	218.6

Income Statement:	Units:	Projected:		Transaction Adjustments:				Projected:					
		2Q 17	3Q 17	Seller	Debit	Credit	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19
Refinancing													
(+) Equity Investment Income:	\$ M	5.0	4.8	Refinancing *reduces* the Interest Rate here, so the first row is positive; new Debt increases the Interest Expense (second row).				5.2	5.6	5.2	5.0	5.5	5.9
(+) Gain / (-) Loss on Real Estate Transactions:	\$ M	-	-					-	-	-	-	-	-
(-) Net Interest Expense:	\$ M	(58.9)	(62.6)					(80.2)	(80.6)	(84.8)	(87.1)	(90.9)	(93.3)
(+/-) Change in Target's Interest from Refinancing:	\$ M							1.8	1.8	1.8	1.8	1.8	1.8
(-) Interest Expense on Transaction Debt:	\$ M			Based on Buyer's Tax Rate, which is nearly 0.				(1.6)	(1.6)	(1.6)	(1.6)	(1.6)	(1.6)
(-) Loss from Debt Extinguishment:	\$ M	-	-					-	-	-	-	-	-
Pre-Tax Income:	\$ M	65.1	66.5	66.5				89.7	103.8	78.4	88.6	116.1	131.2
(-) Income Taxes:	\$ M	(2.8)	(1.1)	(1.1)				(2.1)	(2.7)	(3.3)	(1.5)	(2.7)	(3.4)
Net Income:	\$ M	62.3	65.4	65.4				87.6	101.2	75.1	87.2	113.4	127.9
(-) Net Income Attributable to NCI:	\$ M	(0.9)	(0.7)	(0.7)				(9.2)	(7.6)	(7.1)	(7.9)	(9.7)	(9.7)
Net Income Attributable to Parent:	\$ M	61.4	64.7	64.7				78.4	93.6	68.0	79.3	103.7	118.2
(-) Preferred Stock Dividends:	\$ M	(17.7)	(17.7)	(17.7)				(21.1)	(21.1)	(21.1)	(21.1)	(21.1)	(21.1)
(+/-) Change in Pref. Dividends from Refinancing:	\$ M							0.7	0.7	0.7	0.7	0.7	0.7
(-) Preferred Stock Issuance Costs:	\$ M	-	-	-				-	-	-	-	-	-
Net Income to Common:	\$ M	\$ 43.6	\$ 47.0	\$ 47.0				\$ 58.1	\$ 73.2	\$ 47.7	\$ 58.9	\$ 83.4	\$ 97.8
Weighted Average Diluted Common Shares:													
Weighted Average Diluted Common Shares:	# Millions	160.625	160.703	160.703				160.737	160.940	161.148	161.296	161.324	161.532
(+) Shares Issued to Fund Transaction:	# Millions							49.408	49.408	49.408	49.408	49.408	49.408
Post-Transaction Avg. Diluted Common Shares:	# Millions							210.145	210.349	210.556	210.705	210.732	210.941
Pre-Transaction Diluted EPS:	\$ as Stated	\$ 0.27	\$ 0.29	\$ 0.29				\$ 0.35	\$ 0.44	\$ 0.29	\$ 0.33	\$ 0.41	\$ 0.48
Post-Transaction Diluted EPS:	\$ as Stated							0.28	0.35	0.23	0.28	0.40	0.46
EPS Accretion / (Dilution) - \$:	\$ as Stated							(0.07)	(0.09)	(0.06)	(0.05)	(0.02)	(0.02)
EPS Accretion / (Dilution) - %:	%							(21.1%)	(21.0%)	(22.2%)	(15.0%)	(4.4%)	(3.2%)

Expenses here are shown as negatives, while income sources appear as positives.

Therefore, if an item like Cost Synergies *reduces* the combined company's expenses, then it should have a positive sign *if* it appears in the "Operating Expense" section (as it does here).

Here's the bottom part of the Combined Income Statement:



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Income Statement:	Units:	Projected:		Transaction Adjustments:				Projected:					
		2Q 17	3Q 17	Seller	Debit	Credit	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19
Weighted Average Diluted Common Shares & Units:	# Millions	162.803	162.881				162.881	162.915	163.118	163.326	163.474	163.501	163.710
(+) Shares Issued to Fund Transaction:	# Millions							49.408	49.408	49.408	49.408	49.408	49.408
Post-Trans. Avg. Diluted Common Shares & Units:	# Millions							212.323	212.527	212.734	212.882	212.910	213.119
Pre-Transaction FFO per Diluted Share & Unit:	\$ as Stated	\$ 1.37	\$ 1.45				\$ 1.45	\$ 1.47	\$ 1.66	\$ 1.51	\$ 1.61	\$ 1.63	\$ 1.87
Post-Transaction FFO per Diluted Share & Unit:	\$ as Stated							1.32	1.47	1.35	1.47	1.56	1.75
FFO PS Accretion / (Dilution) - \$:	\$ as Stated							(0.15)	(0.18)	(0.16)	(0.14)	(0.07)	(0.12)
FFO PS Accretion / (Dilution) - %:	%							(10.2%)	(11.0%)	(10.4%)	(8.9%)	(4.6%)	(6.5%)
Pre-Transaction AFFO per Diluted Share & Unit:	\$ as Stated	\$ 1.23	\$ 1.33				\$ 1.33	\$ 1.31	\$ 1.43	\$ 1.37	\$ 1.46	\$ 1.46	\$ 1.62
Post-Transaction AFFO per Diluted Share & Unit:	\$ as Stated							1.35	1.44	1.40	1.51	1.58	1.70
AFFO PS Accretion / (Dilution) - \$:	\$ as Stated							0.04	0.01	0.03	0.05	0.12	0.07
AFFO PS Accretion / (Dilution) - %:	%							3.3%	0.7%	2.1%	3.1%	8.2%	4.5%

When calculating FFO and AFFO per Share, we use the Share + OP/DownREIT Unit count rather than just the Common Share Count (as with EPS) - since FFO and AFFO may also go to these Unitholders.

To calculate the Combined FFO and AFFO, we need the Combined Cash Flow Statement first since many of the components of FFO and AFFO come from there.

On the Combined Cash Flow Statement, we add together the items from the Buyer and Seller's standalone Cash Flow Statements and reverse the new non-cash items created in the deal (the D&A on the Asset Write-Ups, Above- and Below-Market Leases, and In-Place Lease Value):

Cash Flow Statement:	Units:	Projected:		Transaction Adjustments:				Projected:					
		2Q 17	3Q 17	Seller	Debit	Credit	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19
Cash Flow from Operations:													
(+) Net Income to Common:	\$ M	\$ 43.6	\$ 47.0				\$ 47.0	\$ 58.1	\$ 73.2	\$ 47.7	\$ 58.9	\$ 83.4	\$ 97.8
(+) Net Income Attributable to NCI:	\$ M	0.9	0.7				0.7	9.2	7.6	7.1	7.9	9.7	9.7
(-) Equity Investment Income:	\$ M	(5.0)	(4.8)				(4.8)	(5.2)	(5.6)	(5.2)	(5.0)	(5.5)	(5.9)
(+) Distributions from Equity Investments:	\$ M	4.2	4.5				4.5	4.7	4.5	4.4	4.8	4.9	4.7
(+) Depreciation & Amortization:	\$ M	180.1	188.9				188.9	211.9	232.2	233.7	245.7	238.0	265.1
(+) Depreciation of Real Estate Asset Write-Ups:	\$ M							1.1	1.1	1.1	1.1	1.1	1.1
(-) Gain / (+) Loss on Real Estate Transactions:	\$ M	-	-				-	-	-	-	-	-	-
(+/-) Straight-Lining of Rent & Expenses, Net:	\$ M	(1.9)	0.4				0.4	0.2	1.1	2.2	(4.7)	1.5	0.3
(+/-) Elimination of Target's Straight-Lining, Net:	\$ M							0.2	(1.3)	(2.0)	2.4	(1.1)	0.2
(+/-) Above/Below-Market Rent Amortization:	\$ M	(2.0)	(2.0)				(2.0)	(2.1)	(2.2)	(2.1)	(2.1)	(2.1)	(2.2)
(+) Amortization of Acquired Above-Market Leases:	\$ M							10.1	10.1	10.1	10.1	10.1	10.1
(-) Amortization of Acquired Below-Market Leases:	\$ M							(3.8)	(3.8)	(3.8)	(3.8)	(3.8)	(3.8)
(+) Amortization of Acquired In-Place Lease Value:	\$ M							28.4	28.4	28.4	28.4	28.4	28.4
(+/-) Change in Goodwill:	\$ M	-	-				-	-	-	-	-	-	-
(+/-) Other Items, Net:	\$ M	7.6	7.6				7.6	10.4	10.4	10.4	10.4	10.4	10.4
(+/-) Change in Working Capital:	\$ M	(49.0)	41.0				41.0	120.6	(17.5)	(50.4)	32.0	138.3	0.4
Cash Flow from Operations:	\$ M	\$ 178.7	\$ 283.4				\$ 283.4	\$ 443.5	\$ 338.1	\$ 281.4	\$ 386.0	\$ 513.3	\$ 416.1
Cash Flow from Investing:													
(-) Recurring Maintenance CapEx:	\$ M	\$ (18.4)	\$ (16.0)				\$ (16.0)	\$ (22.3)	\$ (35.3)	\$ (21.8)	\$ (19.0)	\$ (24.9)	\$ (40.2)
(-) Capitalized Leasing Commissions:	\$ M	(4.3)	(4.8)				(4.8)	(6.2)	(6.7)	(5.5)	(6.0)	(6.9)	(7.7)
(-) Acquisitions & Developments:	\$ M	(180.0)	(180.0)				(180.0)	(270.0)	(280.0)	(280.0)	(290.0)	(295.0)	(310.0)
(-) Sale of Real Estate Assets:	\$ M	-	-				-	-	-	-	-	-	-
(+/-) Other Items, Net:	\$ M	8.4	8.4				8.4	5.3	5.3	5.3	5.3	5.3	5.3
Cash Flow from Investing:	\$ M	\$ (194.2)	\$ (192.3)				\$ (192.3)	\$ (293.2)	\$ (316.8)	\$ (302.0)	\$ (309.7)	\$ (321.5)	\$ (352.7)
Cash Flow from Financing:													
(+/-) Net Change in Debt:	\$ M	\$ 216.3	\$ 82.7				\$ 82.7	\$ 35.6	\$ 216.4	\$ 220.2	\$ 157.6	\$ 29.1	\$ 221.6
(+/-) Net Change in Preferred Stock:	\$ M	-	-				-	-	-	-	-	-	-
(+) Stock Issuances / (-) Repurchases:	\$ M	24.0	9.2				9.2	4.0	24.0	24.5	17.5	3.2	24.6
(-) Common Dividends:	\$ M	(144.1)	(157.6)				(157.6)	(44.3)	(373.0)	(212.6)	(220.1)	(53.7)	(424.6)
(+/-) Other Financing Activities, Net:	\$ M	(13.9)	(13.9)				(13.9)	(19.4)	(19.4)	(19.4)	(19.4)	(19.4)	(19.4)
Cash Flow from Financing:	\$ M	\$ 82.4	\$ (79.6)				\$ (79.6)	\$ (24.1)	\$ (152.0)	\$ 12.7	\$ (64.4)	\$ (40.8)	\$ (197.9)
(+) Net Change in Cash & Cash-Equivalents:	\$ M	66.9	11.4					126.3	(130.8)	(8.0)	11.9	151.0	(134.5)
(+) Beginning Cash:	\$ M	25.4	92.3					126.0	252.3	121.5	113.5	125.3	276.3
Ending Cash:	\$ M	\$ 92.3	\$ 103.7					\$ 252.3	\$ 121.5	\$ 113.5	\$ 125.3	\$ 276.3	\$ 141.9

We combine the Buyer and Seller's figures in the Straight-Lining line item and then reverse the Seller's figures right below it since its Deferred Rent is written down.

The rest of the CFS is straightforward: combine numbers from each company and reflect the non-cash adjustments from new items created in this deal.



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We eliminate the Seller's Straight-Lining of Rent because the Seller's Deferred Rent is written down in the transaction, so this adjustment on the CFS also goes away.

We do NOT recalculate Dividends or Debt and Equity issuances, even though FFO and AFFO have changed slightly.

If the Combined REIT plans to sell a significant number of assets or otherwise change its portfolio, then we might have to recalculate these numbers.

But this deal is a fairly standard portfolio combination, so we choose to keep FFO and AFFO mostly the same but factor in the new D&A and the elimination of the Seller's Straight-Lining of Rent:

Cash Flow Statement Drivers:	Units:	Projected:		Transaction Adjustments:				Projected:					
		2Q 17	3Q 17	Seller	Debit	Credit	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19
(+) Net Income to Common:	\$ M	\$ 43.6	\$ 47.0			\$ 47.0		\$ 58.1	\$ 73.2	\$ 47.7	\$ 58.9	\$ 83.4	\$ 97.8
(+) Noncontrolling Interests in Op. Partnership:	\$ M	0.9	0.7			0.7		9.2	7.6	7.1	7.9	9.7	9.7
(+) Real Estate-Related Depreciation & Amortization:	\$ M	176.7	185.9			185.9		208.6	228.6	229.7	242.1	234.3	260.9
(+) Depreciation of Real Estate Asset Write-Ups:	\$ M							1.1	1.1	1.1	1.1	1.1	1.1
(+/-) Other Adjustments:	\$ M	-	-			-		-	-	-	-	-	-
(+) Unconsolidated JV Depreciation:	\$ M	2.6	2.4			2.4		2.6	2.8	2.7	2.6	2.7	2.9
(-) Gain / (+) Loss on Real Estate Transactions:	\$ M	-	-			-		-	-	-	-	-	-
Funds from Operations (FFO):	\$ M	223.8	236.0			236.0		279.5	313.2	288.2	312.6	331.2	372.4
Funds from Operations (FFO):	\$ M	223.8	236.0			236.0		279.5	313.2	288.2	312.6	331.2	372.4
(+) Non-RE Depreciation:	\$ M	3.4	3.0			3.0		3.3	3.6	4.1	3.6	3.7	4.1
(+/-) Straight-Lining of Rent & Expenses, Net:	\$ M	(1.9)	0.4			0.4		0.2	1.1	2.2	(4.7)	1.5	0.3
(+/-) Above/Below-Market Rent Amortization:	\$ M	(2.0)	(2.0)			(2.0)		(2.1)	(2.2)	(2.1)	(2.1)	(2.1)	(2.2)
(+/-) Elimination of Target's Straight-Lining, Net:	\$ M							0.2	(1.3)	(2.0)	2.4	(1.1)	0.2
(+) Amortization of Acquired Above-Market Leases:	\$ M							10.1	10.1	10.1	10.1	10.1	10.1
(-) Amortization of Acquired Below-Market Leases:	\$ M							(3.8)	(3.8)	(3.8)	(3.8)	(3.8)	(3.8)
(+) Amortization of Acquired In-Place Lease Value:	\$ M							28.4	28.4	28.4	28.4	28.4	28.4
(-) Recurring Maintenance CapEx:	\$ M	(18.4)	(16.0)			(16.0)		(22.3)	(35.3)	(21.8)	(19.0)	(24.9)	(40.2)
(-) Capitalized Leasing Commissions:	\$ M	(4.3)	(4.8)			(4.8)		(6.2)	(6.7)	(5.5)	(6.0)	(6.9)	(7.7)
Adjusted Funds from Operations (AFFO):	\$ M	\$ 200.7	\$ 216.6			\$ 216.6		\$ 287.3	\$ 307.1	\$ 297.6	\$ 321.4	\$ 336.1	\$ 361.5
Dividends % AFFO:	%	71.8%	72.8%			72.8%		15.4%	121.5%	71.4%	68.5%	16.0%	117.5%
Debt Issued:	\$ M	216.3	82.7			82.7		35.6	216.4	220.2	157.6	29.1	221.6
Equity Issued:	\$ M	24.0	9.2			9.2		4.0	24.0	24.5	17.5	3.2	24.6
Shares Issued:	# Millions	0.204	0.078			0.078		0.034	0.204	0.207	0.148	0.027	0.209

Dividend numbers jump around a lot more; an annual summary would be helpful.

Add Buyer + Seller's figures.

New Shares Issued are based on Buyer's Share Price.

Dividends remain between 65% and 70% of AFFO on an annual basis, so this method seems reasonable.

Step 7: Calculate Metrics, Ratios, and Sensitivities for the Deal

We're now done with the main part of the model, but there are still a few remaining tasks:

- 1) Select and calculate other **key metrics and ratios** to analyze the deal.



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- 2) Create **annual summaries** to make the model easier to read and interpret.
- 3) Build **sensitivity tables** to show the deal outcome under different scenarios.

With the first point, the key metrics and ratios are similar to the ones you've seen in this course: EBITDA / Interest, Debt / EBITDA, Debt / Total Assets, and so on.

The only tricky part here is that we must **annualize** the quarterly numbers when calculating metrics like EBITDA / Net Interest – which, in this case, means multiplying by 4 when required.

Here are the metrics we focus on:

Key Metrics and Ratios:	Units:	Projected:		Transaction Adjustments:				Projected:					
		2Q 17	3Q 17	Seller	Debit	Credit	3Q 17	4Q 17	1Q 18	2Q 18	3Q 18	4Q 18	1Q 19
Annualized EBITDA:	\$ M	\$ 1,196.5	\$ 1,252.6				\$ 1,252.6	\$ 1,648.8	\$ 1,786.4	\$ 1,709.2	\$ 1,807.9	\$ 1,900.4	\$ 2,077.3
EBITDA Growth:	%	3.9%	6.6%				6.6%	31.6%	8.3%	(4.3%)	5.8%	5.1%	9.3%
EBITDA Margin:	%	56.5%	54.7%				54.7%	55.5%	57.5%	56.7%	55.7%	56.7%	58.4%
Annualized Net Operating Income (NOI):	\$ M	1,329.6	1,429.2				1,429.2	1,853.0	1,974.5	1,902.1	2,050.7	2,091.8	2,253.8
NOI Growth:	%	3.6%	5.6%				5.6%	29.7%	6.6%	(3.7%)	7.8%	2.0%	7.7%
Yield on Total Gross Real Estate Assets:	%	11.8%					11.3%	11.9%	11.2%	11.9%	11.9%	12.6%	11.9%
Dividends per Share:	\$ as Stated	\$ 0.90	\$ 0.98					\$ 0.21	\$ 1.77	\$ 1.01	\$ 1.04	\$ 0.25	\$ 2.01
Annualized Dividend Yield:	%	3.0%	3.3%					0.7%	6.0%	3.4%	3.5%	0.9%	6.8%
EBITDA / Net Interest:	x	5.08 x	5.00 x				5.00 x	5.15 x	5.55 x	5.05 x	5.20 x	5.24 x	5.57 x
Minimum Coverage Required:	x	1.50 x											
EBITDA / Fixed Charges:	x	3.90 x	3.90 x				3.90 x	4.10 x	4.43 x	4.07 x	4.21 x	4.28 x	4.57 x
Minimum Coverage Required:	x	1.50 x											
Debt / EBITDA:	x	5.36 x	5.19 x				5.19 x	4.96 x	4.70 x	5.04 x	4.86 x	4.64 x	4.35 x
Net Debt / EBITDA:	x	5.29 x	5.11 x				5.11 x	4.81 x	4.64 x	4.98 x	4.79 x	4.49 x	4.28 x
(Net Debt + Preferred) / EBITDA:	x	6.13 x	5.92 x				5.92 x	5.54 x	5.31 x	5.68 x	5.46 x	5.13 x	4.86 x
Cap Rate Applied to Annualized EBITDA:	%	8.25%											
Total Asset Value:	\$ M	14,503.0	15,183.0					19,985.0	21,652.8	20,717.8	21,913.5	23,034.9	25,178.9
Total Debt / Total Assets:	%	44.3%	42.8%					41.0%	38.8%	41.6%	40.1%	38.2%	35.9%
Maximum Level Allowed:	%	60.0%											

Based on forward, annualized NOI.

From the loan covenants.

"Fixed Charges" include Net Interest Expense and Preferred Dividends.

We see that the Combined Company never comes close to violating its covenants here. If we wanted to stress test something, we would focus on the Total Debt / Total Assets covenant, as that could rise significantly if the company's NOI were to decline in a recession (for example).

To create the annual summaries on the right-hand side of this model, we used a combination of SUMIF, INDEX/MATCH, and manual SUM functions, all using the date rows above each schedule.

Here are a few examples:



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U	V	W	X	Y	Z	AA	AB	AC	AD	AE
2018	2019	2019	2019	2019		2016	2017	2018	2019	
1.25	1.50	1.75	2.00	2.25		2.3%	2.7%	2.7%	2.7%	
2.3%	2.6%	4.2%	1.6%	2.3%						
Projected:						Historical:	Projected:			
4Q 18	1Q 19	2Q 19	3Q 19	4Q 19		FY16	FY17	FY18	FY19	
\$ 485.6	\$ 507.6	\$ 461.1	\$ 480.7	\$ 484.4		\$ 1,542.5	=SUMIF(\$F\$78:\$Y\$78,AB\$78,\$F86:\$Y86)			
110.6	131.9	153.1	175.2	198.1		-	65.5	329.5	658.4	
(10.1)	(10.1)	(10.1)	(10.1)	(10.1)		-	(10.1)	(40.3)	(40.3)	
3.8	3.8	3.8	3.8	3.8		-	3.8	15.4	15.4	
132.4	135.4	130.6	138.4	132.5		355.9	400.2	535.4	536.9	
33.8	39.9	46.3	53.0	59.6		-	18.5	101.4	198.8	
1.9	2.6	1.5	2.1	2.1		6.3	6.4	7.2	8.2	
66.6	69.2	64.2	71.6	70.0		204.3	236.8	260.4	275.0	
6.8	3.1	3.9	1.5	7.0		33.2	6.6	15.1	15.5	
831.4	883.4	854.6	916.2	947.4		2,142.2	2,388.8	3,154.0	3,601.6	

On the Income Statement, we match the years in the annual columns to the years in the quarterly columns and sum up everything like that.

We add up the annual numbers to calculate Total Operating Expenses, Operating Income, Net Income, etc., and we recalculate all the accretion / dilution figures based on the annual numbers.

The same strategy applies to the Cash Flow Statement since it also shows results over a period. The Balance Sheet is different since it represents a snapshot in time:

U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG
Projected:						Historical:	Projected:					
4Q 18	1Q 19	2Q 19	3Q 19	4Q 19		FY16	FY17	FY18	FY19			
\$15,840.0	\$16,162.7	\$16,468.4	\$16,781.3	\$17,106.9		\$10,630.5	=INDEX(\$F158:\$Y158,1,MATCH(AB\$155,\$F\$155:\$Y\$155,0))					
2,046.5	2,076.5	2,111.5	2,136.5	2,161.5		928.0	1,916.5	2,046.5	2,161.5			
17,886.4	18,239.1	18,579.8	18,917.8	19,268.4		11,558.5	16,636.4	17,886.4	19,268.4			
(4,328.8)	(4,594.9)	(4,862.1)	(5,141.5)	(5,412.7)		(2,668.5)	(3,374.9)	(4,328.8)	(5,412.7)			
13,557.6	13,644.2	13,717.8	13,776.3	13,855.7		8,890.0	13,261.5	13,557.6	13,855.7			

We use INDEX/MATCH based on exact dates for the Balance Sheet because "4Q 19" is the same date as "FY 19" (for example).

Finally, the annual summary for the key metrics and ratios differs because we *already* annualized many of these numbers in the quarterly version.

So, we recalculate everything in the annual version instead of summing up the quarterly figures.

The last task is to create a few sensitivities, and the obvious assumptions to sensitize are:

- Purchase Price (driven by the Exchange Ratio)
- Stock vs. Debt Percentage
- Annual Cost Synergies

These assumptions are all impactful and somewhat controllable by the Buyer and Seller. We focus on Year 2 AFFO PS accretion/dilution to capture the results *after* the Year 1 Restructuring goes away:



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Sensitivity Analyses - FY 19 AFFO per Share & Unit Accretion/Dilution

		Exchange Ratio:										
		0.470 x	0.485 x	0.500 x	0.515 x	0.530 x	0.545 x	0.560 x	0.575 x	0.590 x	0.605 x	0.620 x
% Cash Funding via New Debt Issued:	20.0%	12.4%	11.7%	11.0%	10.3%	9.6%	8.9%	8.3%	7.6%	6.9%	6.3%	5.6%
	17.5%	12.2%	11.5%	10.8%	10.1%	9.4%	8.7%	8.0%	7.3%	6.7%	6.0%	5.4%
	15.0%	12.0%	11.2%	10.5%	9.8%	9.1%	8.4%	7.8%	7.1%	6.4%	5.8%	5.1%
	12.5%	11.7%	11.0%	10.3%	9.6%	8.9%	8.2%	7.5%	6.8%	6.2%	5.5%	4.9%
	10.0%	11.5%	10.8%	10.1%	9.4%	8.7%	8.0%	7.3%	6.6%	5.9%	5.3%	4.6%
	7.5%	11.3%	10.5%	9.8%	9.1%	8.4%	7.7%	7.0%	6.4%	5.7%	5.0%	4.4%
	5.0%	11.0%	10.3%	9.6%	8.9%	8.2%	7.5%	6.8%	6.1%	5.5%	4.8%	4.1%
	2.5%	10.8%	10.1%	9.4%	8.7%	8.0%	7.3%	6.6%	5.9%	5.2%	4.6%	3.9%
	0.0%	10.6%	9.9%	9.1%	8.4%	7.7%	7.0%	6.3%	5.7%	5.0%	4.3%	3.7%

		Exchange Ratio:										
		0.470 x	0.485 x	0.500 x	0.515 x	0.530 x	0.545 x	0.560 x	0.575 x	0.590 x	0.605 x	0.620 x
Annualized Cost Synergies:	\$ 24.0	11.0%	10.3%	9.6%	8.9%	8.2%	7.5%	6.8%	6.1%	5.4%	4.8%	4.1%
	21.0	10.8%	10.1%	9.4%	8.6%	7.9%	7.2%	6.6%	5.9%	5.2%	4.5%	3.9%
	18.0	10.6%	9.9%	9.1%	8.4%	7.7%	7.0%	6.3%	5.7%	5.0%	4.3%	3.7%
	15.0	10.4%	9.7%	8.9%	8.2%	7.5%	6.8%	6.1%	5.5%	4.8%	4.1%	3.5%
	12.0	10.2%	9.4%	8.7%	8.0%	7.3%	6.6%	5.9%	5.2%	4.6%	3.9%	3.3%
	9.0	9.9%	9.2%	8.5%	7.8%	7.1%	6.4%	5.7%	5.0%	4.4%	3.7%	3.1%
	6.0	9.7%	9.0%	8.3%	7.6%	6.9%	6.2%	5.5%	4.8%	4.2%	3.5%	2.9%
	3.0	9.5%	8.8%	8.1%	7.4%	6.7%	6.0%	5.3%	4.6%	4.0%	3.3%	2.7%
	-	9.3%	8.6%	7.8%	7.1%	6.4%	5.8%	5.1%	4.4%	3.7%	3.1%	2.5%

The conclusion is that there's a fair amount of "cushion" if the Buyer ends up paying a higher price, synergies come in low, or the Stock vs. Debt percentage changes.

Even with those negative changes, it seems likely that the deal will be accretive to the Buyer's AFFO PS.

However, the Buyer doesn't necessarily care that much about AFFO PS accretion/dilution because this is a REIT M&A deal, where the analyses in the next section can be more meaningful.

Step 8: Complete Contribution and Value Creation Analyses and Summarize the Model

Since most REIT M&A deals are 100% Stock, or close to 100% Stock, the **Contribution Analysis** is a key part of the model.

The setup is similar to any other Contribution Analysis: measure the percentage contributions of EBITDA, NOI, FFO, AFFO, NAV, etc. from the Buyer and Seller and then determine what the Purchase Price *should be* based on those percentages.

Here it is, step-by-step:



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Relative Contribution Analysis:				Contribution (\$):		Contribution (%):	
Equity Value:				Buyer:	Seller:	Pro-Forma:	
Undisturbed Equity Value:				\$ 19,157.6	\$ 5,033.5	\$ 24,191.1	
Pro-Forma Ownership @ 15.8% Premium:				19,157.6	5,831.2	24,988.8	
Net Operating Income (NOI):							
Combined Year 1 - 2018-12-31:				1,546.9	457.8	2,004.8	
Combined Year 1 - 2019-12-31:				1,737.8	543.5	2,281.3	
Net Asset Value (NAV):							
Most Recent Quarter Prior to Deal Announcement:				15,778.8	4,942.5	20,721.2	
EBITDA:							
Combined Year 1 - 2018-12-31:				1,383.4	426.6	1,810.0	
Combined Year 1 - 2019-12-31:				1,553.4	506.5	2,059.9	
Funds from Operations (FFO):							
Combined Year 1 - 2018-12-31:				1,046.1	344.8	1,390.9	
Combined Year 1 - 2019-12-31:				1,179.3	411.5	1,590.8	
Adjusted Funds from Operations (AFFO):							
Combined Year 1 - 2018-12-31:				934.7	336.6	1,271.3	
Combined Year 1 - 2019-12-31:				1,054.5	400.2	1,454.7	
Balance Sheet Items as of Transaction Close Date (Ignores Acquisition Effects):							
Gross Real Estate Assets:				12,255.6	3,796.2	16,051.8	
Total Assets:				12,413.8	3,287.3	15,701.1	
Common Equity:				3,828.1	765.6	4,593.7	

Step 1 is to calculate the % contribution from the Buyer and Seller for all the key metrics. This is basic arithmetic.

Seller - Equity Value and Enterprise Value:				Buyer - Equity Value and Enterprise Value:			
Purchase Equity Value:				Current Equity Value:			
(-) Cash & Cash-Equivalents:				(-) Cash & Cash-Equivalents:			
(-) Equity Investments & Non-Core Assets:				(-) Equity Investments & Non-Core Assets:			
(+ Total Debt:				(+ Total Debt:			
(+ Preferred Stock:				(+ Preferred Stock:			
(+ Non-OP Noncontrolling Interests:				(+ Non-OP Noncontrolling Interests:			
Purchase Enterprise Value:				Current Enterprise Value:			

Next, you calculate the Combined Pro-Forma Enterprise Value and Equity Value for each metric.

To do this, take the Buyer's Current Enterprise Value and divide by its Contribution Percentage, and then reverse the items in the Equity Value to Enterprise Value bridge for both companies to get the Combined Equity Value.

Relative Contribution Analysis:				Contribution (\$):		Contribution (%):		Combined Pro-Forma Value:	
Equity Value:				Buyer:	Seller:	Pro-Forma:		Enterprise Value:	Equity Value:
Undisturbed Equity Value:				\$ 19,157.6	\$ 5,033.5	\$ 24,191.1		\$ 32,837.9	\$ 24,191.1
Pro-Forma Ownership @ 15.8% Premium:				19,157.6	5,831.2	24,988.8		33,635.5	24,988.8
Net Operating Income (NOI):									
Combined Year 1 - 2018-12-31:				1,546.9	457.8	2,004.8	77.2%	25,359.2	
Combined Year 1 - 2019-12-31:				1,737.8	543.5	2,281.3	76.2%	25,800.1	

For Equity value-based metrics such as FFO and AFFO, follow the reverse process and start by dividing the Buyer's Current Equity Value by its Contribution Percentage.



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Seller - Equity Value and Enterprise Value: Buyer - Equity Value and Enterprise Value:

Purchase Equity Value:	\$ 5,831.2
(-) Cash & Cash-Equivalents:	(44.9)
(-) Equity Investments & Non-Core Assets:	-
(+) Total Debt:	1,407.8
(+) Preferred Stock:	201.3
(+) Non-OP Noncontrolling Interests:	-
Purchase Enterprise Value:	\$ 7,395.3

Current Equity Value:	\$ 19,157.6
(-) Cash & Cash-Equivalents:	(25.4)
(-) Equity Investments & Non-Core Assets:	(112.9)
(+) Total Debt:	6,201.3
(+) Preferred Stock:	1,013.0
(+) Non-OP Noncontrolling Interests:	6.6
Current Enterprise Value:	\$ 26,240.2

Then, calculate the Implied Seller Enterprise Value by taking the Combined Enterprise Value and subtracting the Buyer's Current Enterprise Value. Reverse the EV bridge to calculate Implied Equity Value.

Relative Contribution Analysis:	Contribution (\$):			Contribution (%):		Combined Pro-Forma Value:		Implied Seller Value:	
Equity Value:	Buyer:	Seller:	Pro-Forma:	Buyer:	Seller:	Enterprise Value:	Equity Value:	Enterprise Value:	Equity Value:
Undisturbed Equity Value:	\$ 19,157.6	\$ 5,033.5	\$ 24,191.1	79.2%	20.8%	\$ 32,837.9	\$ 24,191.1	\$ 6,597.7	\$ 5,033.5
Pro-Forma Ownership @ 15.8% Premium:	19,157.6	5,831.2	24,988.8	76.7%	23.3%	33,635.5	24,988.8	7,395.3	5,831.2
Net Operating Income (NOI):									
Combined Year 1 - 2018-12-31:	1,546.9	457.8	2,004.8	77.2%	22.8%	34,005.9	25,359.2	=+123-\$13	6,201.6
Combined Year 1 - 2019-12-31:	1,737.8	543.5	2,281.3	76.2%	23.8%	34,446.8	25,800.1	8,206.6	6,642.5

It's the same process for all the metrics in this step.

Relative Contribution Analysis:	Contribution (\$):			Contribution (%):		Combined Pro-Forma Value:		Implied Seller Value:		Implied Figures in 100% Stock Deal:		
Equity Value:	Buyer:	Seller:	Pro-Forma:	Buyer:	Seller:	Enterprise Value:	Equity Value:	Enterprise Value:	Equity Value:	Exchange Ratio:	Purchase Price:	Variance to Offer Price:
Undisturbed Equity Value:	\$ 19,157.6	\$ 5,033.5	\$ 24,191.1	79.2%	20.8%	\$ 32,837.9	\$ 24,191.1	\$ 6,597.7	\$ 5,033.5	0.4706 x	\$ 55.54	(13.7%)
Pro-Forma Ownership @ 15.8% Premium:	19,157.6	5,831.2	24,988.8	76.7%	23.3%	33,635.5	24,988.8	7,395.3	5,831.2	0.5452 x	64.34	0.0%
Net Operating Income (NOI):												
Combined Year 1 - 2018-12-31:	1,546.9	457.8	2,004.8	77.2%	22.8%	34,005.9	25,359.2	7,765.7	6,201.6	0.5798 x	=+L23/Seller_Diluted_Shares	
Combined Year 1 - 2019-12-31:	1,737.8	543.5	2,281.3	76.2%	23.8%	34,446.8	25,800.1	8,206.6	6,642.5	0.6210 x	73.29	13.9%
Net Asset Value (NAV):												
Most Recent Quarter Prior to Deal Announcement:	15,778.8	4,942.5	20,721.2	76.1%	23.9%	33,805.2	25,158.4	7,565.0	6,000.9	0.5610 x	66.21	2.9%

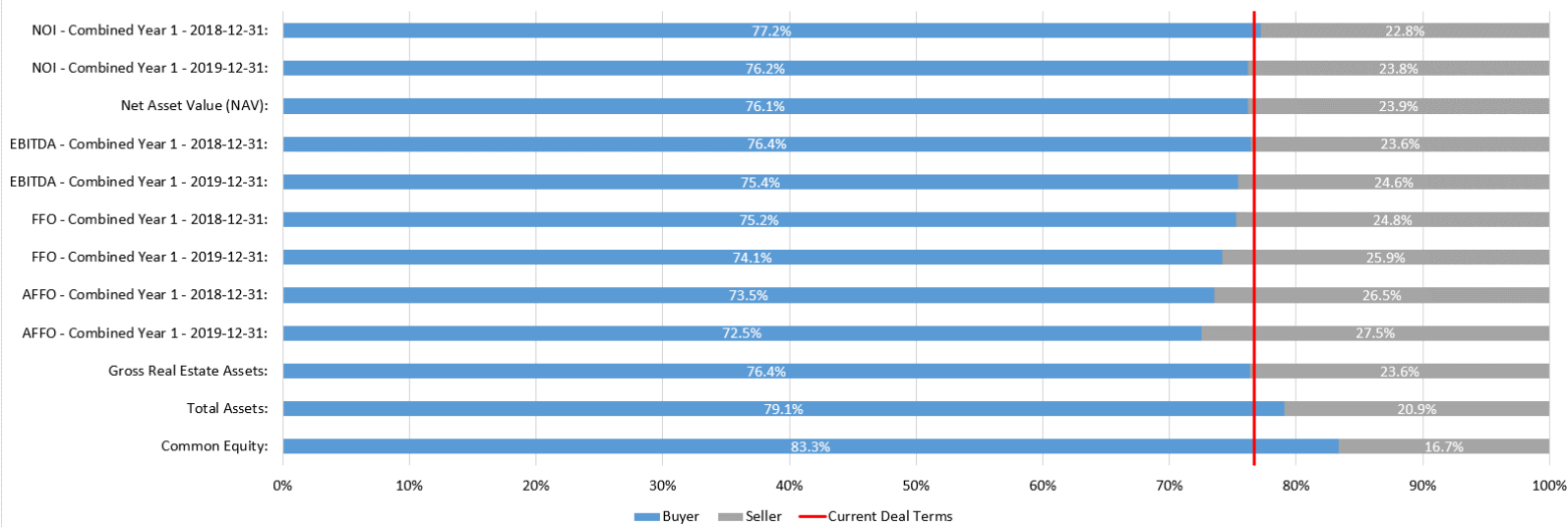
And then you can divide the Seller's Implied Equity Value by its diluted share count to calculate what the Purchase Price "should be" based on its contribution to this metric.

The Exchange Ratio equals this Purchase Price / Buyer's Share Price.

We can now compare these "should be" Purchase Prices to the real Offer Price and create graphs to see what the deal looks like.

According to the Balance Sheet metrics, the Buyer (DLR) should own more, but according to IS/CFS-based metrics like EBITDA, FFO, and AFFO, the Seller (DFT) should own more:

Relative Contribution Analysis:



Overall, it seems like the Seller (DFT) should own *a bit* more of the combined company.

The **Value Creation Analysis** is also important for REIT M&A deals.

This analysis is simple: find a larger peer company that trades at higher multiples than the Buyer or Seller, combine the financial metrics of the Buyer and Seller, and then apply the multiples of that peer company to these combined metrics:

Combined Entity - Post-Transaction and Acquisition Effects:				20,766.7	16,343.2	39.2%	1,338.2	1,801.0	2,077.9	983.1	1,245.2	1,470.2	
Valuation Statistics:				Equity Value	Enterprise Value	Gross Real Estate Assets	Leverage (Debt to Total Assets)	EV / EBITDA			Equity Value / FFO		
Company Name								FY17	FY18	FY19	FY17	FY18	FY19
Standalone Entities:													
Digital Realty Trust, Inc.	DLR	19,157.6	26,240.2	The multiples magically jump up after the deal closes! No, really!				21.1 x	19.0 x	16.9 x	20.3 x	18.3 x	16.2 x
DuPont Fabros Technology, Inc.	DFT	5,033.5	6,597.7					18.5 x	15.5 x	13.0 x	17.5 x	14.6 x	12.2 x
Closest Comparable Public Company for Combined Entity:													
Equinix, Inc.	EQIX	34,596.8	38,833.7					21.0 x	17.4 x	15.3 x	34.9 x	28.6 x	25.2 x
Combined Entity - Post-Transaction and Acquisition Effects:								21.0 x	17.4 x	15.3 x	34.9 x	28.6 x	25.2 x
Combined Entity - Implied Enterprise Value:								28,068.0	31,369.2	31,864.9			
(+/-) Cash & Cash-Equivalents:								126.0	126.0	126.0			
(+/-) Equity Investments & Non-Core Assets:								113.8	113.8	113.8			
(-/-) Total Debt:								(8,149.9)	(8,149.9)	(8,149.9)			
(-/-) Preferred Stock:								(1,208.2)	(1,208.2)	(1,208.2)			
(-/-) Non-OP Noncontrolling Interests:								(8.2)	(8.2)	(8.2)			
Combined Entity - Implied Equity Value:								18,941.5	22,242.8	22,738.5	34,275.2	35,649.9	37,075.3
Combined Entity - Diluted Share & Unit Count:								212.290	212.290	212.290	212.290	212.290	212.290
Combined Entity - Implied Share Price:								\$ 89.22	\$ 104.78	\$ 107.11	\$ 161.46	\$ 167.93	\$ 174.65
Buyer - Share Price Prior to Transaction:								118.02	118.02	118.02	118.02	118.02	118.02
Per-Share Value Created:								(28.80)	(13.24)	(10.91)	43.44	49.91	56.63

The multiples magically jump up after the deal closes! No, really!

And then we recalculate the Implied Enterprise Value for the combined company based on these new, higher multiples, and back into the Implied Equity Value and Share Price.

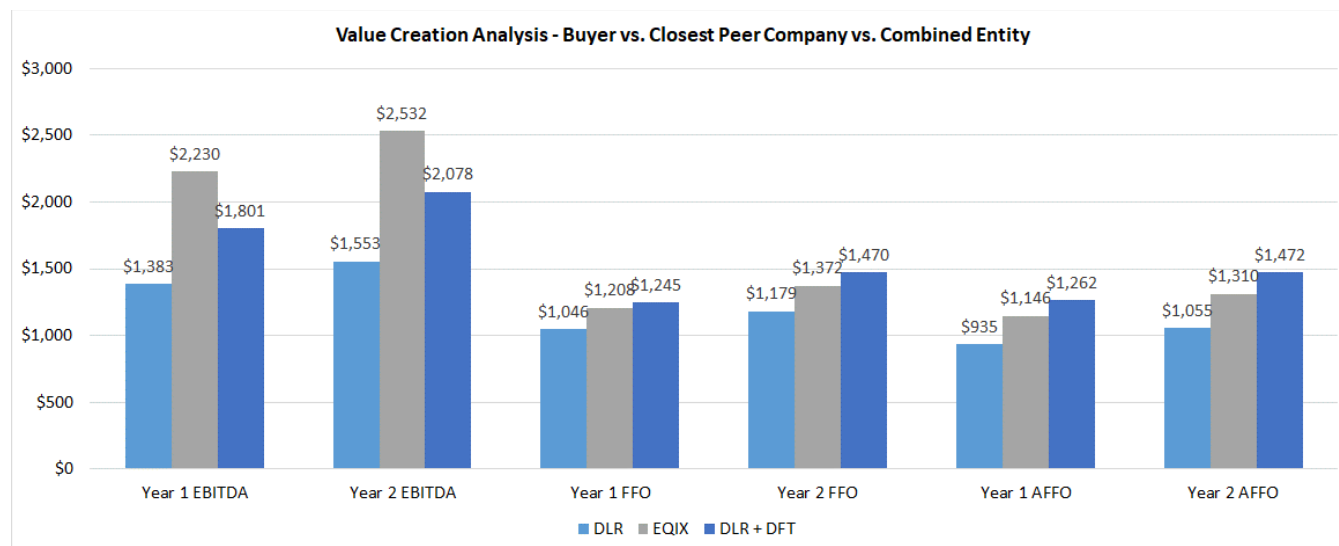
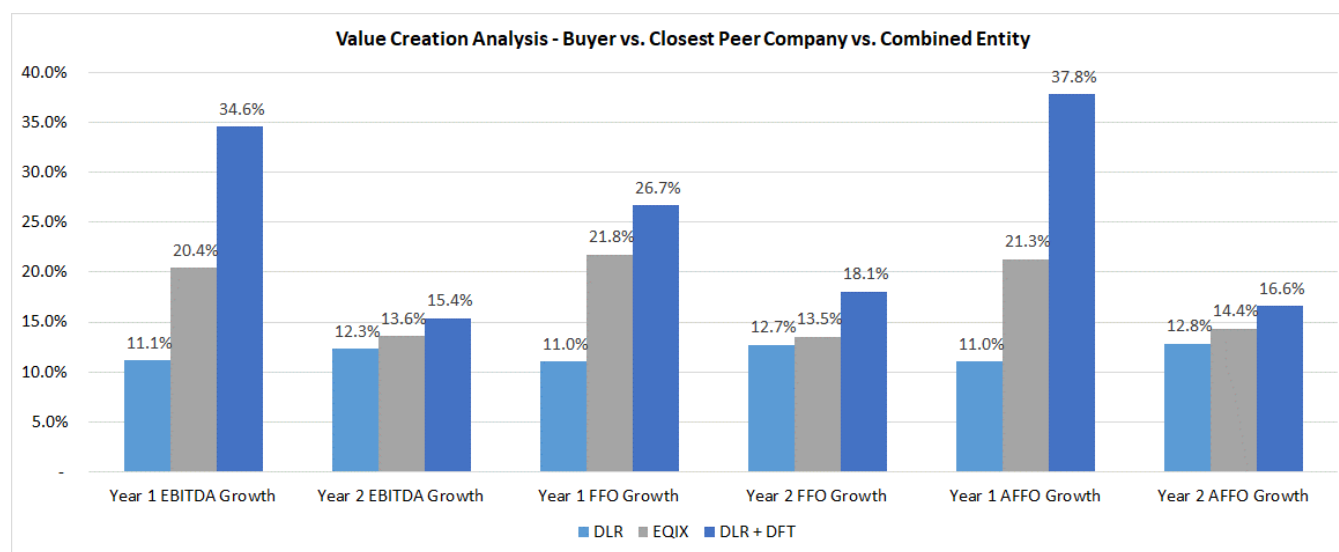
And this new Implied Share Price for the Combined Company should be higher than the Buyer's standalone Share Price, or we have a problem.

The argument is that once the Buyer and Seller combine, they will resemble the more-highly-valued company because they will have a larger, better-diversified property portfolio.

As you can tell by our sarcastic comments, this analysis is *highly speculative* because multiples rarely jump up like this after a deal closes.

This method is also very dependent on the market because there may not necessarily *be* a larger REIT in the same sector that trades at higher multiples.

So, with the Value Creation Analysis, you try to convince your audience that these two REITs combined will equal or exceed the financial performance of the larger, more highly valued REIT:





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The Value Creation Analysis doesn't necessarily "prove" anything if it produces positive results, but if you *cannot* get positive results no matter what, then it might indicate that the deal is not a great idea.

Finally, we like to create a "Deal Summary" page to present everything in an easy-to-understand format for senior professionals. Here's a portion of the one in this model:

Transaction Summary - Digital Realty Trust, Inc. and DuPont Fabros Technology, Inc.

(\$ USD in Millions Except Per Share and Per Unit Amounts in USD as Stated)

Transaction Summary and Key Terms:

Acquirer / Target:	Digital Realty Trust, Inc. / DuPont Fabros Technology, Inc.
Acquisition Close Date:	September 30, 2017
Exchange Ratio:	0.545 x
Implied Offer Price:	\$64.32 Per Share (\$5,831 Million); Undisturbed Share Price Premium of 15.8%

Ownership According to Contribution Analysis (Pre-Synergies and Pre-Acquisition Effects):	Actual: Deal:	NAV MRQ	EBITDA: FY18	FY19	FFO: FY18	FY19	AFFO: FY18	FY19	Gross RE Assets:
Acquirer:	76.7%	76.1%	76.4%	75.4%	75.2%	74.1%	73.5%	72.5%	76.4%
Target:	23.3%	23.9%	23.6%	24.6%	24.8%	25.9%	26.5%	27.5%	23.6%

Value Creation Analysis (Post-Synergies/Acquisition Effects):	FY17	EV / EBITDA FY18	FY19	FY17	P / FFO FY18	FY19	FY17	P / AFFO FY18	FY19
Combined Entity - Implied Share Price:	\$ 89.22	\$ 104.78	\$ 107.11	\$ 161.46	\$ 167.93	\$ 174.65	\$ 158.04	\$ 179.56	\$ 183.09
Buyer - Share Price Prior to Transaction:	118.02	118.02	118.02	118.02	118.02	118.02	118.02	118.02	118.02

Purchase Consideration:	100.0% Common Stock, 0.0% Cash	Cost Synergies:	\$18 million (5.2% of Target's Annual Operating Expenses) 50.0% in Year 1, 100.0% in Year 2, and 100.0% Thereafter
Operating Real Estate Asset Write-Up:	\$167 million (39-Year Depreciation)	Restructuring Charge:	\$27 Million (Pre-Tax), or 150.0% of Full Cost Synergies 100.0% Realized in Year 1; Nothing Thereafter
Land & Construction-in-Progress Write-Up:	\$124 million (20.0% Adjustment)		
In-Place Lease Value:	\$1,590 million (14-Year Amortization)	Above-Market Leases:	\$161 million (4-Year Amortization)
Valuation Multiples:	Current Price:	Offer Price:	Below-Market Leases: \$184 million (12-Year Amortization)

How to Make a Recommendation For or Against the Deal

Once you're done with a REIT merger model, you can use it to make a **deal recommendation**.

In other words, should the Buyer acquire the Seller? Is it a good deal for both sides? Is the Buyer's Offer Price reasonable? Should the Offer Price or other terms change?

Generally, you start with the numbers and say whether or not the Purchase Price makes sense based on the valuation methodologies and the output of the merger model (everything shown above).

You also have to state how realistic the key operating and transaction assumptions are.



Even if a deal looks accretive on paper, it doesn't mean anything if it is based on unrealistically high synergies or unrealistically strong financial performance by either party.

Then, you state whether or not the deal makes sense based on qualitative aspects, such as the companies' customer bases and geographies.

Even if the Buyer is paying a reasonable price for the Seller, the deal might make sense only if it improves the Buyer's portfolio or makes it appear like a more highly valued REIT.

Here's the outline of our recommendation for this DLR / DFT deal:

- We **recommend** pursuing this \$5.8 billion acquisition of DuPont Fabros because the company is appropriately valued at the 0.545x exchange ratio (\$64.32 Offer Price, or a ~16% premium), and the deal is not dependent on unrealistically high Cost Synergies or interest savings from refinancing.
- Despite significantly higher growth than the peer companies, DFT trades at a discount to DLR's multiples as well as the median EBITDA, FFO, and AFFO multiples of the set; this discount still exists even at a ~16% premium.
- The Contribution Analysis suggests that DLR should own less than 77% of the combined company, but the current terms will result in ~77% post-transaction ownership.
- Even if the combined entity does not trade at Equinix's multiples, which are 50-70% higher than DLR's, there is still a solid opportunity for 10-20% value creation for DLR shareholders.
- The deal will be 4-7% accretive to AFFO PS in FY 18 and FY 19; even without the \$18 million in Cost Synergies, the deal would still be neutral-to-accretive.
- The deal will diversify both companies' customer bases and geographies and make the combined entity comparable to Equinix.

In short, we like the deal, but the Buyer should pay a slightly higher price to give the Seller higher percentage ownership in the combined entity.

Even at that higher price, the deal will still be accretive, it will offer potentially higher multiples, and it will deliver qualitative benefits as well.

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REIT Leveraged Buyouts and LBO Models

The first question you probably have about REIT leveraged buyouts is **how they are possible at all**.

After all, REITs in most countries are subject to shareholder and ownership requirements that seem to preclude LBOs. In the U.S., for example:

- REITs must be jointly owned by **100+ people**.
- No more than **50%** of the shares can be held by **5 or fewer people** (the “5/50 rule”).

The short answer is that private equity firms can easily get around these rules with a bit of creativity.

For example, they can bypass the first rule by creating a separate preferred shareholder class where 100+ individuals own a tiny percentage of the REIT, and the lead PE firm owns 99%+.

The 5/50 rule seems trickier to bypass, but it’s actually a pass-through requirement to individual investors – so if a corporate entity owns a REIT, only the *individuals* at the bottom of the ownership structure matter.

For a private equity firm acquiring a REIT, this rule “passes through” and applies to the PE firm’s shareholders – and there’s little chance that 5 or fewer people could own 50%+ of a private equity firm, given that their LPs tend to be pension funds, endowments, and other institutions.

This may seem like a big loophole, but the “5/50 rule” is designed to prevent a few wealthy friends from pooling their funds, investing in real estate, and then avoiding all taxes – the government cares less about what institutional investment firms do.

Finally, in some regions, such as Canada, these same rules do not exist at all, so it’s easier to execute leveraged buyouts.

When buyout deals take place, private equity firms do everything in their power to *maintain* REITs’ tax-exempt status because avoiding a 20%, 30%, or 40% tax rate provides a big boost to the returns.

And in a buyout scenario, the Dividends that REITs issue go directly to the private equity firm, further boosting their returns.

We note this point only because you will see some incorrect/contradictory information online about how leveraged buyouts in this sector work.

REIT Leveraged Buyouts vs. Traditional Leveraged Buyouts



In a traditional leveraged buyout, the private equity firm realizes returns via **EBITDA Growth, Multiple Expansion, and Debt Paydown and Cash Generation.**

A PE firm might buy a \$100 million EBITDA company for 6x EBITDA, use 3x Debt, grow the company's EBITDA to \$150 million over 5 years, sell it for 7x EBITDA, and have the company repay all \$300 million of initial Debt over those 5 years.

That deal would generate a **28% IRR** because the firm contributes \$300 million of Investor Equity and earns back \$150 million * 7 = \$1.05 billion at the end (no remaining Debt), which is a 3.5x multiple.

A 3x multiple over 5 years is a 25% IRR, and a 4x multiple is a 32% IRR, so the 28% is right in the middle.

In a REIT leveraged buyout, the math changes because **Debt Paydown and Cash Generation are far more difficult.**

REITs must distribute 90%+ of their Net Income as Dividends, making it difficult to amass much of a Cash balance – even *before* a buyout takes place.

After the buyout takes place, it's even worse because additional Debt is added, which increases the Net Interest Expense. And then, unlike in a traditional LBO, the Debt continues to increase over time as the REIT draws on it to fund acquisitions and developments.

As a result, REIT leveraged buyouts rely on **EBITDA Growth** and **Multiple Expansion**, which makes them heavily dependent on timing the real estate market cycle.

Dividends help as well, so you could count them as a third returns source. But remember: most REITs have Dividend Yields between **3% and 5%.**

So, Dividends by themselves are not sufficient to boost the IRR in leveraged buyouts up to the 20%+ range that PE firms often target.

One final difference in REIT LBOs is that PE firms cannot add that much Debt in the initial transaction because REITs already tend to be highly leveraged.

So, a dramatic change in the company's capital structure immediately after deal close is less likely.

Taken together, these points mean that REIT LBOs are like traditional LBOs, but with continual Dividend Recaps and Debt-Funded Add-On Acquisitions.



That means the PE firm earns more cash flow during the holding period, but must also repay more Debt upon exit and achieve significant EBITDA Growth or Multiple Expansion by then.

A simple example of the math in a REIT LBO might be the following:

- A PE firm acquires a \$1 billion Enterprise Value REIT for a 10x EBITDA multiple using 50% Debt and 50% Equity. The REIT's Dividend Yield is 4%.
- Each year, the REIT issues $4\% * \$500 \text{ million} = \20 million in Dividends to the PE firm. The holding period is 5 years.
- Over that time, the REIT's EBITDA increases from \$100 million to \$150 million. Its Debt balance also increases from \$500 million to \$800 million because of acquisitions and developments.
- Due to greater scale and better market conditions, the PE firm sells the REIT for 12x EBITDA in Year 5, so the Exit Enterprise Value is $12 * \$150 \text{ million} = \1.8 billion , and the Exit Equity Value is \$1.0 billion (assume no Cash for simplicity).
- That is a 2x multiple of the initial Investor Equity, but the total Dividends of \$100 million over 5 years make this a 2.2x multiple.
- The 5-year IRR is **18%**. This isn't necessarily the result that PE firms look for in LBOs, but for a mature, stabilized REIT, it's a decent outcome that's still ahead of public-market returns.

The **Dividends** contributed about **3%** per year to the IRR, but most of the returns came from EBITDA Growth and Multiple Expansion.

Without any Multiple Expansion, the IRR would have been **10%**, which demonstrates how timing-dependent these deals are.

In the next section, we'll walk through the REIT LBO model for Pure Industrial Real Estate Trust that's used in this course.

Industrial REITs were "hot" at this time because of e-commerce, but PIRET's stock price had already risen by 80% in the 3 years before the deal, and the company had grown mostly via acquisitions.

Cap Rates in the sector had been falling, making growth more expensive but also improving PIRET's valuation.



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So, the question here is simple: Will the strong supply/demand dynamic and overall market growth be enough to overcome falling acquisition yields, rising Debt, and rising interest rates?

To answer that question, we build an LBO model for the deal using the following steps:

Step 1: Set Up the Transaction Assumptions, Sources & Uses, and Purchase Price Allocation Schedules

For a public REIT like PIRET, the Offer Price is still based on a premium to its current share price, and we calculate the purchase multiple(s) based on that:

Pure Industrial Real Estate Trust - Leveraged Buyout Model

(C\$ CAD in Millions Except Per Share and Per Unit Amounts in CAD as Stated)

Transaction Assumptions:			Units:			Exercise			Exercise		
			Name			Dilution:	Number:	Price:	Dilution:		
Company Name:			Pure Industrial Real Estate Trust			Options A:	0.465	\$ 4.29	0.168		
Ticker:			TSX:AAR-UN-T			Restr. Units:	0.826		0.826		
Company Type:			Industrial REIT			Def. Units:	0.033		0.033		
Last Historical Year:			2017-12-31			Total:			1.026		
Valuation Date:			2018-01-09			Diluted Shares Outstanding:			306.906		
Current Share Price:			\$ 6.72			Current Equity Value:			\$ 2,062.4		
Class A Units Outstanding:			# Millions 305.880			(-) Cash & Cash-Equivalents:			(56.6)		
Months in Year:			# 12			(-) Assets Held for Sale, Net of Liabilities:			(14.2)		
Minimum Cash Balance % Cash Operating Expenses:			% 75.0%			(+) Total Debt:			1,211.9		
Purchase Premium:			% 20.5%			(+) Preferred Stock:			-		
Offer Price per Share:			C\$ as Stated \$ 8.10			(+) Noncontrolling Interests:			11.2		
Selected Operating Scenario:			Name Base			Current Enterprise Value:			\$ 3,214.7		
Exit Year:			Date FY22			Current EV / EBITDA Multiple:			20.9 x		
						Purchase EV / EBITDA Multiple:			23.7 x		

Then, we make the key assumptions for the Sources & Uses and Purchase Price Allocation schedules:

Sources & Uses, Goodwill, and Debt Assumptions:

Sources of Funds:				Uses of Funds:			
	\$ M CAD	x EBITDA	% Sources		\$ M CAD	x EBITDA	
Assumed Mortgages Payable & Other Loans:	1,211.9	7.9 x	32.6%	Purchase Equity Value:	\$ 2,486.4	16.2 x	
Subordinated Notes:	307.3	2.0 x	8.3%	Assumed Mortgages Payable & Other Loans:	1,211.9	7.9 x	
Management Rollover:	76.8	0.5 x	2.1%	Transaction Fees:	20.0	0.1 x	
Investor Equity:	2,122.3	13.8 x	57.1%	Total Uses:	\$ 3,718.3	24.2 x	
Total Sources:	\$ 3,718.3	24.2 x	100.0%				
Ownership Details:				Goodwill Calculation:			
	\$ M CAD	% Sponsor:	%		\$ M CAD		
Management:	\$ 76.8		3.5%	Equity Purchase Price:	\$ 2,486.4		
Blackstone Group L.P.:	1,315.8	62.0%	59.8%	(-) Seller's Common Book Value:	(1,895.7)		
Ivanhoe Cambridge, Inc.:	806.5	38.0%	36.7%	(+) Write-Off of Existing Goodwill:	-		
Total Equity:	\$ 2,199.1		100.0%	Total Goodwill:	\$ 590.7		
Management Option Pool:		1.0%					
Threshold Equity Multiple for Ivanhoe Cambridge, Inc.:		2.0 x					
Additional % Exit Equity Proceeds Granted:		20.0%					

These terms incentivize the Developer / Operator to perform by granting additional Equity if the deal does well enough.

We add minimal Debt because PIRET is already highly leveraged.

Simplified calculation due to lack of time and relative unimportance of the PPA here.



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We simplify the Purchase Price Allocation here because there isn't enough information to make the type of detailed schedule we did [in the merger model](#); also, under IFRS, the adjustments would be less significant because Investment Properties are marked to market already.

Step 2: Project the REIT's Business Segments and Income Statement

Next, we create assumptions for PIRET in the Base, Upside, and Downside cases, based on historical data and the company's performance in past recessions and growth periods.

Here are a few of the more significant assumptions:

Income Statement Drivers:	Units:	Historical:			Projected:				
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22
Same-Store Rent and Recovery Growth:	%	(0.7%)	2.6%	2.8%	2.5%	2.5%	2.0%	2.0%	2.0%
Upside	%				3.5%	4.0%	3.5%	3.0%	2.5%
Base	%				2.5%	2.5%	2.0%	2.0%	2.0%
Downside	%				(2.0%)	(2.0%)	4.0%	3.0%	2.0%
All Segments - Expenses % Rent and Recoveries:	%	30.0%	28.9%	27.3%	27.0%	26.5%	26.5%	26.0%	26.0%
Upside	%				26.0%	25.5%	25.0%	24.5%	24.0%
Base	%				27.0%	26.5%	26.5%	26.0%	26.0%
Downside	%				29.0%	31.0%	29.0%	28.0%	27.0%
Acquisition Spending:	\$ M	88.4	289.7	665.6	340.0	350.0	360.0	370.0	380.0
Upside	\$ M				500.0	475.0	450.0	425.0	400.0
Base	\$ M				340.0	350.0	360.0	370.0	380.0
Downside	\$ M				150.0	200.0	250.0	300.0	350.0
Weighted Average Going-In Cap Rate:	%	7.4%	6.5%	5.4%	5.7%	6.0%	6.3%	6.5%	6.8%
Upside	%				5.2%	5.5%	5.8%	6.2%	6.5%
Base	%				5.7%	6.0%	6.3%	6.5%	6.8%
Downside	%				8.0%	7.7%	7.5%	7.3%	7.0%
Incremental Annual NOI from Acquisitions:	\$ M				35.9	19.4	21.0	22.7	24.1
Incremental Annual Revenue from Acquisitions:	\$ M				49.2	26.4	28.6	30.6	32.5

The cases are what you'd expect: stronger revenue growth and margins in the Upside Case, lower numbers in the Base Case, and still lower numbers in the Downside Case.

PIRET is primarily an acquirer, so it spends the most on acquisitions in the Upside Case, but also completes the most expensive deals (lowest Cap Rates).

Development spending and yields are also highest in the Upside Case; Asset Dispositions are highest, and Exit Cap Rates are lowest in the Upside Case, and the Cap Rates used to mark PIRET's portfolio to market value are lowest in the Upside Case.

Based on these assumptions, we can create an Income Statement for PIRET down to Pre-Tax Income (Base Case shown below):



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Income Statement:	Units:	Historical:			Projected:				
		FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22
Revenue:									
(+) Same-Store Rent and Recoveries:	\$ M	\$ 171.4	\$ 186.9	\$ 225.5	\$ 231.1	\$ 236.9	\$ 241.6	\$ 246.5	\$ 251.4
(+) Rent and Recoveries from Acquisitions:	\$ M				49.2	75.6	104.2	134.8	167.3
(+) Rent and Recoveries from Developments:	\$ M				2.2	8.0	13.4	18.4	23.0
(-) Rent and Recoveries from Dispositions:	\$ M				(14.3)	(18.3)	(22.6)	(27.1)	(31.5)
Total Revenue:	\$ M	171.4	186.9	225.5	268.3	302.2	336.6	372.6	410.2
Property-Level Operating Expenses:									
(-) Same-Store Expenses:	\$ M	(51.3)	(54.1)	(61.5)	(62.4)	(62.8)	(64.0)	(64.1)	(65.4)
(-) Expenses from Acquisitions:	\$ M				(13.3)	(20.3)	(27.9)	(35.8)	(44.3)
(-) Expenses from Developments:	\$ M				(0.6)	(2.1)	(3.6)	(4.9)	(6.0)
(+) Expenses from Dispositions:	\$ M				3.9	4.9	6.1	7.2	8.4
Total Property-Level Operating Expenses:	\$ M	(51.3)	(54.1)	(61.5)	(72.4)	(80.3)	(89.4)	(97.5)	(107.3)
Net Operating Income (NOI):	\$ M	120.0	132.8	164.0	195.9	221.9	247.2	275.0	302.9
Other Income / (Expenses):									
(-) General & Administrative:	\$ M	(6.5)	(8.0)	(10.4)	(12.9)	(15.1)	(17.5)	(20.1)	(22.6)
(+/-) Fair Value Adjustments to Investment Properties:	\$ M	97.5	65.7	192.9	71.1	90.7	145.6	69.4	-
(+/-) Fair Value Adjustments to Financial Instruments:	\$ M	(2.4)	2.3	4.4	(1.5)	(1.5)	(1.5)	(1.5)	(1.5)
(+) Net Gain / (-) Loss on Sale of Assets:	\$ M	(2.5)	(0.9)	(4.6)	(1.5)	(1.7)	(1.5)	(1.5)	(1.5)
(-) Goodwill Impairment:	\$ M				-	-	-	-	-
(+/-) Other Items:	\$ M	0.2	3.7	(3.7)	1.5	1.5	1.5	1.5	1.5
Total Other Income / (Expenses):	\$ M	86.2	62.9	178.7	56.8	73.9	126.6	47.8	(24.0)
Net Earnings Before Finance Expense:	\$ M	206.2	195.8	342.7	252.6	295.9	373.8	322.9	278.9

Here are the results across the cases:

- **Upside Case:**
 - Year 5 Revenue: \$454 million
 - Year 5 NOI: \$338 million (16% CAGR)
- **Base Case:**
 - Year 5 Revenue: \$410 million
 - Year 5 NOI: \$303 million (13% CAGR)
- **Downside Case:**
 - Year 5 Revenue: \$366 million
 - Year 5 NOI: \$267 million (10% CAGR)

The cases should be different, but not *that* different since this is still a large, mature REIT.

Step 3: Adjust the Balance Sheet and Project It

The Balance Sheet adjustments are quite simple in this deal since we chose to simplify the Purchase Price Allocation:



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Balance Sheet:	Units:	Historical:			Transaction Adjustments:		
		FY15	FY16	FY17	Debit	Credit	FY17
ASSETS:							
Investment Properties (Inc. Assets Held for Sale):	\$ M	\$ 2,065.6	\$ 2,432.3	\$ 3,129.6	\$ -	\$ -	\$ 3,129.6
Receivables & Other Assets:	\$ M	21.5	24.8	34.7	-	-	34.7
Goodwill:	\$ M				590.7	-	590.7
Cash & Cash-Equivalents:	\$ M	15.3	59.4	56.6	-	-	56.6
TOTAL ASSETS:	\$ M	\$ 2,102.5	\$ 2,516.5	\$ 3,220.8			\$ 3,811.5
LIABILITIES & EQUITY:							
Mortgage Payable & Other Loans:	\$ M	\$ 1,026.2	\$ 1,043.5	\$ 1,211.9	\$ (1,211.9)	\$ 1,211.9	\$ 1,211.9
Subordinated Notes:	\$ M				-	307.3	307.3
Accounts Payable & Accrued Liabilities:	\$ M	29.5	39.5	64.2	-	-	64.2
Other Liabilities:	\$ M	25.6	54.3	37.7	-	-	37.7
Total Liabilities:	\$ M	1,081.3	1,137.3	1,313.9			1,621.2
Unitholders' Equity:	\$ M	1,002.9	1,352.5	1,895.7	(1,915.7)	2,199.1	2,179.1
(+) Noncontrolling Interests (NCI):	\$ M	18.3	26.7	11.2	-	-	11.2
Total Equity:	\$ M	1,021.2	1,379.2	1,906.9			2,190.3
TOTAL LIABILITIES & EQUITY:	\$ M	\$ 2,102.5	\$ 2,516.5	\$ 3,220.8			\$ 3,811.5
Balance Check:		OK!	OK!	OK!			OK!

No write-ups; assume properties are already marked properly.

Standard Goodwill creation; no Cash used for transaction fees.

Debt is "assumed," i.e., replaced with new, identical Debt.

And new Subordinated Notes are put in place; only modest leverage added.

Common Equity written down and replaced with Investor Equity; transaction fees also deducted.

No write-ups; assume properties are already marked properly.

Standard Goodwill creation; no Cash used for transaction fees.

Debt is "assumed," i.e., replaced with new, identical Debt.

And new Subordinated Notes are put in place; only modest leverage added.

Common Equity written down and replaced with Investor Equity; transaction fees also deducted.

Next, we project the Balance Sheet by linking each item to the appropriate driver or Cash Flow Statement line item.

Below, we show the Balance Sheet projections from the *completed* version of this model:

Balance Sheet:	Units:	Projected:					
		FY17	FY18	FY19	FY20	FY21	FY22
ASSETS:							
Investment Properties (Inc. Assets Held for Sale):	\$ M	\$ 3,129.6	\$ 3,579.6	\$ 4,052.6	\$ 4,583.9	\$ 5,047.9	\$ 5,456.5
Receivables & Other Assets:	\$ M	34.7	41.6	47.4	53.5	59.6	65.6
Goodwill:	\$ M	590.7	590.7	590.7	590.7	590.7	590.7
Cash & Cash-Equivalents:	\$ M	56.6	64.0	71.5	80.2	88.2	97.4
TOTAL ASSETS:	\$ M	\$ 3,811.5	\$ 4,275.9	\$ 4,762.2	\$ 5,308.3	\$ 5,786.4	\$ 6,210.2
LIABILITIES & EQUITY:							
Mortgage Payable & Other Loans:	\$ M	\$ 1,211.9	\$ 1,539.6	\$ 1,869.2	\$ 2,196.9	\$ 2,528.3	\$ 2,866.1
Subordinated Notes:	\$ M	307.3	307.3	307.3	307.3	307.3	307.3
Accounts Payable & Accrued Liabilities:	\$ M	64.2	76.8	86.8	98.3	109.4	122.1
Other Liabilities:	\$ M	37.7	45.2	51.5	58.8	65.9	74.0
Total Liabilities:	\$ M	1,621.2	1,968.9	2,314.8	2,661.3	3,010.8	3,369.5
Unitholders' Equity:	\$ M	2,179.1	2,293.4	2,431.0	2,626.5	2,752.5	2,816.1
(+) Noncontrolling Interests (NCI):	\$ M	11.2	13.6	16.4	20.5	23.1	24.6
Total Equity:	\$ M	2,190.3	2,307.0	2,447.4	2,647.0	2,775.6	2,840.7
TOTAL LIABILITIES & EQUITY:	\$ M	\$ 3,811.5	\$ 4,275.9	\$ 4,762.2	\$ 5,308.3	\$ 5,786.4	\$ 6,210.2
Balance Check:		OK!	OK!	OK!	OK!	OK!	OK!

Old balance, Fair Value adjustment, Straight-Lining of Rent, and most of CFI on CFS.

% of revenue.

Old balance - Impairment on CFS.

Bottom of CFS.

Old balances, Fair Value adjustments, and Net Change on CFS.

% of operating expenses.

Old balance, NI to Parent, many items from CFO, and most non-Debt items in CFF.

Old balance, NI to NCI, and Cash Dividends to NCI.

Old balance, Fair Value adjustment, Straight-Lining of Rent, and most of CFI on CFS.

% of revenue.

Old balance - Impairment on CFS.

Bottom of CFS.

Old balances, Fair Value adjustments, and Net Change on CFS.

% of operating expenses.

Old balance, NI to Parent, many items from CFO, and most non-Debt items in CFF.

Old balance, NI to NCI, and Cash Dividends to NCI.

The screenshot above is **NOT** what you will see if you're completing this model sequentially because you won't have a finished Cash Flow Statement yet.

We're walking through the completed version of this model because it's easier to show all the changes and links on the statements like that.

Step 4: Project the Dividends, Funding Sources, and Cash Flow Statement

The Cash Flow Statement, Dividends, and Debt/Equity issuances are all linked together, so we must complete portions of the CFS first, then project the Dividends, Debt, and Equity, and then return to finish linking the CFS.

Here's the finished version of the Cash Flow Statement with a few notes:

Cash Flow Statement:	Units:	FY17	Projected:				
			FY18	FY19	FY20	FY21	FY22
Cash Flow from Operations:							
(+) Net Income Attributable to Parent:	\$ M		\$ 180.3	\$ 203.7	\$ 261.2	\$ 194.9	\$ 139.2
(+) Net Income Attributable to NCI:	\$ M		3.8	4.3	5.5	4.1	2.9
(-/+ Fair Value Adjustments to Investment Properties:	\$ M		(71.1)	(90.7)	(145.6)	(69.4)	-
(-/+ Fair Value Adjustments to Financial Instruments:	\$ M		1.5	1.5	1.5	1.5	1.1
(+) Goodwill Impairment:	\$ M		-	-	-	-	-
(-) Net Gain / (+) Loss on Sale of Assets:	\$ M		1.5	1.7	1.5	1.5	1.5
(-/+ Reversal of Other Items on Income Statement:	\$ M		(1.5)	(1.5)	(1.5)	(1.5)	(1.5)
(-) Straight-Line Rental Adjustment:	\$ M		(5.4)	(6.0)	(6.7)	(7.5)	(8.2)
(+/-) Deferred Income Taxes:	\$ M		4.3	4.8	6.2	4.6	3.3
(+/-) Other Adjustments:	\$ M		6.3	7.1	7.9	8.8	9.7
(+/-) Net Change in Non-Cash Working Capital:	\$ M		13.1	10.4	12.7	12.1	14.8
Cash Flow from Operations:	\$ M		132.8	135.3	142.7	149.0	163.1
Cash Flow from Investing:							
(-) Acquisitions of Investment Properties:	\$ M		(340.0)	(350.0)	(360.0)	(370.0)	(380.0)
(-) Additions to Land & Developments:	\$ M		(65.0)	(60.0)	(55.0)	(50.0)	(50.0)
(-) Recurring Maintenance CapEx:	\$ M		(7.8)	(9.1)	(10.4)	(11.9)	(13.5)
(-) Leasing Costs:	\$ M		(5.6)	(6.3)	(7.4)	(8.2)	(9.4)
(-) Tenant Improvements:	\$ M		(2.7)	(3.3)	(3.7)	(4.5)	(4.9)
(-) Proceeds from Asset Dispositions:	\$ M		50.0	55.0	60.0	60.0	60.0
(+/-) Other Items, Net:	\$ M		(2.5)	(2.5)	(2.5)	(2.5)	(2.5)
Cash Flow from Investing:	\$ M		(373.6)	(376.2)	(379.0)	(387.1)	(400.4)
Cash Flow from Financing:							
(+/-) Net Change in Mortgages Payable:	\$ M		326.2	328.2	326.2	329.9	336.4
(+/-) Net Change in Subordinated Notes:	\$ M		-	-	-	-	-
(+) Stock Issuances / (-) Repurchases:	\$ M		-	-	-	-	-
(-) Cash Dividends to Unitholders:	\$ M		(87.8)	(89.4)	(91.0)	(93.5)	(99.7)
(-) Cash Dividends to Noncontrolling Interests:	\$ M		(1.4)	(1.4)	(1.4)	(1.4)	(1.4)
(+/-) Other Items, Net:	\$ M		11.4	11.4	11.4	11.4	11.4
Cash Flow from Financing:	\$ M		248.3	248.7	245.1	246.3	246.6
FX Rate Effects:							
(+) Net Change in Cash & Cash-Equivalents:	\$ M		7.4	7.5	8.6	8.1	9.2
(+) Beginning Cash:	\$ M		56.6	64.0	71.5	80.2	88.2
Ending Cash:	\$ M		\$ 64.0	\$ 71.5	\$ 80.2	\$ 88.2	\$ 97.4

These are all Income Statement links or reversals.

Projected as %'s of revenue or taxes here; Change in Working Capital flows in from Balance Sheet.

From the segment-by-segment schedule.

%'s of revenue.

From the Debt Schedule.

90% of AFFO.

We assume 0 Stock Issuances here because it's a leveraged buyout deal; doesn't make much sense to issue Stock to the PE owner to fund operations.

One difference is that it's not quite *as simple* as just linking to the Stock Issuances and Debt Issuances on the Dividend/Debt/Equity schedule because this is a leveraged buyout.

We set the Stock Issuances to a hard-coded 0 because they don't make sense in this context – and the Debt Issuance line items link to the separate Debt Schedule instead.

There are mandatory repayments and maturities, additional issuances and draws, and optional repayments for each tranche of Debt, so it's better to link everything via a separate schedule.



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Cash Flow Statement Drivers:	Units:	Projected:				
		FY18	FY19	FY20	FY21	FY22
Straight-Line Rental Adjustment % Revenue:	%	(2.0%)	(2.0%)	(2.0%)	(2.0%)	(2.0%)
Deferred Income Taxes % Book Taxes:	%	75.0%	75.0%	75.0%	75.0%	75.0%
Other Adjustments % Revenue:	%	2.4%	2.4%	2.4%	2.4%	2.4%
Recurring Maintenance CapEx % Revenue:	%	2.9%	3.0%	3.1%	3.2%	3.3%
Leasing Costs % Revenue:	%	2.1%	2.1%	2.2%	2.2%	2.3%
Tenant Improvements % Revenue:	%	1.0%	1.1%	1.1%	1.2%	1.2%
(+) Net Income Attributable to Parent:	\$ M	\$ 180.3	\$ 203.7	\$ 261.2	\$ 194.9	\$ 139.2
(-/+) Fair Value Adjustments to Investment Properties:	\$ M	(71.1)	(90.7)	(145.6)	(69.4)	-
(-) NCI Share of Fair Value Adjustment:	\$ M	1.7	2.1	3.4	1.6	-
(-) Net Gain / (+) Loss on Sale of Assets:	\$ M	1.5	1.7	1.5	1.5	1.5
(+/-) Deferred Income Taxes:	\$ M	4.3	4.8	6.2	4.6	3.3
(-/+) Fair Value Adjustments to Financial Instruments:	\$ M	1.5	1.5	1.5	1.5	1.5
Funds from Operations (FFO):	\$ M	118.2	123.1	128.2	134.7	145.4
NCI Share of FVA % Total FVA:	%	2.4%	2.4%	2.4%	2.4%	2.4%
(-) Recurring Maintenance CapEx:	\$ M	(7.8)	(9.1)	(10.4)	(11.9)	(13.5)
(-) Leasing Costs:	\$ M	(5.6)	(6.3)	(7.4)	(8.2)	(9.4)
(-) Tenant Improvements:	\$ M	(2.7)	(3.3)	(3.7)	(4.5)	(4.9)
(-) Straight-Line Rental Adjustment:	\$ M	(5.4)	(6.0)	(6.7)	(7.5)	(8.2)
(+) NCI Share of Adjustments:	\$ M	0.9	1.0	1.1	1.3	1.4
Adjusted Funds from Operations (AFFO):	\$ M	\$ 97.5	\$ 99.4	\$ 101.1	\$ 103.9	\$ 110.8
NCI Share % of Adjustments:	%	4.0%	4.0%	4.0%	4.0%	4.0%
Dividends % FFO:	%	74.3%	72.6%	70.9%	69.4%	68.6%
Dividends % AFFO:	%	90.0%	90.0%	90.0%	90.0%	90.0%
Planned Dividends:	\$ M	87.8	89.4	91.0	93.5	99.7
Planned Debt Repayments & Maturities:	\$ M	95.4	114.7	239.4	130.7	178.7
Minimum Cash Balance:	\$ M	64.0	71.5	80.2	88.2	97.4
Cash Balance Before Debt Financing:	\$ M	(357.6)	(371.3)	(485.4)	(372.4)	(417.7)
Debt Funding Required:	\$ M	421.6	442.8	565.6	460.6	515.1

Fairly standard calculation, but Canadian REITs also factored in Deferred Taxes and use the Fair Value Adjustment rather than Depreciation.

Also fairly standard; as always, AFFO reflects recurring capital costs and the straight-line rental adjustment to come closer to real cash flow.

Min Cash is a % of Cash OpEx here; Cash Balance Before Debt Financing is Old Cash + everything on the CFS excluding additional Debt draws. Debt Funding Required is MAX(Min Cash - Cash Balance Before Debt Financing, 0).

The “Planned Debt Repayments & Maturities” above comes from the Debt Schedule, which already has the appropriate numbers filled in for these (based on the company’s filings).

Step 5: Set Up the Debt Schedule and the Credit Stats and Ratios

This Debt Schedule is a bit pointless because a company such as PIRET will *never* have enough Cash available to repay Debt optionally.

But the main purpose of a Debt Schedule is to calculate those optional repayments – if the company has limited Cash and can only make the mandatory repayments, then we do not need a full schedule.

The mandatory repayments will happen regardless of the company’s Cash situation, and if the company needs more funding to pay for them, it will simply raise more Debt.

But to make this look more like a traditional LBO model, we set up this schedule anyway:



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Minimum Cash Balance:	\$ M	64.0	71.5	80.2	88.2	97.4
Cash Balance Before Debt Financing:	\$ M	(357.6)	(371.3)	(485.4)	(372.4)	(417.7)
Debt Funding Required:	\$ M	421.6	442.8	565.6	460.6	515.1

Debt Schedule:	Units:	Projected:				
		FY18	FY19	FY20	FY21	FY22
Cash Flow Available for Optional Debt Repayment:						
Cash - Beginning of Period:	\$ M	56.6	64.0	71.5	80.2	88.2
(+) Free Cash Flow After Dividends:	\$ M	(318.8)	(320.6)	(317.6)	(321.8)	(327.2)
(-) Mandatory Debt Repayments:	\$ M	(95.4)	(114.7)	(239.4)	(130.7)	(178.7)
(-) Minimum Cash:	\$ M	(64.0)	(71.5)	(80.2)	(88.2)	(97.4)
Cash Flow Available for Debt Repayment:	\$ M	(421.6)	(442.8)	(565.6)	(460.6)	(515.1)
(-) Optional Repayments of Mortgages Payable:	\$ M	-	-	-	-	-
Cash Flow Available for Subordinated Notes:	\$ M	(421.6)	(442.8)	(565.6)	(460.6)	(515.1)
(-) Optional Repayments of Subordinated Notes:	\$ M	-	-	-	-	-
Cash Generated ABOVE Minimum Cash Balance:	\$ M	(421.6)	(442.8)	(565.6)	(460.6)	(515.1)
Face Value of Mortgages Payable & Other Loans:						
BoP Mortgages Payable & Other Loans:	\$ M	1,211.9	1,538.2	1,866.3	2,192.5	2,522.4
(+) Additional Issuances and Draws:	\$ M	=+L267	442.8	565.6	460.6	515.1
(-) Mandatory Repayments & Maturities:	\$ M	(95.4)	(114.7)	(239.4)	(130.7)	(178.7)
(-) Optional Repayments:	\$ M	-	-	-	-	-
EOp Mortgages Payable & Other Loans:	\$ M	1,538.2	1,866.3	2,192.5	2,522.4	2,858.8
Face Value of Subordinated Notes:						
BoP Subordinated Notes:	\$ M	307.3	307.3	307.3	307.3	307.3
(-) Mandatory Repayments & Maturities:	\$ M	-	-	-	-	-
(-) Optional Repayments:	\$ M	-	-	-	-	-
EOp Subordinated Notes:	\$ M	307.3	307.3	307.3	307.3	307.3

Very traditional Debt Schedule setup, where we calculate the Cash Flow Available for Optional Debt Repayment based on Beginning Cash + FCF After Dividends - Mandatory Repayments - Min Cash.

Optional Repayments: If Cash Flow Available for Debt Repayment (CFADR) is positive, take the MIN between CFADR and the remaining balance at this point, after additional issuances and mandatory repayments.

This Debt Schedule tells us nothing useful because we already calculated the Mortgage Payable issuances required in the previous section, and no optional repayments are possible.

We then calculate the Interest Expense and Interest Income:

Debt Schedule:	Units:	Projected:				
		FY18	FY19	FY20	FY21	FY22
LIBOR:	%	2.00%	2.50%	2.75%	3.00%	3.00%
Interest Rates on Debt:						
Mortgages Payable & Other Loans:	%	3.50%	4.00%	4.25%	4.50%	4.50%
Subordinated Notes:	%	7.00%	7.00%	7.00%	7.00%	7.00%
Interest Paid on Debt:						
Mortgages Payable & Other Loans:	\$ M	42.4	61.5	79.3	98.7	113.5
Subordinated Notes:	\$ M	21.5	21.5	21.5	21.5	21.5
Total Interest Paid:	\$ M	63.9	83.0	100.8	120.2	135.0
Interest Earned on Cash:	\$ M	1.1	1.6	2.0	2.4	2.6

And finally, we calculate the credit stats and ratios:

Credit Stats and Ratios:	Units:	Projected:				
		FY18	FY19	FY20	FY21	FY22
EBITDA:	\$ M	\$ 183.0	\$ 206.8	\$ 229.7	\$ 254.9	\$ 280.3
EBITDA Growth:	%	19.1%	13.0%	11.1%	11.0%	10.0%
EBITDA Margin:	%	68.2%	68.4%	68.2%	68.4%	68.3%
Debt / LTM EBITDA:	x	10.09 x	10.52 x	10.90 x	11.12 x	11.32 x
Net Debt / LTM EBITDA:	x	9.74 x	10.18 x	10.55 x	10.78 x	10.97 x
LTM EBITDA / Interest Expense:	x	2.86 x	2.49 x	2.28 x	2.12 x	2.08 x
Debt Service Coverage Ratio (DSCR):	x	1.15 x	1.05 x	0.68 x	1.02 x	0.89 x
Total Debt / Total Equity:	%	80.1%	88.9%	94.6%	102.2%	111.7%
Total Debt / Total Assets:	%	43.2%	45.7%	47.2%	49.0%	51.1%

DCSR defined as EBITDA / (Interest Expense + Mandatory Debt Repayments).

These credit stats and ratios are concerning – especially the DSCR, which looks dangerously low even in the Base Case. Normally, lenders want DSCRs above 1.0x, and ideally above the 1.3x – 1.5x range.

But since the Debt Repayments & Maturities are fixed, the only solution is to use less Debt to fund the deal, which would reduce the IRR and MoM multiple.

The Debt / Equity and Debt / Total Assets ratios increase substantially, and the Interest Coverage Ratio also falls to nearly 2.0x – and this is only the *Base Case*.

So, just from these simple credit stats and ratios, we're already skeptical of this deal.

Step 6: Calculate the MoM Multiples and IRR

The Returns Calculations in this model are a bit different because we must support multiple scenarios and multiple Exit Dates – it's not as simple as picking one Exit Multiple, or even a set range of Exit Multiples.

To set this up, we observe that *traditionally*, PIRET has traded at an EV / EBITDA multiple that represents a **premium** to the multiple implied by its portfolio-wide Cap Rate.

For example, if its portfolio-wide Cap Rate was 6.4%, the implied EV / EBITDA was $1 / 6.4\% = 15.6x$. If the company traded at a 17.3x multiple at that time, that was a $17.3 / 15.6 - 1 = 11\%$ premium.

Here's the historical data:

Returns Calculations:	Units:	Historical:		
		FY15	FY16	FY17
LTM EV / EBITDA Implied by Portfolio Cap Rate:	x	15.5 x	14.8 x	16.0 x
LTM EV / EBITDA at Year End:	x	17.3 x	18.8 x	20.9 x
Trading Multiple Premium:	%	11.3%	26.8%	31.0%



We assume a **40%** premium in the holding period, making this assumption quite aggressive, and we get these results in the Base Case:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
325																		
326			Returns Calculations:	Units:		FY17	FY18	FY19	Projected:	FY20	FY21	FY22						
327																		
328			LTM EV / EBITDA Implied by Portfolio Cap Rate:	x			16.1 x	16.4 x	16.7 x	16.7 x	16.7 x							1 / Portfolio-Wide Cap Rate from our segment-by-segment projections at the top.
329			LTM EV / EBITDA at Year End:	x			22.6 x	23.0 x	23.3 x	23.3 x	23.3 x							
330			Trading Multiple Premium:	%			40.0%	40.0%	40.0%	40.0%	40.0%							
331																		
332			EBITDA:	\$ M			183.0	206.8	229.7	254.9	280.3							
333			(x) Exit Multiple:	x			22.6 x	23.0 x	23.3 x	23.3 x	23.3 x							
334			Exit Enterprise Value:	\$ M			4,132.0	4,746.8	5,359.8	5,948.1	6,540.7							
335			(-) Net Debt:	\$ M			(1,782.9)	(2,105.0)	(2,424.0)	(2,747.3)	(3,076.0)							
336			Exit Equity Value:	\$ M			2,349.1	2,641.9	2,935.8	3,200.8	3,464.7							
337																		
338			Dividends:	\$ M				87.8	89.4	91.0	93.5	99.7						
339			Total Cash Flows to Equity Investors:	\$ M			(2,199.1)	87.8	89.4	=IF(N326<=Exit_Year,N338,0)+IF(N326>Exit_Year,N336,0)								
340																		
341			Project-Level Returns:															
342			Multiple:	x								1.8 x						
343			IRR:	%								13.0%						

Next, we need to **distribute** the proceeds to each investor group here: Management, Ivanhoé Cambridge, and Blackstone. We'll start with the Returns to Management:

	C	D	K	L	M	O	P	Q	R	S	T	U	V	W
324														
325														
326														
327														
336														
337														
338														
339														
340														
341														
342														
343														
344														
345														
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348														
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A B C D K L M O P Q R S T U V W

Returns Calculations:

Units:

FY17 FY18 FY19 FY20 FY21 FY22

Exit Equity Value: \$ M 2,349.1 2,641.9 2,935.8 3,200.8 3,464.7

Dividends: \$ M 87.8 89.4 91.0 93.5 99.7

Total Cash Flows to Equity Investors: \$ M (2,199.1) 87.8 89.4 91.0 93.5 3,564.4

Project-Level Returns:

Multiple: x 1.8 x

IRR: % 13.0%

Returns to Management:

Initial Investment: \$ M (76.8)

(-) Cash from Management Options: \$ M - - - - = (P326=Exit Year)*IF(P336>Total_Equity,-Mgmt_Option_Pool*Total_Equity,0)

(+) Equity to Management Options: \$ M - - - - 34.5

(+) Dividends: \$ M 3.1 3.1 3.2 3.3 3.5

(+) Exit Equity Proceeds: \$ M - - - - 121.0

Total Cash Flows: \$ M (76.8) 3.1 3.1 3.2 3.3 137.0

Multiple: x 1.9 x

IRR: % 15.1%

If the Exit Equity Value exceeds the Total Equity initially invested, the options are exercisable, and the Management team pays 1% * Total Equity initially invested to exercise them. The 1% is the Option Pool percentage.

If the options are exercisable, this equals 1% / (1 + 1%) * (Exit Equity Value + Cash Paid to Exercise Options). The Cash Paid *expands* the Exit Equity available, so it's not just 1% * Exit Equity Value.

These are straightforward - bring them in if we're in or before the Exit Year, and multiply by Management's Ownership (3.5% in this case).

Moving on, here are the Returns to Ivanhoé Cambridge:



These are linked to the calculations above for Management, but we multiply by Ivanhoe's ownership percentage in the deal.

If the Project Multiple is above 2.0x, we take the portion of the Returns above 2.0x and distribute 20% of them to Ivanhoe; this is 0 otherwise.

Same as above; bring in Dividends if we're in or before the Exit Date, and the Exit Equity Proceeds only on the Exit Date. And multiply by Ivanhoe's ownership percentage.

Returns Calculations:	Units:	FY17	Projected:				
			FY18	FY19	FY20	FY21	FY22
Exit Equity Value:	\$ M		2,349.1	2,641.9	2,935.8	3,200.8	3,464.7
Dividends:	\$ M		87.8	89.4	91.0	93.5	99.7
Total Cash Flows to Equity Investors:	\$ M	(2,199.1)	87.8	89.4	91.0	93.5	3,564.4
Returns to Blackstone Group L.P.:							
Initial Investment:	\$ M	(1,315.8)					
(+) Cash from Management Options:	\$ M		-	-	-	-	13.2
(-) Equity to Management Options:	\$ M		-	-	-	-	(20.7)
(-) Incentive Fee Granted:	\$ M		-	-	-	-	-
(+) Dividends:	\$ M		52.5	53.5	54.4	55.9	59.6
(+) Exit Equity Proceeds:	\$ M		-	-	-	-	2,073.1
Total Cash Flows:	\$ M	(1,315.8)	52.5	53.5	54.4	55.9	2,125.2
Multiple:	x	1.8 x					
IRR:	%	13.0%					

Link to Management's calculations, flip the signs, and multiply by Blackstone's ownership percentage.

Link directly above
to Ivanhoe's Returns
and flip the sign.

The same as always:
check the Exit Date and
multiply by Blackstone's
ownership percentage.

These items are *only* paid to these two investor groups, not to the Management Team, so we should completely exclude Management's ownership from this calculation and only look at the sponsors.



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We missed this problem in the initial model but left it in because we did not want to re-do all the videos and Excel files just to fix it. And it makes a tiny difference in the final output.

Step 7: Build Sensitivities and a Summary (If Applicable)

To make an investment decision, we must look at multiple scenarios and create sensitivity tables to see what happens when the deal assumptions change.

In an LBO model like this one, the most obvious assumptions to sensitize are the Purchase Price, Exit Multiple, Exit Year, Scenario, and Leverage – so that is exactly what we do:

Sensitivity Tables - Sponsor Returns:

		FY22 Exit EV / EBITDA Multiple (Base Scenario):										
Purchase Premium:	20.5%	20.8 x	21.3 x	21.8 x	22.3 x	22.8 x	23.3 x	23.8 x	24.3 x	24.8 x	25.3 x	25.8 x
	18.0%	8.4%	9.4%	10.3%	11.2%	12.1%	12.9%	13.7%	14.5%	15.3%	15.9%	16.4%
	15.5%	9.0%	9.9%	10.9%	11.8%	12.6%	13.5%	14.3%	15.1%	15.8%	16.3%	16.8%
	13.0%	9.5%	10.5%	11.4%	12.3%	13.2%	14.1%	14.9%	15.7%	16.2%	16.7%	17.2%
	10.5%	10.1%	11.1%	12.0%	12.9%	13.8%	14.7%	15.5%	16.1%	16.6%	17.1%	17.6%
	8.0%	10.7%	11.7%	12.7%	13.6%	14.5%	15.3%	16.0%	16.5%	17.1%	17.6%	18.1%
	5.5%	11.3%	12.3%	13.3%	14.2%	15.1%	15.9%	16.4%	17.0%	17.5%	18.1%	18.6%
	3.0%	12.0%	13.0%	13.9%	14.9%	15.8%	16.3%	16.9%	17.5%	18.0%	18.5%	19.1%
	0.5%	12.7%	13.7%	14.6%	15.6%	16.2%	16.8%	17.4%	17.9%	18.5%	19.0%	19.6%
		13.4%	14.4%	15.3%	16.1%	16.7%	17.3%	17.9%	18.5%	19.0%	19.6%	20.1%
		Exit EV / EBITDA Multiple Premium to Multiple Implied by Cap Rate (Base Scenario):										
Exit Year:	FY18	27.5%	30.0%	32.5%	35.0%	37.5%	40.0%	42.5%	45.0%	47.5%	50.0%	52.5%
	FY19	(6.0%)	(2.6%)	0.7%	4.1%	7.4%	10.7%	14.1%	17.4%	20.7%	24.0%	27.3%
	FY20	4.4%	6.3%	8.1%	9.9%	11.6%	13.4%	15.1%	16.7%	18.4%	20.0%	21.6%
	FY21	7.7%	8.9%	10.2%	11.4%	12.6%	13.7%	14.9%	16.0%	17.1%	18.1%	19.2%
	FY22	8.8%	9.7%	10.7%	11.6%	12.5%	13.3%	14.2%	15.0%	15.9%	16.6%	17.4%
		9.3%	10.1%	10.8%	11.5%	12.3%	13.0%	13.6%	14.3%	15.0%	15.6%	16.0%
		Exit EV / EBITDA Multiple Premium to Multiple Implied by Cap Rate (FY22 Exit)										
Operational Scenario:	Upside	27.5%	30.0%	32.5%	35.0%	37.5%	40.0%	42.5%	45.0%	47.5%	50.0%	52.5%
	Base	12.6%	13.4%	14.1%	14.9%	15.6%	16.1%	16.6%	17.1%	17.5%	18.0%	18.4%
	Downside	9.3%	10.1%	10.8%	11.5%	12.3%	13.0%	13.6%	14.3%	15.0%	15.6%	16.0%
		7.0%	7.7%	8.4%	9.1%	9.8%	10.4%	11.1%	11.7%	12.3%	12.9%	13.5%
		Exit EV / EBITDA Multiple Premium to Multiple Implied by Cap Rate (FY22 Exit, Base Scenario)										
Subordinated Notes - Multiple of LTM Purchase EBITDA:	0.0 x	27.5%	30.0%	32.5%	35.0%	37.5%	40.0%	42.5%	45.0%	47.5%	50.0%	52.5%
	1.0 x	9.0%	9.7%	10.4%	11.0%	11.7%	12.3%	12.9%	13.5%	14.1%	14.7%	15.3%
	2.0 x	9.1%	9.9%	10.6%	11.3%	12.0%	12.6%	13.3%	13.9%	14.5%	15.1%	15.7%
	3.0 x	9.3%	10.1%	10.8%	11.5%	12.3%	13.0%	13.6%	14.3%	15.0%	15.6%	16.0%
	4.0 x	9.4%	10.3%	11.1%	11.9%	12.6%	13.4%	14.1%	14.8%	15.5%	15.9%	16.4%
		9.6%	10.5%	11.4%	12.2%	13.0%	13.8%	14.6%	15.3%	15.9%	16.3%	16.8%

In some cases, the ranges for these tables are obvious: for example, the Exit Year and Operational Scenario are both constrained, and we can't add, say, 10x additional leverage on PIRET.

The Purchase Premium of 20.5% is quite high for a REIT, so we choose to *go down* to a near-0% figure.

The Exit Multiple and Exit Multiple Premium are a bit trickier; for the Exit Multiple, we attempt to capture the full range of values across the cases, assuming a Year 5 Exit.

For the Exit Multiple Premium, we put the 40% baseline assumption in the middle of the table and go down to 27.5%, close to the historical premium, and up to 52.5% to be symmetrical.



As shown in the tables, we're extremely unlikely to achieve a 20% IRR in the Base Case, and even in the Upside Case, we'd need significant multiple expansion to reach a 25-30% IRR.

On the positive side, though, we're also unlikely to lose money in the Downside Case, and a 5-10% IRR might be plausible.

How to Make Investment Recommendations for REIT LBOs

To make investment recommendations for LBOs, you assess whether or not you can achieve the targeted IRRs across the different scenarios.

If it seems like the answer is "Yes," then you stress test the model, and if it still seems plausible, you back up your recommendation with qualitative reasoning.

If it seems like the answer is "No," you assess what might change your mind, and then explain your reasoning and back up your decision with qualitative reasoning.

Here, the answer is a clear **"No"** because Blackstone is targeting a 20% IRR, which seems extremely unlikely in the Base Case.

Here's our full recommendation and reasoning:

- We recommend **against** acquiring Pure Industrial Real Estate Trust [PIRET] for a 21% premium (Equity Purchase Price of \$2.5 billion CAD and EV / EBITDA of 23.7x).
- PIRET generates no Cash, and its Net Debt balance grows during the holding period, so the only returns sources in the buyout are Multiple Expansion, Dividends, and EBITDA Growth.
- This purchase multiple is too high relative to the comparables (median EV / EBITDA multiple of 17.9x), and Cap Rates in the Canadian industrial sector are at all-time lows currently, making Multiple Expansion unlikely.
- The Dividend Yield is only ~4%, and Acquisition Yields would have to exceed 10% for the EBITDA to grow enough to support a 20% IRR over the 5-year holding period.
- But historically, Going-In Cap Rates have never exceeded 8% for PIRET – not even in the 2008-2009 period – so achieving a 20% IRR (vs. a 13% IRR in the current Base Case and 16% IRR in the Upside Case) is extremely unlikely.



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- We might change our mind and recommend this deal if Blackstone were to pay a 5-10% premium rather than a 21% one, if PIRET's current trading multiples were 15-20% lower, if we could achieve 20-25% multiple expansion, and if Ivanhoe Cambridge did not receive an incentive fee (which hurts Blackstone quite a bit in the Upside Case).

This deal is problematic not because there's too much risk, but because *there's not enough potential upside*.

It would have made more sense several years ago before the industrial REIT market turned "hot" and valuation multiples increased substantially.

But in the market as of the time of this deal, industrial Cap Rates were already at all-time lows, making further drops (and corresponding EBITDA multiple expansion) unlikely.

PIRET's EBITDA grows by a good amount over this 5-year period, but in this case, it's just not enough to achieve a 20% IRR.

So, in short: PIRET is a fine company, and this deal is unlikely to be a money-losing failure, but it seems like an overpriced transaction done near the peak of the cycle with limited upside potential.

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REIT Interview Questions & Answers

Interview questions about REITs span a wide range of topics, but you're **most likely** to receive questions about the fundamentals: why REITs exist, how they operate, and how the accounting and valuation differ.

We do cover more advanced topics here, but they're far less likely to come up in interviews.

In-depth questions about the nuances of valuation or 3-statement projections are also unlikely; if interviewers want to go into that level of detail, they'll give you a case study or modeling test.

So, aside from a few questions here involving numbers, we focus on **the concepts**.

High-Level REIT Concepts

These are the most important interview questions because they're the most likely to come up in real life. Also, if you can't answer these basic questions correctly, interviewers will conclude that you don't know much about REITs:

1. How do REITs operate, and what are their main requirements?

REITs operate, acquire, develop, and dispose of properties and continually raise Debt and Equity to fund these activities. They are similar to private equity firms, but for properties rather than companies.

REITs must distribute a high percentage of their Net Income as Dividends (90% in the U.S.), must earn a high percentage of revenue from real estate-related sources (often 75%+), and must maintain a high percentage of real estate-related Assets (often 75%+).

If REITs follow these requirements, they pay no corporate income taxes (or very low taxes). In many countries, there are also requirements related to the number of shareholders and the concentration of shareholder ownership.

2. What are the different types of REITs, and how do they differ from each other?

You can divide REITs by **structure** (Traditional vs. UPREIT vs. DownREIT), **strategy** (Equity vs. Mortgage vs. Hybrid), and **sector** (Office vs. Industrial vs. Retail vs. Multifamily vs. Hotels vs. others).

Traditional REITs own their assets directly, while UPREITs and DownREITs form "Operating Partnerships" that own and manage their assets; the methods of asset transfer also differ.



Equity REITs own and manage properties, Mortgage REITs own real estate loans and operate more like commercial banks, earning revenue from net interest income, and Hybrid REITs do both.

Finally, Equity REITs in different sectors differ from each other because of different business models at the property level.

For example, office, retail, and industrial properties tend to have longer-term leases with fixed rental escalations, so REITs in these sectors have more revenue visibility. Hotels have no long-term leases, so hotel REITs are more volatile and tend to fluctuate far more with the overall economy.

3. Which type of Equity REIT is “safest” to invest in, and which type is riskiest?

Hotel REITs tend to be the riskiest because they lack long-term leases and, therefore, revenue visibility, which can sink their fortunes in recessions.

It's harder to say what the “safest” REIT is, but multifamily (apartment) REITs often have the lowest Betas because people always need somewhere to live, even if there's a recession and they lose their jobs. People shift from buying to renting, benefiting multifamily REITs, in recessions.

Office REITs are also safer because they have much longer-term leases (5-10 years rather than the 1-year leases common with apartments), but office properties also tend to have fewer tenants and more varied lease terms, which creates more risk. They're also more sensitive to economic conditions because companies could downsize, move, or shut down in recessions.

Industrial and retail REITs are in a similar category because they also have longer-term leases, but they're also more sensitive to economic conditions; also, industrial properties are often simple to build, so they're less defensible than properties owned by office or multifamily REITs.

4. Walk me through an Equity REIT's financial statements.

The major sections of an Equity REIT's Balance Sheet include Real Estate Assets, Debt, and Equity, with smaller line items for Working Capital.

On the Income Statement, the major line items are Rent and Tenant Reimbursements, Operating Expenses, Property Taxes, Interest Expense, Depreciation, and Gains and Losses; IFRS-based REITs will record the Change in Fair Value of Properties rather than Depreciation.

The Cash Flow Statement starts with Net Income, adjusts for non-cash items such as Depreciation and Gains and Losses, and then includes Developments, Acquisitions, Maintenance CapEx, and Asset Sales in Cash Flow from Investing. Cash Flow from Financing includes Dividends, Equity and Debt Issuances, and Debt Repayments.



There are minor differences on the statements for different types of REITs. For example, hotel REITs' revenue is split into Rooms, Food & Beverage, and Other rather than Rent and Tenant Reimbursements.

5. What are the main differences between REITs that follow U.S. GAAP and ones that follow IFRS?

The biggest difference is that IFRS-based REITs record Fair Value Gains and Losses on their Income Statements and mark their properties to fair market value each year. They do not depreciate their properties at all.

As a result, it is much easier to calculate NAV since it is based directly on the Balance Sheet with a few minor adjustments.

Also, different metrics may be used in place of FFO or AFFO, depending on the region; for example, EPRA Earnings is the equivalent of FFO for European REITs.

Finally, REITs outside the U.S. often use the Direct Method for the Cash Flow Statement, which shows Cash Received and Cash Spent, rather than the Indirect Method, which starts with Net Income and adjusts from there.

6. Why do REITs use Funds from Operations (FFO), and how do you calculate it?

REITs use Funds from Operations in place of the traditional Net Income metric. Net Income is problematic because REITs are constantly buying and selling properties, creating Realized Gains and Losses that make Net Income fluctuate significantly. Also, Depreciation and the Fair Value Gains and Losses are significant non-cash charges that affect REITs' Net Income.

FFO equals Net Income + Real Estate-Related Depreciation & Amortization + Losses / (Gains) + Impairments, and it's a version of Net Income that removes the large, non-recurring, and non-cash items described above.

7. How is Adjusted Funds from Operations (AFFO) different from FFO?

AFFO is a version of FFO that adjusts for recurring maintenance CapEx and other capital costs, such as Leasing Commissions, the amortization of lease intangibles, the straight-lining of rent, and other non-cash items such as non-real-estate-related depreciation.

The calculation varies widely among different REITs, but it is intended to be closer to the recurring cash flow from operations a REIT generates from its same-store properties.



8. How do you value an Equity REIT?

The same way you value any other company: with Comparable Public Companies, Precedent Transactions, and a DCF.

The Net Asset Value (NAV) Model is a very important methodology for U.S.-based REITs as well; in it, you mark the REIT's entire Balance Sheet to fair market value, subtract Liabilities from Assets, and divide by the share count to calculate NAV per Share.

You then compare the REIT's NAV per Share to its current share price to determine if it is valued appropriately.

9. Walk me through Public Comps and Precedent Transactions for an Equity REIT.

You still screen based on geography, industry, and financial size, and "time" as an additional criterion for Precedent Transactions.

But the specific criteria are different, and you often use Gross Real Estate Assets or the Fair Value of Investment Properties to determine "financial size." You must also pay close attention to geography and ensure that you do not mix IFRS-based REITs with U.S. GAAP-based REITs.

You can still use traditional metrics such as EV / EBITDA, but you'll also use FFO, AFFO, P/ AFFO, and P/ AFFO for U.S.-based REITs. For IFRS-based REITs, Book Value and P / BV are common, and if you use FFO and AFFO, you will calculate them a bit differently. Alternative metrics such as EPRA Earnings and EPRA NAV are also common.

As always, you still look at a REIT's growth rates and multiples compared to those of peer companies to make an assessment; if it's growing at the same rate as the peer companies and trades at similar multiples, it's a sign that it may be appropriately valued.

10. How is a DCF different for a REIT?

The main differences in an Unlevered DCF are ongoing CapEx and Disposition activity (from acquisitions, developments, and property sales), even going into the Terminal Period, and ongoing issuances of Debt and Equity.

The first difference changes the Free Cash Flow profile of the REIT and means that CapEx is more complex than a simple percentage of revenue; the second difference affects the Debt / Equity ratio and, therefore, WACC, and the REIT's outstanding share count.



Specifically, you must factor in the future Equity issuances, take their Present Value (including from issuances into the Terminal period), divide by the current share price, and add that estimated number of shares to the company's current share count.

11. Walk me through a NAV Model for a REIT, and explain when it is useful.

The NAV Model assumes that there are pricing discrepancies between local real estate markets and the stock market and that you can quantify these discrepancies by re-valuing a REIT's real estate assets by dividing their NOI by local Cap Rates.

It is very useful for U.S.-based REITs since they do not mark their properties to market value, but it's far less useful for REITs that follow IFRS.

You start by projecting the REIT's 12-month forward NOI. Then, you divide it by an appropriate Cap Rate (or Cap Rates if you separate the NOI into segments) to calculate the Market Value of Gross Real Estate Assets.

Next, you project non-rental income and divide it by a higher Cap Rate to value it, and then you make minor adjustments to the other Assets and add them to everything you previously calculated. This gives you the Market Value of Assets.

Next, you mark the REIT's Liabilities to Market Value, typically adjusting Debt based on current interest rates, and subtract them from the Market Value of Assets to calculate the Net Asset Value.

Finally, you divide this Net Asset Value by the share count to determine the NAV per Share, which you compare to the REIT's current share price.

12. How might the financial statements of an office, retail, industrial, or data center REIT differ from those of a multifamily REIT?

You will see slightly different items on the statements because these REITs tend to have properties with fewer, larger tenants with longer-term, customized leases.

So, you might see "Tenant Receivables" on the Balance Sheet, line items for Tenant Improvements and Leasing Commissions on the Cash Flow Statement, and an adjustment for the Straight-Lining of Rent on the CFS as well (since cash rental rates change over time, but accounting rules require straight-line revenue recognition over the lease term).

Acquired Lease Intangibles and Above/Below-Market Lease items on the Balance Sheet are more common for similar reasons and represent leases on acquired properties.



Finally, the revenue categories might be slightly different, with Tenant Reimbursements as a common item for these REITs; Percentage Rent, Service Fees, and Power Usage Fees might also appear.

13. How are a hotel REIT's financial statements different?

A hotel REIT's financial statements are more similar to the standard hotel Pro-Forma. So, revenue and expenses are split into categories like Rooms, Food & Beverage, and Other, and expenses may be split into "Fixed" and "Variable" categories. There may also be management and incentive fees due to the branding model of many hotels.

The Balance Sheet is similar, but items such as Furniture, Fixtures & Equipment may be listed separately outside of the Real Estate Assets.

The Cash Flow Statement is also similar, but hotels do not have Tenant Improvements or Leasing Commissions, and they often label Recurring Maintenance CapEx the "FF&E Reserve" or something similar.

14. How does EPRA Earnings for European REITs differ from FFO? And how do EPRA NAV and standard NAV differ?

EPRA Earnings is similar to FFO because it starts with Net Income and then adjusts for large non-recurring and non-cash items. But it reverses Fair Value Gains and Losses rather than Depreciation and also adjusts for Deferred Income Taxes.

Many European REITs do pay a small amount of corporate income taxes, so adding back Deferred Income Taxes makes EPRA Earnings closer to the REIT's "cash earnings."

EPRA NAV starts with the REIT's Book Value and then adjust for Unrealized Gains/Losses, the Fair Value of Financial Instruments, and the Deferred Tax Liability, under the logic that most REITs will not end up paying their Deferred Taxes. There may also be minor adjustments for Goodwill and taxes/fees on asset sales and Unrealized Gains/Losses.

By contrast, the NAV Model for U.S.-based REITs requires more assumptions and adjustments because it revalues almost everything on the Balance Sheet and requires forward NOI projections and the selection of Cap Rates.

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REIT Financial Statement Projections



These questions are less important than the high-level ones about REITs above or the REIT valuation questions below. Many of these questions fall into the “If they want to test your knowledge of financial modeling, they’ll give you a modeling test” category:

1. What are the main methods you can use to project a REIT's financial statements and build a 3-statement model?

You normally split up a REIT by business segments, such as Same-Store, Developments, Acquisitions, and Dispositions, and project the revenue, expenses, and capital spending in each of those categories, or you split the REIT into different geographies, such as Spain and France, and project the same items in each region.

If the REIT operates only a few properties, then you might create separate revenue, expense, and capital cost projections for each property.

2. How can you make the appropriate assumptions for Acquisition and Development spending and Yields in a REIT 3-statement model?

You normally link these spending levels to the figures in past market cycles as well as the REIT’s plans, as disclosed in its investor presentations and earnings calls.

For example, if the REIT spent \$1 billion per year on Acquisitions in the last market expansion and \$300 million per year in the last downturn, you might use those figures in the Upside and Downside Cases in the model (perhaps adjusted for inflation). You would also retrieve the REIT’s historical Cap Rates or Yields on this Acquisition spending in both periods and use those as guidelines.

If something fundamental about the REIT’s markets has changed, then you might adjust these numbers up or down. But real estate is a cyclical industry, so you’ll almost always use the numbers and percentages from past cycles when making forecasts.

3. How do you forecast Dividends for a REIT in a projection model?

You almost always make Dividends a percentage of FFO or AFFO, or, for European REITs, a percentage of EPRA Earnings.

The Dividend requirement for REITs is based on Dividends as a percentage of Net Income, but in practice, most REITs distribute Dividends in relation to metrics such as FFO or AFFO to produce more consistent financial results.

Also, you rarely assume huge Gains and Losses in the projections, so Net Income vs. FFO or AFFO does not necessarily matter much.



4. How do you determine the amount of Debt and Equity a REIT raises each year?

You start by adding the REIT's previous Cash balance to everything on the Cash Flow Statement in the current period except for the REIT's Dividends and Debt/Equity line items (so, Previous Cash Balance + CFO + CFI + parts of CFF). Then, you subtract the planned or targeted Dividends, usually projected as a percentage of FFO in the current period.

You then compare this number to the REIT's minimum Cash balance, which is often a percentage of operating expenses or cash operating expenses.

If the REIT needs additional funding to reach this minimum Cash balance, which it almost always does, then you assume that it raises Debt and Equity, based on set percentages. These percentages may be linked to the REIT's targeted Debt/Equity ratio or its historical funding patterns.

5. How do the assumptions and key drivers in a REIT's 3-statement model change in different market scenarios?

Same-Store rental income growth and NOI margins are strongest in the Upside Case, weakest in the Downside Case, and in the middle in the Base Case.

The REIT also tends to dispose of more properties in strong market conditions, and it will get higher prices for its properties, meaning that it loses less NOI relative to these disposal prices.

Acquisition spending and Development/Redevelopment spending depend on the REIT's strategic focus. In strong markets, the REIT tends to spend more on its core activity – so a Development-focused REIT will spend even more on Development in a strong market and comparatively less on Acquisitions. Development Yields will also be highest in the Upside Case and lowest in the Downside Case.

Acquisition Yields or Going-In Cap Rates will follow the opposite pattern – they'll be lowest in strong markets and highest in weak markets because prices and yields move in opposite directions.

6. How might you set up a REIT model where Developments take several years to finish and additional time to stabilize?

You might create a "waterfall schedule" where you assume Development starts each year, amortize the total spending over the average number of years required to complete properties, and sum up the spending from all previous years in each period.

So, Year 4 of the model should include the Development spending from Years 2, 3, and 4 if properties take an average of 3 years to finish and the Year 1 Properties finish up in Year 3.



When each year's Developments finish, you assume a "pre-stabilized" Yield and NOI margin on them to calculate the NOI and revenue contribution in that year, before stabilization.

Once the properties stabilize, you reclassify them to the Stabilized or Same-Store segment and assume higher Yields and NOI margins to reflect this stabilization.

7. When is it most useful to model a REIT by individual city or property?

City-by-city or property-by-property projections are most useful when the REIT operates only a few properties or only in a few cities and does not plan to make major changes.

For example, maybe the REIT operates in 3 cities and does not plan to dispose of any properties over the next few years – it just anticipates a few acquisitions and developments.

Or, maybe the REIT operates only 2 major properties and plans to continue as-is, and it discloses enough information to project revenue and expenses for each one.

8. When are quarterly and half-year models for REITs most useful, and how do the setup and assumptions differ?

Quarterly and half-year models are most useful if the REIT is "seasonal" in some way or if it has high percentages of lease expirations in certain periods but not others. Lenders find quarterly and half-year models particularly useful in these cases because the REIT might violate covenants on a quarterly basis but not an annual basis (for example).

The main difference in these models is that you must use Year-over-Year (YoY) growth rates and slightly different margin, spending, and yield assumptions in each period; also, you must be careful to reflect contributions only from development deliveries *in the period* rather than the entire year.

If the REIT pays Dividends only in certain periods (e.g., Q2 and Q4 but not Q1 or Q3), the model must also reflect that.

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REIT Valuation

Questions about REIT Valuation tend to focus on the **Net Asset Value (NAV) Model**, which is arguably the most important methodology for U.S.-based REITs. You should also understand the DCF and key multiples, such as P / FFO and P / AFFO:



1. What are some different methods of projecting Forward NOI in a NAV Model, and what are the trade-offs?

Ideally, the projections will come from a 3-statement model for the REIT that includes the NOI from each segment, such as Same-Store, Development, and Acquisitions. You can then divide each portion of NOI by a different Cap Rate, even splitting them up by geography if you have the data. This method takes more time and effort but produces more accurate results.

Another option is to take the REIT's NOI from its most recent quarter, adjust for non-cash items such as the Straight-Lining of Rent, adjust for NOI from Acquisitions, Developments, and Dispositions, and then annualize the new figure by multiplying by 4. Then, you can assume a modest growth rate over the next year and divide this Forward NOI by a single Cap Rate to estimate the Market Value of the Gross Real Estate Assets.

With that method, recommended by Green Street, you can calculate Forward NOI more quickly and easily, but you might not properly reflect the REIT's geographies and business segments or the range of market outcomes over the next year.

2. If you split up a REIT's NOI by business segment, how might you pick the appropriate Cap Rate for each portion?

For the Same-Store segment, you pick Cap Rates based mainly on property type and geography; New York City offices should have much lower Cap Rates than hotels in Cincinnati. You can find this data in market reports issued by the large real estate brokerage firms.

The Cap Rates used for NOI Gained from Acquisitions and NOI Lost from Dispositions should match your Cap Rate assumptions in those segments of the 3-statement model.

For the NOI Gained from Developments, you should use Cap Rates that are modestly higher than those for the Same-Store properties because of the added risk associated with new Developments. These rates might be in-line with the assumed Development Yields in the 3-statement model.

3. How might you adjust the NOI figures a REIT discloses in its filings, for use in a NAV Model?

First, if you believe that the REIT has not deducted its Replacement Reserve allocations from its NOI, you might subtract them to reflect the "Economic NOI" that includes the partial impact of capital costs, such as TIs, LCs, and Maintenance CapEx.

There is disagreement about whether the Replacement Reserves should be above or below the NOI line, so the treatment of this item varies among different REITs.



You also adjust for non-cash items that have affected the REIT's NOI, such as the Straight-Lining of Rent and the Amortization of Above and Below-Market Leases. You want the NOI to be as close to the "Cash NOI" as possible.

4. How do you treat Equity Investments in a NAV Model?

Equity Investments are problematic because according to accounting rules, companies record them at historical cost on the Assets side of the Balance Sheet and then adjust them for Net Income from Equity Investments and Dividends from Equity Investments over time.

However, companies do not directly record the Debt and other Liabilities of these partially owned other entities on their Balance Sheets, so in a NAV Model, you usually separate out all the Equity Investments' Assets and Liabilities, re-value them, and multiply each revalued item by the parent company's ownership percentage.

Then, you count each Re-Valued Asset * Parent's Ownership Percentage directly on the Assets side of the NAV Model, and you count each Re-Valued Liability * Parent's Ownership Percentage on the L&E side.

Effectively, you bring these off-Balance Sheet items onto the Balance Sheet for use in a NAV Model.

5. What are the most common adjustments to a REIT's Assets and Liabilities in a NAV Model, outside of the items in Net Real Estate Assets?

On the Assets side, you set the value of Goodwill, Other Intangible Assets, Deferred Rent, Lease Intangibles, and other Intangible Assets created in M&A deals to 0 because their fair market value is 0.

You'll also divide the Management/Development Fees by a fairly high Cap Rate to value them, and you need to factor in the re-valued Assets of the Equity Investments (see explanation above). You will tend to write up Construction in Progress and Land Held for Development slightly as well.

On the Liabilities side, you'll re-value Debt based on prevailing interest rates vs. the coupon rate, include the benefits of any tax-exempt bonds, and factor in the re-valued Liabilities from Equity Investments times the parent company's ownership percentage (see explanation above).

Depending on the NOI projections, you might also factor in the Backlog Completion Costs not already shown on the Balance Sheet (e.g., if your assumed NOI growth rate in the next full year reflects the completion and delivery of this backlog).



6. How do you treat Redeemable Noncontrolling Interests and DownREIT and OP Units in REIT valuation?

Redeemable Noncontrolling Interests are the Balance Sheet representation of these “Units,” so if you **add** the Redeemable NCI when calculating Enterprise Value, you should **not** include the DownREIT and OP Units in the share count when calculating Equity Value.

If you **do not add** the Redeemable NCI when calculating Enterprise Value, then you **should** include the DownREIT and OP Units in the share count when calculating Equity Value.

7. How does a NAV Model differ for different types of REITs?

Besides the different Cap Rates, the main difference lies in the non-cash adjustments in the Forward NOI calculation. Office, industrial, retail, and data center REITs tend to have adjustments for items like the Straight-Lining of Rent and the Amortization of Above and Below-Market Lease Intangibles, while multifamily and hotel REITs tend not to have such adjustments due to shorter or nonexistent leases.

There may also be more of a difference in the NOI adjustments for Dispositions, Acquisitions, and Developments, depending on the REIT’s core strategy.

8. What are the most common non-recurring add-backs and adjustments when creating a set of Comparable Public Companies for REITs?

Asset impairments and provisions for impairments are very common; so are catastrophic charges related to hurricanes and other natural disasters, legal charges, and gains and losses on debt extinguishment.

Many other common non-recurring items appear below the Operating Income line, so you don’t necessarily have to worry about them when calculating metrics such as EBITDA.

9. How would you expect a REIT's Leverage, Portfolio Yield, and Growth to affect its multiples?

Lower Leverage should correlate with higher multiples because REITs with less Debt tend to be less risky and rely more on operational strength to perform (Green Street data also confirms that REITs with less Debt have delivered better historical returns).

Higher Portfolio Yields and higher EBITDA, FFO, and AFFO Growth Rates should correlate with higher multiples because investors are willing to pay more today for higher growth tomorrow.

10. Should you include individual property sales in a set of Precedent Transactions for a REIT?



Ideally, no, because the dynamics around the sale of an individual property are quite different from the dynamics around the sale of a property portfolio or an entire REIT.

For example, an individual property in a region might be valued very highly because of a premium location and long-term leases from blue-chip tenants.

But that doesn't mean that every single property in a REIT's portfolio has the same characteristics – at most, that single sale might tell you something about the general region, but not the *specific* locations, leases, or tenants.

Also, if you include individual property sales, you won't be able to derive metrics such as FFO and AFFO or multiples such as P / FFO and P / AFFO because they do not apply to properties.

That said, if you cannot find *any* relevant Precedent Transactions involving entire REITs in your sector, you might have to use data from property or property portfolio sales and focus on the Cap Rates.

11. How do scenarios affect a REIT DCF?

The main difference is that you need to create different sets of numbers for revenue growth, margins, and re-investment into the business (CapEx and the Change in Working Capital) for the years between the end of the 3-statement model and the end of the projection period (e.g., Years 6 – 10 if you have a 5-year projection model and a 10-year DCF).

For example, in the Upside Case, you might assume 6% growth falling to 3% by the end, while the Base Case might assume 4% growth falling to 2% by the end.

You will also need to create different Terminal Value assumptions because the growth rates and multiples will differ. So, for example, in the Upside Case, you might assume a 2.0% Terminal Growth Rate, but in the Downside Case, you might assume a 0.5% Terminal Growth Rate.

12. What are the advantages and disadvantages of a Dividend Discount Model (DDM) vs. a DCF for a REIT? What about a Levered DCF?

Both the DDM and the Levered DCF take more time and effort to set up than a DCF because you must project more than the REIT's Unlevered Free Cash Flow – you also need to forecast its Debt and Equity balances and Interest Expense so that you can calculate Net Income, FFO, and Dividends.

These analyses won't necessarily produce "better" or more consistent results than the Unlevered DCF, especially since you must add more assumptions to set them up.



However, they may be better in certain cases, such as if the REIT's Dividends fluctuate significantly over time and an Unlevered DCF does not capture this nuance.

The Levered DCF and DDM are nearly identical for REITs; the main difference is that in a DDM, you must also project FFO (or a similar metric like EPRA Earnings) and make Dividends a percentage of that metric instead of stopping at Levered FCF.

13. How does the Replacement Cost methodology work for REITs? What are the advantages and disadvantages?

In the Replacement Cost analysis, you estimate the cost of reconstructing the REIT's entire property portfolio today, including land and construction fees.

As with individual properties, this analysis is useful because it values a REIT based on **what's there right now** instead of future NOI, Dividends, or the potential for capital appreciation.

But the downside is the same as well: if you ask 10 developers for estimates, you'll get 10 different, possibly very different, estimates. The numbers vary so much that Replacement Cost is used as more of a back-of-the-envelope sanity check than anything else.

Also, this analysis often tells you more about market conditions and the merits of acquisitions vs. developments than it does about a specific REIT's value.

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REIT Debt vs. Equity Decisions

These questions move into "unlikely" territory because many of them are not even specific to REITs. However, if you happen to interview with a group that does a lot of capital markets deals, these topics are good to know.

1. At a high level, how does a REIT decide to issue Debt or Equity to fund its operations?

Debt is cheaper than Equity, but costs more in cash and comes with restrictions ("covenants") that limit the REIT's total indebtedness and financial ratios. So, companies tend to raise as much Debt as possible before the costs and covenants require them to switch to Equity.

To make a decision, a REIT will estimate the Costs of Debt and Equity, create operational scenarios to examine compliance with the covenants and assess credit risk, and then recommend percentages of Debt and Equity based on that.



Since REITs issue Debt and Equity constantly, they usually have an ongoing Debt / Total Capital or Debt / Equity target, and they follow that in their fundraising needs. The process above applies more to one-off issuances for large acquisitions or developments.

2. Why do the Cost of Equity and Cost of Debt tend to be closer for REITs than for normal companies?

First, the lack of corporate income taxes means that there's no tax advantage to Debt, so the $(1 - \text{Tax Rate})$ term that normally reduces the Cost of Debt does not exist.

Second, REITs in many sectors, such as multifamily, tend to have low correlations with the overall stock market, with Betas often in the 0.20 – 0.40 range, which makes the Cost of Equity lower than normal.

So, the Cost of Debt is higher than it is for normal companies, and the Cost of Equity is lower, which makes the difference between them smaller.

Note that in some sectors, these rules do not hold up. For example, hotel REITs are more like normal companies, with Betas at or above 1.00, so the difference there is larger.

3. How might the covenants on Debt differ for a REIT?

There will be similar restrictions on ratios such as EBITDA / Interest and Debt / EBITDA, but there may also be restrictions on Balance Sheet-based ratios such as Debt / Total Assets (e.g., a maximum of 60% or 70%).

Also, there may be restrictions on Cap Rate-based metrics. For example, lenders might define a figure like "Capitalization Value" by taking the REIT's EBITDA and dividing it by a Cap Rate, such as 6-7%, and then state that Debt / Capitalization Value cannot exceed a certain percentage.

These metrics make book value less important and make the REIT's financial performance more reflective of market value.

4. Walk me through the Debt vs. Equity decision process for a REIT.

First, you will estimate the After-Tax Cost of Equity for the REIT, using either the CAPM method or by dividing the REIT's Net Income to Common by its Equity Value. Then, you will estimate its After-Tax Cost of Debt via the coupon rates on its issuances or the YTM on its Debt (or Debt from comparable companies).

If the Cost of Equity is somehow lower (unlikely), then the REIT should issue Equity.



If the Cost of Debt is lower, you will then assess the REIT's compliance with possible covenants by creating operational scenarios, such as Base, Downside, and Extreme Downside cases, and evaluating the credit stats and ratios in the projected period across those cases, assuming the REIT raises a certain amount of new Debt.

If, for example, 100% Debt funding results in an EBITDA / Interest Ratio that is too low in the Extreme Downside Case, then you might try 80% or 60% Debt instead until you find numbers that satisfy the requirements while minimizing WACC.

Compromises such as convertible bonds are rarely options for REITs, so the decision is usually limited to tweaking the percentages of Debt and Equity.

5. From a lender's perspective, which types of REITs are the most and least attractive?

Hotel REITs are the riskiest and least attractive because their fortunes vary widely with economic conditions – and the Debt investors receive none of the upside in a strong economy, but all of the downside in a weak economy.

Office, retail, and industrial REITs are more attractive to lenders because of their longer-term leases that lock in rental income regardless of market conditions. *However*, there are still risks because these longer-term leases are meaningful only if the tenants are reliable, mature companies that won't go bankrupt in a recession. Also, poorly staggered lease expirations create risk, so this is a rule of thumb with plenty of exceptions.

Multifamily REITs are somewhere in the middle, with their 1-year leases not providing quite as much of a rental lock-in benefit as office, industrial, and retail REITs, but still far more revenue visibility than hotel REITs. Also, multifamily REITs in many regions are less sensitive to recessions than office, industrial, and retail REITs because people always need a place to live.

6. How do you create scenarios to aid in the Debt vs. Equity decision process?

One common approach is to create Base, Downside, and Extreme Downside scenarios (the Upside scenario is useless for credit analysis because lenders receive none of the benefits).

The Base Case might assume that the market continues "as is," with modest rental growth above the rate of inflation, margins in about the same range, and no major changes to the occupancy rate.

The Extreme Downside Case might be based on the worst-ever outcome for the market, such as the most severe recession over the past several decades; it will reflect lower or negative rental growth, lower margins, higher lease cancellations, a lower occupancy rate, and higher Cap Rates.



The Downside Case will be somewhere in the middle and will reflect a moderate, but not severe, recession, with the same metrics lower or higher.

In these projections, you must also reflect how the Cost of Debt and Equity change as the company uses more Debt; both Costs tend to increase as the Debt percentage increases, which will impact the credit stats and ratios.

7. How does IFRS vs. U.S. GAAP affect the Debt vs. Equity decision for REITs?

The main difference is that since IFRS-based REITs mark their properties to market value, the “Total Assets” figure on the Balance Sheet is more meaningful, and metrics such as Debt / Total Assets can be used more easily as covenants and in credit analysis.

Also, Cap Rate projections are more important under IFRS because you’ll need them to assess the Fair Value of Investment Properties and calculate metrics such as Debt / Total Assets.

8. What are the key risk factors for lenders in REIT credit deals?

The “market cycle” or the possibility of a recession is always the main risk factor because rental income, NOI, and cash flow all decline when the market is weaker, making it more difficult to service Debt and comply with the covenants.

For REITs that operate properties with long-term leases, lease-expiration risk, poorly staggered leases, and the creditworthiness and diversification of tenants are key risks as well.

For example, an office REIT that operates buildings with 5-year leases might appear to be “safe” – but if tenants that occupy 30% of its total space all have lease expirations in the same year, this company will be in trouble if a recession strikes in that same year.

Refinancing risk is also significant, especially if it is paired with lease-expiration risk – if leases for 30% of the space expire in a single year, 50% of the company’s Debt also matures in that year, *and* a recession also starts in the same year, the REIT will suffer.

Finally, some REITs may have misstated or inaccurate numbers for figures such as Debt / Total Assets, which is particularly an issue under IFRS because the REIT must select the appropriate Cap Rate(s).

As a result, a REIT that may not appear to have much credit risk on paper could carry significantly more risk once you dig into the company’s numbers.

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REIT M&A Deals and Merger Models

You should know the basic ideas behind REIT M&A deals, but interviewers are highly unlikely to ask you specific or math-based questions, such as how to calculate real estate-specific Intangible Assets. If you've worked on REIT M&A deals, these questions are more likely, but they're still *far less likely* than everything in the sections above:

1. Why would one REIT acquire another one?

As always, one company might acquire another one if it believes it will be better off afterward; with REITs, different factors explain whether or not they'll be "better off."

The most common deal motivations are geographic expansion, "scale" and value creation, tenant diversification, and synergies. The first three make a REIT more valuable by improving its NOI and EBITDA and making its income sources more diverse, and by making the REIT more comparable to larger REITs that might trade at higher multiples.

Synergies, as always, work by allowing two companies to combine and cut costs, thereby boosting the combined company's margins and NOI and EBITDA.

2. At a high level, how do REIT M&A deals differ from deals involving normal companies?

First, REITs can rarely, if ever, use Cash to fund deals because their Cash balances are extremely low due to the Dividend requirements. They also tend not to use much Debt because they are already highly leveraged, which leaves Stock as the main purchase method. Not all deals are 100% Stock, but the percentage is usually very high.

Second, the Purchase Price Allocation and Balance Sheet adjustments differ because Accumulated Depreciation is eliminated, Real Estate Operating Assets are written up to fair market value, and new Intangible Assets such as Above- and Below-Market Leases and Acquired In-Place Lease Value are created.

Third, the treatment of the Seller's existing Debt and Preferred Stock is very important because if it can be assumed, the combined REIT will continue to pay the same rates. But if it cannot be assumed, the acquirer will have to use a bridge loan, and the rates will change significantly.

Finally, instead of focusing purely on accretion/dilution, you will create a Contribution Analysis and a "Value Creation Analysis" that lets you assess the impact of the deal on the Buyer's share price. Real estate executives often think of deals in these terms, as well as differences in Cap Rates and Replacement Costs, rather than EPS or FFO PS accretion/dilution.



3. Why could the treatment of the Seller's existing Debt make or break a REIT M&A deal?

If the Seller's existing Debt is not assumable, then the Buyer will need to take out bridge loans to fund the deal, and these bridge loans will incur higher interest rates due to the higher risk. That means the deal will become more dilutive across all metrics.

But if the Seller's existing Debt *is* assumable, then the Buyer can keep everything in place and continue to pay the same rates.

In most cases, that should help the Buyer and make the deal more accretive, but in some cases, it might hurt the Buyer if it could attain lower rates by refinancing.

4. What types of Intangible Assets get created in REIT M&A deals?

Besides Goodwill, which is created in almost all M&A deals, items specific to REITs include Intangible Assets for Tenant Relationship Value, Above- and Below-Market Leases, Acquired In-Place Lease Value, and Tenant Origination Costs.

Tenant Relationship Value and Acquired In-Place Lease Value represent the benefit that existing tenants provide by allowing the REIT to renew their leases rather than finding new tenants; they are based on the PV of future NOI from existing tenants and the likely renewal rate.

Above-Market Leases represent the extra rental income the REIT receives beyond market rental rates, and Below-Market Leases are a Liability that represents the opposite. Tenant Origination Costs relate to the cost savings from paying for lower Tenant Improvements and Leasing Commissions if existing tenants renew.

Under IFRS, these items may not exist in the same way, or they may be smaller because the REIT's properties should already be marked to market value.

5. Which items get eliminated in the Purchase Price Allocation process for REITs?

Accumulated Depreciation is written down, and Net Real Estate Assets are written up to fair market value (these adjustments only apply to U.S. REITs).

Deferred Rent and Goodwill are also written down, and the Seller's existing Goodwill is replaced with new Goodwill based on the deal's metrics.



Deferred Tax Assets and Liabilities are often written down and replaced in M&A deals, but they do not exist for most REITs due to the lack of corporate income taxes. However, if the REIT pays a modest tax rate, as some European REITs do, then these items may be adjusted in the deal as well.

6. What are the typical acquisition effects on the Combined Income Statement and Cash Flow Statement in REIT M&A deals?

You add together Rent, Tenant Reimbursements, and Property Expenses for both REITs, as well as all other standard revenue and expense line items.

You record the amortization of the newly created Intangibles within revenue and expenses, and you record cost synergies and restructuring charges, new Depreciation from the asset write-ups, and changes in the Interest Expense and Preferred Dividends from the possible refinancing of Debt and Preferred Stock.

You also record the new shares issued to fund the deal at the bottom of the Income Statement, which affects metrics such as EPS and FFO PS and AFFO PS.

On the Cash Flow Statement, you'll include these new non-cash items as add-backs, and you'll eliminate the Seller's Straight-Lining of Rent to match the write-down of Deferred Rent.

7. How do FFO and AFFO change after a REIT M&A deal?

FFO and AFFO should both increase substantially because you add together the Buyer and Seller's figures. However, it's not as simple as a straight addition because new items are created, and some existing items are eliminated.

For example, the new Depreciation on asset write-ups must be counted in FFO, the amortization of the new Intangibles must be factored into AFFO and possibly FFO, and the Seller's Straight-Lining of Rent must be eliminated in the AFFO calculation.

Also, if Debt was used to fund the deal or the Seller's Debt was refinanced in the deal, the Net Interest Expense will differ from the Buyer's Interest + the Seller's Interest.

8. Should you recalculate the Buyer and Seller's Dividends and Debt and Equity issuances once an M&A deal closes?

It depends on how much the Buyer and Seller change in the deal. If they combine and maintain their property portfolios as-is and continue their development, acquisition, and disposition activities as-is, it is usually not necessary to recalculate these items.



However, if the combined REIT makes major changes, such as divestments of significant properties that were not already planned, then you may need to recalculate Dividends and Debt and Equity issuances based on the ratios and targets for the combined company.

You can check the reasonability of these assumptions by calculating metrics such as Dividends / FFO and Debt / Equity for the combined company and making sure they stay in an appropriate range, i.e., that they do not deviate too far from the Buyer's standalone numbers.

9. How do you set up a Contribution Analysis in a REIT M&A deal, and what does it tell you?

You start by calculating the percentage contribution for each metric you are including; for example, maybe the Buyer contributes 80% of the Combined EBITDA, and the Seller contributes 20%.

Then, you calculate the Combined Pro-Forma Enterprise Value for that metric by taking the Buyer's Current Enterprise Value and dividing by its contribution percentage (so, 80% in the EBITDA example above).

The Seller's Implied Enterprise Value then equals the Combined Pro-Forma Enterprise Value minus the Buyer's Current Enterprise Value, and you back into its Implied Equity Value by following the normal bridge in the reverse direction.

Then, you divide its Implied Equity Value by its diluted share count to determine what its Offer Price "should be," and you divide this Offer Price by the Buyer's Current Share Price to calculate what the Exchange Ratio "should be."

The Contribution Analysis is most useful for determining whether or not the ownership percentages in a 100% Stock M&A deal are appropriate – for example, should the Buyer own 75% of the combined company, or should it own 70% or 80% instead?

You can also use it to determine whether or not the Offer Price is appropriate based on each company's financial contributions.

10. Walk me through a Value Creation Analysis and explain when it is useful.

In the Value Creation Analysis, you find a larger REIT that is similar to the Buyer + Seller as a combined entity, and which ideally trades at higher multiples.

You then add together the Buyer and Seller's financial metrics, factoring in synergies and other acquisition effects, and apply this larger company's multiples to the combined financials.



So, if the larger company trades at 15x EV / EBITDA and the Buyer trades at 12x EV / EBITDA, you multiply the 15x by the Buyer + Seller's EBITDA, adjusted for acquisition effects.

Then, you move from the Implied Enterprise Value to the Implied Equity Value by following the bridge in reverse order, and you calculate the Implied Share Price for the combined entity by dividing by the combined share count.

You then compare this Implied Share Price to the Buyer's standalone Share Price and assess whether or not value has been created for the shareholders.

This analysis is useful if the Buyer + Seller as a combined entity resemble a larger REIT in the same market that trades at higher multiples; if not, or if no such company exists, it's not that useful.

This analysis is very speculative because there's no guarantee that a company's multiples will suddenly change – or even gradually change – after a deal takes place.

It's arguably more useful in the *negative case*, i.e., to show that "value creation" in a deal is *unlikely* if larger companies do not trade at higher multiples.

11. How would you decide to recommend for or against a REIT M&A deal?

You start by verifying that the Seller's valuation is reasonable based on Comparable Public Companies, Precedent Transactions, and the DCF and NAV models.

Then, you check that the ownership percentages are appropriate, based on the Contribution Analysis, and that there's some potential for "value creation" (described above).

In the merger model itself, you check that the synergies seem plausible based on prior deals and that the deal is accretive, or at least neutral, to FFO PS and AFFO PS.

If all those check out, then you assess the qualitative factors, such as geographic expansion and tenant diversification, and if those also seem positive, then you might recommend the deal.

If one or more of these points does not make sense – e.g., the purchase price is too high, or the deal is unlikely to generate value or make the combined REIT more appealing to investors – then you might recommend against the deal.

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REIT Leveraged Buyouts and LBO Models

REIT leveraged buyouts are not that common, so these questions are not that important. As with the M&A section, you should focus on the high-level points about the differences between REIT LBO models and ones for normal companies.

1. How could a private equity firm possibly acquire a REIT in an LBO? Aren't there shareholder and ownership requirements?

Yes, but these rules are easily bypassed with the right legal maneuvering. For example, if the laws in a certain country require 100 or 150 shareholders, the PE firm can create a separate share class and grant small numbers of shares to individuals.

In the U.S., PE firms can bypass the “5/50 rule” (which states that 5 or fewer shareholders cannot own more than 50% of the REIT) because it applies on a pass-through basis – so the firm doing the deal just needs LPs without concentrated ownership.

2. How do REIT LBOs differ from ones of normal companies?

REIT LBOs are like traditional LBOs but with continual Dividend Recaps and Debt-Funded Add-On Acquisitions.

REITs maintain their tax-free structure following a leveraged buyout, which means they must continue to issue high Dividends. And they continue to acquire and develop properties, which means they must continue to raise capital to do so.

However, it doesn't make much sense for REITs to issue Equity to fund these activities because Equity issuances increase the PE firm's investor contributions. So, operational funding tends to become more Debt-based.

Instead of the traditional Returns Sources of Multiple Expansion, EBITDA Growth, and Debt Paydown and Cash Generation, the ones in a REIT LBO are Multiple Expansion, EBITDA Growth, and Dividends – it is virtually impossible for a REIT to repay Debt or generate Cash during the holding period.

As a result, REIT leveraged buyouts are even more dependent on market timing than buyouts of normal companies.

3. How do the transaction assumptions in a REIT LBO differ?



You still assume a purchase EBITDA multiple or purchase premium and a certain amount of Debt and Equity, so the assumptions are fairly similar.

The difference is that PE firms typically cannot add as much leverage since REITs are already highly leveraged.

For example, a normal company might go from 2x Debt / EBITDA to 6x Debt / EBITDA in a leveraged buyout, but tripling the company's Debt would be unusual in a REIT LBO because most REITs already have significantly more than 2x Debt / EBITDA.

A more plausible change might be going from 5x to 7x Debt / EBITDA, or 6x to 8x.

4. How should the assumptions differ in different operational scenarios in a REIT LBO model?

The operational scenarios are the same as in a 3-statement model: revenue growth, margins, and acquisition/development spending should be higher in the optimistic cases, and lower in the others, and Cap Rates should be lower in optimistic cases and higher in pessimistic cases.

5. How do you make assumptions for Dividends and Debt and Equity issuances in a REIT LBO model?

Equity issuances are usually set to 0 because it doesn't make much sense for the PE firm to fund the REIT's operations with its own Equity once the deal closes.

The Debt issuances should be sufficient to meet the company's minimum Cash balance and pay for existing Debt maturities.

Dividends should still be a percentage of FFO to ensure that the REIT maintains its status as a REIT.

6. Why is a Debt Schedule pointless in many REIT LBOs?

Because optional Debt repayments are virtually impossible in most REIT LBOs, given REITs' low Cash balances and their negative cash flow before additional Debt and Equity Issuances.

A Debt Schedule here is only useful for tracking the Debt balances and their mandatory repayments and interest expense over time, and you don't need a full schedule for that.

If there's no variability in the Debt repayments, you can track the balances more simply with direct links on the Balance Sheet.

7. How might you select the range of EBITDA Exit Multiples in a REIT LBO?



The range of Exit Multiples should be linked to the Cap Rates and operational scenarios.

If prevailing Cap Rates are 5% rather than 7%, the Exit Multiples should be higher since valuation multiples are the reciprocals of Cap Rates.

One approach is to take the reciprocal of the Cap Rate and then add a premium to it, such as 20% or 30%, based on the REIT's historical multiples relative to $1 / \text{Cap Rate}$ in each year.

So, if prevailing Cap Rates are 5%, that corresponds to a $1 / 5\%$ or 20x EBITDA multiple.

If the REIT historically traded at EBITDA multiples that were a 20% premium to $1 / \text{Cap Rate}$, then you might assume a $20x * (1 + 20\%) = 24x$ EBITDA multiple as the baseline here.

8. How do the Returns Calculations differ in a REIT LBO?

At a high level, they are the same because you still calculate the Exit Enterprise Value, back into the Exit Equity Value, and calculate the MoM multiple and IRR.

However, you must also track Dividends, distribute them to all the Equity Investor groups in the deal, and make sure they are distributed only in and before the exit year.

Also, there may be a management rollover and management options pool, and incentive fees for the investor group that's more involved in the day-to-day operations, but these features are not specific to REIT LBOs.

An options pool means that you must assume cash payment from one investor group to the others if the options are exercisable upon exit, and then assume an Equity grant from the other investors to that one group.

The incentive fee is similar, but it's a simple redistribution of proceeds if certain IRR or multiple targets are met, with no exercise price and no associated cash payments.

9. How would you make an investment recommendation in a REIT LBO?

The same way you make an investment recommendation in any standard LBO: you assess the IRRs and MoM multiples in different scenarios and compare them to the targeted IRRs and multiples.

For example, the targets might be a 30% IRR in the Upside Case, a 20% IRR in the Base Case, and a 1.0x multiple in the Downside Case, depending on the PE firm and market environment.



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If the deal achieves those targets and the operating assumptions are plausible, then you might lean toward recommending the deal and then back up your argument with qualitative research about the REIT's markets, properties, and tenants.

If not, then you might recommend against the deal.

With REIT LBOs, the key is to assess the potential for EBITDA Growth and Cap Rate compression (AKA Multiple Expansion) – and there's a trade-off between the two because falling Cap Rates mean that future acquisitions will yield less EBITDA.

So, the key is to “buy low” when multiples are lower and Cap Rates are higher and then sell after the market has improved, but not improved *so much* that everyone thinks it's the top.

This explains why REIT leveraged buyouts rarely happen when there's widespread perception that the market is at or near its peak.

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